Strategic Data Regulation in International Trade*

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Abstract

This paper studies how trade openness shapes the incentives to regulate data use. We develop a two-country, two-sector New Trade Theory model in which data is a production input that feeds back from consumption. Firms in a monopolistically competitive Tech sector harvest data from sales to fine-tune products by destination, while consumers incur rising privacy costs as data usage intensifies. We characterise optimal unilateral regulation under a commitment to uniform standards and identify two key distortions under limited extraterritorial reach. First, countries have a strategic disincentive to protect foreign consumers. Second, trade openness creates a regulation–retention trade-off: governments scale back ambition to retain firm presence, manipulate terms of trade, and strengthen oversight. We then analyse a non-cooperative policy game and show that regulatory asymmetries crowd out both regulation and welfare in the symmetric Nash equilibrium. In relevant parameter regions, the model features strategic complementarities that generate regulatory spillovers. Greater privacy awareness supports standards diffusion, while mistrust of foreign data processing can trigger a race to the bottom. Relaxing the non-discrimination constraint, we show how the same strategic distortions underpin incentives for origin-contingent measures, which we link to the growing prevalence of data localisation. Maximal restrictions on outbound data flows emerge as optimal under a wide range of normatively distinct objectives, though typically at the cost of short-run welfare losses. Introducing asymmetric political constraints, we analyse a dynamic policy game and show how weak accountability may allow a strategically motivated government to escalate first and secure lasting gains in consumer surplus, as firm relocation amplifies the cost of retaliation for its opponent.

JEL Classification: L86, O33, K21, L51, F13

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Contents

1 Introduction

Context A nascent literature is giving full weight to the idea that data is a factor of production and is engaged in developing models that address its unique characteristics. Relative to early work in digital economics, a notable departure in this strand lies in its reduced-form treatment of data, typically abstracting from informational nuance to ask new questions about firm behaviour, market structure, and broader outcomes. A foundational contribution is Jones et al. (2020), which incorporates data generation as a by-product of consumption into an endogenous growth model to examine how data's nonrivalry interacts with ownership regimes in shaping aggregate outcomes. Veldkamp et al. (2024) provides a recent survey of these evolving directions.

International considerations remain an underdeveloped facet of this literature, with research still relying on speculative implications of findings for open economies. A common hypothesis, grounded in factor-based reasoning, holds that regulatory choices making data scarcer in one country may crowd out specialisation in data-intensive sectors, while more permissive regimes may enjoy a comparative advantage[1]. This reasoning finds occasional support in empirical work using international comparisons to document relative declines in productivity following regulatory changes (Goldfarb et al. 2011; Ferracane et al. 2020; Quan 2023). More generally, these findings are often interpreted as suggesting that countries face a strategic trade-off between protecting personal information and maintaining a foothold in data-intensive industries. Yet we remain theoretically unarmed to assess such claims, as the few existing models of the international data economy do not engage with trade patterns.

More concerning for the state of the literature is that approaches to data regulation in closed economies *de facto* treat the international environment as exogenous to their core questions. Such treatment appears directly at odds with parallel developments in the legal literature, which have consistently emphasised the role of international inertia in shaping domestic regulatory choices. Specifically, the development of global privacy law is frequently cited as a prime example of the Brussels Effect, whereby EU regulation sets standards that spill over into other jurisdictions (Bradford 2020).

This gap has become increasingly difficult to justify as data governance has adopted a distinctly outward-looking focus. While the first wave of global data regulation featured uniform standards for consumer data processing with potential extraterritorial reach, culminating in the ambitious adoption of the GDPR in 2016, the bulk of newly enacted measures specifically target foreign establishments. These measures require data about a country's citizens to be collected, processed, or stored within the country's borders. Despite their

[1] Variants of this assumption are discussed in the concluding sections of models by Jones et al. (2020) and Beraja et al. (2022), for instance.

heterogeneity, they are often grouped under the umbrella term *data localisation* due to their geographical pull. China's Great Firewall, which played a significant role in bringing these measures to global attention (Gao 2019), stands as the most extreme instance of such frameworks. However, these regulations are becoming increasingly widespread, with their most stringent forms driving much of the recent growth. (Ferracane et al. 2024)

Often justified on privacy grounds, data localisation measures nonetheless differ substantively from earlier frameworks, placing a situated emphasis on risks such as foreign surveillance rather than pursuing consistent regulation of firm–consumer interactions. Because of their *de facto* discriminatory nature, such regulations have been prone to suspicions of weaponisation as strategic trade policies. Yet most attempts to challenge them as unfair practices under existing trade rules have failed, primarily due to the limitations of pre-digital frameworks and the ambiguity in data's dual nature as both personal information and productive input. These efforts have largely originated from the United States in response to Chinese practices[2], alongside sustained lobbying to embed binding digital trade commitments within the WTO. While systematic attempts to quantify the strategic efficacy of data localisation remain limited, available findings are broadly consistent with this possibility[3]. More fundamentally, the widely noted sequence of unprecedented cross-border restrictions targeting foreign presence in China, followed by the emergence of the so-called BAT,[4] as the first instance of firms challenging the lead of U.S. tech giants in both scale and innovative potential, lends intuitive force to this narrative and continues to shape its popular reception.

More generally, the strategic reading of data localisation aligns with a broader policy reorientation, in which data regulation is increasingly subsumed under the logics of technological leadership and economic security. Anecdotal evidence suggests that the recent AI boom has further intensified this trend. On January 28, 2025, Donald Trump framed China's DeepSeek AI as a 'wake-up call' to U.S. industry, urging a 'laser-focused' approach to 'win' the global AI competition (The Guardian 2025). A week prior, the White House had issued an executive order titled *Removing Barriers to American Leadership in Artificial Intelligence*, which outlined a deregulatory agenda directing federal agencies to review existing policies with relevant implications. (The White House 2025). This discourse unfolds within a broader context where involvement in AI is portrayed as nearly existential for national economic positioning, with countries pursuing voluntarist industrial policies to secure a lead that, they believe, will sustain comparative advantage for the years to come.

The pervasive race rhetoric in AI policymaking is reminiscent of Krugman's conceptuali-

[2]Gao (2011) offers a detailed analysis of the challenges surrounding Google's 2010 withdrawal from China following cyberattacks and escalating censorship demands. In response, the USTR explored a WTO case under the GATS, attempting to frame China's filtering regime as a disguised restriction on cross-border services. Despite considerable legal ingenuity, the case never materialised.

[3]See Appendix A.2.2.

[4]Baidu, Alibaba, Tencent.

sation of the *businessman view of trade*, which frames trade between countries as an enlarged version of competition among firms. Krugman (1987b) originally discussed this view to argue that standard trade models offer limited analytical space for the widely held belief that international trade is a strategic contest with high stakes, path dependence, and potentially zero-sum welfare outcomes. In the case of AI, the parallel is also historical to some extent, as Krugman's paper was written in the context of Japan's rise as a challenger to U.S. leadership in high-value-added sectors. Japan, much like China today, was suspected of deliberate attempts to tilt the playing field in favour of its firms through unfair regulatory practices.

At present, the idea of entire countries being edged out of business, much like poorly performing firms, remains unappealing to general equilibrium theorists. Still, there are at least two reasons to seriously consider the businessman's perspective on the data economy.

First, while competition between firms may not provide an accurate analogy for trade between countries, the mere possibility that governments act on this belief renders it an analytical object in its own right. More importantly, it becomes the relevant perspective from which to understand regulatory patterns if there are plausible reasons to believe that they do. Any formal analysis that excludes this motivational layer will necessarily miss crucial interactions that shape observed outcomes. This omission also weakens our ability to anticipate where meaningful departures from social optima may arise and how such patterns might evolve. Grasping the strategic underpinnings of policy is essential for assessing whether, and how, cooperation could steer the system away from collectively inefficient trajectories.

While scholars have credited regulatory spillovers with driving the worldwide expansion of privacy rights, the same forces may now enable initiatives such as the recent US deregulatory push (The White House 2025) may now contribute to their erosion. Yet data regulations differ widely, and despite documented variation in their effects (Goldfarb et al. 2012), no formal analysis has identified the specific features that render some more or less distortionary in a global context. As current industrial trends fuel concerns over an often ill-defined notion of 'economic costs of privacy', this lack of clarity heightens the risk of misguided deregulation that could ultimately prove detrimental to welfare (Acquisti et al. 2023).

Second, the lack of neoclassical support to the businessman view of trade does not, in itself, invalidate its relevance. In the original words of Krugman (1987b), 'businessmen are surely wrong (...), but economists may not have captured the whole of the story either.' Formal models are selective abstractions, designed to isolate a narrow subset of real-world determinants; that some behaviours fall outside their explanatory reach is often irrelevant to assessing their practical significance. Strategic policy is fundamentally about rent capture, yet rents are rendered invisible in neoclassical trade theory not because they do not exist, but because the structure of the model makes them impossible to pursue. If, in a Ricardian world, Portugal can only gain by specialising in wine to exchange for English cloth, it is because the model, by positing exogenous productivity differences, embeds a natural pattern of trade from which any deviation must appear mutually harmful. The fact that *doux commerce* arises in this setting is informative about genuine gains from trade. But it says nothing about how Portugal's welfare might change were it in a position to choose a different relative productivity schedule. Most industrialist motives arise from this latter set of possibilities.

In the long run, it is unclear who stands as the natural candidate to host the next Google in the best interests of the world economy, and there may well be none. What is clear, however, is that whichever country succeeds in doing so will secure an unambiguous rent, in the form of high value added employment and surplus extraction, for the underlying market structure compels some profit to remain beyond the reach of general equilibrium forces. (Krugman 1987a) It is also plausible that whoever achieves this position will be difficult to dislodge by standard competitive forces, not only because of entry costs, but also due to data-powered network effects that generate lock-in dynamics even ambitious investment may struggle to overcome. Surely, countries are not businesses, and the reverse holds true; but if companies can grow to match the scale of entire economies, then firm-level outcomes will carry material implications for national welfare. And if policy choices can raise the odds of joining the lucky few, or lower the odds of others doing so, then it is only natural that governments will begin to think strategically when making their own.

Contribution The purpose of this thesis is to develop a formal analysis of how trade openness reshapes the incentives to regulate data use. We contribute along two dimensions: (i) we develop a tractable two-country trade model of the data economy, suited to the analysis of trade patterns; and (ii) we use this structure to provide a theoretically disciplined account of recent developments in global data governance through strategic incentives.

(i) We develop a two-country, two-sector general equilibrium model of the global data economy that admits a full closed-form characterisation. The framework builds on the asymmetric structure of Helpman et al. (1987). Each country hosts a differentiated Tech sector featuring monopolistic competition, increasing returns, and iceberg frictions, alongside a homogeneous good sector traded freely under perfect competition.

We introduce data as a firm-owned, consumer-generated input that complements labour in Tech production. Following Jones et al. (2020), each unit of consumption yields one unit of data, which firms may reinvest in production subject to regulatory constraints. In equilibrium, firms exhaust the scope of permissible usage, leading to systematic data overuse wherever regulation is absent or insufficient. Meanwhile, consumers incur privacy costs that rise with usage intensity and may vary with the origin of the processing firm, capturing potential biases in perceived risk. Crucially, data is modelled as a consumer-specific factor, usable only in the market where it is collected, thereby inducing destination-specific production technologies. This structure generates a subtle form of firm heterogeneity, as firms competing in the same market with identical fundamentals may exhibit data-driven differences in effective productivity that reflect the regulatory constraints tied to their location.

We derive equilibrium conditions and recast them as generalised market potential equations to show how data regulation shapes aggregate outcomes through familiar trade mechanisms. We show that constraints on a firm's ability to process local data act as selective distortions to market access that crowd out demand through productivity losses. Specifically, a marginal tightening of data rules targeting a given firm location induces two distinct effects, which we interpret dynamically under the assumption that free entry pins down a long-run outcome. In the short run, affected firms raise prices to preserve markups under higher marginal costs. In the long run, these price differentials trigger endogenous *relocation forces* within the Tech sector, generating excess losses wherever firms face tighter constraints, and excess profits elsewhere. Accordingly, consumer substitution prompts opposite industry dynamics that endogenously redirect firm entry towards more permissive environments.

By analogy with the Metzler paradox, we show that the price effect of Tech relocation always dominates, producing a disconnect between consumer data usage and the domestic price index, even if, by assumption, data-driven productivity gains are perfectly segmented by destination. Equilibrium real wages in any location increase with local entrepreneurial capacity to deploy data across destinations, and decline with that of foreign firms. Hence, the so-called *privacy–consumption* trade-off may collapse, as consumers can enjoy both lower digital exposure and higher real wages when restrictions on foreign establishments are tightened. More importantly, we show that the resulting adjustments in the number of active Tech firms across countries sum to zero, rendering the sector a source of pure economic rent accruing to the host country through the terms-of-trade effect. In this context, data regulation acquires natural strategic significance, for it may reallocate this rent across countries without shrinking its global magnitude.

(*ii*) We apply the model to analyse privacy regulation and data localisation as strategic choices. In doing so, we aim to account for recent developments in data governance in light of governments' incentives and constraints.

We begin by examining privacy regulation as a policy instrument designed to enforce uniform data usage standards within a defined scope. Governments choose both the activities subject to regulation and the stringency of constraints, under enforcement frictions that limit the reach of domestic rules abroad. Assuming consumer welfare maximisation, we characterise optimal regulation under internalised relocation dynamics and establish two core results. First, governments face strategic disincentives to protect foreign consumers, as such provisions unilaterally disadvantage domestic firms in foreign markets where local competitors remain exempt for their own market interactions. Second, weak extraterritorial reach gives rise to a novel *regulation–retention trade-off*; as uniform standards fall disproportionately on local firms, governments optimally scale back their ambition to preserve Tech presence, contain real wage losses, and maintain regulatory oversight.

We then examine the implications for global outcomes in a policy game where governments simultaneously set privacy standards under limited extraterritorial reach. We establish three main results. First, the internalisation of relocation forces gives rise to a coordination failure that systematically depresses equilibrium privacy standards in the world economy. This under-regulation generates welfare losses through two distinct channels: governments adopt weaker rules to deter firm relocation, and these diluted protections remain poorly enforced abroad, where local authorities provide no substitute safeguards for foreigners. Second, enforcement frictions tend to generate strategic complementarities, as governments increasingly condition regulation on the extent to which foreign standards align with their own. In equilibrium, these complementarities exacerbate the welfare losses from joint under-regulation, but sustain spillover dynamics consistent with observed patterns of regulatory inertia. Third, increased privacy awareness does not systematically support stronger regulation. While a uniform rise in privacy costs promotes reform through global spillovers, targeted mistrust of foreign firms may instead deter regulation by heightening perceived asymmetries and triggering dumping incentives.

We then turn to data localisation and examine the unilateral incentive to impose origin-contingent restrictions. We demonstrate that a broad set of normatively distinct policy objectives consistently support the optimality of maximal restrictions on foreign data processing. These include short-run producer rents, Tech export promotion, digital sovereignism, and even benevolent long-run welfare maximisation under mild conditions. This convergence of motives may explain the broad diffusion of localisation.

We propose that political accountability may act as a stabilising force by curbing incentives to adopt discriminatory policies. In partial equilibrium, restrictions on foreign firms raise local prices and generate producer rents that are not immediately offset by entry. When consumer surplus constitutes a large share of total welfare, these distortions outweigh any privacy gains and reduce short-run welfare at high restriction levels. We show that a minimal constraint preventing regulation from lowering welfare below its status quo level is sufficient to induce interior solutions. By contrast, we conjecture that weaker constraints on executive discretion may enable governments to adopt more extreme forms of localisation.

To examine how political constraints shape global outcomes, we extend the model to an infinite-horizon policy game with gradual relocation. One government faces no short-run constraints, while the other is bound by a welfare benchmark that limits politically acceptable losses. Assuming both pursue maximal discrimination in every period, a Markov-perfect equilibrium arises in which the unconstrained country imposes full restrictions from the outset, while the constrained government responds gradually as mounting welfare costs limit retaliation. Specifically, relocation towards the unconstrained first mover amplifies the short run real wage losses from further restrictions, progressively narrowing the constrained government's feasible policy space and eventually forcing it to adopt an interior restriction level when its political leeway is exhausted. The resulting steady state features persistent divergence in Tech activity despite identical fundamentals. We show that the risk of such divergence increases with the speed of structural adjustment in Tech, which compresses the window for effective response. Depending on the intensity of retaliation, the first mover in data localisation may secure lasting consumer gains at the expense of others, and thus lacks incentive to cooperate. We interpret this result as a stylised account of tensions between the United States and China, where aggressive localisation measures appear to have supported sustained industrial expansion without prompting commensurate retaliation.

Related literature This paper contributes to the still limited theoretical literature examining the international dimension of the data economy. Chang et al. (2023) develop a two-country DSGE model in which raw data can be transferred across borders and transformed into productive inputs. They show that data mobility gives rise to mutual gains as countries with limited processing capacity export raw data, while those with stronger digital infrastructure convert it into productive inputs, reducing productivity gaps and improving overall welfare. Quan (2023) quantifies the cross-border effects of GDPR enforcement using a general equilibrium model calibrated to firm- and consumer-level data. He shows that US data-intensive firms redirect activity away from the EU and reduce hiring of data-related staff in response to restricted data access, which generates measurable declines in digital welfare for consumers in both the EU and the US. While both papers adopt an open-economy perspective, neither constitutes a trade model in the conventional sense, nor are they suited to address questions such as the determinants of specialisation. Both rely on numerical solution methods, whereas the approach developed in this paper is analytical in nature, with simulations used solely for illustration.

Closer to our paper in analytical focus is Staiger (2021), who develops a reduced-form model to examine whether digital trade alters the rationale for trade agreements. He shows that domestic externalities from digital openness can be addressed within existing WTO rules, but cross border effects introduce new inefficiencies that may require deeper forms of international cooperation. Also related is the prospective contribution of Goldfarb et al. (2018), who outline a forward-looking agenda for trade theory in the age of AI. Building on insights from the literature on strategic trade policy, they reflect on how data regulation may interact with specialisation patterns and formulate several hypotheses about the underlying incentives, many of which are recovered as equilibrium outcomes in our framework. Specifically, they suggest that privacy regulation may disproportionately burden domestic firms, while data localisation could be used to pursue beggar-thy neighbour objectives.

More generally, our paper contributes to a recent wave of theoretical work exploring the aggregate implications of data as a production input, with particular emphasis on its characteristics and implications for industry dynamics, innovation, and growth (Jones et al. 2020; Farboodi et al. 2021; Cong et al. 2021; Cong et al. 2022; Eeckhout et al. 2022; Xie et al. 2022). A related strand of this literature highlights how data access and algorithmic learning shape competition dynamics, giving rise to firm-level asymmetries and market concentration (Begenau et al. 2018; Farboodi et al. 2019; Prüfer et al. 2021; Hagiu et al. 2023). We also build on a long-standing literature examining privacy preferences, behavioural responses to data collection, and the design of privacy regulations (Calzolari et al. 2006; Ichihashi 2020; Bergemann et al. 2022). Our findings are relevant to an expanding empirical literature that documents the economic effects of data regulation on firm performance, innovation, and market structure (Evans 2009; Goldfarb et al. 2011; Johnson 2013; Ferracane et al. 2020; Demirer et al. 2024; Jia et al. 2018; Aridor et al. 2020; Goldberg et al. 2024). Finally, our framework connects to a large body of work examining uncoordinated trade policy in CES-based monopolistic competition models with homogeneous firms (Venables 1987; Helpman et al. 1987; Ossa 2011; Campolmi et al. 2014; Bagwell et al. 2015).

Thesis outline The remainder of the thesis is structured as follows. Section 2 develops a two-country New Trade Theory model of the data economy in which monopolistically competitive Tech firms use consumer-generated data as a productive input. Section 3 uses this structure to analyse the determinants of privacy regulation and data localisation as strategic policy instruments. Sub-section 3.1 focuses on privacy regulation, characterising the optimal unilateral enactment of uniform standards under enforcement constraints and examining the symmetric Nash equilibrium of the uncoordinated policy game. Sub-section 3.2 studies the unilateral incentives to impose origin-contingent restrictions and the role of political accountability in shaping global outcomes, before extending the analysis to a dynamic policy game with gradual firm relocation and asymmetric political constraints. Section 4 concludes. A conceptual overview of global data governance, including regulatory trends across major jurisdictions and working definitions of key instruments, is provided in Appendix A.

2 Model

2.1 Environment

Consider a world with two countries, Home (H) and Foreign (F), and let $\mathcal{C} = \{H, F\}$ denote the set of countries. Throughout, we use i as the reference location and j to denote the other location when relevant. When a statement applies to either, we use $k \in \{i, j\}$ for generality.

The world economy comprises two industries with distinct production processes and market structures. The first, Tech, relies on proprietary data complementing labour to produce differentiated goods, whose trade faces an iceberg friction $\tau > 1$. The second is perfectly competitive and uses only labour to produce a homogeneous good, which is freely traded. Labour is supplied inelastically, with country i's endowment denoted L_i.

Assumption 1 (Outside sector)**.** *The production of the homogeneous good relies on a constant returns technology that is the same worldwide. Specifically, the production function is:* $f(L) = L$.

Assumption 2 (No full specialisation)**.** *Both countries remain diversified. In particular, a sufficient condition to prevent full specialisation in Tech is that the share of expenditure on the homogeneous good,* $1 - \alpha$, *exceeds the relative labour endowment of the larger country:*

$$1 - \alpha > \max\left\{\frac{L_H}{L_H + L_F}, \frac{L_F}{L_H + L_F}\right\}. \tag{1}$$

As in Helpman et al. (1987), the homogeneous good absorbs trade imbalances in Tech and serves as a fallback sector for residual labour, allowing a tractable analysis of specialisation under monopolistic competition. Assumption 1 imposes a common technology for this sector, effectively fixing the relative wage at unity. Assumption 2 further ensures that no country fully specialises in either sector. While slightly stronger than what is strictly required for factor price equalisation, this condition is adopted for convenience. In particular, it guarantees that both countries host a strictly positive mass of Tech firms in equilibrium.

2.1.1 Consumption

Utility function In each country, a representative consumer indexed by its residency i derives utility from consumption but experiences disutility from firms' use of their personal data. Let $\Omega = \bigcup_{k \in \mathcal{C}} \Omega_k$ denote the set of all Tech varieties in the world economy, where Ω_k is the set of firms operating from country k. The upper-tier utility governing consumption of Tech (C_i), a CES aggregate of varieties (ω), and the homogeneous good (Z_i) follows a

Cobb-Douglas specification. The consumer's preferences take the additively separable form:

$$U_i = \chi C_i^{\alpha} Z_i^{1-\alpha} - \Psi_i \tag{2}$$

$$C_i = \left(\int_{\Omega} c_i(\omega)^{\frac{\sigma-1}{\sigma}} \, \mathrm{d}\omega \right)^{\frac{\sigma}{\sigma-1}}, \quad \sigma > 1 \tag{3}$$

where χ is a normalisation constant and Ψ_i is a weighted average of privacy disutility associated with firms' data usage intensity. The privacy cost function is defined as:

$$\Psi_i = \frac{1}{2} \cdot \int_{\Omega} \psi_i(\omega) \, x_i(\omega)^2 \, \mathrm{d}\omega, \quad \psi_i(\omega) = \Big\{ \psi_{ki} \quad \text{if } \omega \in \Omega_k \tag{4}$$

Privacy costs for consumer i increase quadratically with $x_i(\omega)$, the share of their collectable data employed by firm ω. Aggregating over all active Tech firms captures the extensive margin of exposure.

To account for possible biases in perceived privacy risk, we introduce disutility shifters ψ_{ki}, which capture the weight placed by consumer i on data processed by firms located in country k. These parameters distinguish domestic from foreign processing and interact with the spatial distribution of firms to shape aggregate privacy costs.

Optimisation problem Tech firms retain full ownership of the data generated as a by-product of their sales, leaving consumers with no control over its use. Consumer i chooses a consumption schedule $c_i(\cdot)$ for Tech products and a quantity of the homogeneous good Z_i to maximise utility, given income wL. Solving this optimisation problem yields the standard demand function for a typical variety ω in terms of the ideal price index P_i[5]:

$$c_i(\omega) = \left(\frac{p_i(\omega)}{P_i} \right)^{-\sigma} C_i \,, \quad P_i = \left(\int_{\Omega} p_i(\omega)^{1-\sigma} \mathrm{d}\omega \right)^{\frac{1}{1-\sigma}} \tag{5}$$

Welfare As shown in Appendix B.1, normalising the marginal utility of income and exploiting the Cobb-Douglas structure gives the following expression for consumer welfare:

$$\mathscr{W}_i = \underbrace{\left(\frac{w}{P_i} \right)^{\alpha} L_i}_{\triangleq CS_i} - \Psi_i \tag{6}$$

2.1.2 Production

Given that the homogeneous good sector passively employs all labour not allocated to Tech, the analysis can, without loss of generality, focus exclusively on the Tech sector's production.

[5]Formally, P_i denotes the ideal price index over Tech varieties. However, we refer to it interchangeably as the general price index, since the latter is defined as a monotonic transformation $P_i^{\alpha} w^{1-\alpha}$, with w fixed.

Tech setup Each country hosts an unbounded mass of potential Tech firms. Any firm that chooses to operate must incur a fixed labour cost f before producing a non-zero quantity.

When active, Tech firms incorporate proprietary data, passively supplied by consumers through their purchases, as an input in production. Each unit consumed by consumer k generates one unit of data about that consumer, which firms can immediately reinvest. For tractability, we assume that firms fine-tune their products for each destination using only data collected within that market. This feature preserves the standard property in trade models whereby production decisions are analytically separable across destinations.

To serve consumer $k \in \{i, j\}$, a firm $\omega \in \Omega_i$ operating from country i hires labour $l_k(\omega)$ to produce according to the Cobb-Douglas technology:

$$F_k\big(d_k(\omega), l_k(\omega)\big) = \varphi\, d_k(\omega)^{\beta}\, l_k(\omega)^{1-\beta}, \quad \varphi \geq 1. \tag{7}$$

Here, $d_k(\omega)$ denotes the data input employed in production for consumer k, defined as:

$$d_k(\omega) = x_k(\omega)\, c_k(\omega), \tag{8}$$

where $x_k(\omega)$ denotes the share of collectable data that is used in production and $c_k(\omega)$ is the quantity of variety ω consumed by consumer k.

Data intensity of production Firms set data usage intensity through their choice of $x_k(\omega)$ within regulatory limits:

$$x_k(\omega) \in \Big\{[0, \bar{x}_{ik}] \subseteq [0, 1], \quad \text{if } \omega \in \Omega_i \tag{9}$$

We index regulatory caps on data usage by trade dyads. Hence, $\bar{x}_{ik}$ denotes the maximum share of data that a firm $\omega \in \Omega_i$, established in country i, is permitted to use about consumer k per unit of generation. We organise these caps in a global data governance matrix:

$$\mathbf{X} = \begin{pmatrix} \bar{x}_{HH} & \bar{x}_{HF} \\ \bar{x}_{FH} & \bar{x}_{FF} \end{pmatrix} \tag{10}$$

where the rows correspond to the firm-relevant dimension and the columns to the consumer-relevant dimension. Accordingly, for each country $i \in \mathcal{C}$, we further define the vectors:

$$(\mathbf{x}_i^f)' = \big(\bar{x}_{iH},\, \bar{x}_{iF}\big), \quad \mathbf{x}_i^c = \begin{pmatrix} \bar{x}_{Hi} \\ \bar{x}_{Fi} \end{pmatrix} \tag{11}$$

The definition of the data input $d_k(\omega)$ (8) makes clear that firms can amass more data either by expanding sales or by resorting to more invasive data practices. Our formulation for the privacy cost function (4) implies that there is no trade-off between these two margins, meaning that the firm's optimal strategy is to maximise $x_k(\omega)$.

$$x_k^*(\omega) = \begin{cases} \bar{x}_{ik}, & \text{if } \omega \in \Omega_i \end{cases} \tag{12}$$

Another implication of this formulation (8) is that iceberg trade costs impede data collection from foreign sales. Specifically, each exported unit of output yields only $\frac{1}{\tau} < 1$ unit of data, as only that fraction is consumed.

Equilibrium production possibility frontier For clarity, let $q_k(\omega)$ be defined as the gross output of variety ω that must be shipped in order for consumer i to receive $c_i(\omega)$ units:

$$q_k(\omega) \triangleq \begin{cases} \tau_{ik}\, c_k(\omega), & \text{if } \omega \in \Omega_i \end{cases}, \tag{13}$$

$$\tau_{ik} = \begin{cases} 1, & \text{if } k = i, \\ \tau > 1, & \text{otherwise.} \end{cases} \tag{14}$$

The relevance of the Cobb-Douglas specification with constant returns to scale (7) becomes particularly clear when equilibrium production technologies are consolidated. Appendix B.2 shows that equations (8), (12), and (13) jointly imply that the production function of a typical firm ω when supplying consumer k (7) collapses to a linear function of labour (F_k^*), with trade-dyad-specific TFP denoted $\tilde{\varphi}_k(\omega)$.

$$F_k^*\left(l_k(\omega)\right) = \tilde{\varphi}_k(\omega) l_k(\omega), \tag{15}$$

$$\tilde{\varphi}_k(\omega) = \begin{cases} \left(\varphi \left(\frac{\bar{x}_{ik}}{\tau_{ik}}\right)^{\beta}\right)^{\frac{1}{1-\beta}}, & \text{if } \omega \in \Omega_i \end{cases} \tag{16}$$

Notice that, although the model features homogeneous fundamental technology, the equilibrium exhibits effective productivity heterogeneity within each market, as firms differ in their ability to convert sales into data that complement labour depending on their location of establishment.

Pricing schedule Maximising profits subject to the consumption schedule (5) results in only a minor adjustment to the pricing rule, entirely contained within a destination-specific effective productivity parameter $\tilde{\varphi}_k(\omega)$ (16):

$$p_k^*(\omega) = \frac{\sigma}{\sigma - 1} \cdot \frac{w\tau_{ik}}{\tilde{\varphi}_k(\omega)} \tag{17}$$

2.2 Equilibrium

Free entry and market clearing Let $\Pi : \Omega \to \mathbb{R}$ denote the profit function of a typical firm ω. In equilibrium, the mass of firms in each country adjusts endogenously until entry in the Tech sector eliminates rents:

$$\Pi(\omega) = 0 \tag{18}$$

By symmetry, all firms within a given country behave identically. Derivations in Appendix B.3 show that the zero-profit locus (18) can be reformulated as a firm-level, location-specific equilibrium condition in terms of the price index vector $\mathbf{P} = (P_i)_{i \in \mathcal{C}}$:

$$q_0 = \Theta_i(\mathbf{P}) \cdot p_0^{-\sigma} \tag{19}$$

where q_0 and p_0 respectively denote firm size and the free on board price in a standard New Trade Theory model.

$$q_0 \triangleq \varphi f(\sigma - 1), \quad p_0 \triangleq \frac{\sigma}{\sigma - 1} \cdot \frac{w}{\varphi}, \tag{20}$$

We use the subscript '0' to indicate that, as the output elasticity of data β (15) approaches zero, both output and prices tend to these baseline quantities, and the resulting equilibrium mirrors that of Helpman et al. (1987). Meanwhile, Θ_i captures an augmented index of market potential for a firm located in country $i \in \mathcal{C}$:

$$\Theta_i(\mathbf{P}) \triangleq \varphi^{\frac{\beta(\sigma-1)}{1-\beta}} \sum_{k \in \mathcal{C}} \left(E_k \, P_k^{\sigma-1} \, \bar{x}_{ik}^{\frac{\beta(\sigma-1)}{1-\beta}} \, \tau_{ik}^{\frac{1-\sigma}{1-\beta}} \right), \quad E_k \triangleq \alpha w L_k \tag{21}$$

Here, E_i denotes the expenditure on Tech goods by the representative consumer residing in country i. Recasting firm-level market-clearing conditions using quantities from a standard Krugman model frames the outcomes in our framework as deviations from the baseline equilibrium, driven by data-induced shifts in market access.

Data-augmented market access index Θ_i (21) aggregates market contributions into an augmented market access index that determines the supplier capacity of firms established in country i. In line with the literature[6], this index is defined as an expenditure-weighted measure of relative accessibility, augmented by the role of data in production.

[6] Market access (Redding et al. 2004) and market potential (Head et al. 2004) are foundational concepts in economic geography. Both capture how proximity to large markets, combined with scale economies, shapes firm performance and the spatial distribution of activity. Profit-maximising firms facing decreasing average costs are drawn to locations with high expenditure, low trade costs, and dense supplier networks.

- The factor $\varphi^{\frac{\beta(\sigma-1)}{1-\beta}} \geq 1$ is a homogeneous shifter that captures the expansion of firms' market access in all destinations due to data-driven productivity gains. This amplification increases with the importance of data (β) and the substitutability across varieties (σ), with their interaction reflecting how technological advantages, reinforced by feedback from sales, can translate into more than proportional market share gains, hinting at the kind of effects one would expect in a Melitz-type extension.

- The term $\tau_{ik}^{\frac{1-\sigma}{1-\beta}}$ captures the dampening effect of trade costs on market access in our model. Relative to the standard Krugman formulation, where trade costs enter as $\tau_{ik}^{1-\sigma}$, the exponent in our setting implies a stronger penalty when production relies more heavily on consumption-generated data. Intuitively, trade frictions undermine both the volume of sales and the productivity gains that firms derive from these transactions.

- The term $\bar{x}_{ik}^{\frac{\beta(\sigma-1)}{1-\beta}}$ shows that caps on data usage act as policy-induced shifts that selectively modulate market access across destinations. Since firms always operate at the regulatory ceiling (12), any change in $\bar{x}_{ik}$ directly alters their capacity to convert sales into market-specific productivity gains. Tighter limits in location k (lower $\bar{x}_{ik}$) thus crowd out market potential in that jurisdiction. The strength of this transmission depends on both the elasticity of substitution (σ) and the role of data in production (β), which together determine how sales and productivity gains are mutually linked.

World equilibrium The Home and Foreign market-clearing conditions (19) constitute a system of two equations in the two-dimensional price vector $\mathbf{P}$. An equilibrium is fully characterised by a pair of price indices (P_H, P_F) that jointly satisfy these conditions, so that all firms in the world economy operate at zero profit. For notational convenience, we define:

$$i \in \mathcal{C}, \quad j \neq i, \qquad \Delta_i \triangleq \bar{x}_{ii}^{\frac{\beta(\sigma-1)}{1-\beta}} - \tau^{\frac{1-\sigma}{1-\beta}} \bar{x}_{ji}^{\frac{\beta(\sigma-1)}{1-\beta}} \tag{22}$$

$$\Delta \triangleq \bar{x}_{HH}^{\frac{\beta(\sigma-1)}{1-\beta}} \bar{x}_{FF}^{\frac{\beta(\sigma-1)}{1-\beta}} - \bar{x}_{HF}^{\frac{\beta(\sigma-1)}{1-\beta}} \bar{x}_{FH}^{\frac{\beta(\sigma-1)}{1-\beta}} \tau^{\frac{2(1-\sigma)}{1-\beta}} \tag{23}$$

Assumption 3 (Home market effect)**.** *Data caps $(\bar{x}_{ij})_{k\in\mathcal{C}}$ and the trade cost τ satisfy:*

$$\frac{\bar{x}_{ii}}{\bar{x}_{ji}} > \tau^{-\frac{1}{\beta}} \quad \Longleftrightarrow \quad \Delta_i > 0 \quad \Longrightarrow \quad \Delta > 0 \tag{24}$$

Proposition 2.1 (Existence and Uniqueness of the World Equilibrium)**.** *There exists a unique world equilibrium such that the vector of price indices $\mathbf{P} = (P_H, P_F)$ clears the good market in both countries (19), all firms $\omega \in \Omega$ set destination-specific prices $\big(p_k^*(\omega)\big)_{k\in\mathcal{C}}$ (17), and firm masses adjust to ensure that price indices satisfy their defining equations (5).*

Assumptions. 1, 2 and 3.

Proof. See Appendix C.1. □

Assumption 3 imposes the home market effect by ensuring that domestic firms depend more on their local market than do foreign entrants. It also constitutes the key condition for equilibrium determinacy, since requiring that Δ (23) be strictly positive is equivalent to:

$$\left| \left(\frac{\mathrm{d}P_F^{\sigma-1}}{\mathrm{d}P_H^{\sigma-1}} \right)_{\Theta_H} \right| > \left| \left(\frac{\mathrm{d}P_F^{\sigma-1}}{\mathrm{d}P_H^{\sigma-1}} \right)_{\Theta_j} \right| \tag{25}$$

In words, the market-clearing locus for Home firms is always more downward-sloping than that of Foreign firms in the $P_F^{\sigma-1}P_H^{\sigma-1}$-plane.[7] When the Home market matters more to Home firms than it does to Foreign firms, a given decline in P_H depresses the former's sales more sharply. Restoring their profitability thus requires a larger compensating increase in P_F than would be needed for Foreign firms; hence the steeper slope (25).

Assumption 4. Ω_i *is an interval of the form* $[0, N_i]$.

Figure 1 provides a graphical representation of how equilibrium is reached. As in Helpman et al. (1987), moving south-east along either schedule in the $P_F^{\sigma-1}P_H^{\sigma-1}$-plane corresponds to a reallocation of Tech production towards Foreign that hurts Home's terms-of-trade.

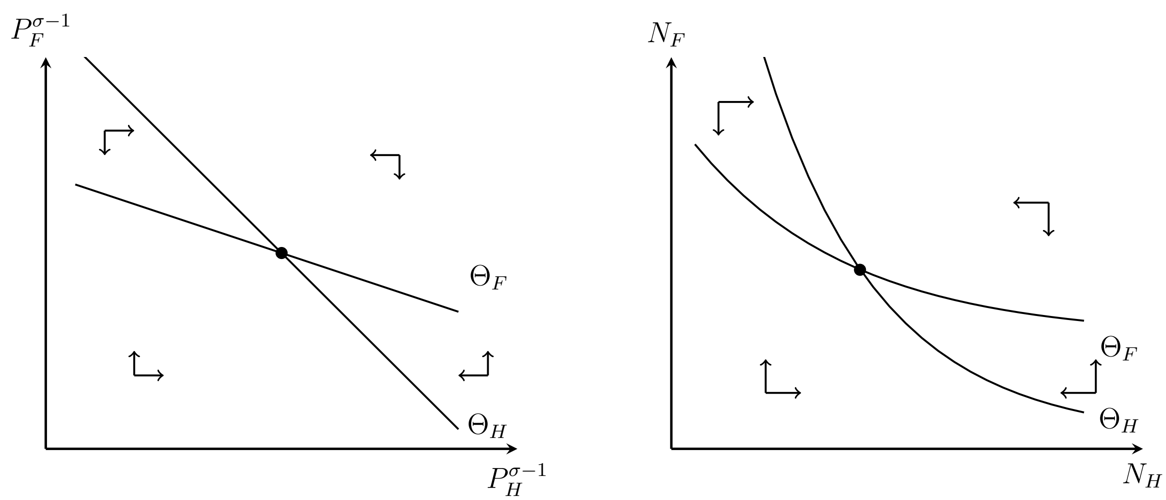


Figure 1: An illustration of world equilibrium determination

Borrowing from economic geography, we adopt a long-run perspective on equilibrium and use arrows of motion to suggest how disequilibrium might resolve over time.

For example, at any point $(P_H^{\sigma-1}, P_F^{\sigma-1})$ lying below the Θ_F-locus and above the Θ_H-locus, Foreign firms incur losses while Home firms earn excess profits. Restoring equilibrium

[7] The stronger condition $\Delta_i > 0$ ensures that the two curves intersect in the positive quadrant.

requires easing pressure on Foreign firms and tempering profitability in Home. Moreover, because wages are pinned down by the outside good sector, adjustment must occur through firm entry and exit. Specifically, the mass of Foreign firms must shrink, while entry by Home entrepreneurs dissipates local rents. These dynamics are reflected in a rise in P_F and a fall in P_H, which move inversely with the intensity of local competition.

While our model remains static, the arrows in Figure 1 serve as a heuristic for a dynamic interpretation of comparative statics. In Section 3, we use this device to distinguish between the short- and long-run consequences of policy interventions.

Closed-form characterisation Because the system of firm market-clearing conditions is linear in the transformed vector $(P_F^{\sigma-1}, P_H^{\sigma-1})$, the model admits a full analytical solution. Fixing a global regulatory environment $\mathbf{X}$ (10), a direct application of matrix algebra yields the following solution for the price index in country i:

$$P_i = \left(\frac{q_0 p_0^{\sigma} \Delta_j}{\varphi^{\frac{\beta(\sigma-1)}{1-\beta}} E_i \Delta} \right)^{\frac{1}{\sigma-1}} \tag{26}$$

Country firm masses can then be determined from the definition of the price indices (5):

$$N_i = \frac{\left(E_i \Delta_i \bar{x}_{jj}^{\frac{\beta(\sigma-1)}{1-\beta}} - E_j \Delta_j \tau^{\frac{1-\sigma}{1-\beta}} \bar{x}_{ji}^{\frac{\beta(\sigma-1)}{1-\beta}} \right)}{\sigma w f \Delta_i \Delta_j} \tag{27}$$

A closed-form characterisation of other remaining equilibrium quantities, including firm sizes and labour allocation, is provided in Appendix B.4.

2.3 Comparative Statics

Towards the subsequent policy analysis, this section presents the comparative statics of how changes in data usage caps affect price indices and the spatial allocation of Tech activity. In each case, the exact differential expressions are provided in the corresponding appendix.

Price indices Under our heuristic interpretation of general equilibrium adjustments as capturing long-run dynamics, comparative statics of Tech prices may be analysed either in the short run, holding the spatial distribution of firms fixed; or in the long run, as firms gradually enter, exit, or relocate in response to changing profitability conditions.

For clarity, we introduce the following notation for the Tech price indices faced by consumers in country $i \in \mathcal{C}$, under the data governance matrix $\mathbf{X}$ (10):

$$P_i^s(\mathbf{X}) \triangleq P_i(\mathbf{X}; N_H^s, N_F^s), \qquad P_i^\ell(\mathbf{X}) \triangleq P_i(\mathbf{X}, N_H^\ell(\mathbf{X}), N_F^\ell(\mathbf{X})) \tag{28}$$

where the superscript s denotes the short run, with fixed firm masses (N_H^s, N_F^s), and ℓ the long run, where the distribution $(N_H^\ell(\mathbf{X}), N_F^\ell(\mathbf{X}))$ adjusts endogenously. Formally, these capture the partial and general equilibrium responses to changes in data regulation, respectively.

Proposition 2.2 (Short-Run Price Adjustments)**.** *In the short run, consumers in country $i \in \mathcal{C}$ face a privacy-consumption trade-off, as the local price level P_i^s declines with the ability of* all *Tech firms in the world economy to exploit their data. Since data usage is segmented by market, P_i^s is unaffected by firms' access to data on consumers in the other location.*

$$\nabla_{\mathbf{x}_k^c} P_i^s \triangleq \left(\frac{\partial P_i^s}{\partial \bar{x}_{ik}}, \frac{\partial P_i^s}{\partial \bar{x}_{jk}}\right)' = \begin{cases} < \mathbf{0} & k = i, \\ = \mathbf{0} & k = j. \end{cases} \tag{29}$$

Assumptions. 1, 2 and 3.
Proof. See Appendix C.2. □

Proposition 2.2 implies that, in the short run, consumer i's wage in terms of Tech goods moves inversely with their data usage vector $\mathbf{x}_i^c$, as the ability of any firm in the global economy to exploit their data raises effective productivity in the local market and reduces related variety prices. In this sense, the *privacy-consumption trade-off* extends to the world economy, encompassing all firms irrespective of origin.

Yet this outcome reflects only a partial equilibrium artefact in our framework. We now show how data-driven productivity wedges may induce long-run relocation forces that fundamentally alter the nature of data governance trade-offs in an open economy.

Proposition 2.3 (Long-Run Price Adjustments)**.** *In the long run, the price index P_i in country $i \in \mathcal{C}$ falls with domestic firms' data-processing capacity across all markets, but rises with foreign firms' ability to harness data globally, including that of local consumers.*

$$\nabla_{\mathbf{x}_k^f} P_i^\ell \triangleq \left(\frac{\partial P_i^\ell}{\partial \bar{x}_{ki}}, \frac{\partial P_i^\ell}{\partial \bar{x}_{kj}}\right) = \begin{cases} < \mathbf{0} & k = i, \\ > \mathbf{0} & k = j. \end{cases} \tag{30}$$

Assumptions. 1, 2 and 3.
Proof. See Appendix C.3. □

Proposition 2.3 is notable in several ways. First, it implies that the privacy-consumption trade-off may selectively break down in an open economy, as tighter restrictions on foreign firms can simultaneously raise real wages and reduce privacy exposure in equilibrium[8]. Specifically, our dynamic interpretation of equilibrium adjustments suggests that the trade-off may persist during a transition period, as the fixed spatial distribution of Tech firms compels consumers to purchase higher-priced varieties from foreign suppliers. In the meantime, the rising productivity wedge also triggers a substitution effect that shifts demand towards domestic varieties, generating temporary rents in the local market as domestic firms grow abnormally large. With fixed wages, these rents are ultimately dissipated through endogenous entry, which distributes surplus to consumers and lowers prices via a terms-of-trade effect that outweighs the direct price impact of data-driven productivity changes.

Second, the result generalises the Metzler paradox[9], by drawing an isomorphism with a setting in which countries may impose restrictions that reduce the effective productivity of foreign firms operating in their market. As the definition of Θ_i (21) makes clear, destination-specific productivity reductions are analytically equivalent to restrictions on market access through higher trade costs. The key implication for our model is that, in the long run, purchasing power rises with the global data usage capacity of domestic firms and falls with that of foreign firms, even though data-driven productivity improvements are perfectly segmented by market (8).

Firm masses Knowing the exact local changes in the price indices, one can pin down the implied responses of firm masses as profit converges to zero everywhere.

Proposition 2.4 (Long-Run Firm Masses Adjustments). *In any country $i \in \mathcal{C}$, the long-run mass of firms N_i^ℓ increases with the capacity of domestic firms to use data in any market, and decreases with the capacity of non-domestic firms to use data in any market.*

$$\nabla_{\mathbf{x}_k^f} N_i^\ell = \left(\frac{\partial N_i^\ell}{\partial \bar{x}_{ki}}, \ \frac{\partial N_i^\ell}{\partial \bar{x}_{kj}} \right) = \begin{cases} > \mathbf{0} & k = i, \\ < \mathbf{0} & k = j. \end{cases} \tag{31}$$

Assumptions. 1, 2, 3, and 4.
Proof. See Appendix C.4. □

Proposition 2.4 formalises the intuition that the evolution of long-run prices depends on implicit adjustments in the geographic distribution of Tech firms. The vector of firm-level data usage capacity is inherently linked to market potential (21) and, by extension, to the mass of zero-profit Tech firms that a given location can jointly sustain. In particular, local

[8] The exact total effect on aggregate privacy costs would also depend on an extensive margin effect.

[9] The Metzler Paradox refers to the counterintuitive situation where imposing a tariff lowers the domestic price of the imported good, rather than raising it, due to a strong terms-of-trade effect. Helpman et al. (1987) shows that New Trade Theory can accommodate this paradox through the home market effect.

industry always expands in response to any regulatory change that increases the capacity of domestic firms to exploit data in any given market, and contracts when foreign firms gain a comparative data advantage, everything else given.

Unlike in neoclassical models, implications for purchasing power follow directly, as gains from trade do not arise from differences in autarky relative prices that generate natural patterns of specialisation. Instead, the Tech sector behaves as a source of location-specific rents, making it systematically beneficial to attract a larger share of activity at the expense of the trading partner, whose labour is reallocated towards the lower value-added outside good. In this sense, our framework captures the strategic logic underpinning industrial policy in the data economy, even though it features a continuum of monopolistic competitors rather than a discrete Tech giant that countries would compete to host. For our purposes, this modelling choice preserves tractability by convexifying outcomes and avoiding the need to model strategic interactions between individual producers on top of those between governments.

Proposition 2.5 (Beggar-Thy-Neighbour Equilibrium Adjustments)**.** *The number of firms in the world economy, N, is independent of data usage restrictions and is given by:*

$$N = \frac{E_H + E_F}{\sigma w f} \tag{32}$$

Accordingly, changes in data usage caps induce zero-sum adjustments in firm masses:

$$\mathrm{d}N_H + \mathrm{d}N_F = 0 \tag{33}$$

Assumptions. 1, 2, 3, and 4.
Proof. See Appendix C.5. □

Proposition 2.5 sheds light on a structural feature of the model which, though implicit in Helpman et al. (1987), is crucial for deriving monotonic comparative statics in world equilibria under factor price equalisation. Specifically, the total measure of Tech firms in the global economy is pinned down by fundamental industry parameters that govern the degree of increasing returns to scale; namely, the fixed labour cost of production f, the elasticity of substitution among varieties σ, and total expenditure on Tech varieties $\sum_i E_i$.

Since expenditure is fixed, shifts in supplier capacity stemming from data usage restrictions result solely in a reallocation of Tech activity across locations, rather than in an expansion or contraction of the global industry. An implication of this structure for our purposes, is that, holding fundamentals constant, real wage gains must arise from reallocating labour towards the higher value-added Tech sector, with beggar-thy-neighbour implications for the trading partner. This feature gives a role to the dynamics underlying the policy debate on strategic data regulation, which would be absent in a more neoclassical setting.

3 Policy Analysis

In this section, we endogenise global data governance. Policymakers set restrictions on data usage to advance an objective, subject to their territorially bounded regulatory capacity. We examine the determinants of unilateral policy choices and characterise the resulting equilibrium patterns. Sub-section 3.1 analyses privacy regulation under a commitment to non-discriminatory standards, while Section 3.2 examines cross-border data restrictions, building on the broader conceptual framework developed in Appendix A.2.

Simplifying restrictions Throughout this section, we make two parameter restrictions that greatly streamline the analysis, gathered in Assumption 5. While more stringent than strictly necessary to establish most results, these conditions greatly contribute to preserving tractability without altering the economic intuition or the core mechanisms of the model.

Assumption 5 (Parameter Restrictions)**.**

$$\sigma > 1 + \alpha \quad \textit{and} \quad \beta < \frac{1}{\sigma} = \bar{\beta} \tag{34}$$

The first inequality is largely unrestrictive, as $\sigma > 1 + \alpha$ holds under standard parameter values.[10] The second condition restricts the elasticity of output with respect to data, β, which we interpret as a small parameter.[11] Together, these restrictions simplify the analysis by reducing the need for case-by-case distinctions or cumbersome analytical arguments.

For simplicity, we conduct most of the analysis under the restriction that privacy preferences do not vary with firm location.[12] This assumption is relaxed in Sub-section 3.1.2.2.

Assumption 6 (Homogeneous Privacy Preferences)**.** *Consumer i is indifferent to the location of data processing: $\psi_i \triangleq \psi_{ii} = \psi_{ji}$.*

Calibration We rely on numerical simulations for illustrative purposes. Labour endowments are symmetric across countries ($L_i = 100$), and both wages and fundamental productivity are normalised to unity ($w = \varphi = 1$). The iceberg trade cost is set to $\tau = 1.2$, and the fixed cost of production to $f = 5$. The elasticity of substitution is $\sigma = 6$, the output elasticity with respect to data is $\beta = 0.1$, and the expenditure share on Tech is $\alpha = 0.1$.

[10] α is a share, while most empirical estimates place the elasticity of substitution between goods around 6.

[11] See Jones et al. (2020) for a discussion of plausible values for β. Using a calibration based on the share of GDP spent on data and empirical evidence from machine learning error rates, they estimate that β (denoted by η in their model) plausibly ranges from 0.03 to 0.12.

[12] None of our qualitative results rely on this assumption, but they require $\psi_{ji} \geq \psi_{ii}$, so that consumers weakly prefer local data processing. The concern that individuals may fear surveillance by their own government is irrelevant here, as such government would not pursue welfare maximisation.

3.1 Privacy Regulation

We begin by analysing privacy law as a policy instrument aimed at enforcing consistent data minimisation within a regulated domain. Our analysis emphasises the *intention* of privacy regulation to uphold uniform standards irrespective of firm origin. This ideal-typical benchmark constitutes the natural extension of privacy policy analyses in closed economies and distinguishes such regulation from trade-specific instruments targeting cross-border flows.

Our aim is to show that uniformity is not without cost when enforcement is territorially bounded, as regulation may fall disproportionately on domestic firms. Absent coordination, the interaction between asymmetric regulatory constraints and endogenous relocation generates structural trade-offs that systematically depreciate regulatory ambition in equilibrium.

Scope of privacy regulation A privacy policy is defined by two core components: (*i*) its *scope of application*, which delineates the set of firm-consumer interactions subject to regulation; and (*ii*) the *stringency of restrictions* imposed within that domain. In our model, a regulation is more or less stringent depending on the share of data that firms are permitted to use for covered processing activities, mimicking a data minimisation requirement.

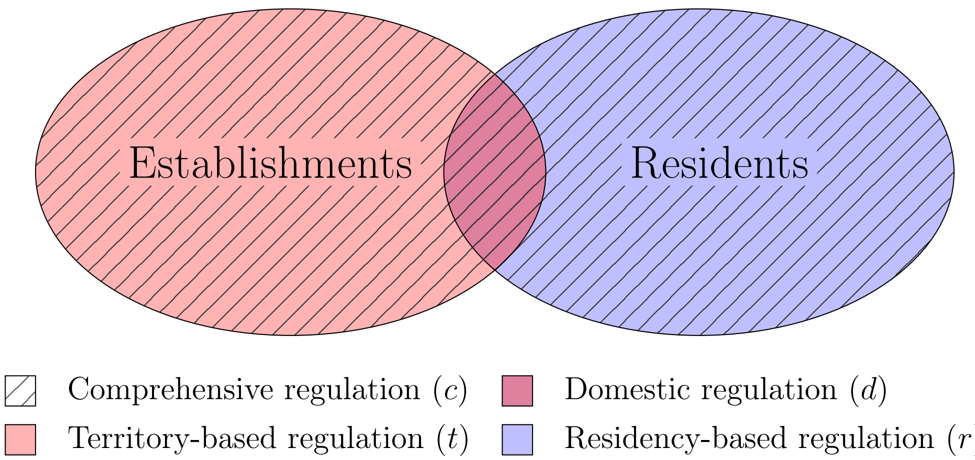
Scope variation is a structuring dimension of global data governance that remains significantly underexplored in the literature, particularly in studies assessing the economic impacts of privacy regulation. Most post-GDPR regimes now incorporate an extraterritorial component that extends obligations to all foreign firms when serving the local market, although enforcement challenges are frequently reported, as discussed in Appendix A.2.1. Unlike the GDPR, however, many regimes do not extend protection to foreign individuals, either *de jure* (e.g., California's CCPA or the proposed US ADPPA) or *de facto* (e.g., China's PIPL).

Our model features two types of data subjects (i.e Home and Foreign consumers) and two types of data handlers (i.e. Home and Foreign Tech firms), suggesting a simple typology:

Regulation Type	Description	Domain
Domestic (d)	Applies only to interactions between domestic firms and domestic consumers.	$\{d_{ii}\}$
Residency-based (r)	Applies to all data collected from domestic consumers, including by foreign firms. No protection for foreigners.	$\{d_{ii}, d_{ji}\}$
Territory-based (t)	Applies to all data processing by domestic firms, including data on foreign consumers. No extraterritoriality.	$\{d_{ii}, d_{ij}\}$
Comprehensive (c)	Applies to all data by domestic firms and all data on residents, including when handled by foreign firms.	$\{d_{ii}, d_{ij}, d_{ji}\}$

Figure 2 provides a visual representation of the typology. Residency-based regulations extend protection to residents in their interactions with any data handler, whereas territory-based regulations impose obligations on local establishments in their dealings with all data subjects. A comprehensive framework, like the GDPR, encompasses both dimensions.

Figure 2: A typology of privacy regulation



3.1.1 Determinants of Unilateral Optimal Privacy Regulation

Setup In each country, a planner seeks to maximise national consumer welfare, either from a short-run perspective or from a longer-term one that accounts for the relocation effects of regulation. Consistent with earlier notation (28), we index the policy horizon by $h \in \{s, \ell\}$, where s denotes the short run, with a fixed firm distribution, and ℓ the long run, where firm location is endogenous and react to changes in profitability conditions in each country.

A government operating under a short-run horizon ($h = s$) maximises welfare within the prevailing equilibrium. By contrast, a forward-looking government ($h = \ell$) internalises relocation effects and the general equilibrium implications of regulation. Such a stance may reflect strategic or industrial policy motives, or simply a non-naïve approach to the trade-offs inherent in regulatory design.

Both decision-making modes serve as useful benchmarks for illustrating how trade openness shapes the design of optimal privacy regulation.

Government problem Evaluating consumer welfare (6) in equilibrium, the objective function of the government in country i, deciding on policy with a planning horizon $h \in \{s, l\}$, can be expressed as:

$$\mathscr{W}_i^h\left(\bar{x}_{ii}, \bar{x}_{ij}, \bar{x}_{ji}; \bar{x}_{jj}\right) = \underbrace{\left(\frac{w}{P_i^h}\right)^{\alpha} L_i}_{CS_i^h} - \frac{\psi_i}{2} \underbrace{\left(N_i^h \bar{x}_{ii}^2 + N_j^h \bar{x}_{ji}^2\right)}_{\Psi_i^h(\mathbf{x}_i^c)} \tag{35}$$

Given exogenous enforcement constraints, the planner's problem consists in determining both the scope of data processing activities to be covered by the regulation and the intensity of regulation applied within that scope to maximise consumer welfare (6)[13]. Formally, the government in country i solves the following optimisation program:

$$\max_{s_i,\, g_i} \quad \mathcal{W}_i^h\left(\bar{x}_{ii},\, \bar{x}_{ij},\, \bar{x}_{ji}; \bar{x}_{jj}\right) \tag{36}$$

$$\begin{aligned} \text{s.t.} \quad & \bar{x}_{ii} \;=\; 1 - \mathbb{1}\{s_i \in \{d, r, t, c\}\} \cdot g_i \\ & \bar{x}_{ij} \;=\; 1 - \mathbb{1}\{s_i \in \{t, c\}\} \cdot g_i \\ & \bar{x}_{ji} \;=\; 1 - \mathbb{1}\{s_i \in \{r, c\}\} \cdot \pi_i g_i \\ & s_i \in \{d,\, r,\, t,\, c\}, \\ & g_i \in [0, \bar{g}_i] \end{aligned} \tag{37}$$

Building on the preceding taxonomy, we model the government's regulatory decision as a discrete choice over regulation types, $s_i \in \{d, r, t, c\}$, paired with a continuous intensity parameter $g_i \in [0, \bar{g}_i]$ that governs the stringency of data minimisation. Higher values of g_i entail stricter limits on data usage within the regulated scope.

To capture frictions in extending domestic rules to foreign firms, we introduce the parameter $\pi_i \in (0, 1]$, which measures the effective enforcement coverage of country i's regulations on data handled by foreign firms serving domestic consumers. That is, when the government sets a restriction intensity g_i, only a share $\pi_i g_i$ is effectively imposed on the data practices of foreign firms targeting local consumers. A value of $\pi_i = 1$ corresponds to full extraterritorial enforcement, while $\pi_i < 1$ reflects legal or institutional barriers that constrain the international reach of domestic standards. Imposing $\pi_i > 0$ is without loss of generality, as the case $\pi_i = 0$ would render residency-based regulation equivalent to domestic regulation.

The upper bound $\bar{g}_i$ is an exogenous constant that limits the government's ability to minimise data collection. Economically, it may capture an inherent constraint on the government's ability to reduce data collection, since a minimum amount of data is required to ensure basic service functionality. Technically, it contributes to the well-posedness of the problem. Specifically, we assume that it is always sufficiently low for Assumption 3 to hold, thereby guaranteeing equilibrium determinacy throughout the analysis.[14]

In the analytical characterisation, this constraint further ensures that the government's optimisation occurs over a concave region of the welfare function, thus precluding ambiguity in the policy solution. We formalise the relevant restrictions in Assumption 7.

[13] We abstract from monopoly-related distortions and focus on constrained efficient outcomes.

[14] When extraterritorial enforcement is imperfect ($\pi_i < 1$), tightening regulation gradually erodes the competitive advantage of domestic firms in their home market. The upper bound $\bar{g}_i$ rules out cases in which regulation becomes so stringent that firms would prefer serving country i from abroad.

Assumption 7 (Maximum Regulation Level)**.** *The data minimisation capacity of country i is bounded above by a constant $\bar{g}_i$ such that:*

$$0 < \bar{g}_i < \min\left\{1 - \tau^{-\frac{1}{\beta}}, \frac{1}{1+\pi_i}\right\} \tag{38}$$

Optimal scope of privacy regulation Problem (36) is best approached as a two-layered decision problem. For any regulatory scope s_i, define the second-stage value as:

$$V_i^h(s_i) \triangleq \max_{g_i \in [0,\bar{g}_i]} \mathcal{W}_i\Big(\bar{x}_{ii}\big(s_i, g_i\big),\ \bar{x}_{ij}\big(s_i, g_i\big),\ \bar{x}_{ji}\big(s_i, 1-g_i\big);\ \bar{x}_{jj}\Big) \tag{39}$$

The monotonicity of comparative statics enables an immediate characterisation of the optimal scope of unilateral privacy regulation, as formalised in the following proposition.

Proposition 3.1 (Optimal Scope of Privacy Regulation)**.** *Consider a pair $\left(s_i^h, g_i^h\right)$ that solves Problem (36) at planning horizon $h \in \{s, \ell\}$. At any $g_i^h > 0$, the following must hold:*

- *If $h = s$, then the regulation must cover the extraterritorial processing of resident data by foreign firms. In particular, it is always optimal to choose a residency-based scope, and a short-sighted planner may be only indifferent to a comprehensive regulation that further extends in the territorial dimension.*

$$s_i^s \in \{c, r\}$$

- *If $h = \ell$, then the policy must be residency-based. Any alternative scope yields strictly lower welfare for consumers in country i.*

$$s_i^\ell = r. \tag{40}$$

Assumptions. 1, 2, 3, 4, 5, 6 and 7.
Proof. See Appendix C.6. □

Proposition 3.1 provides theoretical foundations for two empirical patterns discussed in Appendix A.2.1: (i) the increasing extraterritorial scope of privacy regulations,[15] and (ii) the frequent restriction of protections to residents. More importantly, it identifies a first mechanism through which international trade shapes privacy regulation, by creating adverse incentives for strategic governments to extend protections to foreigners. In the absence of

[15]The result continues to hold in a variant of the model where the government endogenously chooses the degree of extraterritorial enforcement π_i. In such a setting, the enforcement constraint would bind. Our interchangeable use of extraterritorial reach and regulatory capacity thus implies no loss of generality.

international coordination, the model predicts that data interactions involving foreign firms will generally be governed by the extraterritorial enforcement of domestic rules, provided that countries internalise general equilibrium effects to some extent.

In the short run, extraterritorial enforcement is a key determinant of welfare gains from privacy regulation, as the convexity of privacy costs renders the omission of foreign firms inherently costly. Conditional on such enforcement, however, the government remains indifferent as to whether regulation should extend to the processing of foreign data by domestic firms, as this activity bears no direct effect on the domestic price index or privacy costs.

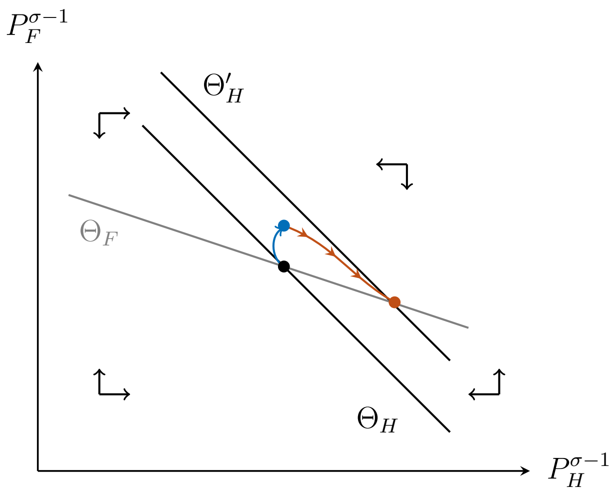


Figure 3: Extending protections to Foreign consumers reduces long run real wages in Home.

Note: The red dot denotes the long-run equilibrium, reached after firms adjust their entry and exit decisions in both countries. The blue dot indicates the short-run outcome if the government were to unilaterally extend existing data minimisation standards ($g_i > 0$) to foreign consumers. This short-run outcome corresponds to a vertical shift from the initial equilibrium, leaving the Home price index unchanged, as data usage is segmented by market. The point lies below the new zero-profit locus for Home firms, which can no longer clear under the revised regulatory environment. To restore profitability, the Home Tech sector must contract for surviving firms to regain market share, having permanently lost foreign demand to local competitors not subject to the regulation in their own market. Since this policy delivers no privacy benefit to Home consumers yet reduces their real wage, such a policy would never be optimal for a government maximising long-run national welfare.

In the long run, this indifference breaks down. Unilaterally extending regulation to protect foreign consumers imposes strictly positive welfare costs on the enacting country, as it weakens the competitive position of domestic firms abroad, where local rivals are not subject to the regulation for their own market interactions. Any binding constraint induces an effective productivity differential in the foreign market, permanently reducing domestic firms' market potential as foreign consumers shift towards local varieties. As general equilibrium adjustments unfold, Tech production gradually relocates to the foreign country.

This reallocation depresses domestic welfare through two distinct channels. First, the domestic price index rises due to the standard terms-of-trade effect, as illustrated in Figure 3. Second, and more subtly, it is the welfare gains from the regulation themselves that are ultimately eroded under imperfect extraterritorial reach ($\pi_i < 1$), as domestic firms are gradually replaced by foreign competitors that are only imperfectly regulated.

The result speaks to the empirical observation that many GDPR-inspired regulations fail to provide effective protections to non-residents, as discussed in Appendix A.2.1. More fundamentally, the preceding analysis suggests that empirical studies seeking to quantify the 'economic costs of privacy' through international comparisons of firm performance before and after the enactment of privacy laws should more carefully account for this dimension of regulatory heterogeneity when evaluating the external validity of their findings. Notably, much of the existing literature focuses on European regulations, which tend to extend protections to foreign consumers and may therefore yield poor estimates in other settings.

Optimal intensity of regulation By Proposition 3.1, it is without loss of generality to restrict attention to residency-based regulation when characterising optimal data usage minimisation[16]. Fix $s_i^h = r$, and consider an environment in which country i determines its optimal regulatory intensity g_i^h to protect residents in their digital interactions, subject to its enforcement capacity π_i.

The problem of the government in country i (36) then reduces to:

$$\begin{aligned} \max_{g_i \in [0, \bar{g}_i]} \quad & \mathscr{W}_i^h \left(\bar{x}_{ii}, \bar{x}_{ij}; \bar{x}_{ji}, \bar{x}_{jj}\right) \\ \text{s.t.} \quad & \bar{x}_{ii} \; = \; 1 - g_i \\ & \bar{x}_{ji} \; = \; 1 - \pi_i g_i \end{aligned} \tag{41}$$

To prepare for the intuition underlying the results, it is useful to examine an infinitesimal tightening of the data minimisation policy, $\mathrm{d}g_i > 0$. A total differentiation of the welfare function (35) suggests the following re-expression for the first-order change in welfare, $\mathrm{d}\mathscr{W}_i^h$:

$$\begin{aligned} \mathrm{d}\mathscr{W}_i^h &= \left(MB_i^h - MC_i^h\right) \mathrm{d}g_i, \\ MC_i^h &\triangleq \frac{\partial CS_i}{\partial P_i} \left[\underbrace{\frac{\partial P_i}{\partial g_i}}_{\text{Direct price change}} + \underbrace{\frac{\partial P_i}{\partial N_i} \frac{\mathrm{d}N_i^h}{\mathrm{d}g_i}}_{\text{Relocation effect on prices}} \right] \end{aligned} \tag{42}$$

[16] In the long run, residency-based regulation is uniquely optimal whenever $g_i^\ell > 0$. In the short run, the government may be indifferent between residency-based and comprehensive regulation, but both induce the same set of optimal choices, as restrictions on foreign consumers have no effect on domestic welfare.

$$MB_i^h \triangleq \underbrace{\frac{\partial \Psi_i}{\partial g_i}}_{\text{Direct privacy gain}} + \underbrace{\frac{\partial \Psi_i}{\partial N_i}\frac{\mathrm{d}N_i^h}{\mathrm{d}g_i}}_{\text{Relocation effect on exposure}}$$

In partial equilibrium, regulation operates through two direct channels. On the one hand, it reduces consumers' exposure to data collection, as all incumbent firms must comply with stricter data usage standards. These privacy gains form the basis of the marginal benefit of regulation, MB_i^h at horizon h. On the other hand, it induces productivity losses for affected firms, which are passed through to consumers in the form of higher prices to preserve markups. These pass-through effects form the basis of the marginal cost of regulation, MC_i^h.

The comparison of these two forces at the margin defines the so-called privacy-consumption trade-off, a central focus of the existing literature. However, as hinted in Section 2.3, this account is incomplete in our trade framework. In the long run, both the marginal benefit MB_i^h and the marginal cost MC_i^h of data minimisation are further mediated by general equilibrium forces arising from the endogenous relocation forces triggered by the regulation.

We begin with a technical result that establishes the existence and uniqueness of the optimal regulation level in each decision horizon.

Proposition 3.2 (Existence and Uniqueness of an Optimal Regulation Level). [17] *For any decision horizon $h \in \{s, \ell\}$, the government's problem (41) is strictly concave. In particular:*

- *The marginal benefit function $MB_i^h : [0, \bar{g}] \to \mathbb{R}$ is strictly decreasing in g_i;*
- *The marginal cost function $MC_i^h : [0, \bar{g}] \to \mathbb{R}_{>0}$ is strictly increasing in g_i.*

Assumptions. 1, 2, 3, 4, 5, 6 and 7.
Proof. See Appendix C.7. □

Proposition 3.2 casts the government's problem in marginal terms and characterises the optimal regulation intensity through an intuitive comparison of marginal benefit and marginal cost of data minimisation. The marginal benefit MB_i^h decreases with tighter regulation g_i, as disutility from firm data usage is convex, so that consumers derive diminishing gains from further restrictions. The marginal cost MC_i^h is an increasing function of g_i because reduced data usage lowers firm productivity and raises prices. As regulation tightens, this loss becomes more severe, reflecting the growing difficulty Tech firms face in maintaining efficiency when data is scarce.

Given an horizon $h \in \{s, \ell\}$, the unique optimal policy g_i^h balances these forces. In the interior case, the optimum is found at the unique point where marginal benefit equals

[17] As heterogeneity in privacy sensitivity is not central to the benchmark analysis, assuming symmetric disutility does not affect the main insights.

marginal cost. Otherwise, corner solutions arise naturally given the compactness of the choice set. No regulation is optimal when even a small restriction results in a cost that outweighs the privacy gain. Full regulation is optimal when privacy gains remain dominant throughout the range of admissible policies.

The preceding discussion abstracts from relocation effects to focus on the common structure of optimal regulation across decision modes. We now turn to the interesting question of how the internalisation of these general equilibrium equilibrium forces reshapes the planner's incentives to regulate.

Proposition 3.3 (Regulation-Retention Trade-off)**.** *Suppose country i is initially unregulated, $g_i^0 = 0$. At any interior solution to Problem (41), the optimal regulation levels in the short and long run coincide if and only if enforcement abroad is perfect:*

$$g_i^\ell = g_i^s \quad \iff \quad \pi_i = 1. \tag{43}$$

Any limit on extraterritorial reach creates a wedge in regulatory incentives by raising the marginal cost and lowering the marginal benefit of regulation. Accordingly, optimal long-run regulation is strictly weaker than in the short run if interior whenever $\pi_i \in (0,1)$:

$$\pi_i < 1 \quad \implies \quad MB_i^\ell(g_i) < MB_i^s(g_i), \quad \textit{and} \quad MC_i^\ell(g_i) > MC_i^s(g_i), \tag{44}$$

$$\implies \quad g_i^\ell < g_i^s, \quad \textit{whenever } g_i^s \textit{ or } g_i^\ell \textit{ belongs to } (0, \bar{g}_i). \tag{45}$$

Assumptions. 1, 2 and 3.
Proof. See Appendix C.8. □

Proposition 3.3 introduces a second distortion in the design of optimal privacy regulation that arises specifically in open economies. Whereas Proposition 3.1 showed that forward-looking governments tend to limit the scope of protection by excluding foreign consumers, the present result reveals that they also adopt systematically weaker regulation whenever domestic rules cannot be enforced symmetrically across foreign firms.

In other words, trade openness and limited extraterritorial reach interact to endogenously depress regulatory ambition in privacy matters. Because regulatory power is territorially anchored, any uniform regulation disproportionately burdens domestic establishments, which gradually erodes their position in the local market as productivity differentials lead consumers to shift consumption towards more efficient foreign competitors.

As illustrated in Figure 4, the resulting relocation forces induce a wedge between short- and long-run incentives to regulate. On the marginal cost side, the decline in domestic firm presence compounds the direct productivity effect on prices with an indirect terms-of-trade deterioration, as the share of domestic Tech in the consumption bundle shrinks. On the marginal benefit side, relocation reduces the effective reach of regulation, as any tightening applies to a shrinking set of domestic firms, replaced by foreign producers that are only partially subject to national rules. Anticipating this replacement, the planner optimally trades off regulation intensity to *retain* both production scale and preserve oversight.

(a) Perfect enforcement $\big(\pi_H = 1,\ h \in \{s, \ell\}\big)$ (b) Imperfect enforcement $\big(\pi_H \in (0,1)\big)$

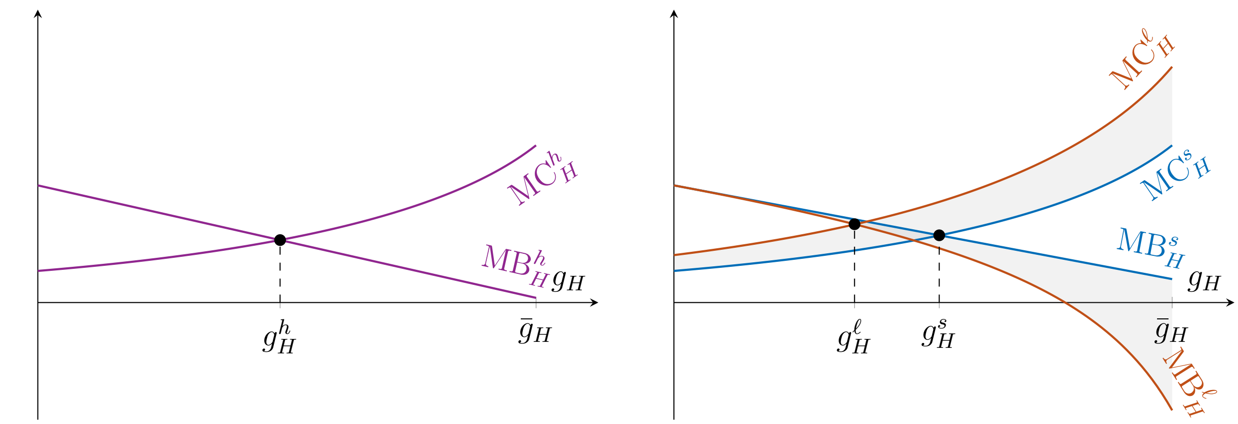


Figure 4: Imperfect extraterritorial reach drives a wedge in regulatory incentives

Note: The left panel illustrates the case of perfect extraterritorial enforcement ($\pi_H = 1$), where short- and long-run regulatory incentives are aligned for the Home government. Because regulation applies symmetrically to all firms in their interactions with consumers in the Home market, it does not distort competition or firm location decisions. The right panel depicts the case of imperfect enforcement abroad ($\pi_H \in (0,1)$), where domestic rules cannot be fully imposed on Foreign firms. Internalising firm relocation in response to asymmetric enforcement, the planner downweights the benefits of privacy regulation and places greater weight on its costs. As a result, the optimal regulation level is strictly lower in the long run.

The welfare implication is that realistic extraterritorial frictions do not merely generate direct privacy losses through weak compliance by foreign firms, but also induce additional, indirect losses by dampening the planner's incentive to regulate in the first place.

As illustrated in Figure 5, the forward-looking government, indexed by ℓ, optimises on a deflated welfare locus that incorporates the relocation effects of regulation. Under imperfect enforcement, this planner selects a suboptimally low level of regulation in the short run as a second-best strategy to balance privacy protection with the retention of Tech activity, thereby maximising welfare in the long-run equilibrium. By contrast, a short-sighted planner would adopt stricter regulation without anticipating these adjustments, only to incur significant welfare losses as the spatial distribution of firms shifts in response.

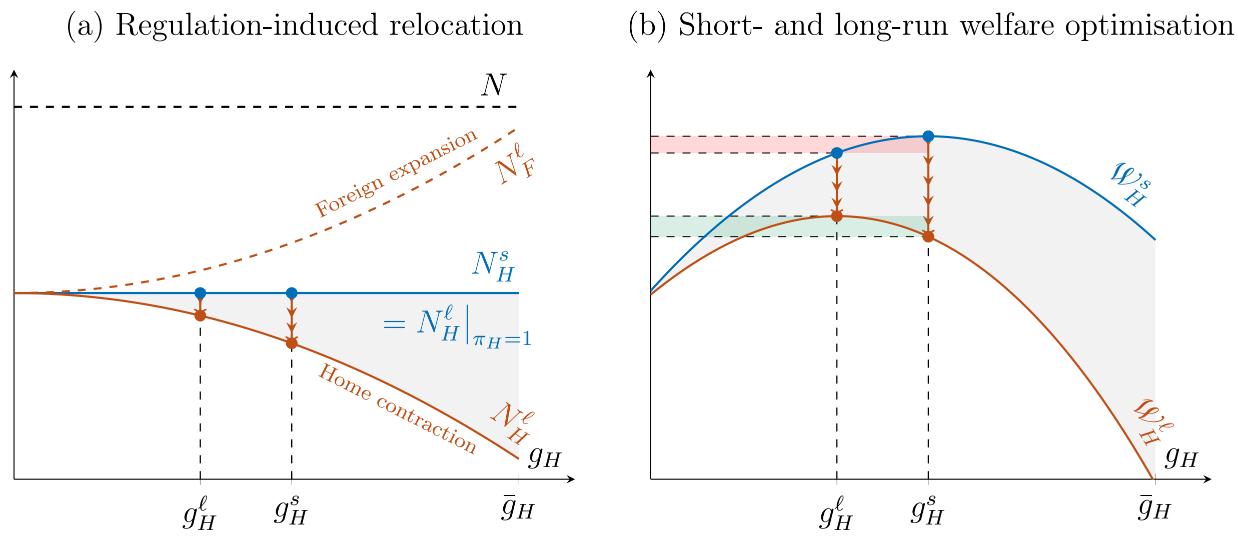


Figure 5: Regulation-retention trade-off in privacy regulation when enforcement abroad is imperfect

Note: The left panel shows how privacy regulation in Home affects the spatial distribution of Tech activity. Starting from a symmetric initial equilibrium, the blue line plots the mass of firms in Home in the short run, which remains constant by assumption and also coincides with the long-run outcome under perfect enforcement, where regulation applies symmetrically to all firms. By contrast, the red line shows the long-run firm mass under imperfect enforcement ($\pi_H \in (0,1)$). As shown in Appendix (see Lemma 1), domestic Tech presence is a decreasing and strictly concave function of regulation when extraterritoriality is imperfect. Accordingly, each additional tightening triggers increasingly larger firm exits from Home. Since the total number of firms in the world economy is fixed (see Proposition 2.4), this decline implies a one-for-one expansion of the Foreign Tech sector. The right panel shows the short-run and long-run welfare loci, with the gray shaded area highlighting the wedge induced by these relocation effects. The long-run locus lies strictly below the short-run one, reflecting indirect welfare losses from the inability to prevent domestic Tech contraction for any strictly positive g_H. A short-sighted Home government would adopt stricter regulation that raises welfare in the short run but leads to substantial losses as equilibrium adjusts. By contrast, a forward-looking planner internalises these effects and opts for a suboptimally low level of regulation, accepting weaker data protection in exchange for long-run welfare savings from retaining domestic industry.

Privacy regulation has increasingly extended beyond national borders, and the challenges of enforcing extraterritorial provisions are well documented, as discussed in Appendix A.2.1. Strengthening enforcement has become a central focus of recent developments in data governance. The preceding analysis shows that effective enforcement is necessary not only to achieve greater welfare gains in the short run, but also to preserve those gains over time. In the absence of stronger enforcement or international coordination, there is a genuine concern that such frictions may constrain long-term regulatory ambition in privacy matters.

3.1.2 Strategic Policy-Making in the Uncoordinated Privacy Regulation Game

Game specification We now consider the strategic case in which forward-looking governments simultaneously choose the intensity of privacy regulation within a residency-based scope. The purpose of this extension is to examine how the retention incentives identified in the study of optimal policy-making interact across jurisdictions when each government internalises the long-run consequences of its policy in a non-cooperative setting.

Notice that Problem (41) already characterises the relevant optimisation problem for each government $i \in \mathcal{C}$, once the data usage caps concerning foreign residents are understood as determined by the other country's regulatory choice. In particular, $\bar{x}_{ij}$ and $\bar{x}_{jj}$ are pinned down by the policy g_j implemented in country $j \neq i$, given enforcement capacity π_j.

$$\bar{x}_{jj} = 1 - g_j \quad \text{and} \quad \bar{x}_{ij} = 1 - \pi_j g_j, \quad \text{with} \quad g_j \in [0, \bar{g}_j]. \tag{46}$$

Hence, all preceding results extend to the strategic setting, and the strict concavity of the government's problem established in Proposition 3.2 directly yields the following result.

Proposition 3.4 (Existence of a Best Response Function in Privacy Regulation)**.** *Let $\bar{x}_{jj}(g_j)$ and $\bar{x}_{ij}(g_j)$ as in (46). Then, Problem (41) gives rise to a continuous best response function that maps any regulation level in country j, g_j, to a unique optimal data usage policy $\mathcal{B}_i(g_j)$.*

$$\mathcal{B}_i : [0, \bar{g}_j] \mapsto [0, \bar{g}_i] \tag{47}$$

Assumptions. 1, 2, 3, 4, 5, 6 and 7.
Proof. See Appendix C.9. □

Nash equilibrium A Nash equilibrium of the privacy regulation game is a profile of regulation choices $(g_i^*)_{i \in \mathcal{C}}$ such that each government's policy is optimal given the others'.

$$g_i^* = \mathcal{B}_i(g_j^*) \quad \text{for each } i \in \mathcal{C},\ j \neq i \tag{48}$$

Solving Problem (41) numerically for each value of the opponent's policy, it is possible to approximate the best response function $\mathcal{B}_i(g_j)$. A Nash equilibrium is then found at the intersection of the two best response curves in the g_i, g_j-plane.

When analysing the game analytically, we restrict attention to symmetric equilibria, under the assumption that countries are identical in all structural parameters, including enforcement capacity π_i. This simplification enables a fixed-point characterisation of equilibrium policy in terms of a single variable, g^*. In some cases, we examine local deviations from symmetry by perturbing a structural parameter for one country while holding it fixed in the other. The following proposition establishes existence and uniqueness of such a symmetric equilibrium.

Proposition 3.5 (Existence and Uniqueness of a Symmetric Nash Equilibrium in Privacy Regulation)**.** *There exists a unique symmetric Nash equilibrium policy* $g^* \in [0, \bar{g}]$ *satisfying the fixed-point condition:*

$$g^* \triangleq \mathcal{B}(g^*) \tag{49}$$

Assumptions. 1, 2, 3, 4, 5, 6 and 7.
Proof. See Appendix C.10. □

3.1.2.1 Imperfect Extraterritorial Enforcement

Enforcement capacity and regulatory ambition Proposition 3.3 showed that limited extraterritorial reach systematically weakens regulatory incentives for a long-term welfare maximiser, driven by the internalisation of relocation effects. This *regulation–retention trade-off* is a distinctive feature of open economies and points to a source of inertia in privacy regulation, as countries may compete for Tech specialisation. We now turn to a formal analysis of implications for equilibrium regulatory patterns.

Proposition 3.6 (Effect of Extraterritorial Enforcement on Best Responses)**.** *The best response mapping* $\mathcal{B}_i^*(\cdot\,;\pi_i)$ *is increasing in country i's enforcement capacity* π_i*, in the pointwise order. For any policy* $g_j \in [0, \bar{g}_j]$ *implemented by country j, we have:*

$$\pi_i < \pi_i' \leq 1 \quad \Longrightarrow \quad \mathcal{B}_i^*(g_j\,;\pi_i') > \mathcal{B}_i(g_j\,;\pi_i), \quad \textit{if}\ \mathcal{B}_i(g_j\,;\pi_i) \in (0, \bar{g}_i).$$

Assumptions. 1, 2, 3, 4, 5, 6 and 7.
Proof. See Appendix C.11. □

Proposition 3.6 formalises that the regulation–retention trade-off induces a monotonic relationship between enforcement capacity and regulatory ambition. Everything else equal, a government facing tighter enforcement constraints must have a best response function that lies everywhere below that of a government with stronger extraterritorial reach. In this sense, limited enforcement not only constrains the effectiveness of regulation but also lowers the set of policies that a government may rationally sustain in equilibrium.

Underlying this complementarity is the fact that the mass of domestic firms is a decreasing, concave function of the regulation level, with a slope that becomes steeper as π_i falls.[18] Intuitively, the capacity parameter π_i is inversely related to the degree of regulatory asymmetry. A lower value of π_i thus magnifies the relocation effect of any given tightening.

[18] See Lemma 1.

While the preceding result characterises how unilateral incentives vary with own enforcement capacity, it does not suffice to deduce the Nash response. This outcome reflects the full structure of strategic interdependence, including how a shift in one country's enforcement capacity alters the other's best response. For instance, an increase in Home's capacity allows it to retain more firms at any regulation level, which in turn reshapes Foreign's incentives by shifting the relocation implications of its own policy choices. We now show that these *a priori* non-trivial interactions do not prevent the qualitative comparative statics from carrying over to the symmetric Nash equilibrium.

Proposition 3.7 (Weak Extraterritorial Reach Crowds Out Privacy)**.** *In the symmetric formulation of the uncoordinated privacy regulation game, the interior Nash equilibrium regulation level $g^*(\pi)$ is strictly increasing in extraterritorial enforcement capacity π:*

$$\pi' > \pi \quad \Longrightarrow \quad g^*(\pi') > g^*(\pi). \tag{50}$$

In the limiting case of perfect enforcement ($\pi = 1$), best response functions are constant and the game becomes non-strategic. Each government's optimal regulation level converges to the unique policy that solves the country's privacy–consumption trade-off in Problem (41) for each horizon $h \in \{s, \ell\}$, regardless of the other's policy g_j:

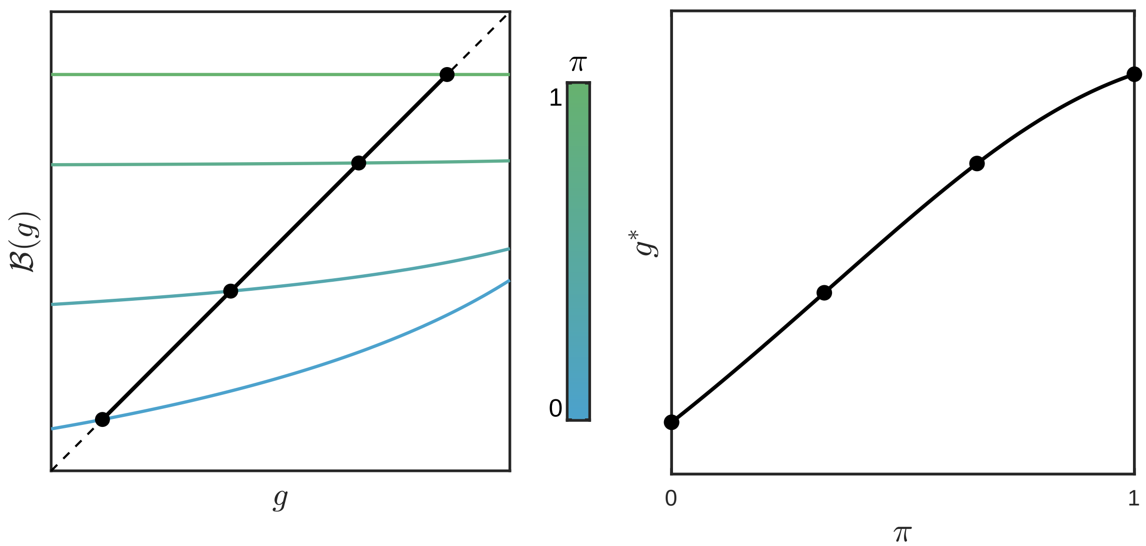
$$\mathscr{B}_i(g_j; \pi = 1) = g_i^{\ell*} = g_i^{s*} \quad \textit{for all } g_j \in [0, \bar{g}_j]. \tag{51}$$

Assumptions. 1, 2, 3, 4, 5, 6 and 7.
Proof. See Appendix C.12. □

Proposition 3.7 formalises how trade openness distorts privacy regulation patterns by linking regulatory incentives to countries' limited capacity to bind foreign firms. Figure 6 complements this result by illustrating, through a calibrated example, how strategic distortions shape the Nash equilibrium as enforcement capacity varies for both symmetric countries.

The knife-edge case of perfect enforcement offers a useful benchmark. When all consumer–firm interactions can be regulated symmetrically, the privacy–consumption trade-off mirrors that of a closed economy. Given local preferences for privacy, each government can implement its preferred policy uniformly across all firms serving its residents, as if regulatory authority were jointly exercised. Since all firms face identical regulatory constraints, the spatial distribution of Tech remains unchanged, and privacy regulation is decoupled from strategic considerations.

This strategic independence breaks down under imperfect enforcement abroad. Since regulatory authority is territorially bounded, governments cannot enforce domestic privacy rules on foreign firms processing resident data abroad with equal effectiveness. As a result, uniform standards tend to constrain domestic firms more stringently. As regulation tightens, this asymmetry distorts the spatial allocation of Tech activity, and governments' internalisation of this distortion tends to crowd out regulatory incentives. The unfolding of these strategic interactions gives rise to a monotonic relationship between the difficulty of subjecting foreign firms to domestic data usage standards and the level of privacy sustained in equilibrium.



(a) Symmetric equilibrium determination (b) Nash equilibrium as a function of π

Figure 6: Weak extraterritorial reach crowds out privacy in the symmetric Nash outcome

Note: This figure presents results from a numerical simulation of the symmetric policy game under varying levels of enforcement capacity π. The left panel plots the best response function of a representative government as a function of the regulation level set by its symmetric opponent. Symmetric Nash equilibria correspond to intersections with the 45-degree line, marking fixed points of the best response mapping. At these points, neither government has an incentive to deviate. As extraterritorial enforcement weakens for both countries, best responses shift downward. The right panel traces the resulting Nash equilibrium regulation level $g^*(\pi)$ as a function of π. As extraterritorial reach improves, relocation effects weaken and higher privacy standards become sustainable. At low π, regulation becomes more conditional on foreign policies, and under-regulation emerges as the equilibrium outcome.

Figure 6 also highlights a feature of the game that numerical simulations suggest is robust across the parameter space. While best response functions are flat under perfect extraterritorial enforcement, they tend to slope upward as enforcement weakens. In other words, limited enforcement capacity gives rise to strategic complementarities in privacy regulation. The broad intuition is that, while a domestic tightening shifts activity abroad and thereby dampens regulatory incentives, stronger foreign standards exert the opposite effect by drawing Tech activity inward. As a result, comparable foreign regulation becomes a key determinant of unilateral choices. Yet, in the absence of coordination, these complementarities fail to yield welfare-improving outcomes and instead depress privacy standards globally.

We conjecture that it would be possible to derive sufficient conditions under which the symmetric best response function is weakly increasing, with a slope that rises as enforcement capacity π decreases. However, this property is not required to characterise the symmetric Nash equilibrium, and a formal derivation would involve substantial algebraic complexity due to intricate interactions in regulatory incentives, with limited additional insight. For completeness, we summarise the outcome of our exploration of the underlying mechanism in the following proposition, which characterises how the slope of the best response function depends on the relative sensitivity of marginal regulatory costs and benefits[19].

Proposition 3.8 (Slope of the Symmetric Best Response Function)**.** *In the symmetric privacy regulation game with mild extraterritorial reach ($\pi < 1$), a change $\mathrm{d}g_j > 0$ shifts both the marginal benefit $MB_i^\ell(\cdot\,; g_j)$ and marginal cost $MC_i^\ell(\cdot\,; g_j)$ upward in the point-wise order. The slope of the best response function $\mathscr{B}_i(\cdot)$ therefore depends on the relative sensitivity of these two forces.*

Assumptions. 1, 2 and 3.
Proof. See Appendix C.13. □

Where relevant, we may invoke strategic complementarity in privacy regulation as a feature of our benchmark calibration and a robust regularity observed across numerical simulations to reflect on its implications. In such cases, we refer to the following assumption.

Assumption 8 (Weak Strategic Complementarities in the Symmetric Game)**.** *When $\pi \in (0,1)$, the partial effect of foreign regulation g_j on the marginal benefit of domestic regulation MB_i^ℓ always exceeds its effect on the marginal cost MC_i^ℓ:*

$$\pi \in (0,1) \quad \Longrightarrow \quad \frac{\partial MB_i^\ell(g_i; g_j)}{\partial g_j} > \frac{\partial MC_i^\ell(g_i; g_j)}{\partial g_j}, \quad \forall\, (g_i, g_j) \in [0, \bar{g}]^2 \tag{52}$$

Enforcement capacity and welfare Proposition 3.7 show that limited extraterritorial reach systematically lowers equilibrium privacy standards in the symmetric Nash equilibrium. Symmetry also makes the resulting welfare losses transparent, as we now establish formally.

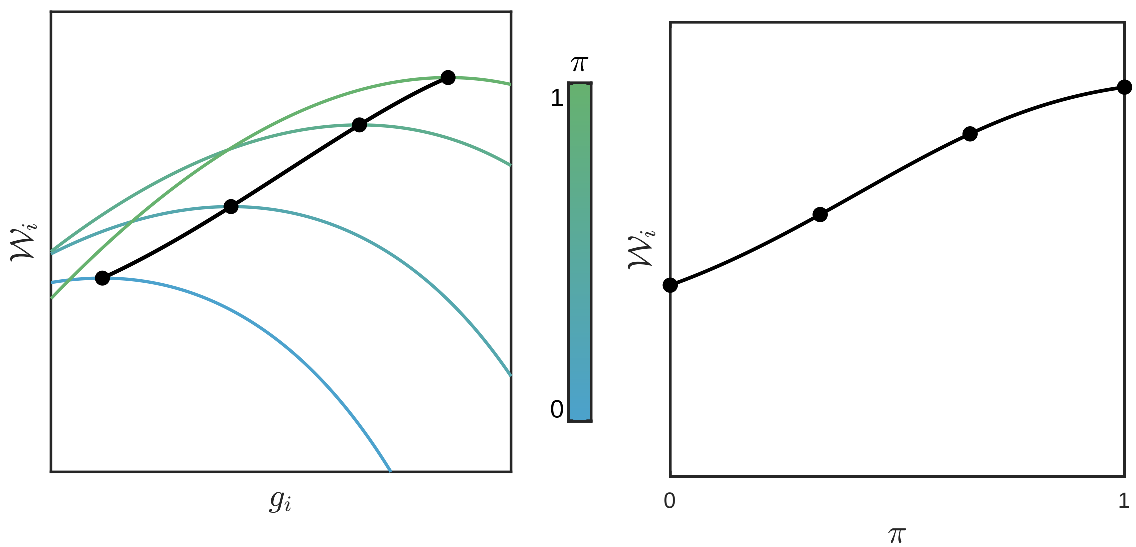
[19] A Foreign regulatory expansion has ambiguous implications for Home's regulatory incentives when extraterritorial enforcement is imperfect ($\pi < 1$). By crowding in Tech activity, it raises the marginal benefit of regulation, as Home firms account for a larger share of consumer interactions and thus privacy costs. At the same time, the marginal cost increases point-wise, as greater domestic activity enhances regulatory efficiency and amplifies the extent to which a given regulation level affects equilibrium prices. In our benchmark calibration, the former effect dominates the latter.

Proposition 3.9 (Weak Extraterritorial Reach Crowds Out Welfare)**.** *Stronger extraterritorial reach π strictly raises welfare at any interior symmetric Nash equilibrium.*

$$\pi' > \pi \quad \Longrightarrow \quad \mathscr{W}_i\big(g^*(\pi'); \pi'\big) > \mathscr{W}_i\big(g^*(\pi); \pi\big), \quad \textit{if } g^*(\pi) \in (0, \bar{g}). \tag{53}$$

Assumptions. 1, 2, 3, 4, 5, 6 and 7.
Proof. See Appendix C.14. □

The intuition underlying Proposition 3.9 follows directly from Proposition 2.5, which demonstrated that the total number of Tech firms active in the world economy is fixed and independent of data governance. At any symmetric Nash equilibrium, each country must therefore host the same number of firms. Yet governments choose suboptimally low regulation levels as they internalise relocation effects, and do so even more markedly as π falls, In equilibrium, welfare increases monotonically with π, as illustrated in Figure 7.



(a) Welfare loci (b) Equilibrium welfare as a function of π

Figure 7: Weak extraterritorial reach crowds out welfare in the symmetric Nash equilibrium

Note: The left panel shows the welfare loci of government i for different levels of enforcement π, when the other country plays the Nash equilibrium policy. When enforcement is weak, the locus attains a lower maximum at a smaller value of g_i, leading to under-regulation and lower equilibrium welfare. As π increases, the maximum attainable rises and the locus shifts rightward, enabling the world economy to sustain stronger privacy protection. The right panel traces the resulting equilibrium welfare as a function of π, which increases monotonically with enforcement capacity.

The gains from cooperation are particularly clear when extraterritorial enforcement is imperfect. Both countries would benefit from mutualising their regulatory capacity by extending protections to non-domestic consumers. With symmetric preferences, the structure of such an arrangement is straightforward: if both governments had full oversight over all firms, they would choose the same regulation level for all interactions involving their residents. With asymmetric preferences, the model could support cooperative schemes in which each country commits to enforcing the other's preferred standards, based on the optimal trade-off between privacy and real wages implied by consumer preferences. In both cases, each country would achieve the same welfare as if it had full regulatory reach over all firms.[20]

Such an arrangement would likely require binding commitments, since, as shown in Proposition 3.1, countries face strategic disincentives to protect foreigners through unilateral action. Still, several governments have expanded their privacy regimes territorially, applying domestic standards to all data processing by local establishments, regardless of consumer origin. As noted in Appendix A.2.1, many have done so to secure an EU adequacy decision under the GDPR, which permits data to flow freely to third countries whose privacy regimes are deemed 'essentially equivalent.' Our framework helps reinterpret these developments as institutional steps aimed at overcoming the inefficiencies of decentralised policymaking by promoting mutual recognition of standards and shared territorial regulatory capacity.

3.1.2.2 (Heterogeneous) Privacy Preferences

The privacy calculus is contextual. Beyond a general valuation of personal information, the perceived cost of data sharing reflects a subjective assessment of risk that depends on the specific interaction. Accordingly, privacy preferences are plausibly heterogeneous across consumers, and also correlated with firm characteristics. Moreover, these preferences are likely to be moving objects, shaped by individual experiences or broader narratives that affect the perceived terms of the trade-off.[21]

We extend the analysis in this direction by examining the strategic implications of a range of (possibly asymmetric) shocks to privacy preferences. In doing so, we have in mind two broad developments shaping data governance; namely, (i) the growing public awareness of digital privacy, and (ii) the increasing prominence of national security concerns.

(i) Unbiased preference shocks We define an *unbiased* preference shock as a uniform increase in privacy disutility applied by a consumer i to all firms, regardless of origin. Recall that, in the baseline privacy regulation game, consumers in country i apply an identical penalty ψ_i to data use by both domestic and foreign firms. An unbiased shock is thus one that preserves this symmetry. Our starting point is a nearly mechanical result that formalises the effect of such a shock on best response functions and the symmetric Nash equilibrium.

[20] In the asymmetric case, confirming the individual rationality of such an agreement would further require establishing that no country could attain higher welfare by acting unilaterally.

[21] For instance, Chen et al. (2021) show that Alipay users' privacy preferences evolve with platform use, with more active users increasingly likely to revoke data.

Proposition 3.10 (Effects of Unbiased Privacy Preference Shocks)**.** *The following hold:*

(i) For any $g_j \in [0, \bar{g}_j]$ such that $\mathcal{B}_i(g_j; \psi_i) \in (0, \bar{g}_i)$, the best response $\mathcal{B}_i(g_j; \psi_i)$ is strictly increasing in ψ_i, in the point-wise order.

$$\psi_i' > \psi_i \quad \Longrightarrow \quad \mathcal{B}_i(g_j\,; \psi_i') > \mathcal{B}_i(g_j\,; \psi_i) \quad \textit{if} \quad \mathcal{B}_i(g_j\,; \psi_i) \in (0, \bar{g}_i)$$

(ii) The symmetric equilibrium regulation level $g^*(\psi)$ is strictly increasing in ψ if interior.

$$\psi' > \psi \quad \Longrightarrow \quad g^*(\psi') > g^*(\psi) \quad \textit{if} \quad g^*(\psi) \in (0, \bar{g})$$

Assumptions. 1, 2, 3, 4, 5, 6 and 7.
Proof. See Appendix C.15. □

The intuition behind Proposition 3.10 is straightforward. At any interior solution, the optimal regulation level equates the marginal cost and marginal benefit of data minimisation. An unbiased increase in privacy disutility ψ_i raises the marginal benefit of regulation at every point but leaves the marginal cost unchanged, reinforcing the government's willingness to trade off consumption in favour of privacy. In unilateral terms, stricter privacy standards emerge for any given policy implemented abroad. In the symmetric case, where all governments respond similarly, mutual adjustments lead to a higher regulation level in the Nash equilibrium.

More interesting is the case where the preference shock is asymmetric, with one country experiencing a rise in privacy costs that breaks the *ex ante* symmetry and positions it as a first mover in regulatory innovation. One possible interpretation is that privacy becomes politically salient in that country, perhaps due to the politicisation of digital rights or a viral scandal. We now show how the model can account for patterns of regulatory spillovers discussed in Appendix A.2.1 through strategic complementarities in privacy regulation.

Proposition 3.11 (Asymmetric Unbiased Preference Shock and the Brussels Effect)**.** *Start from an interior symmetric equilibrium $g^*(\psi, \pi) \in (0, \bar{g})$, with $\pi \in (0, 1)$, and suppose that country i experiences an unbiased increase in privacy preferences to $\psi' > \psi$, while country j remains at ψ. Further assume that Assumption 8 holds, so that the best response functions $\mathcal{B}_k : [0, \bar{g}] \to [0, \bar{g}]$ are strictly increasing for each $k \in \{i, j\}$. Then, for small enough shocks, there exists an interior Nash equilibrium $(g_i', g_j') \in (0, \bar{g})^2$ of the modified game in which:*

(i) The privacy regulation level rises in country i:

$$g_i' > g^*. \tag{54}$$

(*ii*) *Strategic spillovers raise regulation in country j as well, though to a lesser extent:*

$$g_i' > g_j' > g^*. \tag{55}$$

(*iii*) *In the long run, Tech activity reallocates from country i towards country j:*

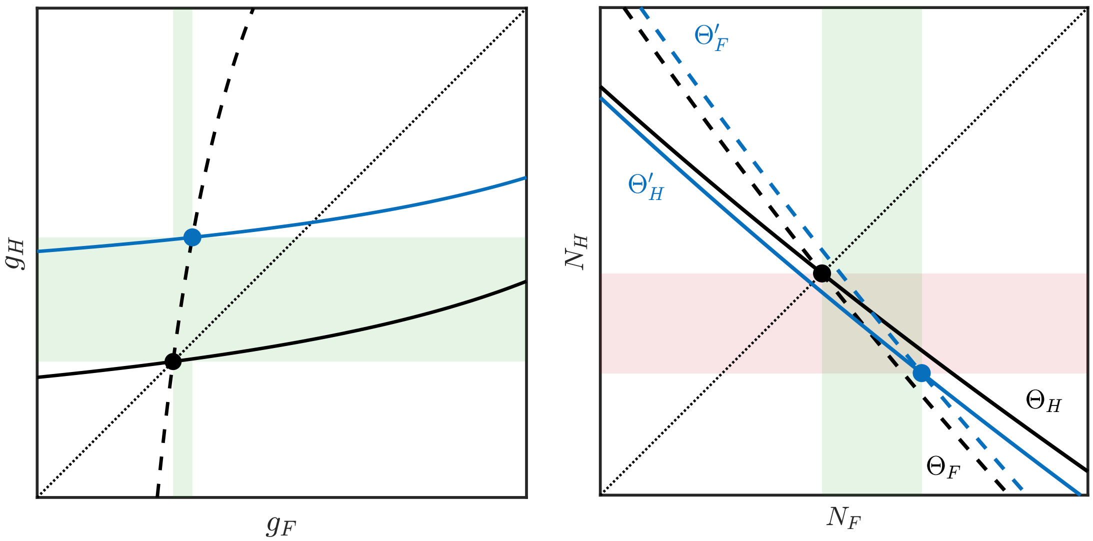
$$N_i^{\ell} < \frac{N}{2} < N_j^{\ell}. \tag{56}$$

Assumptions. 1, 2, 3, 4, 5, 6, 7 and 8.
Proof. See Appendix C.16. □

Proposition 3.11 formalises how a shift in privacy concerns within a single country can generate regulatory spillovers that raise privacy standards globally when privacy is underprovided in the baseline equilibrium. In doing so, it offers a theoretical foundation for observed developments in global data governance, highlighting the inertia of privacy regulation across borders. As discussed in Appendix A.2.1, the EU's adoption of the GDPR coincided with a worldwide expansion of privacy laws, which legal scholars have linked to the so-called 'Brussels effect' (Bradford 2020). While this literature has emphasised the innovative content of EU regulation, our model offers a complementary perspective, showing how such spillovers can arise endogenously from the economic incentives that sustain regulatory equilibria.

The mechanism behind this result is the possibility of strategic complementarities in privacy regulation when enforcement is territorially bounded. With such limited reach, government intervention entails a trade-off between the benefits of data minimisation and the costs of Tech sector contraction driven by regulatory asymmetry, via both reduced real wages and weakened oversight. While these forces tend to crowd out regulatory ambition in equilibrium, they may also generate positive regulatory spillovers when one country, prompted by an idiosyncratic rise in privacy preferences, accepts to shoulder greater economic costs. By proactively tightening its own privacy standards, this country alleviates the regulatory trade-off faced by others, making them more willing to strengthen their own regulations.

As illustrated in Figure 8, this complementarity feeds back onto the incentives of the initial mover, pushing the world economy towards a new Nash equilibrium with stricter regulation globally, even though preferences have shifted in only one country. The adjustment, however, is asymmetric. The initiating country, Home, remains structurally more privacy-conscious and finds that Foreign ultimately stops short of fully aligning with its stricter standards. As a result, Home bears the adjustment costs for the global economy, as the persistent regulatory gap induces a structural relocation of Tech activity towards Foreign. Foreign, by contrast, benefits unambiguously. Not only does it adopt stricter privacy standards *per se*, but the relocation is itself a vector of regulatory efficiency along the extensive margin, as the set of entities subject to full territorial enforcement widens. In addition, Foreign consumers also enjoy greater purchasing power through the terms-of-trade effect.



(a) Nash equilibrium adjustment

(b) Long-run firm masses adjustments

Figure 8: Unbiased privacy shock in Home drives positive regulatory spillovers in Foreign

Note: The figure illustrates the strategic and economic implications of an unbiased privacy preference shock in Home for the world economy (blue), starting from the symmetric equilibrium (black). The left panel shows how the new equilibrium is reached in the best response plane. The upward shift in Home's best response curve generates a sequence of best response updates that leads both countries to adopt stricter privacy regulations, although Home must ultimately outbid Foreign due to stronger concerns for privacy. The right panel shows the long-run implications of this persistent asymmetry for the spatial distribution of Tech activity. After adjustments, the symmetric equilibrium lies above the market-clearing locus that determines the number of firms in Home that can collectively sustain non-negative profits. As a result, Home's Tech sector must contract, and the new equilibrium is found by moving downward and to the right, as Foreign expands. The calibration matches that used in preceding figures, with privacy disutility initially set at $\psi = 5$ for both countries, and a shock raising Home's disutility to $\psi' = 10$. Extraterritorial reach is fixed at $\pi = 0.2$.

The model thus accounts for the empirical observation that, although the GDPR triggered a global expansion of privacy, it remains a gold standard, which more recent frameworks, while often inspired by it, often do not equal. This persistent gap is well documented in the legal literature, where it has become common practice to critically examine claims that new laws constitute 'local GDPRs' and to find that such characterisations often reflect political marketing rather than substantive regulatory convergence. (Greenleaf 2022)

(ii) **Biased Preference Shocks** We define a *biased* preference shock as a rise in privacy disutility selectively applied to foreign firms, leaving attitudes towards domestic data processing unchanged. While the preceding case captured a generalised shift in consumer sensitivity

to data use, we interpret the biased variant as reflecting a targeted mistrust of cross-border data flows, possibly triggered by geopolitical trends or broader cultural anxieties. Such a shock breaks the anonymity of origin embedded in the baseline penalty structure and induces a wedge in the perceived privacy cost of domestic versus foreign services.

Recall that the baseline model allows for location-specific privacy preferences indexed by trade dyad (4). In this context, ψ_{ii} denotes the penalty applied by consumers in country i to local firms, while ψ_{ji} captures the penalty applied to firms operating from country j. Since the existence and uniqueness of equilibrium regulation in Proposition 2.1 relied on the simplifying Assumption 6 whereby $\psi_{ii} = \psi_{ji} \equiv \psi_i$, we restrict attention to sufficiently small deviations from this benchmark. By continuity of the second derivative of the government's objective, global concavity is preserved in a neighbourhood of the origin-neutral case.

Assumption 9. *Consumer i's preference profile lies in a neighbourhood of the origin-neutral benchmark where Problem (41) remains strictly concave:*

$$(\psi_{ii}, \psi_{ji}) = (\psi_i,\, \psi_i + b_i), \quad b_i \in [0, \varepsilon). \tag{57}$$

Because it ultimately shifts a privacy cost, this case may initially appear as a minor variation of the previous one, with similar implications for strategic incentives. Yet, by relaxing the assumption of origin-neutrality, it brings into focus the role that homogeneity played in sustaining privacy standards. Our first result established how a biased preference shock may lead a government to regulate *less* in the presence of regulatory asymmetries.

Proposition 3.12 (Effect of Biased Privacy Preference Shocks)**.** *There exists a threshold $\pi^* \in (0,1)$ such that the following is true:*

(i) For any $\pi_i \in (0, \pi^*)$, and every $g_j \in [0, \bar{g}_j]$ such that $\mathcal{B}_i(g_j; b_i, \pi_i) \in (0, \bar{g}_i)$, the best response $\mathcal{B}_i(g_j; b_i, \pi_i)$ is strictly decreasing in b_i, in the point-wise order:

$$\pi_i \in (0, \pi^*), \quad b_i > 0 \quad \Longrightarrow \quad \mathcal{B}_i(g_j\,; b_i, \pi_i) < \mathcal{B}_i(g_j\,; 0, \pi_i), \quad \textit{if } \mathcal{B}_i(g_j\,; b_i, \pi_i) \in (0, \bar{g}_i).$$

(ii) For any $\pi \in (0, \pi^*)$, the symmetric equilibrium regulation level $g^*(b, \pi)$ is strictly decreasing in b if interior:

$$\pi \in (0, \pi^*), \quad b > 0 \quad \Longrightarrow \quad g^*(b, \pi) < g^*(0, \pi), \quad \textit{if } g^*(b, \pi) \in (0, \bar{g}). \tag{58}$$

Assumptions. 1, 2, 3, 4, 5, 7, and 9.
Proof. See Appendix C.17. ☐

Proposition 3.12 shows that stronger privacy concerns may paradoxically weaken regulatory ambition when specifically directed at foreign firms. Interestingly, while the expansion of standards under a uniform rise in privacy aversion reflects mechanical forces, the reversal identified here arises entirely from the endogenous costs of regulation in the model.

As foreign data processing becomes more salient in consumers' disutility, aggregate privacy costs rise in a biased manner. All else equal, the planner then faces two opposing incentives, whose balance depends on its ability to bind foreign firms effectively, π. On one hand, stronger privacy aversion raises the marginal value of regulation, pushing towards tighter standards. On the other hand, the bias heightens the welfare losses from crowding out domestic Tech due to regulatory asymmetries, weakening the case for high standards.

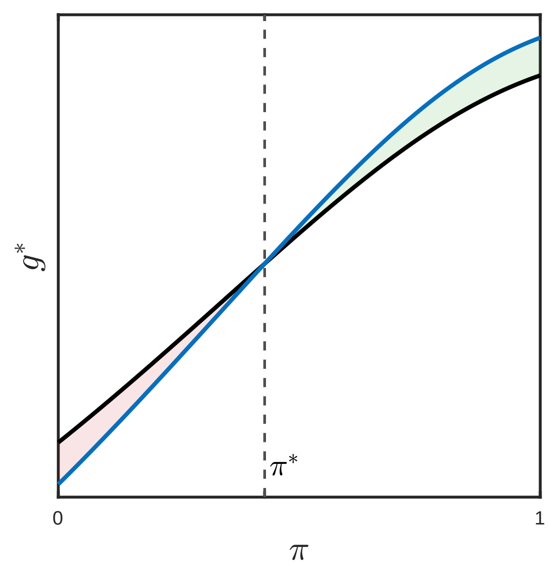


Figure 9: Biased preference shock induces dumping under mild extraterritorial reach.

Note: The figure shows the symmetric Nash equilibrium regulation level $g^*(b, \pi)$ as a function of enforcement capacity π, under origin-neutral preferences ($b = 0$, black) and with a foreign-bias penalty ($b = 1.5$, blue). In the presence of regulatory asymmetries, a rise in privacy aversion targeting foreign firms amplifies the welfare cost of sustaining high standards that primarily constrain domestic Tech. For $\pi < \pi^*$, this effect dominates the direct increase in the marginal value of regulation due to higher exposure disutility. In this region, governments undercut one another to regain Tech presence, triggering a race to the bottom in privacy standards. As enforcement improves, this effect fades and the privacy bias instead supports stricter regulation. In this simulation, a unique threshold $\pi^* \in (0, 1)$ marks the point around which the equilibrium locus rotates counter-clockwise.

As illustrated in Figure 9, there exists a critical threshold $\pi^* \in (0, 1)$ such that for all $\pi < \pi^*$, the second effect dominates. In this regime, symmetric best-response functions shift downward, triggering a strategic race to the bottom, as as each government undercuts the other's standards in an attempt to attract a larger share of Tech activity. Since symmetry ensures that both countries ultimately host the same number of firms, the resulting decline in privacy protection entails unequivocal welfare losses.

We now derive the implications for the world economy when the privacy bias shock is limited to a single country, under the assumption that best response functions display strategic complementarities.

Proposition 3.13 (Asymmetric Biased Preference Shock and the Race to the Bottom)**.** *Start from an interior symmetric equilibrium $g^*(0, \pi) \in (0, \bar{g})$, with origin-neutral privacy preferences and $\pi \in (0, \pi^*)$. Suppose country i experiences a biased increase in privacy concerns specifically directed at foreign firms, so that the consumer preference profile shifts to $(\psi, \psi{+}b)$ with $b > 0$, while country j remains at (ψ, ψ). Further assume that Assumption 8 holds, so that the best response functions $\mathscr{B}_k(\cdot; b', \pi) : [0, \bar{g}] \to [0, \bar{g}]$ are strictly increasing for each $k \in \{i, j\}$. Then, for small enough shocks, there exists a Nash equilibrium (g_i', g_j') of the modified game in which:*

(i) Despite higher privacy costs, regulation weakens in country i:

$$g_i' < g^*. \tag{59}$$

(ii) Strategic spillovers lower regulation in country j as well, though to a lesser extent:

$$g_i' < g_j' < g^*. \tag{60}$$

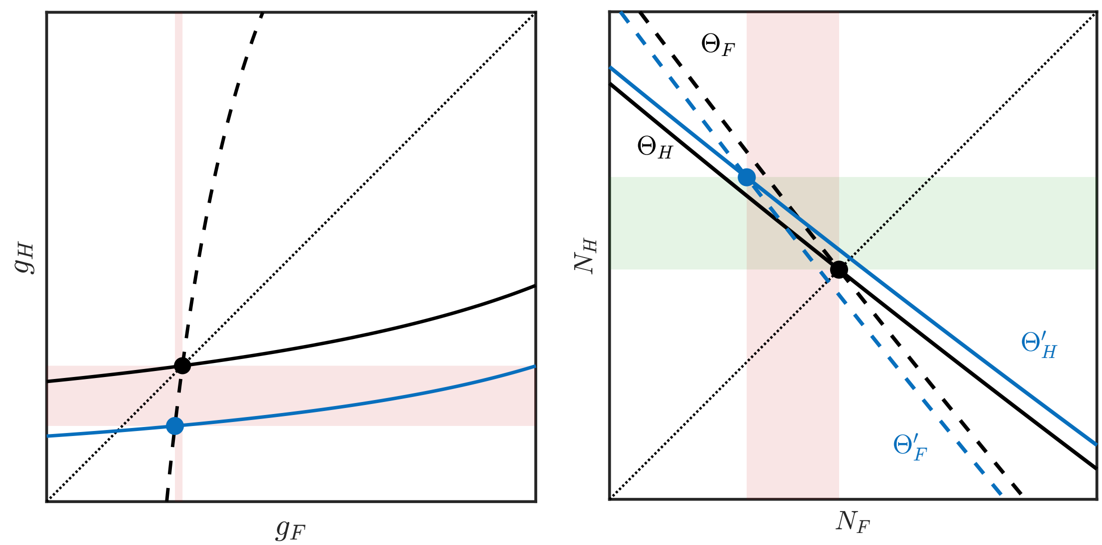
(iii) In the long run, Tech activity reallocates from country j towards country i:

$$N_i^\ell > \frac{N}{2} > N_j^\ell. \tag{61}$$

Assumptions. 1, 2, 3, 4, 5, 7, 9 and 8.
Proof. See Appendix C.18. □

Proposition 3.13 mirrors the regulatory spillover analysis of the preceding section to show how a localised mistrust shock, biased against foreign data processing, can propagate internationally and erode privacy standards worldwide in the presence of strong enough regulatory asymmetries.

Upon impact, the country experiencing the shock engages in regulatory dumping, as illustrated in Figure 10. When strategic complementarities are active due to imperfect enforcement, this distortion is transmitted abroad, prompting weaker standards in the other country as it reoptimises to stem the outflow of Tech activity. The result is a sequence of best response updates that transforms geographically-targeted privacy concerns into a self-reinforcing race to the bottom, driven by uncoordinated policymaking and leaving all countries worse off.



(a) Nash equilibrium adjustment

(b) Long-run firm masses adjustments

Figure 10: Biased privacy preference shock in Home crowds out privacy standards worldwide

Note: The figure illustrates the strategic and economic effects of a biased privacy preference shock in Home (blue), starting from the symmetric equilibrium (black). The left panel shows how the downward shift in Home's best response curve leads both countries to adopt weaker privacy regulations, despite stronger privacy concerns in Home. By selectively raising aversion to Foreign data processing, the bias creates incentives for the Home government to force Tech activity to relocate domestically through regulatory dumping. In response, Foreign loosens its own standards to narrow the policy gap and retain firms. The right panel shows the resulting long-run adjustments in the spatial distribution of firms. Policy changes shift the zero-profit loci in both countries, now intersecting northwest of the original Nash equilibrium. Home's stronger dumping gives its firms a regulatory edge, reducing Foreign firm profitability. As a result, the post-shock equilibrium features a higher number of firms operating in Home. Entry there drives down local rents, while exit in Foreign continues until all firms in the world economy once again break even. The simulation uses baseline parameters, with $b = 5$ in Home for readability.

This dynamic points to another role for international cooperation in sustaining regulatory ambition. It is plausible that uncoordinated policymaking not only weakens standards through relocation incentives, but also creates fertile ground for the emergence and spread of such mistrust shocks, thereby amplifying its welfare costs. Insecurities around foreign data processing do not arise in a vacuum, but are fuelled by the absence of credible frameworks for mutual enforcement or the recognition of equivalent standards. As privacy becomes entangled with broader narratives of sovereignty and control, governments may come to favour weaker rules that preserve domestic oversight and industrial presence over stronger regulation that risks hollowing out the local Tech sector. In this respect, the model suggests a possible role for international dialogue on data governance in re-establishing a shared basis of trust on which more ambitious privacy frameworks might be built.

3.2 Data Localisation

Our treatment of privacy regulation assumes a commitment to uniform standards across all digital interactions between consumers and firms, irrespective of firm origin. We have shown that, under territorially anchored authority, such a commitment can backfire. Domestic firms bear the full weight of compliance, while foreign competitors remain only partially constrained. Over time, this asymmetry erodes the domestic Tech sector and magnifies regulatory costs, including diminished regulatory efficiency and adverse terms-of-trade effects. These forces, in turn, undermine the government's incentive to sustain ambitious standards.

Unpacking these distortions points to a natural logic behind the adoption of discriminatory measures in data governance. If uniform regulation imposes structural disadvantages, what compels states to maintain it? Confronted with the challenge of sustaining both regulatory ambition and firm retention, governments may resort to origin-contingent measures, selectively tightening rules on foreign firms to compensate for limited oversight. As Goldfarb et al. (2018) put it, data localisation may function as a form of "privacy policy that could favour domestic firms." And once regulation begins to serve national advantage, the boundary between corrective intervention and strategic opportunism becomes increasingly blurred. If rules can be bent to contain a disadvantage, why not to secure a gain?

Such departures from non-discrimination become easier to pursue given the institutional vacuum surrounding cross-border data regulation. Existing trade frameworks offer limited grounds for legal challenge, while the dual nature of data as both economic and personal information allows states to invoke sovereign prerogatives with minimal friction. The USTR's repeated but ultimately unsuccessful efforts to contest China's Great Firewall under WTO provisions, discussed in Appendix A.2.2, are instructive in this regard. In such an environment, where binding commitments are absent, states are likely to adopt discriminatory regulation whenever it serves their interests.

This section examines the unilateral determinants of cross-border data handling regulation. We show that a broad range of objectives commonly associated with such policies can support arbitrarily large restrictions on foreign data handling. We further advance the hypothesis that short-run welfare constraints may curb these incentives in practice, particularly in more politically accountable settings, and explore the strategic implications when such constraints are asymmetric across countries.

Simplifying restrictions For tractability, we restrict attention to a setting in which domestic data usage standards are treated as exogenous. While the model could naturally be extended to allow governments to set distinct standards by firm location, such an extension would introduce a second dimension of policy choice and complicate the analysis without much additional insight. Hence, we fix domestic regulation at some level $\bar{x}_{ii}$ to permit a focused analysis of the incentives to impose targeted restrictions on foreign firms.

Recall that Assumption 3 already imposes a lower bound on any such domestic regulation of resident data handling to ensure equilibrium determinacy. It is further relevant to the present analysis to consider the following upper bound:

Assumption 10 (Optimality of Local Processing)**.** *Data usage in domestic firm-resident interactions is capped at a level* $\bar{x}_{ii}$ *such that:*

$$\bar{x}_{ii} < \sqrt{\frac{\psi_{ji}}{\psi_{ii}}} \cdot (1 - \pi_i) \tag{62}$$

Assumption 10 constrains domestic data usage standards to lie within a parameter range where local data processing is socially desirable, either due to more effective enforcement or politically salient consumer preferences. The latter is relevant to the analysis of data localisation incentives, as it echoes the stated motives commonly invoked by policymakers and, by construction, delineates the region of the parameter space in which such claims might be taken seriously. The model will ultimately allow us to isolate this channel from alternative drivers, such as real wage or industrial policy considerations.[22]

3.2.1 Determinants of Unilateral Discriminatory Incentives

Setup Consider a planner located in country i, who aims to regulate cross-border data flows in pursuit of a general policy objective $\mathbb{O}_i$. As in the previous section, we distinguish between the short run and the long run, and endow the planner with a policy horizon $h \in \{s, \ell\}$. In the short run ($h = s$), the planner selects the optimal policy conditional on the prevailing spatial distribution of Tech firms. In the long run ($h = \ell$), the planner adopts a forward looking perspective that accounts for general equilibrium adjustments, which we interpret as unfolding over time through dynamic firm entry and exit across countries. If F denotes the target that the planner seeks to optimise under horizon h, we thus write $\mathbb{O}_i = F_i^h$.

Contrary to the preceding analysis, the objective function $\mathbb{O}_i$ need not coincide with national consumer welfare. This more general formulation reflects the broader range of policy objectives commonly associated with cross-border data regulation. While privacy regulation is generally acknowledged to be a welfare-motivated intervention, data localisation is often linked to alternative or opaque motives beyond official government statements, including industrial development or strategic control over data, particularly for national security

[22] From a technical standpoint, Assumption 10 eliminates second-order welfare effects that are essentially specific to our one-dimensional reformulation. If domestic standards are highly permissive and consumers are nearly indifferent to data processing location, a local Tech expansion may reduce welfare through increased privacy exposure via a composition effect. Accordingly, this extensive-margin effect would lead a planner to internalise that privacy is better protected when firms operate from abroad for exogenous reasons.

purposes. In doing so, we also extend the analysis of data regulation to institutional environments in which welfare maximisation cannot be presumed to guide policymaking.

Government problem Given a policy objective $\mathbb{O}_i$, the government in country i selects a level of cross-border data restrictions $g_i \in [0,1]$ to solve:

$$\begin{aligned} \max_{g_i \in [0,1]} \quad & \mathbb{O}_i \left(\bar{x}_{ji}(g_i); \bar{x}_{ij}, \bar{x}_{ii}, \bar{x}_{jj} \right) \\ \text{s.t.} \quad & \bar{x}_{ji} \; = \; 1 - \pi_i g_i \\ & \pi_i \in (0,1), \; \bar{x}_{ii}, \; \bar{x}_{jj} \; \text{ and } \bar{x}_{ij}(g_j) \; \text{ given.} \end{aligned} \tag{63}$$

As in Section A.2.1, the parameter $\pi_i \in (0,1)$ denotes the extraterritorial enforcement capacity of country i, such that a policy targeting a restriction g_i on foreign firms' use of resident data results in an effective constraint of $\pi_i g_i$. The upper bound $g_i = 1$ corresponds to the intention to prohibit entirely the use of domestic data by foreign establishments.[23]

Optimality of maximal discrimination Building, in particular, on the monotonicity properties of comparative statics in Section 2.3, we now show that Problem (63) gives rise to non-satiable incentives to restrict foreign data usage across a broad class of objectives. Towards the formalisation of this result, we first introduce a set of policy-relevant metrics.

Recalling that Ω_k^h denote the set of Tech firms operating from country k under horizon $h \in \{s, \ell\}$, define the stock of data on residents of country i held by firms from country j as:

$$D_{ji}^h \triangleq \int_{\Omega_j^h} d_i(\omega)\, \mathrm{d}\omega. \tag{64}$$

Let $\hat{X}_i^h$ denote the net exports of Tech from country i, defined as the value of exports to country j net of shipment frictions, minus the value of imports from country j:

$$\hat{X}_i^h \triangleq \int_{\Omega_i^h} p_j(\omega) c_j(\omega)\, \mathrm{d}\omega - \int_{\Omega_j^h} p_i(\omega) c_i(\omega)\, \mathrm{d}\omega. \tag{65}$$

We also recall the expression for consumer surplus CS_i^h, and define producer surplus PS_i^h in country i as the aggregate profits of locally established Tech firms:

$$CS_i^h \triangleq \left(\frac{w}{P_i^h} \right)^{\alpha} L_i, \qquad PS_i^h \triangleq \int_{\Omega_i^h} \Pi(\omega)\, \mathrm{d}\omega. \tag{66}$$

[23] Since $\bar{x}_{ii}$ is exogenous, it suffices to choose it large enough that Assumption 3 holds for a world equilibrium to exist. Moreover, $\pi_i \in (0,1)$ ensures that $\bar{x}_{ji} > 0$ for all $g_i \in [0,1]$, so all expressions are well-defined and the objective is continuous on a compact domain. No further restriction is required.

Proposition 3.14 (Multiple Rationales for Maximal Cross-Border Restrictions)**.** *Suppose that government i solves Problem (63) to pursue any one of the following policy objectives:*

(*i*) *Maximisation of short-run producer surplus or rents:* $\mathcal{O}_i = PS_i^s$;

(*ii*) *Minimisation of foreign-held resident data, in the short- or long-run:* $\mathcal{O}_i = -D_{ji}^h$, $\forall h \in \{s, \ell\}$;

(*iii*) *Maximisation of net Tech exports, in the short- or long-run:* $\mathcal{O}_i = \hat{X}_i^h$, $\forall h \in \{s, \ell\}$;

(*iv*) *Maximisation of the long-run number of locally established Tech firms:* $\mathcal{O}_i = N_i^\ell$;

(*v*) *Maximisation of long-run consumer surplus:* $\mathcal{O}_i = CS_i^\ell$;

(*vi*) *Maximisation of long-run national welfare, under Assumption 10:* $\mathcal{O}_i = \mathcal{W}_i^\ell$.

Then, the government's objective is strictly increasing in the policy instrument g_i. Accordingly, the optimal policy is always corner and corresponds to a full restriction on the cross-border processing of resident data:

$$\frac{\mathrm{d}\mathcal{O}_i}{\mathrm{d}g_i} > 0 \quad \textit{for all} \;\; g_i \in [0,1] \quad \Longrightarrow \quad g_i^* = 1. \tag{67}$$

Assumptions. 1, 2, 3, 4, 5, and 10 if $\mathcal{O}_i = \mathcal{W}_\ell^s$.
Proof. See Appendix C.19. □

Proposition 3.14 establishes that maximal discrimination is uniquely optimal under a wide range of government objectives commonly associated with cross-border data restrictions.

As discussed in Section 2.3, restrictions that specifically target data handling by foreign establishments generate both a direct effect on data collection and a permanent depreciation of those firms' market potential. The latter operates through general equilibrium forces, which we interpret as unfolding gradually via firm entry and exit decisions. Specifically, reduced productivity of foreign competitors in the local market shifts demand towards domestic producers. This substitution is reflected in opposite movements along the zero-profit loci, inducing losses abroad and excess profits at home that must be eventually dissipated through domestic Tech expansion, with beggar-thy-neighbour implications for the trade partner.

It is notable that the class of objective functions for which these dynamics generate strictly monotonic incentives to discriminate is not defined by any particular normative criterion. The immediate substitution effect may benefit a government seeking to maximise domestic rents, or one motivated by national security concerns about foreign data collection. What unites these otherwise divergent aims is their shared capacity to mobilise the same underlying economic forces in a manner that systematically advances the planner's objective. In the case of long-run welfare, for example, maximal discrimination may prove triply optimal: it raises consumer surplus through improved terms of trade, reduces privacy costs along the intensive margin, and, under Assumption 10, improves the composition of data use along the extensive margin by favouring locally processed data.

Two distinct connections emerge between this result and contemporary discussions on data localisation, as surveyed in Appendix A.2.2.

First, the multiplicity of objectives that can rationalise aggressive restrictions on outbound data flows may help explain both the growing diffusion of such policies and the diversity of their adopters. While some covariates (e.g. regime type, regulatory tradition, or industrial orientation) are correlated[24], much of the policy literature has sought to organise this heterogeneity by focusing on variation in implementation. For instance, the *data models* of Ferracane et al. (2021) highlight the arbitrariness of exemptions, the role of licensing regimes, and the degree of procedural transparency as key distinctions between Chinese and European regulatory approaches. What our analysis suggests is that such institutional features may be largely orthogonal to the deeper economic forces at work.

Second, the result underscores a deeper challenge for international cooperation, especially amid ongoing efforts to strengthen global disciplines on cross-border data regulation. When identical regulatory outcomes can stem from fundamentally different policy objectives, *ex post* rationalisation of intent becomes structurally fragile. Rather than a transient pathology pending updates to trade rules, suspicions of strategic manipulation are likely to remain a stable and enduring feature of global trade discussions on data governance.

The moderating role of political constraints While data localisation has become increasingly widespread, near-total restrictions on the cross-border processing of resident data remain rare. Outside a handful of non-democratic regimes, few states implement full-scale discrimination, even in the absence of binding international commitments. What limits the unilateral pursuit of such policies? We now argue that short-run political accountability (e.g. electoral cycles, institutional checks, or public opinion) is sufficient to curb the escalation of restrictions, even when the pursuit of the government's objective would otherwise favour maximal restrictions against foreign firms.

To formalise this moderating effect, suppose that the government pursues some general objective $\mathcal{O}_i$, but can only implement those reforms that do not reduce short-run welfare relative to the status quo. Letting $g_i^0 \in [0, 1)$ denote the initial level of restrictions on foreign firms' access to resident data in the prevailing equilibrium, the feasibility constraint writes:

$$\mathcal{W}_i^s(g_i) \geq \mathcal{W}_i^s(g_i^0). \tag{68}$$

[24]See Ferracane et al. (2018) for an investigation of cross-country variation in outbound data restrictions, suggesting associations with ICT readiness, regulatory assertiveness, and constraints on civil liberties.

Our next result establishes that, for sufficiently bounded aversion for foreign data processing, this political constraint binds at an interior point whenever the government would otherwise have non-satiated incentives to tighten restrictions.

Proposition 3.15 (Short-Run Welfare Constraints Curb Discrimination)**.** *Suppose that the government in country i pursues any of the policy objectives listed in Proposition 3.14, and consider augmenting Problem (63) with the political feasibility constraint (68). Then, for any reference policy level $g_i^0 \in [0,1)$, there exists a threshold $\psi_{ji}^* > 0$ such that, whenever $\psi_{ji} < \psi_{ji}^*$, the political feasibility constraint binds necessarily:*

$$0 < \psi_{ji} < \psi_{ji}^* \quad \Longrightarrow \quad g_i^* \in [0,1) \quad \textit{with} \quad \mathcal{W}_i^s(g_i^*) = \mathcal{W}_i^s(g_i^0). \tag{69}$$

Assumptions. 1, 2, 3, 4, 5, and 10 if $\mathcal{O}_i = \mathcal{W}_i^\ell$.
Proof. See Appendix C.20. □

Proposition 3.15 shows that political accountability, proxied by short-run welfare constraints, can curb the escalation of cross-border data restrictions when, absent this constraint, the government's objective would support maximal discrimination.

The mechanism is intuitive. Restrictions on the cross-border handling of data by foreign firms reduce their efficiency and raise prices for domestic consumers, who bear the full pass-through of the resulting marginal cost increase. With product differentiation, substitution towards local producers is inherently limited, and consumer surplus declines immediately. Whether incremental tightening remains politically viable then depends on the strength of privacy preferences in particular. If aversion to foreign data use is sufficiently strong, the associated savings in privacy costs may offset the real wage loss. But when privacy concerns are weaker than some threshold value ψ_{ji}^*, the constraint must eventually bind, preventing the government from implementing full discrimination on foreign firms.

Crucially, the result does not rely on any substantive conflict between the consumer and the government, since, as shown in Proposition 3.14, a wide range of normatively distinct policy objectives can rationalise maximal discrimination incentives. In particular, the constraint may (and in fact must under Assumption 10) bind even when the planner is fully benevolent and seeks to maximise long-run national welfare. The reason is that, while a marginal tightening of cross-border data restrictions must raise consumer surplus in the long run, these gains only materialise once entry dissipates the short run rents initially captured by local firms, leaving consumers worse off in the interim.

The possibility of political resistance draws a sharp contrast with optimal privacy regulation, as studied in Section 3.1. In that case, a forward-looking planner would optimise over a depreciated welfare locus that accounts for the endogenous relocation costs of regulation, yet still select a point on the upward-sloping portion of short-run welfare, as shown in Figure 5.

Starting from a legal vacuum, the immediate welfare gains from regulation are unambiguous, though not fully realised due to limited regulatory ambition. By contrast, political feasibility becomes a binding constraint in the present setting precisely because the planner's preferred policy could reduce short-run welfare below the status quo level set when consumer surplus carries significant weight in the welfare function. An illustration is provided in Figure 11.

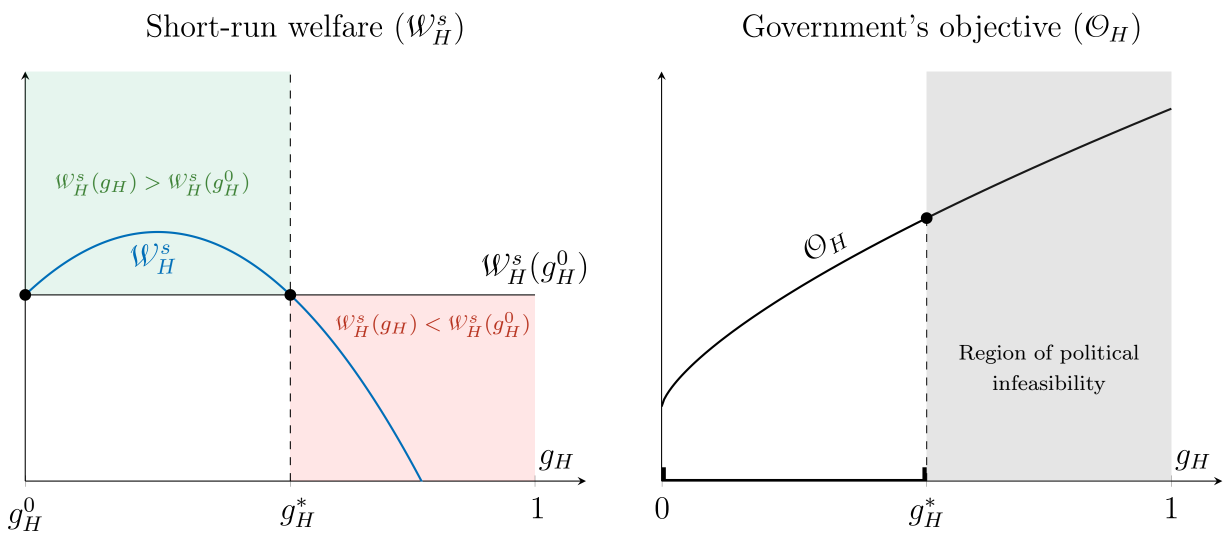


Figure 11: Short-run welfare constraints can curb maximal discrimination incentives

Note: The figure illustrates how short-run political accountability can curb otherwise non-satiable incentives to discriminate in Home, assuming an initial regulation level $g_H^0 = 0$ for simplicity. The left panel shows short-run welfare as a function of cross-border restrictions. For moderate exposure disutility, real wage losses eventually outweigh privacy cost savings, causing welfare to cross the status quo level curve at some interior $g_H^* < 1$. The right panel depicts the implications for the pursuit of the government's objective, which is assumed to rise monotonically with g_H, as any of those listed in Proposition 3.14. The optimal solution is found at the politically feasible corner g_H^*.

Two key insights into the political economy of data localisation emerge. First, since the acceptability of high restrictions on outbound data flows ultimately depends on the perceived disutility of foreign data processing, governments may have a rational incentive to politicise the issue in pursuit of their specific objectives. In particular, while the model treats privacy preferences as exogenous, the mechanism it uncovers suggests that the salience of foreign data risks may be actively cultivated to relax domestic political constraints and enable more assertive forms of cross-border regulation. In this light, the rise of national security narratives need not be seen as a mere product of structural inertia in international relations.

Second, the result offers a natural explanation for the empirical regularity that the most restrictive cross-border data policies are concentrated in countries with weak political accountability. Given that maximal discrimination emerges as optimal across a wide class of policy objectives, governments unconstrained by short-run welfare considerations are more likely to adopt aggressive forms of data localisation. This perspective complements prevailing accounts that frame such measures as instruments of state surveillance or authoritarian control (Han 2024), but also highlights their potential limitations in cases where state control is better understood as an extension of industrial strategy rather than coercive intent. As discussed in Appendix A.1.3, China's Great Firewall is a complex institutional object whose evolution reflects shifting strategic incentives. The development of its data localisation component is arguably less the product of an initial drive for censorship than a trajectory increasingly aligned with China's broader turn towards indigenous innovation.

3.2.2 Strategic Data Localisation with Asymmetric Political Constraints

We now leverage the preceding characterisation of unilateral incentives to discriminate against foreign firms' data handling to examine how these forces shape the long-run trajectory of the world economy. Our main motivation is the widespread perception that cross-border data restrictions function as instruments of industrial or security policy, with lasting consequences for the spatial distribution of economic activity.

In this context, we have in mind policy debates linking China's rapid technological catch-up to extreme forms of data restriction targeting foreign presence without commensurate retaliation by their most affected partners, notably the United States. We advance the hypothesis that less politically constrained governments may gain a first-mover advantage in regulatory escalation. By imposing maximal cross-border data restrictions early, they can trigger expansion in their domestic Tech sector, which increases the short-run welfare costs that any further retaliation would impose on their more politically accountable opponent, even if the latter is ultimately willing to escalate. The result is a persistent asymmetry in regulatory outcomes, with lasting consequences for the geography of Tech.

Game setup To give formal content to our interpretation of equilibrium adjustments as dynamic relocation processes, we extend the model to an infinite-horizon setting in which the spatial distribution of Tech activity adjusts gradually over time. Each period, the short-run mass of Tech firms established in country i, $N^s_{i,t}$, partially adjusts towards its long-run level, $N^\ell_i(\mathbf{X}_t)$, consistent with the current policy environment. Specifically, the spatial distribution of Tech firms evolves according to the deterministic law:

$$N^s_{i,t+1} - N^s_{i,t} = \rho \cdot \left(N^\ell_i(\mathbf{X}_t) - N^s_{i,t}\right), \quad \rho \in (0,1). \tag{70}$$

The adjustment speed ρ can be interpreted as an inverse measure of frictions in firm mobility, such as fixed costs, regulatory delays, uncertainty over market outcomes, or organisational inertia. For our purposes, such sequential adjustment is instrumental in analysing how relocation dynamics interact with political constraints over time.

We assume that the world economy initially rests in a steady state, characterised by a data governance matrix $\mathbf{X}_0 \in [0,1]^{2\times 2}$ (10), which encodes the effective data usage restrictions applied to each firm-consumer dyad. This matrix uniquely determines the equilibrium firm distribution via the long-run solution (27), which in turn pins down the short-run mass of firms at the beginning of the game:

$$N^s_{i,1} = N^\ell_i(\mathbf{X}_0). \tag{71}$$

Government problem Given a per-period policy objective $\mathcal{O}_i$, each government $i \in \mathcal{C}$ chooses a sequence of cross-border data restrictions $\{g_{i,t}\}_{t\geq 1}$ to maximise a discounted stream of payoffs, taking as given the initial firm distribution $(N^s_{i,1})_{i\in\mathcal{C}}$ (71), and anticipating the endogenous evolution of Tech activity via the law of motion (70).

For simplicity, we assume that both governments' per-period objectives feature insatiable incentives to discriminate against foreign firms.

Assumption 11 (Insatiable Discrimination Incentives)**.** *The per-period objective function $\mathcal{O}_i$ is continuous, strictly increasing in outbound restrictions $g_{i,t}$, non-increasing in inbound restrictions $g_{j,t}$, and non-decreasing in local Tech presence $N^s_{i,t}$.*

Assumption 11 can be heuristically motivated by Proposition 3.14, which shows that a wide class of objectives in our static model sustain maximal discrimination as unilaterally optimal. However, care must be taken when extending these rationales to the dynamic setting, since long-run variables now become steady-state outcomes, and several objectives listed there do not exhibit the appropriate monotonicity properties in the short run.

Nevertheless, some interesting cases remain compatible with Assumption 11. For instance, we may set $\mathcal{O}_i = -D^s_{ji}$ to capture a government with securist concerns aiming to minimise the stock of domestic data held by foreign firms in each period. Alternatively, setting $\mathcal{O}_i = \hat{X}^s_i$ would capture a government pursuing technological dominance through maximisation of net Tech exports. For our purposes, endowing $\mathcal{O}_i$ with these simple monotonicity properties allows for a straightforward characterisation of the dynamic game that focuses on the role of political constraints.[25]

Players We consider an extreme case of asymmetry in which Foreign faces no constraint on policy, while Home remains politically accountable and cannot impose arbitrarily large welfare losses in the short run. We formalise below the modalities through which this tension limits the space of feasible policies for the Home government.

[25] Both objectives also satisfy the monotonicity conditions in g_j and N^s_i.

Assumption 12 (Political Conflict over Cross-Border Data in Home)**.** *Maximal discrimination of Foreign firms faces short-run political resistances in Home. Specifically, we impose:*

(i) Political accountability of the Home government. *In each period $t \geq 1$, the Home government faces political resistance that limits the decline in short-run consumer welfare to at most a fraction $\gamma_t \in [0,1)$ relative to the status quo. Given a sequence of accountability thresholds $\{\gamma_t\}_{t\geq 1}$, the chosen restriction level $g_{H,t} \in [0,1]$ must satisfy the political feasibility constraint:*

$$\hat{\mathcal{W}}^s_{H,t} \equiv \frac{\mathcal{W}^s_H(g_{H,t}, N^s_{H,t})}{\mathcal{W}^s_H(g_{H,t-1}, N^s_{H,t})} - 1 \geq -\gamma_t \tag{72}$$

(ii) Bounded aversion to Foreign data uses. *The foreign exposure parameter ψ_{FH} lies in an open interval $(0, \psi^*_{FH})$ for which the short-run welfare function $\mathcal{W}^s_H$ is positive-valued, attains a unique global minimum at $g_H = 1$, and is strictly submodular in (g_H, N^s_F):*

$$\frac{\partial^2 \mathcal{W}^s_H}{\partial g_H \, \partial N^s_F} < 0 \quad \textit{and} \quad g_H < 1 \quad \Longrightarrow \quad \mathcal{W}^s_H(g_H, N^s_H) > \mathcal{W}^s_H(1, N^s_H) > 0 \tag{73}$$

Assumption 12 formalises political frictions in Home along two dimensions. First, we relax the constraint from the previous section by allowing short-run welfare losses up to a bounded fraction $\gamma_t > 0$, where the sequence $\{\gamma_t\}_{t\geq 1}$ is unrestricted and may vary with political cycles or shocks. Second, we assume that consumer aversion to Foreign data use is sufficiently limited for maximal discrimination to lack political support at the time of implementation.[26] For regularity, we assume that short-run welfare remains strictly positive.

Gathering all elements, the dynamic policy problem faced by government i may be compactly written as follows:

$$\max_{\{g_{i,t}\}_{t\geq 1}} \quad \sum_{t=1}^{\infty} \delta^{t-1} \, \mathcal{O}_i\big(g_{i,t}, N^s_{i,t}; g_{j,t}\big), \quad \delta \in (0,1) \tag{74}$$

$$\begin{aligned} & g_{i,t} \in [0,1] \quad \text{for all } t, \\ & N^s_{i,t+1} = f\Big(N^s_{i,t}, N^\ell_i\big(\mathbf{X}_t(g_{i,t})\big)\Big), \quad N^s_1(\mathbf{X}_0) \text{ given}, \\ \text{and} \quad & \mathcal{W}^s_i(g_{i,t}, N^s_{i,t}) \geq (1-\gamma_t) \cdot \mathcal{W}^s_i(g_{i,t-1}, N^s_{i,t}), \quad \text{if } i = H. \end{aligned}$$

Equilibrium characterisation We focus on Markov perfect equilibria, where each government's decision depends only on the current state of the system. Since the total number of firms is exogenous by Proposition 2.1, the distribution of Tech activity is fully captured

[26]This assumption could be weakened by simply requiring that it holds at the initial condition $N^s_{H,1}$.

by the short-run mass in Home, $N^s_{F,t}$. Together with the previous regulation level $g_{H,t-1}$, which determines the status quo welfare benchmark, and the current political constraint γ_t, these variables define the state. We represent the state as a tuple $\mathscr{s} = (N^s_F, g_{H,-1}, \gamma) \in \mathscr{S}$, and define a strategy as a mapping $s_i : \mathscr{S} \to [0,1]$ assigning a restriction level to each state.

Proposition 3.16 (Existence of an Asymmetric Markov Perfect Equilibrium)**.** *Suppose that both governments $i \in \mathcal{C}$ solve Problem (74) in otherwise identical countries. Fix any initial restriction level $g_0 < 1$, and let the global data governance matrix at time $t = 0$ be:*

$$\mathbf{X}_0 = \begin{bmatrix} \bar{x}_{ii,0} \\ 1 - \pi g_0 \end{bmatrix} \begin{bmatrix} 1 & 1 \end{bmatrix}, \quad \textit{so that} \quad N^s_{i,1} = \frac{N}{2} \quad \textit{for each } i. \tag{75}$$

Then, a Markov perfect equilibrium exists in which:

(i) Foreign enjoys a first-mover advantage and implements full discrimination in every state, and thus in every period:

$$s_F(\mathscr{s}) = 1, \quad \textit{for all } \mathscr{s} \in \mathscr{S}. \tag{76}$$

(ii) Home follows a monotonically increasing retaliation path, but faces growing political resistance to tighten restrictions as Foreign Tech expands:

$$g_H = s_H(\mathscr{s}) \geq g_{H,-1}, \quad \tilde{N}^s_F < N^s_F \implies s_H(\tilde{N}^s_F, g_{H,-1}, \gamma) \leq s_H(N^s_F, g_{H,-1}, \gamma). \tag{77}$$

(iii) There exists a finite intertemporal political budget $S(g_0) < \infty$ such that, for any sequence $\{\gamma_t\}_{t \geq 1}$ satisfying:

$$\sum_{t=1}^{\infty} \gamma_t < S(g_0), \tag{78}$$

the Home retaliation path $\{g_{H,t}\}$ converges to a level strictly below full discrimination:

$$\lim_{t \to \infty} g_{H,t} \equiv g_{H,\infty} < 1. \tag{79}$$

In the corresponding steady state, Foreign commands the majority of Tech activity:

$$N^\ell_F(\mathbf{X}_\infty) > \frac{N}{2} > N^\ell_H(\mathbf{X}_\infty). \tag{80}$$

Assumptions. 1, 2, 3, 4, 5, 10, 11 and 12.
Proof. See Appendix C.21. □

Proposition 3.16 shows that Foreign may leverage weak political accountability to secure long-run industrial dominance in Tech by imposing aggressive cross-border data restrictions that Home may be unable to reciprocate, even asymptotically. Despite identical fundamentals, Home's retaliation path may remain bounded away from full discrimination due to short-run political constraints that tend to tighten as local Tech presence declines.

We stress that this result is not trivial, for at least two reasons.

First, the objective function is not imposed arbitrarily. As shown in Proposition 3.14, the monotonicity properties that sustain corner outcomes emerge endogenously across a range of plausible and microfounded short-run objectives, each of which would support the same qualitative dynamics. The objective is left unspecified in Proposition 3.16 precisely because its specific form is ultimately irrelevant to the underlying mechanism.

Second, the optimality of operating persistently at the margin of political feasibility in Home reflects a genuinely strategic feature of the intertemporal problem. Since the political constraint ensures that the status quo is always implementable, the planner could, in principle, adopt more sophisticated paths if current inaction were expected to expand the feasible set over time. For instance, if consumer support for cross-border restrictions rises as the Foreign Tech sector expands, a forward-looking government might choose to under-regulate today in order to escalate later and lock in a stricter regime for the remainder of the game.

What rules out the optimality of such deviations in the present setting is a structural property of the short-run welfare function within the parameter region defined by Assumption 12; namely, its *submodularity* in (g_H, N_F^s). In other words, cross-border data restrictions and Foreign Tech presence act as *substitutes* from the perspective of a consumer whose preferences place greater weight on consumption than on privacy. As Foreign's industry expands, a larger share of firms becomes subject to regulation, amplifying the price effect of any given tightening through the extensive margin. For sufficiently bounded privacy aversion, the resulting decline in consumer surplus outweighs marginal privacy gains, thereby aggravating the short-run welfare losses from stricter regulation.

In our story, this substitutability interacts with Foreign's ability to move first. With genuine political constraints, Home cannot match Foreign's restrictions immediately, which shifts the long-run distribution of Tech activity abroad and gradually embeds Foreign firms more deeply into domestic consumption. Regulatory asymmetry thus becomes self-sustaining and rules out any strategic path through which Home might wait out resistance or hope for shifting preferences. The government instead enters a race against time, as each period of divergence narrows its political space and locks in lasting losses in industrial presence, real income, and oversight. Whether this spiral can be avoided depends on its ability to overreach resistance, which we model as a finite political budget that, if too limited, forces convergence to an interior policy.

Role of adjustment speed The last characterisation echoes Krugman's (1987b) attempt to give formal footing to the "businessman's view" that frames trade as a high-stakes contest between countries rather than firms, as noted in the introduction. He conjectures that, while nations "cannot be driven altogether out of business," they may still "be driven out of some businesses, so that in fact temporary shocks can have permanent effects on trade." By introducing dynamics, our extension offers an account of how temporary regulatory asymmetries can generate path-dependent Tech specialisation and lasting divergence between otherwise identical countries. More broadly, this logic may help explain why such concerns are often framed with a heightened sense of urgency, grounded in the belief that present inaction could permanently foreclose future strategic opportunities. We develop this logic further in the next proposition.

Proposition 3.17 (Relocation Speed and Long-Run Retaliation Capacity in Home)**.** *Suppose that Home's intertemporal political budget satisfies $\sum_t \gamma_t \in (0, S(g_0))$, so that the constraint supports an interior steady-state solution $g_{H,\infty} < 1$. Suppose further, for regularity, that at least one γ_t is strictly positive for $t \geq 2$, and, for simplicity, that the sequence $\{\gamma_t\}$ has only finitely many non-zero terms[27]. Then, everything else equal, the steady-state level of outbound data restrictions in Home varies strictly negatively with the relocation speed ρ:*

$$\rho' > \rho \quad \Longrightarrow \quad g'_{H,\infty} < g_{H,\infty}. \tag{81}$$

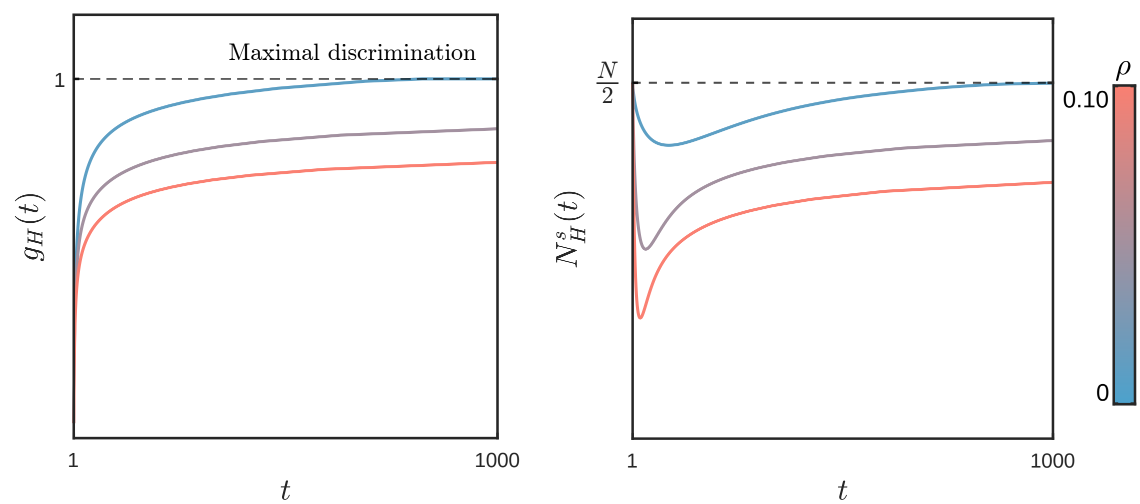
Assumptions. 1, 2, 3, 4, 5, 10,11 and 12.
Proof. See Appendix C.22. □

Proposition 3.17 clarifies the role of relocation forces in magnifying political constraints by linking the speed of adjustment in the Tech sector to the steady-state level of retaliation in Home. Specifically, faster structural transformation compresses the set of dynamically politically feasible responses, ultimately forcing the accountable government to settle on a strictly lower long-run level of outbound restrictions.

As the game opens, Foreign immediately implements maximal discrimination against Home firms, which permanently depresses their market potential in the country. Foreign consumers are accordingly redirected towards local alternatives, and demand for Foreign varieties increases. Meanwhile, the politically constrained government in Home can only afford partial retaliation, implying that domestic consumers do not substitute away from Foreign varieties to the same extent. Home firms are left operating at unprofitably low scale and are pushed towards the exit, while Foreign firms earn excess rents that support new entry by local entrepreneurs.

[27] This assumption avoids the need for cumbersome arguments to ensure that the inequality in Proposition 3.17 remains strict in the limit as $t \to \infty$, and does not affect the underlying intuition.

The short-run consequence of this regulatory asymmetry is a gradual replacement of Home varieties by Foreign ones in the domestic consumption bundle, with both the speed and extent of displacement increasing in the structural transformation rate ρ. Specifically, the trajectory of firm masses in Home generally follows two phases, as shown in Figure 12. The first is a contraction driven by firm exit due to insufficient scale, while the second involves a slower phase of partial regrowth as the government tightens restrictions within politically acceptable bounds. The extent to which retaliation can ultimately match Foreign's depends on the government's political leeway and its interaction with the endogenous costs of retaliation. Early reliance on Foreign Tech heightens consumer exposure, which amplifies surplus losses from further tightening and narrows the feasible policy space. Given a fixed sequence of political constraints γ_t, long-run industry recovery in Home becomes increasingly unlikely as structural transformation is fast and the initial shock is large.



(a) Retaliation path $g_H(t)$ (b) Home Tech path $N_H^s(t)$

Figure 12: Faster structural adjustment narrows the space of politically feasible retaliation in Home

Note: The figure shows the transition paths of retaliation $g_H(t)$ and Home Tech presence $N_H^s(t)$ over 1000 periods, for three structural adjustment rates $\rho \in \{0.01, 0.05, 0.10\}$. The simulation begins from symmetric fundamentals and fully unregulated cross-border data flows. The Home government adjusts gradually, subject to a time-varying political feasibility threshold $\gamma_t = \gamma_1/t^p$, with $\gamma_1 = 0.001$ and $p = 1.2$. This hyperbolic decay, assumed for simplicity, ensures summability when $p > 1$. When adjustment is sufficiently slow ($\rho = 0.01$), the Home government eventually catches up and restores full Tech presence. For faster adjustment rates ($\rho = 0.05$ and $\rho = 0.10$), retaliation remains imperfect and the steady state features persistent losses of activity to Foreign.

Anecdotal evidence supports the mechanism outlined in the preceding analysis. While Chinese authorities have long enforced sweeping restrictions that formally or effectively exclude U.S. tech firms from their domestic market, comparable initiatives in the United States have repeatedly stalled amid institutional, legal, and public resistance. A prominent and ongoing illustration is the case of TikTok, whose attempted bans have consistently failed to materialise despite bipartisan narratives portraying the platform as a national security threat. In August 2020, President Trump issued an executive order to prohibit the app, but federal courts blocked the measure on the grounds that the executive lacked statutory authority (Chander 2022). In April 2024, Congress passed legislation requiring ByteDance to divest from TikTok by January 2025. Upon returning to office in January 2025, Trump twice extended the divestiture deadline. Despite continued scrutiny, the platform remains active under successive temporary reprieves.

Welfare implications of escalation Escalation in trade barriers is often depicted as a lose–lose dynamic, where mutual restrictions erode gains from trade and reduce welfare globally. Yet this logic may break down when one side enjoys stronger institutional capacity to commit. We conclude by examining the welfare implications of the resulting asymmetric equilibrium for the world economy.

Proposition 3.18 (Welfare Implications of Asymmetric Escalation)**.** *Relative to period* 0*, long-run welfare in Home is always strictly lower under strategic escalation. By contrast, welfare in Foreign may rise or fall depending on the extent of retaliation. Specifically, if* $\mathscr{W}_F$ *satisfies the properties of Assumption 12, then for any initial restriction level* $g_0 < 1$*, there exists a threshold retaliation level* $g_H^*(g_0) \in (g_0, 1)$ *such that Foreign welfare increases if and only if* $g_{H,\infty} < g_H^*(g_0)$*.*

$$\mathscr{W}_{H,\infty} < \mathscr{W}_{H,0} \qquad \textit{and} \qquad \mathscr{W}_{F,\infty} \gtrless \mathscr{W}_{F,0} \iff g_{H,\infty} \lessgtr g_H^*(g_0). \tag{82}$$

Assumptions. 1, 2, 3, 4, 5, 10, 11, and 12.
Proof. See Appendix C.23. □

Proposition 3.18 formalises how Foreign may secure a unilateral long-run welfare gain from a strategic escalation that entails a net global loss.[28] Critically, the outcome hinges on the retaliatory capacity of the politically accountable government in Home. If such retaliation is sufficiently credible, the symmetric allocation is ultimately restored, and Foreign incurs a lasting loss in gains from trade with no compensating benefit. By contrast, if Home's response remains sufficiently weak, the resulting steady state asymmetry in firm distribution may be large enough to raise Foreign's welfare, at the expense of Home, whose welfare

[28] The result relies on Assumption 10, which presumes a consumer preference for local data processing. While analytically convenient, this assumption is difficult to justify in regimes lacking political accountability, where relocation may plausibly serve state surveillance. A similar conclusion, however, could be derived without this assumption by focusing solely on consumer surplus.

declines unambiguously in any scenario. A critical retaliation threshold identifies the level below which this latter effect dominates.

This result yields two broader insights into the global political economy of cross-border data restrictions. First, governments may be more prone to escalate towards maximal discrimination when they expect limited retaliation from their trading partners, particularly where institutional or political constraints abroad undermine the credibility of reciprocal responses. Second, even when such escalation entails a net global loss, cooperation may fail to arise if some countries stand to gain unilaterally from the resulting asymmetry.

4 Conclusion

This thesis has examined how trade openness reshapes unilateral incentives to regulate data and the regulatory patterns that result. While the analysis is not exhaustive, the framework suffices to identify several plausible mechanisms.

First, trade openness induces a geographic unmooring of regulatory authority from data usage. As consumers engage with firms across borders, their data are processed globally, while regulatory power remains largely confined to national territory. Uniform data usage standards therefore weigh disproportionately on domestic firms, eroding their market potential relative to foreign competitors and gradually displacing local industry as equilibrium adjustments trigger relocation pressures. As foreign Tech expands within the consumption bundle, real wages decline and regulatory effectiveness deteriorates. Internalising these frictions confronts governments with a novel *regulation–retention trade-off*, whereby the desire to preserve domestic Tech presence and institutional oversight systematically depresses regulatory ambition in equilibrium. The same relocation dynamics dissuade unilateral protections for foreign consumers, even though reciprocal commitments could pool territorial capacity and improve privacy enforcement for all. The result is a persistent decline in regulatory ambition, which further undermines welfare as consumers are left to rely solely on the limited reach of their own government's enforcement of suboptimally lax standards abroad.

Second, trade openness enables a new class of instruments that redirect the focus of regulation. Data localisation measures allow governments to selectively constrain foreign firms while exempting domestic ones, encouraging a shift away from the broader aim of regulating firm–consumer interactions on equal terms. Maximal discrimination becomes a unilaterally rational strategy under a broad set of normatively distinct objectives, ranging from rent extraction to the long-run pursuit of national welfare when foreign data processing fails to meet acceptable safeguards. In such settings, only short-run political accountability may provide a credible check on escalation, when aggressive restrictions on cross-border data flows significantly reduce immediate consumer surplus. Without a structured international response, the resulting dynamic threatens to erode the gains from trade globally and to entrench a structural bias in favour of governments that, facing fewer institutional constraints, can move first and advance industrial or securist aims at the expense of others.

In principle, cooperation could offer a natural response to the inefficiencies of decentralised policymaking. Mutual recognition of standards, supported by binding commitments, could help realign incentives across jurisdictions.

On the optimistic side, recent experience suggests that some governments may be willing to engage in regulatory cooperation on privacy matters when appropriately incentivised. The European Union's adequacy regime, which grants deeper market access to countries offering protections deemed equivalent to the GDPR, has proved modestly effective in this respect. By 2024, eleven countries had obtained adequacy status, often after adopting comprehensive privacy frameworks that extend protections to foreign residents with the specific objective of securing such recognition (Appendix A.2.1).

On the pessimistic side, recent developments suggest that incentives to coordinate may be highly contingent on evolving economic positions. Yet, in beggar-thy-neighbour settings, there is always one party for whom the moment to engage is never quite right. The shifting stances of China and the United States offer a suggestive illustration of these tensions. In the early 2010s, China resisted the liberalisation of cross-border data flows, while the US actively promoted binding commitments at the WTO. Today, China has grown more open to dialogue as its technology sector has matured, just as the US has adopted a more defensive posture (Appendix A.1). These reversals suggest that structural obstacles may preclude the emergence of any broad international consensus. When such strategic frictions are compounded by rising geopolitical tensions, intensifying competition for technological leadership, and the dual nature of data as both economic resource and bearer of meaning, the prospects for sustained policy alignment become even more fragile. It is doubtful that existing trade frameworks are equipped to address these challenges. (Staiger 2021)

The framework developed here provides a tractable structure for organising inquiry into these issues. The current analysis points to several natural directions for extension.

The first strand of extensions concerns our treatment of the data economy. In its current form, the model represents data as an exclusive productive asset held by firms, which offers a plausible benchmark given their typical control over access, but abstracts from the international implications of data's non-rivalry. In practice, even proprietary datasets are not perfectly siloed, and both trade frictions and policy choices may shape the extent to which data circulate across firms in geographically meaningful ways.

A natural extension would link the present framework to approaches such as Jones et al. (2020) by allowing multiple uses of data within the global economy. Such a generalisation would support an analysis of how the integration of national data ecosystems influences aggregate outcomes. It could also enrich the model's equilibrium structure by introducing agglomeration forces that are absent from the baseline setting. For example, if data sharing intensifies among proximate firms, relocating a single firm to one country could generate disproportionate productivity gains nearby and sustain endogenous winner-takes-all patterns. In that context, measures that affect the integration of national data ecosystems, such as data localisation, may carry particular importance for aggregate outcomes. Governments may also find unilateral value in enacting such measures if they enable the partial appropriation of spillovers generated by data's non-rivalry.

Although such an extension would reduce analytical tractability, it would allow for a more comprehensive study of current policy instruments. For instance, Beraja et al. (2023) show that subsidies in the form of public data access have material effects on the performance of Chinese AI firms, and conjecture that such policies might be used to influence trade patterns. Other relevant instruments include recent legal innovations such as data portability, which allows consumers to transfer their personal data across firms. More broadly, it remains to develop a theory of the global data economy suited to studying governance approaches that assign individuals claims over their personal data, rather than directly regulating firm behaviour.

A second avenue concerns the connection with more recent developments in trade theory. Among our assumptions, firm homogeneity in fundamental technology is the least tenable, particularly in the context of highly concentrated, data-intensive industries. While introducing Melitz-type productive heterogeneity would not resolve the deeper mismatch between our large-group structure and the oligopolistic nature of Big Tech, it could nonetheless provide a tractable way to explore selection dynamics and their interaction with the determinants of data regulation examined here. For instance, if territorially bounded enforcement causes regulation to fall disproportionately on domestic activity, it may magnify Melitz-type reallocations by placing greater regulatory pressure on the data processing of firms that do not export. Similarly, cross-border data restrictions could prove especially damaging by targeting exporters directly, amplifying productivity losses in the affected location and potentially increasing the strategic relevance of such measures as beggar-thy-neighbour instruments.

References

Acquisti, A. et al. (2023). "The economics of privacy at a crossroads". In: *Economics Of Privacy. University Of Chicago Press.*

Aridor, G., Y.-K. Che, and T. Salz (2020). *The economic consequences of data privacy regulation: Empirical evidence from GDPR.* National Bureau Of Economic Research Cambridge, Ma, Usa.

Azzi, A. (2018). “The challenges faced by the extraterritorial scope of the General Data Protection Regulation”. In: *J. Intell. Prop. Info. Tech. & Elec. Com. L.* 9, p. 126.

Bagwell, K. and R. W. Staiger (2015). “Delocation and trade agreements in imperfectly competitive markets”. In: *Research In Economics* 69.2, pp. 132–156.

Begenau, J., M. Farboodi, and L. Veldkamp (2018). “Big data in finance and the growth of large firms”. In: *Journal Of Monetary Economics* 97, pp. 71–87.

Beraja, M., D. Y. Yang, and N. Yuchtman (Aug. 2022). “Data-intensive Innovation and the State: Evidence from AI Firms in China”. In: *The Review of Economic Studies* 90.4, pp. 1701–1723. ISSN: 0034-6527. DOI: 10.1093/restud/rdac056. eprint: https://academic.oup.com/restud/article-pdf/90/4/1701/50825780/rdac056.pdf.

— (2023). “Data-intensive innovation and the state: Evidence from AI firms in China”. In: *The Review Of Economic Studies* 90.4, pp. 1701–1723.

Bergemann, D., A. Bonatti, and T. Gan (2022). “The economics of social data”. In: *The Rand Journal Of Economics* 53.2, pp. 263–296.

Bradford, A. (2020). *The Brussels Effect: How the European Union Rules the World*. Oxford University Press.

Bundesverfassungsgericht (1983). *Judgment of 15 December 1983 - 1 BvR 2089/83*. https://www.bundesverfassungsgericht.de/SharedDocs/Entscheidungen/EN/1983/12/rs19831215_1bvr020983en.html.

CAC (2024). *Notice on the Implementation of Cross-Border Data Transfer Security Assessment Measures*. URL: https://www.cac.gov.cn/2024-03/22/c_1712776611775634.htm.

Calzolari, G. and A. Pavan (2006). “On the optimality of privacy in sequential contracting”. In: *Journal Of Economic Theory* 130.1, pp. 168–204.

Campolmi, A., H. Fadinger, and C. Forlati (2014). “Trade Policy: Home Market Effect versus Terms-of-Trade Externality”. In: *Journal Of International Economics* 93.1, pp. 92–107. DOI: 10.1016/j.jinteco.2013.12.010.

Chander, A. (2022). “Trump v. TikTok”. In: *Vand. J. Transnat’l L.* 55, p. 1145.

Chang, Q., L. W. Cong, L. Wang, and L. Zhang (2023). “Production, Trade, and Cross-Border Data Flows”. In: *Available At Ssrn 4672185*.

Chen, L., Y. Huang, S. Ouyang, and W. Xiong (2021). *The data privacy paradox and digital demand*. Tech. rep. National Bureau Of Economic Research.

Cong, L. W., W. Wei, D. Xie, and L. Zhang (2022). “Endogenous growth under multiple uses of data”. In: *Journal Of Economic Dynamics And Control* 141, p. 104395.

Cong, L. W., D. Xie, and L. Zhang (2021). “Knowledge accumulation, privacy, and growth in a data economy”. In: *Management Science* 67.10, pp. 6480–6492.

Demirer, M., D. J. J. Hernández, D. Li, and S. Peng (2024). *Data, privacy laws and firm production: Evidence from the GDPR*. Tech. rep. National Bureau Of Economic Research.

Eeckhout, J. and L. Veldkamp (2022). “Data and Market Power”. Cepr Discussion Paper No. Dp17272.

European Commission (2024). *Communication from the Commission to the European Parliament and the Council – First review of the adequacy decisions adopted under Article 25(6) of Directive 95/46/EC*. Accessed May 2025. URL: https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX%3A52024DC0007.

European Union (2022). *Regulation (EU) 2022/1925 of the European Parliament and of the Council*. Accessed: 2025-02-12. URL: https://eur-lex.europa.eu/eli/reg/2022/1925/oj/eng.

Evans, D. S. (2009). "The online advertising industry: Economics, evolution, and privacy". In: *Journal Of Economic Perspectives* 23.3, pp. 37–60.

Fang, M. (2024). "Mathematical modelling, observation and identification of epidemiological models with reinfection". PhD thesis. Sorbonne Universite.

Farboodi, M., R. Mihet, T. Philippon, and L. Veldkamp (2019). "Big data and firm dynamics". In: *AEA papers and proceedings*. Vol. 109. American Economic Association 2014 Broadway, Suite 305, Nashville, TN 37203, pp. 38–42.

Farboodi, M. and L. Veldkamp (2021). *A model of the data economy*. Tech. rep. National Bureau Of Economic Research Cambridge, Ma, Usa.

Ferracane, M. F. and S. GONZÁLEZ (2024). *Data localisation: global trends*. Tech. rep. Working Paper. European University Institute.

Ferracane, M. F., J. Kren, and E. Van Der Marel (2020). "Do data policy restrictions impact the productivity performance of firms and industries?" In: *Review Of International Economics* 28.3, pp. 676–722.

Ferracane, M. F., H. Lee-Makiyama, and E. Van Der Marel (2018). "Digital trade restrictiveness index". In: *European Center For International Political Economy* 5.

Ferracane, M. F. and E. van der Marel (2021). "Regulating Personal Data". In: *World Development Report*.

Gao, H. (2011). "Google's China problem: A case study on trade, technology and human rights under the GATS". In: *Asian J. Wto & Int'L Health L & Pol'Y* 6, p. 349.

— (2019). "Data regulation with Chinese characteristics". In: *Smu Centre For Ai & Data Governance Research Paper* 2019/04, pp. 245–267.

Goldberg, S. G., G. A. Johnson, and S. K. Shriver (2024). "Regulating privacy online: An economic evaluation of the GDPR". In: *American Economic Journal: Economic Policy* 16.1, pp. 325–358.

Goldfarb, A. and D. Trefler (2018). *AI and international trade*. Tech. rep. National Bureau Of Economic Research.

Goldfarb, A. and C. Tucker (2012). "Privacy and innovation". In: *Innovation Policy And The Economy* 12.1, pp. 65–90.

Goldfarb, A. and C. E. Tucker (2011). "Privacy regulation and online advertising". In: *Management Science* 57.1, pp. 57–71.

Greenleaf, G. (2012). "The influence of European data privacy standards outside Europe: implications for globalization of Convention 108". In: *International Data Privacy Law* 2.2, pp. 68–92.

— (2020). *California's CCPA 2.0: Does the US Finally Have a Data Privacy Act?*

Greenleaf, G. (2022). *Proposed US federal data privacy law offers strong protections but only to US Residents.*

— (2023). *Global data privacy laws 2023: 162 national laws and 20 Bills.*

Greze, B. (2019). "The extra-territorial enforcement of the GDPR: a genuine issue and the quest for alternatives". In: *International Data Privacy Law* 9.2, pp. 109–128.

Gupta, S., P. Ghosh, and V. Sridhar (2022). "Impact of data trade restrictions on IT services export: A cross-country analysis". In: *Telecommunications Policy* 46.9, p. 102403.

Hagiu, A. and J. Wright (2023). "Data-enabled learning, network effects, and competitive advantage". In: *The Rand Journal Of Economics* 54.4, pp. 638–667.

Han, S. (2024). "Data and statecraft: why and how states localize data". In: *Business and politics* 26.2, pp. 263–288.

Head, K. and T. Mayer (2004). "Market potential and the location of Japanese investment in the European Union". In: *Review Of Economics And Statistics* 86.4, pp. 959–972.

Helpman, E. and P. Krugman (1987). *Market structure and foreign trade: Increasing returns, imperfect competition, and the international economy.* Mit Press.

Ichihashi, S. (2020). "Online privacy and information disclosure by consumers". In: *American Economic Review* 110.2, pp. 569–595.

Islam, M. T. and M. E. Karim (2020). "Extraterritorial application of the EU General Data Protection Regulation: An international law perspective". In: *Iiumlj* 28, p. 531.

Jia, J., G. Z. Jin, and L. Wagman (2018). *The short-run effects of GDPR on technology venture investment.* Tech. rep. National Bureau Of Economic Research.

Johnson, G. (2013). *The Impact of Privacy Policy on the Auction Market for Online Display Advertising.* Simon School Working Paper No. Fr 13-26. DOI: 10.2139/ssrn.2333193.

Jones, C. I. and C. Tonetti (2020). "Nonrivalry and the Economics of Data". In: *American Economic Review* 110.9, pp. 2819–2858.

Kastlová, H. (Apr. 2024). *Report on Extraterritorial Enforcement of GDPR.* European Data Protection Board (EDPB) Support Pool of Experts. URL: https://edpb.europa.eu.

Krugman, P. (1987a). "Is free trade passé?" In: *Journal Of Economic Perspectives* 1.2, pp. 131–144.

— (1987b). "The narrow moving band, the Dutch disease, and the competitive consequences of Mrs. Thatcher: Notes on trade in the presence of dynamic scale economies". In: *Journal Of Development Economics* 27.1-2, pp. 41–55.

Li, W. and J. Chen (2024). "From brussels effect to gravity assists: Understanding the evolution of the GDPR-inspired personal information protection law in China". In: *Computer Law & Security Review* 54, p. 105994.

Lundvall, B.-Å. and C. Rikap (2022). "China's catching-up in artificial intelligence seen as a co-evolution of corporate and national innovation systems". In: *Research Policy* 51.1, p. 104395.

Ma, S., S. Huang, and P. Wu (2024). "Data policy restrictions and cross-border E-commerce: Evidence from China". In: *Journal Of Asian Economics* 95, p. 101826.

Ossa, R. (2011). "A "New Trade" Theory of GATT/WTO Negotiations". In: *Journal Of Political Economy* 119.1, pp. 122–152. DOI: 10.1086/659371.

Prüfer, J. and C. Schottmüller (2021). "Competing with big data". In: *The Journal Of Industrial Economics* 69.4, pp. 967–1008.

Quan, J. (2023). "Tracing Out International Data Flow: The Value of Data and Privacy". In: *Available At Ssrn 4185169*.

Redding, S. and A. J. Venables (2004). "Economic geography and international inequality". In: *Journal Of International Economics* 62.1, pp. 53–82.

Solove, D. J. and P. M. Schwartz (2024). "Privacy law fundamentals". In: *D. Solove & P. Schwartz, Privacy Law Fundamentals, International Association Of Privacy Professionals*.

Staiger, R. W. (2021). *Does digital trade change the purpose of a trade agreement?* Tech. rep. National Bureau Of Economic Research.

Tarski, A. (1955). "A lattice-theoretical fixpoint theorem and its applications". In: *Pacific Journal Of Mathematics* 5.2, pp. 285–309. DOI: 10.2140/pjm.1955.5.285.

The Guardian (Jan. 2025). "First Thing: Donald Trump Calls China's DeepSeek AI Chatbot a Wake-Up Call". In: *The Guardian*. URL: https://www.theguardian.com/us-news/2025/jan/28/first-thing-donald-trump-calls-chinas-deepseek-ai-chatbot-a-wake-up-call.

The White House (Mar. 2024). *Preventing Access to Americans' Bulk Sensitive Personal Data and United States Government-Related Data by Countries of Concern*. Executive Order 14117. URL: https://www.federalregister.gov/documents/2024/03/01/2024-04573/preventing-access-to-americans-bulk-sensitive-personal-data-and-united-states-government-related.

— (Jan. 2025). *Removing Barriers to American Leadership in Artificial Intelligence*. Executive Order Issued By The President Of The United States. URL: https://www.whitehouse.gov/presidential-actions/2025/01/removing-barriers-to-american-leadership-in-artificial-intelligence/.

TrustArc (July 2018). *GDPR compliance status: A comparison of US, UK and EU companies*. Technical report. Trustarc.

United States Trade Representative, O. of the (Oct. 2023). *USTR Statement on the WTO E-Commerce Negotiations*. URL: https://ustr.gov/about-us/policy-offices/press-office/press-releases/2023/october/ustr-statement-wto-e-commerce-negotiations.

Veldkamp, L. and C. Chung (2024). "Data and the aggregate economy". In: *Journal Of Economic Literature* 62.2, pp. 458–484.

Venables, A. J. (1987). "Trade and trade policy with differentiated products: A Chamberlinian-Ricardian model". In: *The Economic Journal* 97.387, pp. 700–717.

Whitman, J. Q. (2003). "The two western cultures of privacy: Dignity versus liberty". In: *Yale Lj* 113, p. 1151.

WTO (2021). *Cross-Border Data Flows and Privacy Protection: Background Note by the WTO Secretariat*. URL: https://docs.wto.org/dol2fe/Pages/SS/directdoc.aspx?filename=q:/INF/ECOM/19.pdf&Open=True.

Xie, D. and L. Zhang (2022). "A generalized model of growth in the data economy". In: *Available At Ssrn 4033576*.

Appendices

Contents

A Global Data Regulation: A Conceptual Framework

This appendix develops a structured framework for global data regulation to support the analysis in the main text. Rather than offering a comprehensive legal overview, it adopts a contextual and instrumental perspective, focusing on regulatory dimensions most likely to generate distinct economic mechanisms from a trade standpoint. In doing so, it abstracts from both the heterogeneity of national regimes and the inherently bundled nature of data laws. Specific regulations are referenced where relevant.

The first subsection situates international data governance in historical context, tracing the emergence and evolution of key regulatory models. The second subsection clarifies core concepts of privacy regulation and data localisation, laying the conceptual groundwork for the policy analysis in Section 3.

A.1 Trends in Global Data Governance and Regulatory Models

Policy briefs often classify data regulations into three 'data models', presented in Table 1. This taxonomy, developed by European researchers, arguably reflects a distinctly European perspective. For example, the US-based model is defined primarily by what it lacks; namely, "a comprehensive framework on personal data" (Ferracane et al. 2021), suggesting a relatively situated approach to regulatory differences. Such cultural tensions are well documented in the privacy law literature: "To people accustomed to the continental way of doing things, American law seems to tolerate relentless (...) violations of privacy." (Whitman 2003)

Model	**Example**	**Description**
Open	US	Relies on firm self-regulation according to general privacy principles and minimal regulation of data flows.
Conditional	EU	Stresses a rights-based approach, establishes a comprehensive regulatory regime for data handling, and sets pre-transfer conditions for data flows.
Limited	China	Prioritises national security, allows for exemptions granting government access to personal data, and enforces strict restrictions on cross-border data transfers.

Table 1: Data Regulation Models (Ferracane et al. 2021)

What is uncontested, however, is that global data governance has been shaped by the emergence and diffusion of three regulatory approaches: one originating in the US, one in Europe, and one in China. Each reflects distinct legal traditions, political priorities, and economic strategies that have influenced policymaking far beyond their regions of origin.

Interestingly, each of these regulatory approaches has entered a period of noticeable evolution, driven by rising national security concerns and the global competition for tech leadership.

A.1.1 United States: Persistent Fragmentation and the 'Free Flows' of Data

Fragmented legal landscape Upon international comparison, the US approach stands out as a patchwork system, where states play a significant role and regulation has historically followed a *sector-by-sector* approach. As a result, the regulation applicable to different types of data, if any, vary significantly based on the industry and the specific location where, within the US, the data is both generated and processed.

Solove et al. (2024) mentions that, while the Children's Online Privacy Protection Act (COPPA) offers privacy protection for children online, there is no federal law that regulates privacy for adults on the web. While health (HIPAA) and financial (GLBA) data are subject to strict federal regulations, workplace surveillance rules, web browsing history or location data often fall under state laws, leaving room for significant heterogeneity in standards across jurisdictions. Adding another layer of fragmentation, different statutes regulate data usages separately for public and private entities.

Privacy as a reactive consumer right In the United States, privacy has deep roots in tort law, a primarily state-based branch of law enabling individuals to seek compensation for harm caused by others. State courts historically protected individuals through tort actions such as *intrusion upon seclusion* and *public disclosure of private facts.* Over time, privacy law evolved into information privacy law, which encompasses data regulation as a subset. The state-level origins of tort law were a determining factor in the decentralised nature of US data governance, as privacy protections developed through dispersed judicial rulings rather than a unified regulatory framework. At the same time, the case-by-case enforcement of tort claims positioned privacy as a reactive consumer right, exercised in response to harm rather than as a proactively enforced fundamental protection.

Current debates More recently, consistent frameworks have emerged at the state level, partly influenced by European developments, such as the 2020 California Consumer Privacy Act (CCPA). Paradoxically, these enactments have complicated federal attempts to establish a unified law by exacerbating legal disparities. In 2022, the US moved closer than ever to adopting a nationwide regulation for data handling with the American Data Privacy and Protection Act (ADPPA), but the bill failed due to state preemption concerns and resistance from states with stronger protections (Greenleaf 2022).

Global lobbying efforts Big Tech corporations have been influential stakeholders in US data policy, actively shaping both domestic regulations and the USTR's approach to data-related trade issues (Gao 2011). These companies have been instrumental in establishing the US as a global leader in data governance, advocating for international standards at the WTO and supporting cross-border data flow provisions in trade agreements. (Gao 2019).

The notion of 'free flow of data' was largely popularised by USTR lobbying efforts to secure binding commitments by trade partners and represents the country's historical position. Emerging developments suggest that this long-standing stance is undergoing revisions, seemingly shaped by national security concerns and broader strategic considerations. In 2023, the US took an important step back by withdrawing its support for WTO e-commerce provisions related to cross-border data flows, citing the need for "policy space" (United States Trade Representative 2023). While historically an opponent of data localisation mandates, the country has also recently moved towards restricting outbound data flows in sensitive areas. (The White House 2024).

A.1.2 Europe: Data Protection, Brussels Effect and Digital Market Regulation

Data protection Using the United States as a comparison is pedagogical in highlighting the central elements of the European regulatory approach to data governance. While US privacy law mostly responds to violations after harm occurs, the European model is centered on *data protection*. This concept is rooted in the recognition of personal data as a human right requiring proactive safeguards. It is mirrored by an *omnibus* (as opposed to sectoral) approach, with a single regulation applying broadly to both public and private sectors alike. Sector-specific laws may add additional requirements, but unlike in the US, they build on a general framework that ensures all contexts are regulated by default (Solove et al. 2024).

Unlike the US, where digital privacy grew from older tort law concepts, the emergence of data protection in Europe is integral to the development of information technologies. In 1970, the Hessian Parliament passed world's first comprehensive data protection law, quickly followed in 1983 by the German Constitutional Court's recognition of the "fundamental right" attached to any individual "against the unlimited collection, storage, use and sharing of their personal data" (Bundesverfassungsgericht 1983). At the continental level, the EU's 1995 Data Protection Directive established the basic framework for personal data processing, later replaced by the General Data Protection Regulation (GDPR) in 2018.

Global influence Scholars have attributed significant regulatory spillover effects to European innovations in data governance, a phenomenon known as the Brussels Effect (Bradford 2020). Early, Greenleaf (2012) noted that "something reasonably described as 'European standard' privacy laws" was "becoming the norm in most parts of the world." More than a decade later, this trend persists, with many countries drafting or implementing GDPR-inspired data protection frameworks. Notable examples include Brazil's LGPD (2020), Japan's APPI (effective 2022), South Korea's PIPA (revised 2023), India's DPDP (2023), Thailand's PDPA (2022), and Canada's proposed CPPA.

Data as market power More recently, the EU's data governance has undergone a significant shift, evolving from a privacy-first framework to one that increasingly addresses the economic implications of data. The EU had long recognised the economic value of data, aiming to balance growth and privacy by complementing its strong regulatory frameworks with proactive actions to stimulate indigenous innovation. In particular, the Digital Single Market Strategy (2015) promoted internal digital market integration, while the Open Data Directive (2019) facilitated public sector data reuse.

What is relatively new is the recent engagement with market structure, with data increasingly seen as a major determinant of entrenched inertia in economic outcomes. Big Tech's ever growing market power and its location outside Europe, notably in the US and China, are increasingly seen as significant threats to Europe's long-term role in digital markets and its capacity to enforce its standards. The EU-US Privacy Shield (2016) was struck down by the EU Court of Justice in 2020 due to concerns that US surveillance laws violated EU privacy standards, disrupting cross-border data flows and further intensifying trade tensions.

In 2023, the European Commission for the first time designated six non-EU companies[29] as *gatekeepers* under the Digital Market Act (DMA). EU officials noted that these actors are "structurally extremely difficult to challenge" by existing or new market operators, "irrespective of how innovative and efficient those market operators may be." It is revealing that, in assessing weak contestability, the Commission now points to these gatekeepers' control over "key inputs in the digital economy, such as data." (European Union 2022)

A.1.3 China: The Great Firewall, National Security and Indigenous Innovation

Origins of the Great Firewall China's Great Firewall originated from a 1996 cybersecurity law that transformed the country's internet from an almost lawless space into a tightly monitored system, imposing mandatory state-controlled gateways for all international traffic. Within a short span, this initially rudimentary system expanded into a highly sophisticated control apparatus, integrating advanced content filtering, surveillance mechanisms, and strict data regulations.

While the Great Firewall's rapid growth has often outpaced understanding, it is now generally agreed that political control was the primary motivation behind its development

[29]The six gatekeepers designated by the European Commission are Alphabet, Amazon, Apple, ByteDance, Meta, and Microsoft.

(Gao 2019). A first strand of extensions came as hardware controls proved inadequate for managing the growing complexity of online content, particularly with the rise of social media and real-time interactions. With software-related vulnerabilities contained, attention turned to cybersecurity and foreign interference, prompting stricter data flow restrictions and the establishment of a more centralised oversight structure. The Cyberspace Administration of China (CAC) was finally established in 2011 and gained sweeping authority by 2014 as digital governance became central to national sovereignty.

Trade tensions With what is typically regarded as the world's most stringent digital regulations (Ferracane et al. 2018), China has been the defining case in international debates over when data policies become disguised trade restrictions. US tech giants, supported by the USTR, have pushed the case that aspects of the Great Firewall are intentionally designed to erect barriers against foreign competitors. With commitments on cross-border data flows lacking, these firms have struggled to navigate outdated WTO rules to mount legal challenges. The USTR went as far as to argue that censorship should be considered a market access restriction *per se*, but the argument failed to gain legal recognition.

A handful of US tech firms continue to operate in China (e.g., Microsoft, Apple, Tesla) under strict conditions, while most have been banned (e.g., Facebook, Twitter, YouTube) or withdrawn voluntarily (e.g., Google, LinkedIn, Airbnb), sometimes even citing threats to intellectual property as in Google's case (Gao 2011).

Indigenous innovation Officially, China has continuously framed its data regulations as a matter of sovereignty. Still, the recent rise of Baidu, Alibaba, and Tencent (BAT) as the only non-US tech giants of comparable scale, now actively engaged in AI leadership, is often portrayed in public discourse as evidence that these policies helped break extreme path dependency. Lundvall et al. (2022) provide qualitative evidence that the Great Firewall was part of a broader shift towards indigenous innovation, reflecting a reassessment of openness in Chinese policymaking after the export-led model failed to foster innovation in highly valued sectors like automobiles and telecommunications.

Clear evidence of deliberate strategy remains elusive, yet recent shifts in China's stance on cross-border data flows could align with an infant industry maturing after reaching sufficient scale. While resisting the traditional American approach to "freeing" data flows, Chinese officials have increasingly acknowledged that the "trade-related aspects of data flows are crucial to trade development" (WTO 2021). These developments gained significance in 2024, marked by China's unilateral easing of restrictions on certain cross-border data transfers (CAC 2024) and the launch of an EU-China communication mechanism to discuss data flow regulations and their trade implications. This shift is all the more significant as it unfolds while the US scales back its own data openness agenda.

A.2 Data Policy: Definitions, Stylised Facts and Economic Perspectives

A.2.1 Privacy Regulation

Definition A *privacy regulation* is a legal framework that structures the relationship between individuals and entities handling personal data to address exposure concerns. It may proceed by introducing formal requirements at any stage of data use, empowering individuals with rights, or incorporating both strategies in tandem. When such regulation has extraterritorial reach, it applies to entities based on their interactions with covered individuals with the primary objective of ensuring uniform standards. The lack of provisions intended to shape the geography of data flows, combined with an emphasis on structuring micro-interactions between individuals and data handlers, distinguishes privacy law from data localisation.

A privacy regulation can be seen as assigning a specific regulatory status (or none at all) to each combination of a data subject, data handler, and data type. How many combinations fall under regulation defines the *scope*, which legal scholars typically characterise along three dimensions: personal (who is covered), territorial (where it applies), and material (what data or activities are regulated).

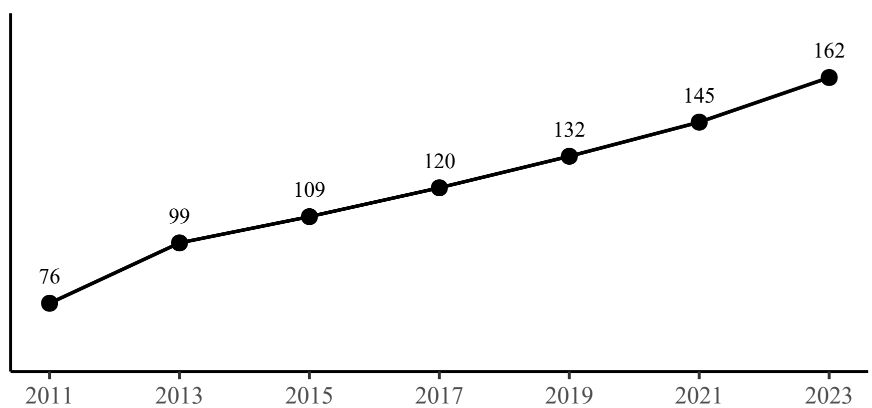


Figure 13: Number of privacy laws enacted in the world (2011-2023).

Note: Figures were collected from Greenleaf (2023).

Scope heterogeneity Substantive differences in scope are often overlooked in international comparisons of privacy regulations. For instance, the GDPR applies to all data processing by EU establishments, regardless of consumer location, and to non-EU entities when handling data of residents. As such, it exemplifies a *comprehensive* privacy regulation as categorised in Figure 2, for it encompasses all interactions involving both EU-based firms and EU-based consumers.

Interestingly, neither the recent attempt at 'America's GDPR' (Greenleaf 2022), which failed to be enacted in 2023, nor China's 'GDPR-inspired' regulation (Li et al. 2024) uphold this maximalist standard.

The US ADPPA was designed to govern data concerning US residents regardless of processor location, but excluded foreign individuals even when their data was processed in the US. A comparable limitation characterises the 2020 California's PRA, which remains confined to residents of California. More subtly, China's PIPL, effective since 2021, may appear comparable to the GDPR in scope, but is in fact significantly narrower. While it governs data related to Chinese residents and data processed within China, it allows Chinese firms to process foreign data abroad without restriction. This omission creates an obvious loophole that renders the PIPL functionally closer to a residency-based framework than to a genuinely comprehensive regime (see Figure 2).

Greenleaf (2020) notes that "many data privacy laws limit protections to their own citizens or residents", while "some extend them to any person." The typical motivation for this broader scope is to secure EU *adequacy* status under the GDPR, whereby the European Commission recognises that a non-EU country affords an "equivalent" level of data protection to EU consumers. This designation simplifies compliance, removes the need for additional transfer mechanisms, and facilitates deeper integration into the European market.

Though unilateral in nature, the EU's adequacy system has proved relatively effective in promoting regulatory convergence and fostering *de facto* cooperation. By 2024, eleven jurisdictions had obtained and maintained adequacy status under the original framework. According to the European Commission, "many of said countries and territories have modernised and strengthened their privacy legislation through comprehensive or partial reforms (...) prompted amongst other by the need to ensure the continuity of the adequacy decisions" (European Commission 2024).

Challenges of extraterritorial enforcement Since the GDPR's adoption, extraterritoriality has become a hallmark of modern privacy law. While only a fraction of GDPR-inspired regulations extend comparable protections to foreigners, most emulate its ambition to govern the conduct of *all* data handlers interacting with domestic residents. Yet the enforceability of such provisions remains constrained by both legal and practical obstacles. On the legal side, scholars have questioned their viability from the outset, citing the absence of meaningful jurisdictional anchors or bilateral agreements that would secure the consent of the foreign state (Azzi 2018; Greze 2019; Islam et al. 2020). In practice, enforcement of the GDPR itself has revealed significant compliance gaps among non-EU firms. For instance, by mid-2018, only 12% of US-based companies reported full compliance with the GDPR, compared to 27% within the EU (TrustArc 2018).

As of 2024, little has improved. A recent expert report for the European Data Protection Board underscores that fines imposed under the GDPR remain unenforceable in the United States, where courts refuse to recognise foreign administrative sanctions on principle (Kastlová 2024). Even when upheld by a European court, such decisions are routinely denied enforcement, and national authorities lack any formal mechanism to compel compliance from firms without a substantive presence in the EU. In the absence of binding commitments, cooperation with US regulators remains strictly voluntary, rendering most extraterritorial enforcement efforts effectively symbolic. In our model, these limitations are captured through an explicit parameter governing extraterritorial enforcement capacity.

A.2.2 Data Localisation

Definition *Data localisation* consists of policies that effectively narrow the scope for handling data on a country's citizens or residents from abroad. These measures vary widely, from soft requirements to store local copies before transfer to conditional frameworks that impose strict legal scrutiny on cross-border flows, up to near-total restrictions that require all data to be stored and processed within national borders with few exceptions. Enforcement also differs, with some regimes relying on private-sector mechanisms like contracts and certifications, while others impose direct government control through approvals or case-by-case authorisation. Notwithstanding this heterogeneity, framing these policies around their outcome highlights how localisation may arise both from explicit mandates and from an indirect domestic pull shaped by legal uncertainty and compliance burdens.

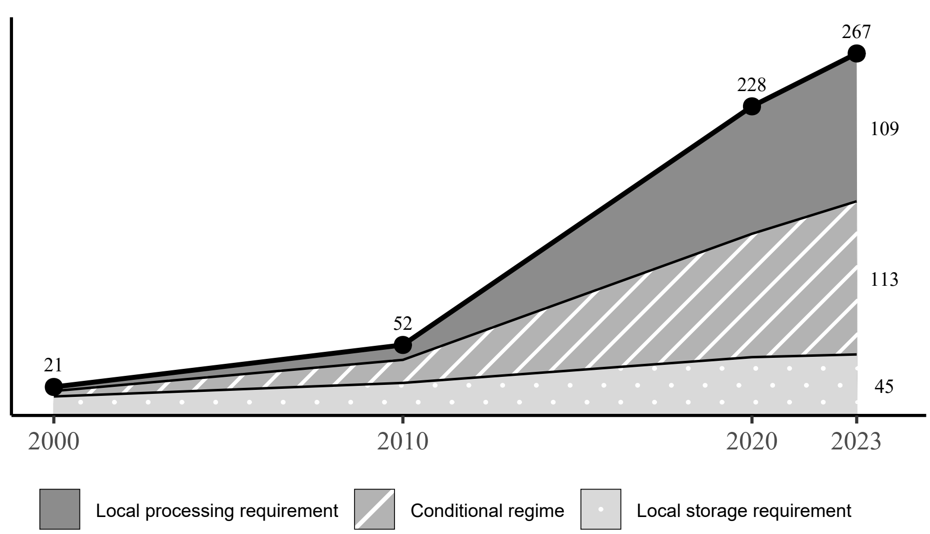


Figure 14: Number of data localisation measures in the world (2000-2023).

Note: Figures were collected from Ferracane et al. (2024).

Rising trend A recent survey by Ferracane et al. (2024) documents that the number of data localisation measures has grown fivefold in just 15 years, reaching 267 in 2024. Local processing requirements, the strictest form of regulation, account for most of this increase, suggesting a gradual erosion of the possibility for firms to operate a centralised data management model across borders. Conditional transfer regimes have also grown significantly, but to a lesser extent.

Data localisation versus privacy Because it is often justified on privacy grounds, data localisation is frequently misconceived as a subset of privacy regulations, though the two should be kept distinct analytically. While typical privacy regulations focus on structuring the micro-interaction between individuals and firms to reduce exposure concerns, data localisation is generally grounded within the realm of national interests, aiming to address foreign surveillance risks and strengthen government oversight. As such, the tendency to frame localisation as a privacy measure rests on the implicit assumption that it naturally complements privacy protections by ensuring foreign firm accountability. Yet this perspective assumes both the existence of enforceable privacy rights and legal safeguards to prevent localisation from merely expanding state surveillance. Where these conditions are lacking, localisation does little to reduce exposure in ways that matter to individuals.

More relevant to our purpose, a crucial distinction between privacy regulation and data localisation lies in their divergent implications for the spatial distribution of control, access, and value creation in data-intensive industries. In our framework, a defining feature of privacy regulation is its minimal discriminative content toward foreign firms, with any extraterritorial reach aimed solely at ensuring consistent standards for all covered individuals. While this feature does not preclude location-based disparities in burdens, it implies that any such differences stem from factors such as enforcement constraints or asymmetric exposures to the regulation's scope.

Non-tariff trade barrier Data localisation is expected to disproportionately burden foreign firms (Goldfarb et al. 2018). This feature has sparked debates over potential to tilt competitive dynamics in favour of domestic firms in data intensive industries. Though evidence remains scarce, initial findings are consistent with this hypothesis. Ferracane et al. (2020) use sectoral differences in data intensity to show that stronger data localisation requirements are associated with lower imports in data-intensive services. Similarly, Gupta et al. (2022) exploit variations in cross-border data restrictions among trading partners to show that stricter partner regulations are significantly linked to reduced ICT exports. More recently, Ma et al. (2024) report that stricter data policies in importing countries significantly reduce China's cross-border e-commerce exports, with stronger negative effects for differentiated and high-tech products, especially in high-income countries.

B Derivations

B.1 Consumer Problem

Firms own the data generated as a by-product of their sales. Accordingly, the privacy cost function (4) is independent of consumer choice. The consumer problem is thus equivalent to:

$$\max_{\{c_i(\cdot), Z_i\}} \quad \chi C_i^{\alpha} Z_i^{1-\alpha} \tag{B-1}$$

$$\text{s.t.} \quad \int_{\Omega} p_i(\omega) c_i(\omega) \mathrm{d}\omega + p_Z Z_i \leq w L_i \tag{B-2}$$

$$C_i = \left[\int_{\Omega} c_i(\omega)^{\frac{\sigma-1}{\sigma}} \mathrm{d}\omega \right]^{\frac{\sigma}{\sigma-1}} \tag{B-3}$$

The FOCs with respect to $c_i(\omega)$ and Z_i are:

$$\chi \alpha C_i^{\alpha-1} Z_i^{1-\alpha} \cdot \frac{\partial C_i}{\partial c_i(\omega)} = \lambda p_i(\omega) \tag{B-4}$$

$$\chi (1-\alpha) C_i^{\alpha} Z_i^{-\alpha} = \lambda p_Z \tag{B-5}$$

Pose $P_i \triangleq p_i(\omega) \left(\frac{\partial C_i}{\partial c_i(\omega)} \right)^{-1}$. Combining the two FOCs yields the relative expenditure share equation:

$$\frac{\alpha}{1-\alpha} = \frac{C_i p_i(\omega) \left(\frac{\partial C_i}{\partial c_i(\omega)} \right)^{-1}}{p_Z Z_i} \equiv \frac{P_i C_i}{p_Z Z_i} \tag{B-6}$$

P_i is clearly a constant since (B-6) holds for all ω. Using (B-6) into the budget constraint (B-2) further establishes the standard result that consumer i allocates a share α of their income to Tech goods, while the remaining share $1-\alpha$ is spent on the outside good:

$$P_i C_i = \alpha w L_i, \quad p_Z Z_i = (1-\alpha) w L_i \tag{B-7}$$

To derive the c_i schedule, write $C_i = \left[\int_{\Omega} c_i(\omega)^{\frac{\sigma-1}{\sigma}} \mathrm{d}\omega \right]^{\frac{\sigma}{\sigma-1}} \equiv S^{\frac{\sigma}{\sigma-1}}$. It is immediate that:

$$\frac{\partial C_i}{\partial c_i(\omega)} = S^{\frac{1}{\sigma-1}} c_i(\omega)^{-\frac{1}{\sigma}} = \left(S^{\frac{\sigma}{\sigma-1}} \right)^{\frac{1}{\sigma}} c_i(\omega)^{-\frac{1}{\sigma}} = C_i^{\frac{1}{\sigma}} c_i(\omega)^{-\frac{1}{\sigma}} \tag{B-8}$$

Using the definition of P_i, one recovers the residual demand function for a typical variety ω:

$$P_i = p_i(\omega) \left(\frac{\partial C_i}{\partial c_i(\omega)} \right)^{-1} \implies c_i(\omega) = \left(\frac{p_i(\omega)}{P_i} \right)^{-\sigma} C_i \tag{B-9}$$

Note that the consumption schedule c_i can be equivalently expressed in terms of consumer i's fixed nominal expenditure on Tech varieties (B-2):

$$\begin{aligned} c_i(\omega) &= \left(\frac{p_i(\omega)}{P_i} \right)^{-\sigma} \frac{\alpha w L_i}{P_i} \\ &\equiv E_i \cdot P_i^{\sigma-1} \cdot p_i(\omega)^{-\sigma}, \quad E_i \triangleq \alpha w L_i \end{aligned} \tag{B-10}$$

Using the demand schedule (B-9) into the definition of C_i (B-3) and simplifying, one recovers the standard expression for the ideal price index:

$$P_i = \left(\int_{\Omega} p_i(\omega)^{1-\sigma} \mathrm{d}\omega \right)^{\frac{1}{1-\sigma}} \tag{B-11}$$

Pose $\chi = \alpha^{-\alpha}(1-\alpha)^{-(1-\alpha)}$. Assumptions 2 and 1 together imply that $w = p_Z$. Eliminating C_i and Z_i from the utility function expenditure equations (B-7), one obtains the welfare function:

$$\begin{aligned} \mathscr{W}_i \triangleq U_i(C^*, Z^*) - \Psi_i &= \chi \left(\frac{\alpha w L_i}{P_i} \right)^{\alpha} \left(\frac{(1-\alpha) w L_i}{p_Z} \right)^{1-\alpha} - \Psi_i \\ &= \left(\frac{w}{P_i} \right)^{\alpha} L_i - \Psi_i \end{aligned} \tag{B-12}$$

B.2 Firm Problem

Because firms own the data generated as a by-product of their sales, they choose how much data to use for production. Let $\Pi_i : \Omega_i \to \mathbb{R}$ denote the profit function of a typical firm ω established in country i. Hence, one may write:

$$\Pi(\omega) \triangleq \Big\{ \Pi_i(\omega), \quad \text{if } \omega \in \Omega_i \tag{B-13}$$

For $i, j \in \mathcal{C}$, $k \in \{i, j\}$, define:

$$\tau_{ik} \triangleq \begin{cases} 1, & \text{if } k = i, \\ \tau, & \text{if } k \neq i, \end{cases} \tag{B-14}$$

The optimisation problem facing a typical firm ω established in country i writes:

$$\Pi_i(\omega) = \max_{\substack{k \in \mathcal{C}, \\ p_k(\omega), q_k(\omega), \\ l_k(\omega), x_k(\omega)}} \left\{ \sum_{k \in \mathcal{C}} \Big(p_k(\omega) c_k(\omega) - w l_k(\omega) \Big) - w f \right\} \tag{B-15}$$

$$\begin{aligned} \text{s.t. } \forall k \in \mathcal{C}, \quad & p_k(\omega) = P_k \left(\frac{c_k(\omega)}{C_k} \right)^{-\frac{1}{\sigma}} \\ & q_k(\omega) = \varphi d_k(\omega)^{\beta} l_k(\omega)^{1-\beta} \\ & d_k(\omega) = x_k(\omega) c_k(\omega) \\ & c_k(\omega) = \frac{q_k(\omega)}{\tau_{ik}} \\ & x_k(\omega) \geq 0, \quad x_k(\omega) \leq \bar{x}_{ik} \end{aligned}$$

with $(\bar{x}_{ik})_{k \in \mathcal{C}}$ specifying the upper bound on data usage intensity faced by a firm $\omega \in \Omega_i$ with respect to the representative consumer in country k. The global regulatory environment is summarised in the matrix:

$$\mathbf{X} = \begin{pmatrix} \bar{x}_{HH} & \bar{x}_{HF} \\ \bar{x}_{FH} & \bar{x}_{FF} \end{pmatrix} \tag{B-16}$$

When serving market destination k, a typical firm $\omega \in \Omega_i$ has access to the equilibrium production technology:

$$\begin{aligned} q_k(\omega) &= \varphi d_k(\omega)^{\beta} l_k(\omega)^{1-\beta} \\ &= \varphi \big(x_k(\omega) c_k(\omega)\big)^{\beta} l_k(\omega)^{1-\beta} \\ &= \varphi \left(\frac{x_k(\omega) q_k(\omega)}{\tau_{ik}}\right)^{\beta} l_k(\omega)^{1-\beta} \end{aligned} \tag{B-17}$$

Hence, for any $q_k(\omega) > 0$:

$$q_k(\omega) = \underbrace{\left(\varphi \left(\frac{x_k(\omega)}{\tau_{ik}}\right)^{\beta}\right)^{\frac{1}{1-\beta}}}_{\triangleq \tilde{\varphi}_k(\omega)} l_k(\omega), \tag{B-18}$$

where $\tilde{\varphi}_k(\omega)$ reads as the effective TFP of firm ω when producing for the representative consumer in country k.

Let μ_k^u and μ_k^ℓ be the Lagrange multipliers associated with the upper and lower bounds of the interval constraining the choice of $x_k(\omega)$. Treating $p_k(\omega)$ and $c_k(\omega)$ as functions of $x_k(\omega)$ and using the chain rule, the FOC with respect to $x_k(\omega)$ is:

$$\begin{aligned} & \frac{\partial}{\partial x_k(\omega)} \Big[p_k(x_k(\omega)) \cdot c_k(x_k(\omega)) \Big] = \mu_k^u - \mu_k^\ell \\ \iff \quad & \frac{\partial p_k(\omega)}{\partial c_k(\omega)} \frac{\partial c_k(\omega)}{\partial q_k(\omega)} \frac{\partial q_k(\omega)}{\partial \tilde{\varphi}_k(\omega)} \frac{\partial \tilde{\varphi}_k(\omega)}{\partial x_k(\omega)} c_k(\omega) + \frac{\partial c_k(\omega)}{\partial q_k(\omega)} \frac{\partial q_k(\omega)}{\partial \tilde{\varphi}_k(\omega)} \frac{\partial \tilde{\varphi}_k(\omega)}{\partial x_k(\omega)} p_k(\omega) = \mu_k^u - \mu_k^\ell \end{aligned} \tag{B-19}$$

Similarly, the FOC with respect to $l_k(\omega)$ is:

$$\frac{\partial p_k(\omega)}{\partial c_k(\omega)} \frac{\partial c_k(\omega)}{\partial q_k(\omega)} \frac{\partial q_k(\omega)}{\partial l_k(\omega)} c_k(\omega) + \frac{\partial c_k(\omega)}{\partial q_k(\omega)} \frac{\partial q_k(\omega)}{\partial l_k(\omega)} p_k(\omega) = w \tag{B-20}$$

Defining $K_k(\omega) \triangleq \frac{\partial c_k(\omega)}{\partial q_k(\omega)} \Big[\frac{\partial p_k(\omega)}{\partial c_k(\omega)} c_k(\omega) + p_k(\omega) \Big]$, these two optimality conditions might be rewritten as:

$$K_k(\omega) \frac{\partial q_k(\omega)}{\partial \tilde{\varphi}_k(\omega)} \frac{\partial \tilde{\varphi}_k(\omega)}{\partial x_k(\omega)} = \mu_k^u - \mu_k^\ell \tag{B-21}$$

$$K_k(\omega) \frac{\partial q_k(\omega)}{\partial l_k(\omega)} = w \tag{B-22}$$

For any $q_k(\omega) > 0$, we must have $x_k(\omega) > 0$ and thus $\mu_k^\ell = 0$, since data is an essential input. Meanwhile, $w > 0$ implies $K_k(\omega) > 0$, while $\frac{\partial q_k(\omega)}{\partial \tilde{\varphi}_k(\omega)} \frac{\partial \tilde{\varphi}_k(\omega)}{\partial x_k(\omega)} > 0$ whenever $x_k(\omega) > 0$. Hence, the LHS of equation (B-21) is strictly positive and it follows that:

$$\mu_k^u > 0 \quad \Longrightarrow \quad x_k^*(\omega) = \bar{x}_{ik} \tag{B-23}$$

Accordingly, the equilibrium productivity of a firm $\omega \in \Omega_i$ when serving consumer k is:

$$\tilde{\varphi}_k^*(\omega) = \left(\varphi \left(\frac{\bar{x}_{ik}}{\tau_{ik}}\right)^{\beta}\right)^{\frac{1}{1-\beta}} \triangleq \tilde{\varphi}_{ik} \tag{B-24}$$

Using that $\frac{\partial y}{\partial x} = \frac{\partial \ln y}{\partial \ln x}\frac{y}{x}$, the FOC for $l_k(\omega)$ (B-20) might be rearranged as:

$$p_k(\omega)\frac{\partial c_k(\omega)}{\partial q_k(\omega)}\frac{\partial q_k(\omega)}{\partial l_k(\omega)}\left(1 + \frac{\partial \ln p_k(\omega)}{\partial \ln c_k(\omega)}\right) = w \tag{B-25}$$

Substituting $\frac{\mathrm{d}\ln p_k(\omega)}{\mathrm{d}\ln c_k(\omega)} = -\frac{1}{\sigma}$, $\frac{\partial q_k(\omega)}{\partial l_k(\omega)} = \tilde{\varphi}_k(\omega)$ from (B-18), $\frac{\partial c_k(\omega)}{\partial q_k(\omega)} = \frac{1}{\tau_{ik}}$ yields:

$$\begin{aligned} p_k(\omega)\frac{1}{\tau_{ik}}\tilde{\varphi}_k(\omega)\left(1 - \frac{1}{\sigma}\right) = w \\ \iff \quad p_k^*(\omega) = \frac{\sigma}{\sigma-1}\cdot\frac{w\tau_{ik}}{\tilde{\varphi}_k(\omega)} \end{aligned} \tag{B-26}$$

Or, more explicitly:

$$p_k^*(\omega) = \frac{\sigma}{\sigma-1}\cdot\frac{w\,\tau_{ik}^{\frac{1}{1-\beta}}}{\left(\varphi\,\bar{x}_{ik}^{\beta}\right)^{\frac{1}{1-\beta}}} \triangleq p_{ik}, \quad \text{for all } \ \omega \in \Omega_i \tag{B-27}$$

Given the pricing rule (B-26), and knowing that, in equilibrium, $l_k(\omega) = \frac{q_k(\omega)}{\tilde{\varphi}_k(\omega)} = \frac{c_k(\omega)}{\tau_{ik}\tilde{\varphi}_k(\omega)}$ using (B-18), the profit function can be expressed as:

$$\begin{aligned} \Pi_i^*(\omega) &= \sum_{k\in\mathcal{C}}\Big(p_k^*(\omega)c_k(\omega) - wl_k(\omega)\Big) - wf \\ &= \sum_{k\in\mathcal{C}}\left(\frac{\sigma}{\sigma-1}\cdot\frac{c_k(\omega)w\tau_{ik}}{\tilde{\varphi}_k(\omega)} - \frac{c_k(\omega)w\tau_{ik}}{\tilde{\varphi}_k(\omega)}\right) - wf \\ &= \frac{w}{\sigma-1}\cdot\sum_{k\in\mathcal{C}}\left(\frac{c_k(\omega)\tau_{ik}}{\tilde{\varphi}_k(\omega)}\right) - wf \end{aligned} \tag{B-28}$$

B.3 Equilibrium

In equilibrium, free entry in Tech must dissipate rents.

$$\Pi(\omega) = 0 \tag{B-29}$$

In particular, equation (B-28) implies that for any firm $\omega \in \Omega_i$:

$$\sum_{k\in\mathcal{C}}\left(\frac{c_k(\omega)\tau_{ik}}{\tilde{\varphi}_k(\omega)}\right) = f(\sigma-1) \tag{B-30}$$

Define:

$$q_0 \triangleq \varphi f(\sigma-1) \quad \text{and} \quad p_0 \triangleq \frac{\sigma}{\sigma-1}\cdot\frac{w}{\varphi} \tag{B-31}$$

Using equations (B-24) and (B-26), the pricing schedule can be expressed as a transformation of p_0:

$$\begin{aligned} \frac{p_k^*(\omega)}{p_0} &= \frac{\frac{\sigma}{\sigma-1}\cdot\frac{w}{\tilde{\varphi}_k^*(\omega)}}{\frac{\sigma}{\sigma-1}\cdot\frac{w}{\varphi}} \\ p_k^*(\omega) &= p_0\cdot\frac{\tau_{ik}\varphi}{\tilde{\varphi}_k^*(\omega)}, \quad \omega \in \Omega_i \end{aligned} \tag{B-32}$$

$$\iff \quad p_{ik} = p_0 \cdot \left(\frac{\tau_{ik}}{\varphi^{\beta}\, \bar{x}_{ik}^{\beta}} \right)^{\frac{1}{1-\beta}} \tag{B-33}$$

Plugging the demand functions facing firm $\omega \in \Omega_i$ (B-10) into the zero-profit condition and using the above rewriting then yields:

$$\begin{aligned}
\sum_{k \in \mathcal{C}} \left(\frac{c_k(\omega)\tau_{ik}}{\tilde{\varphi}_k(\omega)} \right) &= f(\sigma - 1) \\
\iff \quad \sum_{k \in \mathcal{C}} \left(\frac{\varphi E_k \tau_{ik} P_k^{\sigma-1}\, p_k(\omega)^{-\sigma}}{\tilde{\varphi}_k(\omega)} \right) &= \varphi f(\sigma - 1) \\
\iff \quad p_0^{-\sigma} \cdot \sum_{k \in \mathcal{C}} \left[\frac{\varphi E_k \tau_{ik} P_k^{\sigma-1}}{\tilde{\varphi}_k(\omega)} \cdot \left(\frac{p_k(\omega)}{p_0} \right)^{-\sigma} \right] &= q_0 \\
\iff \quad p_0^{-\sigma} \cdot \sum_{k \in \mathcal{C}} \left(E_k P_k^{\sigma-1} \varphi^{1-\sigma} \tau_{ik}^{1-\sigma} \tilde{\varphi}_k(\omega)^{\sigma-1} \right) &= q_0 \\
\iff \quad p_0^{-\sigma} \cdot \sum_{k \in \mathcal{C}} \left[E_k P_k^{\sigma-1} \varphi^{1-\sigma} \tau_{ik}^{1-\sigma} \cdot \varphi^{\frac{\sigma-1}{1-\beta}} \left(\frac{\bar{x}_{ik}}{\tau_{ik}} \right)^{\frac{\beta(\sigma-1)}{1-\beta}} \right] &= q_0 \\
\iff \quad p_0^{-\sigma} \cdot \varphi^{\frac{\beta(\sigma-1)}{1-\beta}} \sum_{k \in \mathcal{C}} \Big[E_k P_k^{\sigma-1} \tau_{ik}^{\frac{1-\sigma}{1-\beta}} \bar{x}_{ik}^{\frac{\beta(\sigma-1)}{1-\beta}} \Big] &= q_0 \\
\iff \quad p_0^{-\sigma} \cdot \Theta_i(\mathbf{P}) &= q_0
\end{aligned} \tag{B-34}$$

Hence, the zero-profit condition (18) can be restated as a firm-level, location-specific equilibrium condition in terms of an augmented market access index, denoted $\Theta_i(\mathbf{P})$, that incorporates firms' data usage capabilities:

$$\Theta_i(\mathbf{P}) = \varphi^{\frac{\beta(\sigma-1)}{1-\beta}} \cdot \sum_{k \in \mathcal{C}} \Big[E_k P_k^{\sigma-1} \tau_{ik}^{\frac{1-\sigma}{1-\beta}} \bar{x}_{ik}^{\frac{\beta(\sigma-1)}{1-\beta}} \Big] \equiv \sum_{k \in \mathcal{C}} \theta_{ik} P_k, \tag{B-35}$$

An equilibrium in the model is fully characterised by a vector of price indices, $\mathbf{P}^* = (P_k^*)_{k \in \mathcal{C}}$, which satisfies equation (B-34) in every country $i \in \mathcal{C}$, ensuring goods market clearing under zero-profit conditions. Given a regulatory environment for data usage practices, Tech firms set their prices according to the constant rule (B-11), and equilibrium quantities are uniquely determined from the demand functions (B-10).

For any equilibrium price vector $\mathbf{P}^*$, the mass of Tech firms in each country is determined from the definition of the price indices (B-11). Under Assumption 4, $\Omega_i = [0, N_i]$ and a spatial allocation of Tech firms is a vector $\mathbf{N} = (N_i^*)_{i \in \mathcal{C}}$. Given N_i^*, the price vector $\mathbf{P}^*$, and equilibrium quantities, the labour allocated to the Tech sector in country i is by obtained inverting the production function (B-28):

$$L_{C,i} \triangleq \int_{\Omega_i} \Big(f + \sum_{k \in \mathcal{C}} l_k^*(\omega) \Big)\, \mathrm{d}\omega \tag{B-36}$$

Enforcing labour market clearing, the remaining workforce allocated to the production of the outside good is given by:

$$L_{z,i} \triangleq L_i - L_{C,i} \tag{B-37}$$

B.4 Closed-Form Characterisation

Price indices The firm market-clearing conditions (B-34) can be written as a system of linear equations in the transformed price vector $(P_H^{\sigma-1}, P_F^{\sigma-1})'$:

$$\underbrace{\begin{pmatrix} \theta_{HH} & \theta_{HF} \\ \theta_{FH} & \theta_{FF} \end{pmatrix}}_{\mathbf{A}} \begin{pmatrix} P_H^{\sigma-1} \\ P_F^{\sigma-1} \end{pmatrix} = q_0 \begin{pmatrix} 1 \\ 1 \end{pmatrix} \tag{B-38}$$

where the bilateral market access coefficients are given by:

$$\theta_{ik} \triangleq p_0^{-\sigma} \varphi^{\frac{\beta(\sigma-1)}{1-\beta}} E_k \, \tau_{ik}^{\frac{1-\sigma}{1-\beta}} \, \bar{x}_{ik}^{\frac{\beta(\sigma-1)}{1-\beta}}, \quad i \in \mathcal{C}, \; k \in \{i, j\} \tag{B-39}$$

From Proposition 2.1, $\mathbf{A}$ (B-38) is non-singular and thus admits a unique solution that can be explicitly solved using Cramer's rule:

$$P_i^{\sigma-1} = \frac{q_0(\theta_{jj} - \theta_{ij})}{\theta_{HH}\theta_{FF} - \theta_{HF}\theta_{FH}}, \quad i, j \in \mathcal{C}, \;\; j \neq i \tag{B-40}$$

Substituting the coefficients (B-39):

$$P_i^{\sigma-1} = \frac{q_0 \left(p_0^{-\sigma} \varphi^{\frac{\beta(\sigma-1)}{1-\beta}} \right) E_j \left(\bar{x}_{jj}^{\frac{\beta(\sigma-1)}{1-\beta}} - \tau^{\frac{1-\sigma}{1-\beta}} \bar{x}_{ij}^{\frac{\beta(\sigma-1)}{1-\beta}} \right)}{\left(p_0^{-\sigma} \varphi^{\frac{\beta(\sigma-1)}{1-\beta}} \right)^2 E_i E_j \left(\bar{x}_{HH}^{\frac{\beta(\sigma-1)}{1-\beta}} \bar{x}_{FF}^{\frac{\beta(\sigma-1)}{1-\beta}} - \tau^{\frac{2(1-\sigma)}{1-\beta}} \bar{x}_{HF}^{\frac{\beta(\sigma-1)}{1-\beta}} \bar{x}_{FH}^{\frac{\beta(\sigma-1)}{1-\beta}} \right)} \tag{B-41}$$

Simplifying and using the definitions of (Δ_i) (22) and Δ (23), we obtain:

$$P_i^{\sigma-1} = \frac{q_0 p_0^{\sigma} \Delta_j}{E_i \varphi^{\frac{\beta(\sigma-1)}{1-\beta}} \Delta} \quad \Longrightarrow \quad P_i^* = \left[\frac{q_0 p_0^{\sigma} \Delta_j}{E_i \varphi^{\frac{\beta(\sigma-1)}{1-\beta}} \Delta} \right]^{\frac{1}{\sigma-1}} \tag{B-42}$$

Accordingly, the relative price index is:

$$\frac{P_i}{P_j} = \left(\frac{\Delta_j \, E_j}{\Delta_i \, E_i} \right)^{\frac{1}{\sigma-1}} \tag{B-43}$$

Firm masses By symmetry, $p_i^*(\omega) \equiv p_{ik}^*$ (B-27) for all $\omega \in \Omega_i$. Using the definitional equation of the price indices (P_i) (B-11):

$$\begin{aligned} P_i^{\sigma-1} &= \left(\sum_{k \in \mathcal{C}} N_k (p_{ki}^*)^{1-\sigma} \right)^{-1} \\ &= \left(\frac{p_0}{\varphi^{\frac{\beta}{1-\beta}}} \right)^{\sigma-1} \left(\sum_{k \in \mathcal{C}} N_k \tau_{ki}^{\frac{1-\sigma}{1-\beta}} \bar{x}_{ki}^{\frac{\beta(\sigma-1)}{1-\beta}} \right)^{-1} \end{aligned} \tag{B-44}$$

In equilibrium, equations (B-42) and (B-44) coincide. Equating them leads to:

$$\frac{q_0 p_0^{\sigma} \Delta_j}{E_i \varphi^{\frac{\beta(\sigma-1)}{1-\beta}} \Delta} = \left(\frac{p_0}{\varphi^{\frac{\beta}{1-\beta}}} \right)^{\sigma-1} \left(\sum_{k \in \mathcal{C}} N_k \tau_{ki}^{\frac{1-\sigma}{1-\beta}} \bar{x}_{ki}^{\frac{\beta(\sigma-1)}{1-\beta}} \right)^{-1}$$

$$\iff \quad \sum_{k\in\mathcal{C}} N_k\, \tau_{ki}^{\frac{1-\sigma}{1-\beta}}\, \bar{x}_{ki}^{\frac{\beta(\sigma-1)}{1-\beta}} = \frac{E_i\,\Delta}{\sigma w f \Delta_j}, \quad \text{for all } i,j\in\mathcal{C}, \quad j\neq i \tag{B-45}$$

Hence, we may write:

$$\begin{pmatrix} \bar{x}_{HH}^{\frac{\beta(\sigma-1)}{1-\beta}} & \tau^{\frac{1-\sigma}{1-\beta}}\,\bar{x}_{FH}^{\frac{\beta(\sigma-1)}{1-\beta}} \\ \tau^{\frac{1-\sigma}{1-\beta}}\,\bar{x}_{HF}^{\frac{\beta(\sigma-1)}{1-\beta}} & \bar{x}_{FF}^{\frac{\beta(\sigma-1)}{1-\beta}} \end{pmatrix} \begin{pmatrix} N_H \\ N_F \end{pmatrix} = \frac{\Delta}{\sigma w f} \begin{pmatrix} E_H \Delta_F^{-1} \\ E_F \Delta_H^{-1} \end{pmatrix} \tag{B-46}$$

A straightforward application of Cramer's rule gives:

$$N_i = \frac{1}{\sigma w f}\left(\frac{E_i\,\bar{x}_{jj}^{\frac{\beta(\sigma-1)}{1-\beta}}}{\Delta_j} - \frac{\tau^{\frac{1-\sigma}{1-\beta}}\,\bar{x}_{ji}^{\frac{\beta(\sigma-1)}{1-\beta}}\,E_j}{\Delta_i} \right) \tag{B-47}$$

Bringing the terms to a common denominator results in:

$$N_i^* = \frac{\left(E_i \Delta_i \bar{x}_{jj}^{\frac{\beta(\sigma-1)}{1-\beta}} - E_j \Delta_j \tau^{\frac{1-\sigma}{1-\beta}} \bar{x}_{ji}^{\frac{\beta(\sigma-1)}{1-\beta}} \right)}{\sigma w f \Delta_i \Delta_j} \tag{B-48}$$

Quantities and firm sizes Using the pricing schedule (B-33) and the solution for the price index (B-42) into the residual demand function (B-10), we obtain that, for $k\in\{i,j\}$:

$$\begin{aligned} c_{ik} \triangleq c_k^*(\omega) &= (p_{ik}^*)^{-\sigma} P_k^{\sigma-1} E_k \\ &= p_0^{-\sigma}\left(\frac{\varphi^\beta\,\bar{x}_{ik}^\beta}{\tau_{ik}}\right)^{\frac{\sigma}{1-\beta}} \cdot \frac{q_0 p_0^\sigma \Delta_{-k}}{E_k \varphi^{\frac{\beta(\sigma-1)}{1-\beta}} \Delta} \cdot E_k \\ &= q_0 \cdot \frac{\varphi^{\frac{\beta}{1-\beta}}\,\Delta_{-k}}{\Delta} \cdot \left(\frac{\bar{x}_{ik}^\beta}{\tau_{ik}}\right)^{\frac{\sigma}{1-\beta}} \end{aligned} \tag{B-49}$$

Define the firm size as the sum of gross output across all markets:

$$q(\omega) \triangleq \sum_{k\in\mathcal{C}} q_k(\omega) \tag{B-50}$$

Substituting equilibrium consumption (B-49), it is immediate that, for a typical firm $\omega\in\Omega_i$

$$\begin{aligned} q_i \triangleq q(\omega) &= c_{ii} + \tau c_{ij} \\ &= q_0 \times \frac{\varphi^{\frac{\beta}{1-\beta}}}{\Delta} \cdot \left(\bar{x}_{ii}^{\frac{\beta\sigma}{1-\beta}}\,\Delta_j + \tau^{\frac{1-\beta-\sigma}{1-\beta}}\,\bar{x}_{ij}^{\frac{\beta\sigma}{1-\beta}}\,\Delta_i \right) \end{aligned} \tag{B-51}$$

Labour allocation Inverting the equilibrium production function (B-28), using the definition of $\tilde{\varphi}_{ik}$ (B-24) and substituting equilibrium quantities (B-49), one obtains that:

$$\begin{aligned} L_{C,i} &\triangleq N_i\left(f + \sum_{k\in\mathcal{C}} l_k^*(\omega) \right) \\ &= N_i\left[f + \sum_{k\in\mathcal{C}} \left(\frac{\tau_{ik} c_{ik}}{\tilde{\varphi}_{ik}} \right) \right] \end{aligned}$$

$$= N_i \left[f + \frac{q_0}{\varphi \Delta} \left(\Delta_j \, \bar{x}_{ii}^{\frac{\beta(\sigma-1)}{1-\beta}} + \Delta_i \, \bar{x}_{ij}^{\frac{\beta(\sigma-1)}{1-\beta}} \, \tau^{\frac{1-\sigma}{1-\beta}} \right) \right] \tag{B-52}$$

Using the definitions of (Δ_i) (22) and Δ (23), it can be shown that:

$$\begin{aligned} \Delta &= \bar{x}_{ii}^{\frac{\beta(\sigma-1)}{1-\beta}} \, \bar{x}_{jj}^{\frac{\beta(\sigma-1)}{1-\beta}} - \tau^{\frac{2(1-\sigma)}{1-\beta}} \, \bar{x}_{ji}^{\frac{\beta(\sigma-1)}{1-\beta}} \, \bar{x}_{ij}^{\frac{\beta(\sigma-1)}{1-\beta}} \\ &= \Delta_i \bar{x}_{jj}^{\frac{\beta(\sigma-1)}{1-\beta}} + \Delta_j \bar{x}_{ji}^{\frac{\beta(\sigma-1)}{1-\beta}} \, \tau^{\frac{1-\sigma}{1-\beta}} \end{aligned} \tag{B-53}$$

Hence:

$$\begin{aligned} L^*_{C,i} &= N_i \left(f + \frac{q_0}{\varphi} \right) \\ &= N_i f \sigma \end{aligned} \tag{B-54}$$

One can then substitute the expression for N_i (B-48) into the above to obtain the closed form solution:

$$L^*_{C,i} = \frac{f \left(E_i \Delta_i \bar{x}_{jj}^{\frac{\beta(\sigma-1)}{1-\beta}} - E_j \Delta_j \tau^{\frac{1-\sigma}{1-\beta}} \bar{x}_{ji}^{\frac{\beta(\sigma-1)}{1-\beta}} \right)}{w \Delta_i \Delta_j} \tag{B-55}$$

C Proofs of Propositions

C.1 Proof of Proposition 2.1 (Existence and Uniqueness of the World Equilibrium)

The system of firm market-clearing conditions (B-34) is linear in the transformed price vector $(P_i^{\sigma-1})_{i \in \mathcal{C}}$ (B-38):

$$\underbrace{\begin{pmatrix} \theta_{HH} & \theta_{HF} \\ \theta_{FH} & \theta_{FF} \end{pmatrix}}_{\mathbf{A}} \begin{pmatrix} P_H^{\sigma-1} \\ P_F^{\sigma-1} \end{pmatrix} = q_0 \begin{pmatrix} 1 \\ 1 \end{pmatrix} \tag{C-1}$$

Define $\tilde{P}_i \triangleq P_i^{\sigma-1}$. Note that the function $x \mapsto x^{\sigma-1}$ maps $\mathbb{R}_{>0}$ into itself when $\sigma > 1$. Therefore, any candidate solution where $\tilde{P}_i < 0$ would imply $P_i \leq 0$, contradicting $P_i^* > 0$. Therefore, the relevant equilibrium solutions in the transformed system are precisely those where $\tilde{P}_i > 0$.

Furthermore, since the inverse mapping $y \mapsto y^{\frac{1}{\sigma-1}}$ is one-to-one on $\mathbb{R}_{>0}$, the equilibrium price vector $\mathbf{P}^*$ is unique if and only if the system (C-1) has a unique solution in $\tilde{\mathbf{P}}$. By linearity, the system admits zero, one, or infinitely many solutions, with uniqueness occurring precisely when $\det \mathbf{A} \neq 0$. In that event, the solution is:

$$\tilde{P}_i^* = \frac{q_0(\theta_{jj} - \theta_{ij})}{\theta_{HH}\theta_{FF} - \theta_{HF}\theta_{FH}}, \qquad j \neq i \tag{C-2}$$

For $\tilde{P}_i^* > 0$, the numerator and denominator must share the same sign. Hence, a well-defined (and unique) equilibrium exists if and only if either:

$$\begin{aligned} &(i) \quad \theta_{FF} > \theta_{FH}, \quad \theta_{HH} > \theta_{HF}, \quad \text{and} \quad \theta_{HH}\theta_{FF} - \theta_{HF}\theta_{FH} > 0 \\ \text{or} \quad &(ii) \quad \theta_{FF} < \theta_{FH}, \quad \theta_{HH} < \theta_{HF}, \quad \text{and} \quad \theta_{HH}\theta_{FF} - \theta_{HF}\theta_{FH} < 0 \end{aligned} \tag{C-3}$$

Since $\theta_{ik} > 0$ for all $i \in \mathcal{C}$, $k \in \{i, j\}$ the third condition is redundant. Thus, $\mathbf{P}^* \in \mathbb{R}^2_{>0}$ if and only if:

$$(i) \quad \theta_{ii} > \theta_{ji} \quad \text{or} \quad (ii) \quad \theta_{ii} < \theta_{ji} \tag{C-4}$$

In words, a unique world equilibrium exists if and only if supplier capacity coefficients exhibit a systematic bias that grants each location preferential access to some market. This bias may take the form of either a home market effect (case (i)) or an anti-home market effect (case (ii)). Only case (i) is empirically relevant. Therefore, in any meaningful equilibrium:

$$\begin{aligned} & \theta_{ii} > \theta_{ji} \\ \iff \quad & p_0^{-\sigma} \varphi^{\frac{\beta(\sigma-1)}{1-\beta}} E_i \, \tau_{ii}^{\frac{1-\sigma}{1-\beta}} \, \bar{x}_{ii}^{\frac{\beta(\sigma-1)}{1-\beta}} > p_0^{-\sigma} \varphi^{\frac{\beta(\sigma-1)}{1-\beta}} E_j \, \tau_{ji}^{\frac{1-\sigma}{1-\beta}} \, \bar{x}_{ji}^{\frac{\beta(\sigma-1)}{1-\beta}} \\ \iff \quad & \Delta_i := \bar{x}_{ii}^{\frac{\beta(\sigma-1)}{1-\beta}} - \tau^{\frac{1-\sigma}{1-\beta}} \, \bar{x}_{ji}^{\frac{\beta(\sigma-1)}{1-\beta}} > 0 \end{aligned} \tag{C-5}$$

$\square$

C.2 Proof of Proposition 2.2 (Short-Run Price Adjustments)

Using equation (B-44), P_i can be solved for as a function of firm masses to give:

$$P_i = \left(\frac{p_0}{\varphi^{\frac{\beta}{1-\beta}}} \right) \left(\sum_{m \in \mathcal{C}} N_m \tau_{mi}^{\frac{1-\sigma}{1-\beta}} \, \bar{x}_{mi}^{\frac{\beta(\sigma-1)}{1-\beta}} \right)^{\frac{1}{1-\sigma}} \iff \ln P_i = \ln \left(\frac{p_0}{\varphi^{\frac{\beta}{1-\beta}}} \right) - \frac{1}{\sigma - 1} \cdot \ln \left(\sum_{m \in \mathcal{C}} N_m \, \tau_{mi}^{\frac{1-\sigma}{1-\beta}} \, \bar{x}_{mi}^{\frac{\beta(\sigma-1)}{1-\beta}} \right) \tag{C-6}$$

Straightforward differentiation then yields that, for any origin country $k \in \{i, j\}$:

$$\frac{\partial P_i}{\partial \bar{x}_{ki}} = P_i \cdot \frac{\partial \ln P_i}{\partial \bar{x}_{ki}} = -P_i \cdot \frac{\beta}{1-\beta} \cdot \frac{N_k \, \tau_{ki}^{\frac{1-\sigma}{1-\beta}} \, \bar{x}_{ki}^{\frac{\beta(\sigma-1)}{1-\beta} - 1}}{\sum_{m \in \mathcal{C}} N_m \, \tau_{mi}^{\frac{1-\sigma}{1-\beta}} \, \bar{x}_{mi}^{\frac{\beta(\sigma-1)}{1-\beta}}}, \quad \text{with } \tau_{ki} \triangleq \begin{cases} 1, & \text{if } k = i, \\ \tau, & \text{if } k \neq i, \end{cases} \tag{C-7}$$

Moreover, for any destination country $k \in \{i, j\}$:

$$\frac{\partial P_i}{\partial \bar{x}_{kj}} = 0 \tag{C-8}$$

In particular notice that, defining $P_i^s(\mathbf{X}) \triangleq P_i(\mathbf{X}; N_H^s, N_F^s)$, we may write, compactly:

$$\nabla_{\mathbf{x}_k^c} P_i^s = \left(\frac{\partial P_i^s}{\partial \bar{x}_{ik}}, \, \frac{\partial P_i^s}{\partial \bar{x}_{jk}} \right)' = \begin{cases} < \mathbf{0} & k = i, \\ = \mathbf{0} & k = j. \end{cases}' \tag{C-9}$$

$\square$

C.3 Proof of Proposition 2.3 (Long-Run Price Adjustments)

Use firm market clearing conditions (B-34) to define the implicit functions $(F_i)_{i \in \mathcal{C}}$ as:

$$F_i(\mathbf{P}) := \theta_{ii}(\bar{x}_{ii}) P_i^{\sigma-1} + \theta_{ij}(\bar{x}_{ij}) P_j^{\sigma-1} - q_0 = 0, \quad j \neq i \in \mathcal{C} \tag{C-10}$$

with bilateral market access coefficients $\theta_{ik},\ k \in \{i,j\}$:

$$\theta_{ik} \triangleq p_0^{-\sigma} \varphi^{\frac{\beta(\sigma-1)}{1-\beta}}\, E_k\, \tau_{ik}^{\frac{1-\sigma}{1-\beta}}\, \bar{x}_{ik}^{\frac{\beta(\sigma-1)}{1-\beta}} \tag{C-11}$$

By symmetry, it is enough to consider two perturbations $\mathrm{d}\bar{x}_{ii} > 0$, $\mathrm{d}\bar{x}_{ij} > 0$ to characterise all comparative statics. The relevant systems of total differential are:

$$\mathbf{J}_P \begin{pmatrix} \frac{\mathrm{d}P_i}{\mathrm{d}\bar{x}_{ii}} \\ \frac{\mathrm{d}P_j}{\mathrm{d}\bar{x}_{ii}} \end{pmatrix} = - \begin{pmatrix} \frac{\partial F_i}{\partial \bar{x}_{ii}} \\ \frac{\partial F_j}{\partial \bar{x}_{ii}} \end{pmatrix} \quad \text{and} \quad \mathbf{J}_P \begin{pmatrix} \frac{\mathrm{d}P_i}{\mathrm{d}\bar{x}_{ij}} \\ \frac{\mathrm{d}P_j}{\mathrm{d}\bar{x}_{ij}} \end{pmatrix} = - \begin{pmatrix} \frac{\partial F_i}{\partial \bar{x}_{ij}} \\ \frac{\partial F_j}{\partial \bar{x}_{ij}} \end{pmatrix}, \quad \text{with Jacobian } \mathbf{J}_P \triangleq \begin{pmatrix} \frac{\partial F_i}{\partial P_i} & \frac{\partial F_i}{\partial P_j} \\ \frac{\partial F_j}{\partial P_i} & \frac{\partial F_j}{\partial P_j} \end{pmatrix} \tag{C-12}$$

Straightforward differentiation of equation (C-11) yields:

$$\frac{\partial F_k}{\partial \bar{x}_{ik}} = \begin{cases} \frac{M\beta}{1-\beta}\, E_k\, \tau_{ik}^{\frac{1-\sigma}{1-\beta}}\, \bar{x}_{ik}^{\frac{\beta(\sigma-1)}{1-\beta}-1}\, P_k^{\sigma-1}, & k = i, \\ 0, & k = j, \end{cases} \qquad \text{with } M = p_0^{-\sigma} \varphi^{\frac{\beta(\sigma-1)}{1-\beta}} (\sigma - 1) \tag{C-13}$$

Also, for any $k \in \{i,j\}$, and any $m \in \{i,j\}$:

$$\frac{\partial F_m}{\partial P_k} = (\sigma-1)\, \theta_{mk} \bar{x}_{mk}\, P_k^{\sigma-2} = M E_k P_k^{\sigma-2} \bar{x}_{mk}^{\frac{\beta(\sigma-1)}{1-\beta}}\, \tau_{mk}^{\frac{1-\sigma}{1-\beta}} \tag{C-14}$$

Hence:

$$\det \mathbf{J}_P = \frac{\partial F_i}{\partial P_i} \frac{\partial F_j}{\partial P_j} - \frac{\partial F_i}{\partial P_j} \frac{\partial F_j}{\partial P_i} = M^2 E_i E_j P_i^{\sigma-2} P_j^{\sigma-2} \Delta, \quad \text{with } \Delta = \left[\bar{x}_{HH}^{\frac{\beta(\sigma-1)}{1-\beta}}\, \bar{x}_{FF}^{\frac{\beta(\sigma-1)}{1-\beta}} - \bar{x}_{HF}^{\frac{\beta(\sigma-1)}{1-\beta}}\, \bar{x}_{FH}^{\frac{\beta(\sigma-1)}{1-\beta}}\, \tau^{\frac{2(1-\sigma)}{1-\beta}} \right]$$

Under Assumption 3, $\Delta > 0$, and it follows that $\det \mathbf{J}_P \neq 0$ in particular. Applying Cramer's rule yields:

$$\begin{aligned} \frac{\mathrm{d}P_i}{\mathrm{d}\bar{x}_{ii}} &= \left[\frac{\partial F_i}{\partial \bar{x}_{ii}} \frac{\partial F_j}{\partial P_j} - \frac{\partial F_i}{\partial \bar{x}_{ii}} \frac{\partial F_j}{\partial P_j} \right] \cdot (\det \mathbf{J}_P)^{-1} = - \frac{\left[M \frac{\beta}{1-\beta}\, E_i\, \overbrace{\tau_{ii}^{\frac{1-\sigma}{1-\beta}}}^{=1}\, \bar{x}_{ii}^{\frac{\beta(\sigma-1)}{1-\beta}-1}\, P_i^{\sigma-1} \right] \left[M\, E_j\, \overbrace{\tau_{jj}^{\frac{1-\sigma}{1-\beta}}}^{=1}\, \bar{x}_{jj}^{\frac{\beta(\sigma-1)}{1-\beta}}\, P_j^{\sigma-2} \right]}{M^2\, E_i\, E_j\, P_i^{\sigma-2}\, P_j^{\sigma-2}\, \Delta} \\ &= - \frac{\beta}{1-\beta} \cdot P_i \cdot \frac{\bar{x}_{ii}^{\frac{\beta(\sigma-1)}{1-\beta}-1}\, \bar{x}_{jj}^{\frac{\beta(\sigma-1)}{1-\beta}}}{\Delta} < 0 \end{aligned} \tag{C-15}$$

With similar derivations, and using index inversion to deduce $\frac{\mathrm{d}P_i}{\mathrm{d}\bar{x}_{ij}}$ from $\frac{\mathrm{d}P_j}{\mathrm{d}\bar{x}_{ji}}$, and $\frac{\mathrm{d}P_i}{\mathrm{d}\bar{x}_{jj}}$ from $\frac{\mathrm{d}P_j}{\mathrm{d}\bar{x}_{ii}}$ we obtain:

$$\frac{\mathrm{d}P_i}{\mathrm{d}\bar{x}_{ij}} = - \frac{\beta}{1-\beta} \cdot P_i \cdot \frac{\Delta_i}{\Delta_j} \cdot \frac{\bar{x}_{ij}^{\frac{\beta(\sigma-1)}{1-\beta}-1}\, \bar{x}_{jj}^{\frac{\beta(\sigma-1)}{1-\beta}}\, \tau^{\frac{1-\sigma}{1-\beta}}}{\Delta} < 0 \tag{C-16}$$

$$\frac{\mathrm{d}P_i}{\mathrm{d}\bar{x}_{ji}} = \frac{\beta}{1-\beta} \cdot P_i \cdot \frac{\bar{x}_{ji}^{\frac{\beta(\sigma-1)}{1-\beta}-1}\, \bar{x}_{ij}^{\frac{\beta(\sigma-1)}{1-\beta}}\, \tau^{\frac{2(1-\sigma)}{1-\beta}}}{\Delta} > 0 \tag{C-17}$$

$$\frac{\mathrm{d}P_i}{\mathrm{d}\bar{x}_{jj}} = \frac{\beta}{1-\beta} \cdot P_i \cdot \frac{\Delta_i}{\Delta_j} \cdot \frac{\bar{x}_{jj}^{\frac{\beta(\sigma-1)}{1-\beta}-1}\, \bar{x}_{ij}^{\frac{\beta(\sigma-1)}{1-\beta}}\, \tau^{\frac{1-\sigma}{1-\beta}}}{\Delta} > 0 \tag{C-18}$$

Hence, defining $P_i^\ell(\mathbf{X}) \triangleq P_i(\mathbf{X}, N_H^\ell(\mathbf{X}), N_F^\ell(\mathbf{X}))$, we may compactly write:

$$\nabla_{\mathbf{x}_k^f} P_i^\ell = \left(\frac{\partial P_i^\ell}{\partial \bar{x}_{ki}},\ \frac{\partial P_i^\ell}{\partial \bar{x}_{kj}} \right) = \begin{cases} < \mathbf{0} & k = i, \\ > \mathbf{0} & k = j. \end{cases} \tag{C-19}$$

C.4 Proof of Proposition 2.4 (Long-Run Firm Masses Adjustments)

Consider a reference country $i \in \mathcal{C}$, and let $k \in \{i, j\}$ with $j \neq i$. By symmetry, it is sufficient to examine two perturbations, $\mathrm{d}\bar{x}_{jj}$ and $\mathrm{d}\bar{x}_{ij}$ to characterise all comparative statics. Equation (B-44) can be rearranged as:

$$N_i \bar{x}_{ii}^{\frac{\beta(\sigma-1)}{1-\beta}} + N_j \tau^{\frac{1-\sigma}{1-\beta}} \bar{x}_{ji}^{\frac{\beta(\sigma-1)}{1-\beta}} = \left(\frac{p_0}{\varphi^{\frac{\beta}{1-\beta}}}\right)^{\sigma-1} P_i^{1-\sigma} \tag{C-20}$$

From the forthcoming Proposition 2.5, the total number of firms in the world economy is independent of data governance. In particular, for any perturbation $\mathrm{d}\bar{x}_{kj}$, we must have:

$$\mathrm{d}N_i = -\mathrm{d}N_j \tag{C-21}$$

Taking the total differential of equation (C-20) and using fact (C-21), we get that:

$$\begin{aligned}
&\frac{\mathrm{d}N_i}{\mathrm{d}\bar{x}_{kj}} \bar{x}_{ii}^{\frac{\beta(\sigma-1)}{1-\beta}} + \frac{\mathrm{d}N_j}{\mathrm{d}\bar{x}_{kj}} \tau^{\frac{1-\sigma}{1-\beta}} \bar{x}_{ji}^{\frac{\beta(\sigma-1)}{1-\beta}} = -\frac{\mathrm{d}P_i}{\mathrm{d}\bar{x}_{kj}} (\sigma-1) \left(\frac{p_0}{\varphi^{\frac{\beta}{1-\beta}}}\right)^{\sigma-1} P_i^{-\sigma} \\
\iff\quad &\frac{\mathrm{d}N_i}{\mathrm{d}\bar{x}_{kj}} \underbrace{\left(\bar{x}_{ii}^{\frac{\beta(\sigma-1)}{1-\beta}} - \tau^{\frac{1-\sigma}{1-\beta}} \bar{x}_{ji}^{\frac{\beta(\sigma-1)}{1-\beta}}\right)}_{=\Delta_i \text{ by } (23)} = -\frac{\mathrm{d}P_i}{\mathrm{d}\bar{x}_{kj}} (\sigma-1) \left(\frac{p_0}{\varphi^{\frac{\beta}{1-\beta}}}\right)^{\sigma-1} P_i^{-\sigma} \\
\iff\quad &\frac{\mathrm{d}N_i}{\mathrm{d}\bar{x}_{kj}} = -\frac{1}{P_i} \frac{\mathrm{d}P_i}{\mathrm{d}\bar{x}_{kj}} \frac{(\sigma-1)}{\Delta_i} \left(\frac{p_0}{\varphi^{\frac{\beta}{1-\beta}}}\right)^{\sigma-1} P_i^{1-\sigma}
\end{aligned} \tag{C-22}$$

From Proposition 2.3, and letting $-k = \mathcal{C} \setminus k$:

$$\frac{1}{P_i} \frac{\mathrm{d}P_i}{\mathrm{d}\bar{x}_{kj}} = \frac{\beta}{1-\beta} \frac{\Delta_i}{\Delta_j} \frac{\bar{x}_{kj}^{\frac{\beta(\sigma-1)}{1-\beta}-1} \bar{x}_{-kj}^{\frac{\beta(\sigma-1)}{1-\beta}} \tau^{\frac{1-\sigma}{1-\beta}}}{\Delta} (-1)^{\mathbb{I}\{k=i\}} \tag{C-23}$$

Substituting equation (C-23) and the price index solution (B-42) into (C-22) and simplifying then yields:

$$\begin{aligned}
\frac{\mathrm{d}N_i}{\mathrm{d}\bar{x}_{ij}} &= \frac{\beta}{1-\beta} \frac{\Delta_i}{\Delta_j} \frac{\bar{x}_{ij}^{\frac{\beta(\sigma-1)}{1-\beta}-1} \bar{x}_{jj}^{\frac{\beta(\sigma-1)}{1-\beta}} \tau^{\frac{1-\sigma}{1-\beta}}}{\Delta} \cdot \frac{(\sigma-1)}{\Delta_i} \left(\frac{p_0}{\varphi^{\frac{\beta}{1-\beta}}}\right)^{\sigma-1} \cdot \frac{E_i \varphi^{\frac{\beta(\sigma-1)}{1-\beta}} \Delta}{q_0 p_0^{\sigma} \Delta_j} \\
&= \frac{\beta(\sigma-1)E_i}{(1-\beta)\sigma w f} \cdot \frac{\bar{x}_{ij}^{\frac{\beta(\sigma-1)}{1-\beta}-1} \bar{x}_{jj}^{\frac{\beta(\sigma-1)}{1-\beta}} \tau^{\frac{1-\sigma}{1-\beta}}}{\Delta_j^2} > 0,
\end{aligned} \tag{C-24}$$

$$\text{and}\quad \frac{\mathrm{d}N_i}{\mathrm{d}\bar{x}_{jj}} = -\frac{\beta(\sigma-1)E_i}{(1-\beta)\sigma w f} \cdot \frac{\bar{x}_{jj}^{\frac{\beta(\sigma-1)}{1-\beta}-1} \bar{x}_{ij}^{\frac{\beta(\sigma-1)}{1-\beta}} \tau^{\frac{1-\sigma}{1-\beta}}}{\Delta_j^2} < 0 \tag{C-25}$$

Since we have already shown that $\frac{\mathrm{d}N_i}{\mathrm{d}\bar{x}_{ij}} = -\frac{\mathrm{d}N_j}{\mathrm{d}\bar{x}_{ij}}$ and $\frac{\mathrm{d}N_i}{\mathrm{d}\bar{x}_{jj}} = -\frac{\mathrm{d}N_j}{\mathrm{d}\bar{x}_{jj}}$ and the problem is symmetric, inverting indices is sufficient to conclude that:

$$\frac{\mathrm{d}N_i}{\mathrm{d}\bar{x}_{ji}} = -\frac{\beta(\sigma-1)E_j}{(1-\beta)\sigma w f} \cdot \frac{\bar{x}_{ji}^{\frac{\beta(\sigma-1)}{1-\beta}-1} \bar{x}_{ii}^{\frac{\beta(\sigma-1)}{1-\beta}} \tau^{\frac{1-\sigma}{1-\beta}}}{\Delta_i^2} < 0 \tag{C-26}$$

$$\frac{\mathrm{d}N_i}{\mathrm{d}\bar{x}_{ii}} = \frac{\beta(\sigma-1)E_j}{(1-\beta)\sigma w f} \cdot \frac{\bar{x}_{ii}^{\frac{\beta(\sigma-1)}{1-\beta}-1} \bar{x}_{ji}^{\frac{\beta(\sigma-1)}{1-\beta}} \tau^{\frac{1-\sigma}{1-\beta}}}{\Delta_i^2} > 0 \tag{C-27}$$

Writing $N_i^\ell(\mathbf{X})$, we might compactly conclude that:

$$\nabla_{\mathbf{x}_k^f} N_i^\ell = \left(\frac{\partial N_i^\ell}{\partial \bar{x}_{ki}}, \frac{\partial N_i^\ell}{\partial \bar{x}_{kj}}\right) = \begin{cases} > \mathbf{0} & k = i, \\ < \mathbf{0} & k = j. \end{cases} \tag{C-28}$$

$\square$

C.5 Proof of Proposition 2.5 (Beggar-Thy-Neighbour Equilibrium Adjustments)

Given the solution for country firm mass (B-48), straightforward algebraic manipulations show that the equilibrium mass of Tech firms in the world economy is:

$$\begin{aligned} N &\triangleq N_H + N_F \\ &= \frac{\left(E_H \Delta_H \bar{x}_{FF}^{\frac{\beta(\sigma-1)}{1-\beta}} - E_F \Delta_F \tau^{\frac{1-\sigma}{1-\beta}} \bar{x}_{FH}^{\frac{\beta(\sigma-1)}{1-\beta}}\right) + \left(E_F \Delta_F \bar{x}_{HH}^{\frac{\beta(\sigma-1)}{1-\beta}} - E_H \Delta_H \tau^{\frac{1-\sigma}{1-\beta}} \bar{x}_{HF}^{\frac{\beta(\sigma-1)}{1-\beta}}\right)}{\sigma w f \Delta_H \Delta_F} \\ &= \frac{E_H + E_F}{\sigma w f} \end{aligned} \tag{C-29}$$

$\square$

C.6 Proof of Proposition 3.1 (Optimal Scope of Privacy Regulation)

The government's problem writes:

$$\begin{aligned} \max_{s_i,\, g_i} \quad & \mathcal{W}_i^h\left(\bar{x}_{ii}, \bar{x}_{ij}, \bar{x}_{ji}; \bar{x}_{jj}\right) \\ \text{s.t.} \quad & \bar{x}_{ii} \;=\; 1 - \mathbb{1}\{s_i \in \{d, r, t, c\}\} \cdot g_i \\ & \bar{x}_{ij} \;=\; 1 - \mathbb{1}\{s_i \in \{t, c\}\} \cdot g_i \\ & \bar{x}_{ji} \;=\; 1 - \mathbb{1}\{s_i \in \{r, c\}\} \cdot \pi_i g_i \\ & s_i \in \{d,\, r,\, t,\, c\} \\ & g_i \in [0, \bar{g}_i] \end{aligned} \tag{C-30}$$

with $\mathcal{W}_i^h$ defined as:

$$\mathcal{W}_i^h\left(\bar{x}_{ii}, \bar{x}_{ij}, \bar{x}_{ji}; \bar{x}_{jj}\right) \triangleq \left(\frac{w}{P_i^h}\right)^\alpha L_i - \frac{\psi_i}{2}\left(N_i^h \bar{x}_{ii}^2 + N_j^h \bar{x}_{ji}^2\right) \tag{C-31}$$

$$\text{s.t.} \quad \forall\, k \in \mathcal{C}, \quad P_i^h = \left(\frac{p_0}{\varphi^{\frac{\beta}{1-\beta}}}\right)\left(\sum_{k \in \mathcal{C}} N_k^h \tau_{ki}^{\frac{1-\sigma}{1-\beta}} \bar{x}_{ki}^{\frac{\beta(\sigma-1)}{1-\beta}}\right)^{\frac{1}{1-\sigma}}$$

$$N_k^h = \begin{cases} \bar{N}_k & \text{if } h = s \\ \dfrac{E_k \Delta_k\, \bar{x}_{-k-k}^{\frac{\beta(\sigma-1)}{1-\beta}} - E_{-k} \Delta_{-k}\, \tau^{\frac{1-\sigma}{1-\beta}}\, \bar{x}_{-kk}^{\frac{\beta(\sigma-1)}{1-\beta}}}{\sigma w f\, \Delta_k\, \Delta_{-k}} & \text{if } h = \ell \end{cases}$$

For any regulatory scope $s_i \in \{d, r, t, c\}$, define the second-stage value as:

$$V_i^h(s_i) \triangleq \max_{g_i \in [0, \bar{g}_i]} \mathcal{W}_i^h\Big(\bar{x}_{ii}(s_i, g_i),\ \bar{x}_{ij}(s_i, g_i),\ \bar{x}_{ji}(s_i, g_i);\ \bar{x}_{jj}\Big) \tag{C-32}$$

Define the set of maximisers as:

$$\mathcal{G}_i^h(s_i) \triangleq \Big\{g_i \in [0, \bar{g}_i] : \mathcal{W}_i^h\Big(\bar{x}_{ii}(s_i, g_i), \bar{x}_{ij}(s_i, g_i), \bar{x}_{ji}(s_i, g_i);\ \bar{x}_{jj}\Big) = V_i^h(s_i)\Big\}. \tag{C-33}$$

By Weierstrass' theorem, $\mathcal{G}_i^h(s_i)$ is non-empty. We let $g_{i,s_i}^h \in \mathcal{G}_i^h(s_i)$ denote a typical element of this set.

C.6.1 Long-run Welfare Maximisation

We claim that:

$$\forall s_i \in \{d, t, c\}, \quad \Big(\exists\, g_{i,s_i}^\ell > 0 \ \text{ s.t. } \ g_{i,s_i}^\ell \in \mathcal{G}_i^\ell(s_i)\Big) \implies V_i^\ell(r) > V_i^\ell(s_i) \tag{C-34}$$

In words, whenever the government would find it optimal to regulate under a scope other than residency-based, it must strictly prefer the residency-based scope in value terms. It follows that any optimal pair (s_i^ℓ, g_i^ℓ) with $g_i^\ell > 0$ must be of the form (r, g_i^ℓ).

$\underline{\exists g_{i,d}^\ell > 0 \implies V_i^\ell(r) > V_i^\ell(d):}$ Assume that there exists $g_{i,d}^\ell > 0$ such that $g_{i,d}^\ell \in \mathcal{G}_i^\ell(d)$. Using the expression for the consumer welfare function (B-12) and the result from Proposition 2.5 that the total mass of Tech firms in the world economy $N = N_H + N_F$ is independent of data usage restrictions, we may write:

$$\begin{aligned}
V_i^\ell(d) &= \mathcal{W}_i^\ell\big(1 - g_{i,d}^\ell, 1, 1; \bar{x}_{jj}\big) \\
&= \left[\frac{w}{P_i^\ell(1 - g_{i,d}^\ell, 1, 1; \bar{x}_{jj})}\right]^\alpha L_i - \frac{\psi_i}{2}\left\{N_i^\ell(1 - g_{i,d}^\ell, 1, 1; \bar{x}_{jj}) \cdot (1 - g_{i,d}^\ell)^2 + N_j^\ell(1 - g_{i,d}^\ell, 1, 1\,; \bar{x}_{jj})\right\} \\
&= \left[\frac{w}{P_i^\ell(1 - g_{i,d}^\ell, 1, 1; \bar{x}_{jj})}\right]^\alpha L_i - \frac{\psi_i}{2}\left\{N + \underbrace{\Big[(1 - g_{i,d}^\ell)^2 - 1\Big]}_{\leq 0} N_i^\ell(1 - g_{i,d}^\ell, 1, 1\,; \bar{x}_{jj})\right\} \\
&< \left[\frac{w}{P_i^\ell(1 - g_{i,d}^\ell, 1, 1 - \pi_i g_{i,d}^\ell\,; \bar{x}_{jj})}\right]^\alpha L_i \\
&\qquad - \frac{\psi_i}{2}\left\{(1 - \pi_i g_{i,d}^\ell)^2 N + \underbrace{\Big[(1 - g_{i,d}^\ell)^2 - (1 - \pi_i g_{i,d}^\ell)^2\Big]}_{\leq 0} N_i^\ell(1 - g_{i,d}^\ell, 1, 1 - \pi_i g_{i,d}^\ell\,; \bar{x}_{jj})\right\} \\
&= \mathcal{W}_i^\ell\big(1 - g_{i,d}^\ell,\ 1,\ 1 - \pi_i g_{i,d}^\ell\,; \bar{x}_{jj}\big)
\end{aligned} \tag{C-35}$$

The final inequality therefore follows from Propositions 2.3 and 2.4, which establish that P_i^ℓ increases and N_i^ℓ decreases in their third argument. By definition of the maximum, we then have:

$$V_i^\ell(r) = \max_{g_i \in [0, \bar{g}_i]} \mathcal{W}_i^\ell\big(1 - g_i,\ 1,\ 1 - \pi_i g_i\,; \bar{x}_{jj}\big) \geq \mathcal{W}_i^\ell\big(1 - g_{i,d}^\ell,\ 1,\ 1 - \pi_i g_{i,d}^\ell\,; \bar{x}_{jj}\big) > V_i^\ell(d) \tag{C-36}$$

$\underline{\exists g_{i,c}^{\ell} > 0 \implies V_i^{\ell}(r) > V_i^{\ell}(c):}$ Fixing $g_{i,c}^{\ell} > 0$ and proceeding with a similar argument as above, we have:

$$\begin{aligned}
V_i^{\ell}(c) &= \mathscr{W}_i^{\ell}\big(1-g_{i,c}^{\ell}, 1-g_{i,c}^{\ell}, 1-\pi_i g_{i,c}^{\ell}; \bar{x}_{jj}\big) && \text{(C-37)}\\
&= \left[\frac{w}{P_i^{\ell}(1-g_{i,c}^{\ell}, 1-g_{i,c}^{\ell}, 1-\pi_i g_{i,c}^{\ell}\,;\, \bar{x}_{jj})}\right]^{\alpha} L_i \\
&\quad - \frac{\psi_i}{2}\left\{ (1-\pi_i g_{i,c}^{\ell})^2\, N + \underbrace{\Big[(1-g_{i,c}^{\ell})^2 - (1-\pi_i g_{i,c}^{\ell})^2\Big]}_{\leq 0} N_i^{\ell}(1-g_{i,c}^{\ell}, 1-g_{i,c}^{\ell}, 1-\pi_i g_{i,c}^{\ell}\,;\, \bar{x}_{jj})\right\}\\
&< \left[\frac{w}{P_i^{\ell}(1-g_{i,c}^{\ell}, 1, 1-\pi_i g_{i,c}^{\ell}\,;\, \bar{x}_{jj})}\right]^{\alpha} L_i \\
&\quad - \frac{\psi_i}{2}\left\{ (1-\pi_i g_{i,c}^{\ell})^2\, N + \underbrace{\Big[(1-g_{i,c}^{\ell})^2 - (1-\pi_i g_{i,c}^{\ell})^2\Big]}_{\leq 0} N_i^{\ell}(1-g_{i,c}^{\ell}, 1, 1-\pi_i g_{i,c}^{\ell}\,;\, \bar{x}_{jj})\right\}\\
&= \mathscr{W}_i^{\ell}\big(1-g_{i,c}^{\ell},\, 1,\, 1-\pi_i g_{i,c}^{\ell}\,;\, \bar{x}_{jj}\big) && \text{(C-38)}
\end{aligned}$$

where the last inequality uses that P_i^{ℓ} and N_i^{ℓ} are respectively increasing and decreasing in their second argument, by Propositions 2.3 and 2.4. Thus, we obtain:

$$V_i^{\ell}(r) = \max_{g_i \in [0,\bar{g}_i]} \mathscr{W}_i^{\ell}\big(1-g_i,\, 1,\, 1-\pi_i g_i\,;\, \bar{x}_{jj}\big) \geq \mathscr{W}_i^{\ell}\big(1-g_{i,c}^{\ell},\, 1,\, 1-\pi_i g_{i,c}^{\ell}\,;\, \bar{x}_{jj}\big) > V_i^{\ell}(c) \qquad \text{(C-39)}$$

$\underline{\exists g_{i,t}^{\ell} > 0 \implies V_i^{\ell}(r) > V_i^{\ell}(t):}$ Also using that P_i^{ℓ} and N_i^{ℓ} are respectively increasing and decreasing in their second argument, by Propositions 2.3 and 2.4, it is immediate that for any $g_{i,t}^{\ell} > 0$, we have:

$$\begin{aligned}
V_i^{\ell}(t) &= \left[\frac{w}{P_i^{\ell}(1-g_{i,t}^{\ell}, 1-g_{i,t}^{\ell}, 1\,;\, \bar{x}_{jj})}\right]^{\alpha} L_i - \frac{\psi_i}{2}\left\{N + \Big[(1-g_{i,t}^{\ell})^2 - 1\Big] N_i^{\ell}(1-g_{i,t}^{\ell}, 1-g_{i,t}^{\ell}, 1\,;\, \bar{x}_{jj})\right\}\\
&< \left[\frac{w}{P_i^{\ell}(1-g_{i,t}^{\ell}, 1, 1\,;\, \bar{x}_{jj})}\right]^{\alpha} L_i - \frac{\psi_i}{2}\left\{N + \Big[(1-g_{i,t}^{\ell})^2 - 1\Big] N_i^{\ell}(1-g_{i,t}^{\ell}, 1, 1;\, \bar{x}_{jj})\right\}\\
&= \mathscr{W}_i^{\ell}\big(1-g_{i,t}^{\ell},\, 1,\, 1\,;\, \bar{x}_{jj}\big)\\
&\leq V_i^{\ell}(d)\\
&< V_i^{\ell}(r) && \text{(C-40)}
\end{aligned}$$

where the last inequality follows from the fact that $g_{i,t}^{\ell} > 0$ rules out $0 \in \mathcal{G}_i^{\ell}(d)$, for otherwise one would have both $V_i^{\ell}(t) > \mathscr{W}_i^{\ell}\big(1,\, 1,\, 1\,;\, \bar{x}_{jj}\big) \equiv W_0$ and $V_i^{\ell}(d) = W_0 > V_i^{\ell}(t)$, a contradiction. Hence $\mathcal{G}_i^{\ell}(d) \subset (0, \bar{g}_i]$ and thus $V_i^{\ell}(r) > V_i^{\ell}(d)$, as established before whenever $g_{i,d}^{\ell} > 0$.

C.6.2 Short-Run Welfare Maximisation

By definition, a government maximising short-run welfare treats firm masses as given. Hence, it is a trivial observation from the definition of the consumer welfare function (B-12) that such a government must be indifferent to any regulation affecting the processing of foreign consumer data ($\bar{x}_{ij}$), for such activity neither influences the domestic price level in the short run nor enters the privacy cost function of domestic consumers. Specifically, one must have $V_i^s(r) = V_i^s(c)$ and $V_i^s(d) = V_i^s(t)$.

Hence, without loss of generality, the task reduces to showing that, when extraterritorial enforcement is available, not choosing it can only reduce welfare whenever some regulation level is optimal. Or, formally:

$$\exists g_{i,d}^s > 0 \quad \Longrightarrow \quad V_i^s(r) > V_i^s(d) \tag{C-41}$$

It follows that any optimal pair (g_i^s, s_i^s) must be of the form (g_i^s, r) or (g_i^s, c), as claimed.

Observe that, by definition, we may write:

$$V_i^s(d) = \mathcal{W}_i^s\big(1-g_{i,d}^s,\, \bar{x}_{ij},\, 1;\, \bar{x}_{jj}\big) = \mathcal{W}_i^s\big(1-g_{i,d}^s,\, \bar{x}_{ij},\, 1-\pi_i g_{i,d}^s;\, \bar{x}_{jj}, \pi_i\big)\big|_{\pi_i=0} = V_i^s(r;\pi_i)\big|_{\pi_i=0} \tag{C-42}$$

Hence, a sufficient condition for inequality (C-41) to hold is:

$$\frac{\partial \mathcal{W}_i^s}{\partial \pi_i}\big(\bar{x}_{ii}(g_{i,d}^s),\, \bar{x}_{ij},\, \bar{x}_{ji}(g_{i,d}^s;\pi_i);\, \bar{x}_{jj}\big) > 0, \quad \forall\, \pi_i \in [0,1] \tag{C-43}$$

Differentiating $\mathcal{W}_i^s$ with respect to its first argument yields:

$$\begin{aligned}\frac{\partial \mathcal{W}_i^s}{\partial \bar{x}_{ii}} &= -\,\alpha\, L_i\, w^{\alpha}\, (P_i^s)^{-\alpha-1}\,\frac{\partial P_i^s}{\partial \bar{x}_{ii}} \;-\; \psi_i\, \bar{x}_{ii}\, N_i^s \\ &= N_i^s \cdot \left[\frac{\alpha\beta L_i w^{\alpha}}{1-\beta}\cdot \frac{(P_i^s)^{-\alpha}\, \bar{x}_{ii}^{\frac{\beta(\sigma-1)}{1-\beta}-1}}{\sum_{k\in\mathcal{C}} N_k\, \tau_{ki}^{\frac{1-\sigma}{1-\beta}}\, \bar{x}_{ki}^{\frac{\beta(\sigma-1)}{1-\beta}}} - \psi_i\, \bar{x}_{ii}\right]\end{aligned} \tag{C-44}$$

Hence, for any $g_{i,d}^s \in (0, \bar{g}_i]$ that achieves $V^s(d)$, one must have:

$$\frac{\partial \mathcal{W}_i^s}{\partial \bar{x}_{ii}}\underbrace{\frac{\partial \bar{x}_{ii}}{\partial g_i}}_{<0}(\bar{x}_{ii}(g_{i,d}^s), \bar{x}_{ij}, 1) \geq 0 \quad \Longrightarrow \quad \psi_i\, \bar{x}_{ii} - \frac{\alpha\beta L_i w^{\alpha}}{1-\beta}\cdot \frac{[P_i^s(\bar{x}_{ii}(g_{i,d}^s), \bar{x}_{ij}, 1)]^{-\alpha}\, \bar{x}_{ii}^{\frac{\beta(\sigma-1)}{1-\beta}-1}}{N_i^s \bar{x}_{ii}\big(g_{i,d}^s\big)^{\frac{\beta(\sigma-1)}{1-\beta}} + N_j^s \tau^{\frac{1-\sigma}{1-\beta}}} \geq 0 \tag{C-45}$$

Meanwhile, similar derivations yield:

$$\begin{aligned}\frac{\partial \mathcal{W}_i^s}{\partial \pi_i} = \frac{\partial \mathcal{W}_i^s}{\partial \bar{x}_{ji}}\cdot\frac{\partial \bar{x}_{ji}}{\partial \pi_i} &= \left[-\alpha w^{\alpha} L_i (P_i^s)^{\alpha-1}\frac{\partial P_i^s}{\partial \bar{x}_{ji}} - \psi_i\, \bar{x}_{ji}\cdot N_j^s\right]\cdot(-g_{i,d}^s) \\ &= N_j^s g_i \cdot \left[\psi_i\, \bar{x}_{ji} - \frac{\alpha\beta L_i w^{\alpha}}{1-\beta}\cdot\frac{(P_i^s)^{-\alpha}\ \bar{x}_{ji}^{\frac{\beta(\sigma-1)}{1-\beta}-1}}{\sum_{k\in\mathcal{C}} N_k^s\, \tau_{ki}^{\frac{1-\sigma}{1-\beta}}\ \bar{x}_{ki}^{\frac{\beta(\sigma-1)}{1-\beta}}}\right]\end{aligned} \tag{C-46}$$

Under Assumption 5, $\frac{\beta(\sigma-1)}{1-\beta} - 1 < 0$, so that $y \mapsto y^{\frac{\beta(\sigma-1)}{1-\beta}-1}$ is strictly decreasing. Moreover, at any $g_i > 0$, and for each $\pi_i \leq 1$, we have $\bar{x}_{ji}(g_i;\pi_i) \leq \bar{x}_{ji}(g_i;1) = \bar{x}_{ii}(g_i) < 1$, while $(P_i^s)^{-\alpha}$ is increasing in its strictly third argument in the short run by Proposition 2.2, which gives:

$$\bar{x}_{ji}^{\frac{\beta(\sigma-1)}{1-\beta}-1} \geq \bar{x}_{ii}^{\frac{\beta(\sigma-1)}{1-\beta}-1} \quad \text{and} \quad P_i^s(\bar{x}_{ii}(g_{i,d}^s), \bar{x}_{ij}, 1) \geq P_i^s\big(\bar{x}_{ii}(g_{i,d}^s), \bar{x}_{ij}, \bar{x}_{ji}(g_{i,d}^s;\pi_i)\big) \tag{C-47}$$

In particular, we deduce from the preceding optimality condition (C-45) that:

$$\frac{\partial \mathcal{W}_i^s}{\partial \pi_i}\big(\bar{x}_{ii}(g_{i,d}^s), \bar{x}_{ij}, \bar{x}_{ji}(g_{i,d}^s;\pi_i)\big) \propto \psi_i \bar{x}_{ji}(g_{i,d}^s;\pi_i) - \frac{\alpha\beta L_i w^{\alpha}}{1-\beta}\frac{[P_i^s\big(\bar{x}_{ii}(g_{i,d}^s), \bar{x}_{ij}, \bar{x}_{ji}(g_{i,d}^s;\pi_i)\big)]^{-\alpha}[\bar{x}_{ji}(g_{i,d}^s;\pi_i)]^{\frac{\beta(\sigma-1)}{1-\beta}-1}}{N_i^s \bar{x}_{ii}\big(g_{i,d}^s\big)^{\frac{\beta(\sigma-1)}{1-\beta}} + N_j^s \tau^{\frac{1-\sigma}{1-\beta}} \bar{x}_{ji}\big(g_{i,d}^s;\pi_i\big)^{\frac{\beta(\sigma-1)}{1-\beta}}}$$

$$
\begin{aligned}
&> \psi_i \bar{x}_{ii}(g_{i,d}^s;1) - \frac{\alpha\beta L_i w^\alpha}{1-\beta}\frac{[P_i^s(\bar{x}_{ii}(g_{i,d}^s),\bar{x}_{ij},1)]^{-\alpha}[\bar{x}_{ii}(g_{i,d}^s)]^{\frac{\beta(\sigma-1)}{1-\beta}-1}}{N_i^s\bar{x}_{ii}\left(g_{i,d}^s\right)^{\frac{\beta(\sigma-1)}{1-\beta}}+N_j^s\tau^{\frac{1-\sigma}{1-\beta}}} \\
&\geq 0 \qquad \text{(C-48)}
\end{aligned}
$$

where the strict inequality follows from the denominator. Collecting our results, we can safely conclude that:

$$
\begin{aligned}
g_{i,d}^s > 0 \quad \Longrightarrow \quad V_i^s(r)\left(=V_i^s(c)\right) &= \max_{g_i\in[0,\bar{g}_i]} \mathscr{W}_i^s\left(1-g_i,\,1,\,1-\pi_i g_i;\,\bar{x}_{jj}\right) \\
&\geq \mathscr{W}_i^s\left(1-g_{i,d}^s,\,1,\,1-\pi_i g_{i,d}^s\,;\,\bar{x}_{jj},\pi_i\right), \quad \forall\,\pi_i\in(0,1] \\
&> \mathscr{W}_i^s\left(1-g_{i,d}^s,\,1,\,1-\pi_i g_{i,d}^s\,;\,\bar{x}_{jj},\pi_i\right)\big|_{\pi_i=0} \\
&= V_i^s(d)\left(=V_i^s(t)\right) \qquad \text{(C-49)}
\end{aligned}
$$

Hence, any optimal pair $(s_i^s,\, g_i^s)$ with $g_i^s > 0$ must be of the form $(r,\, g_i^s)$ or $(c,\, g_i^s)$ with strict indifference.

□

C.7 Proof of Proposition 3.2 (Existence and Uniqueness of an Optimal Regulation Level)

Conditional on a residency-based scope of applicability, the problem of the government might be written as:

$$
\begin{aligned}
\max_{g_i\in[0,\bar{g}_i]} \quad \mathscr{W}_i^h &= \left(\frac{w}{P_i^h\left(N_i^h,\bar{x}_{ii},\bar{x}_{ji}\right)}\right)^\alpha L_i - \frac{\psi_i}{2}\left[\bar{x}_{ji}^2\,N + \left(\bar{x}_{ii}^2-\bar{x}_{ji}^2\right)N_i^h\left(\bar{x}_{ii},\bar{x}_{ji}\right)\right] \qquad \text{(C-50)} \\
\text{s.t.} \quad \bar{x}_{ii} &= 1-g_i, \quad \bar{x}_{ji} = 1-\pi_i g_i, \quad P_i^h,\ N_i^h \text{ as in (C-31).}
\end{aligned}
$$

Treating all constraints as functions of g_i and differentiating $\mathscr{W}_i^h$, we obtain that:

$$
\frac{\mathrm{d}\mathscr{W}_i^h}{\mathrm{d}g_i} = -\alpha L_i w^\alpha (P_i^h)^{-\alpha-1}\frac{\mathrm{d}P_i^h}{\mathrm{d}g_i} + \psi_i\left(\bar{x}_{ii}\,N_i^h + \pi_i\,\bar{x}_{ji}\,(N-N_i^h) - \frac{1}{2}\left(\bar{x}_{ii}^2-\bar{x}_{ji}^2\right)\frac{\mathrm{d}N_i^h}{\mathrm{d}g_i}\right) \qquad \text{(C-51)}
$$

Hence, at any interior optimum such that $\frac{\mathrm{d}\mathscr{W}_i^h}{\mathrm{d}g_i}=0$:

$$
\underbrace{\alpha L_i w^\alpha (P_i^h)^{-\alpha}\cdot\frac{1}{P_i^h}\frac{\mathrm{d}P_i^h}{\mathrm{d}g_i}}_{MC_i^h\equiv\text{Marginal cost of regulation}} = \underbrace{\psi_i\left(\bar{x}_{ii}\,N_i^h + \pi_i\,\bar{x}_{ji}\,(N-N_i^h) - \frac{1}{2}\left(\bar{x}_{ii}^2-\bar{x}_{ji}^2\right)\frac{\mathrm{d}N_i^h}{\mathrm{d}g_i}\right)}_{MB_i^h\equiv\text{Marginal benefit of regulation}} \qquad \text{(C-52)}
$$

We want to show that a solution to the government's problem (C-50) must exist and is unique.

Existence Existence of at least one maximiser follows directly from the continuity of $\mathscr{W}_i^h$ over the compact set $[0,\bar{g}_i]$, by Weierstrass' theorem.

Uniqueness As a pedagogical device for the main text, we establish that the marginal benefit and marginal cost functions (C-52) exhibit strict monotonicity in the regulation level g_i. Specifically, for each horizon $h\in\{s,\ell\}$, the mapping $g_i\mapsto MB_i^h(g_i)$ is strictly decreasing, while $g_i\mapsto MC_i^h(g_i)$ is strictly increasing.

Notice that this condition is sufficient for $\mathscr{W}_i^h$ to be strictly concave, since for $0 \le g_i' < g_i'' \le \bar{g}_i$, we have:

$$\frac{\mathrm{d}\mathscr{W}_i^h}{\mathrm{d}g_i}(g_i'') = MB_i^h(g_i'') - MC_i^h(g_i'') < MB_i^h(g_i') - MC_i^h(g_i') = \frac{\mathrm{d}\mathscr{W}_i^h}{\mathrm{d}g_i}(g_i'). \tag{C-53}$$

Accordingly, by continuity, the implicit equation $\frac{\mathrm{d}\mathscr{W}_i^h}{\mathrm{d}g_i}(g_i) = 0$ (C-52) then admits at most one solution in $(0, \bar{g}_i)$. If no such root exists, $\frac{\mathrm{d}\mathscr{W}_i^h}{\mathrm{d}g_i}$ retains a constant sign on $[0, \bar{g}_i]$, and the unique maximiser occurs at $g_i = 0$ if it is negative throughout, or at $g_i = \bar{g}_i$ if it is positive throughout.

It is analytically convenient to let $r_i(g_i)$ denote the effective data usage capacity of local establishments relative to foreign firms in market i. Observe that the function r_i decreases with the regulation level g_i, and is strictly decreasing whenever $\pi_i < 1$:

$$\begin{aligned} & r_i(g_i) \triangleq \frac{\bar{x}_{ii}(g_i)}{\bar{x}_{ji}(g_i)} = \frac{1-g_i}{1-\pi_i g_i} \le 1 \;\text{ with equality if and only if }\; \pi_i = 1 \text{ or } g_i = 0 \\ \Longrightarrow\quad & \frac{\mathrm{d}r_i}{\mathrm{d}g_i} = \frac{-(1-\pi_i)}{\left(1-\pi_i\, g_i\right)^2} \le 0 \;\text{ for all }\; g_i \in [0, \bar{g}_i] \;\text{ with equality if and only if }\; \pi_i = 1 \end{aligned}$$

C.7.1 Short Run Welfare Maximisation

$MC_i^s : [0, \bar{g}_i] \mapsto \mathbb{R}_{>0}$ is an increasing function of g_i: Using the comparative statics results from Proposition 2.2, one obtains:

$$\frac{\mathrm{d}P_i^s}{\mathrm{d}g_i} = (-1)\cdot\left[\frac{\partial P_i^s}{\partial \bar{x}_{ii}} + \pi_i \frac{\partial P_i^s}{\partial \bar{x}_{ji}}\right] \quad\Longrightarrow\quad \frac{1}{P_i^s}\frac{\mathrm{d}P_i^s}{\mathrm{d}g_i} = \frac{\beta}{1-\beta}\cdot\left[\frac{N_i^s\,\bar{x}_{ii}^{\frac{\beta(\sigma-1)}{1-\beta}-1} + \pi_i \tau^{\frac{1-\sigma}{1-\beta}} N_j^s \bar{x}_{ji}^{\frac{\beta(\sigma-1)}{1-\beta}-1}}{N_i^s\,\bar{x}_{ii}^{\frac{\beta(\sigma-1)}{1-\beta}} + \tau^{\frac{1-\sigma}{1-\beta}} N_j^s \bar{x}_{ji}^{\frac{\beta(\sigma-1)}{1-\beta}}}\right] \tag{C-54}$$

A rewriting in terms of r_i (C-54) then yields:

$$\frac{1}{P_i^s}\frac{\mathrm{d}P_i^s}{\mathrm{d}g_i} = \frac{\beta}{1-\beta}\cdot\left[N_i^s\, r_i^{\frac{\beta(\sigma-1)}{1-\beta}-1} + \pi_i\,\tau^{\frac{1-\sigma}{1-\beta}}\, N_j^s\right]\cdot \bar{x}_{ji}^{-1}\cdot\left[N_i^s\, r_i^{\frac{\beta(\sigma-1)}{1-\beta}} + \tau^{\frac{1-\sigma}{1-\beta}}\, N_j^s\right]^{-1} \tag{C-55}$$

Moreover, a straightforward rewriting of the price index (C-31) in terms of r_i gives:

$$[P_i^s]^{-\alpha} = \left[\frac{p_0}{\varphi^{\frac{\beta}{1-\beta}}}\right]^{-\alpha} \bar{x}_{ji}^{\frac{\alpha\beta}{1-\beta}}\left[N_i^s r_i^{\frac{\beta(\sigma-1)}{1-\beta}} + N_j^s \tau^{\frac{1-\sigma}{1-\beta}}\right]^{\frac{\alpha}{\sigma-1}} \tag{C-56}$$

Hence using (C-55) and (C-56), the short-run marginal cost of regulation MC_i^s (C-52) might be written as:

$$\begin{aligned} MC_i^s(g_i) &= \alpha L_i w^\alpha (P_i^s)^{-\alpha}\cdot\frac{1}{P_i^s}\frac{\mathrm{d}P_i^s}{\mathrm{d}g_i} \\ &= \frac{\alpha\beta L_i\, w^\alpha}{1-\beta}\left[\frac{p_0}{\varphi^{\frac{\beta}{1-\beta}}}\right]^{-\alpha}\left[N_i^s[r_i(g_i)]^{\frac{\beta(\sigma-1)}{1-\beta}} + N_j^s\,\tau^{\frac{1-\sigma}{1-\beta}}\right]^{\frac{\alpha}{\sigma-1}-1}\left[N_i^s[r_i(g_i)]^{\frac{\beta(\sigma-1)}{1-\beta}-1} + \pi_i\, N_j^s\,\tau^{\frac{1-\sigma}{1-\beta}}\right][\bar{x}_{ji}(g_i)]^{\frac{\alpha\beta}{1-\beta}-1} \\ &> 0 \end{aligned} \tag{C-57}$$

where the last inequality stresses that $MC_i^s : [0, \bar{g}_i] \mapsto \mathbb{R}_{>0}$, as claimed in the statement of Proposition 2.1.

Notice that, under Assumption 5, and using that $\frac{\mathrm{d}r_i}{\mathrm{d}g_i} \leq 0$, we have that:

$$\sigma > 1+\alpha \quad \Longrightarrow \quad \frac{\alpha}{\sigma-1} - 1 < 0 \quad \Longrightarrow \quad \frac{\mathrm{d}}{\mathrm{d}g_i}\left\{\left[N_i^s\big(r_i(g_i)\big)^{\frac{\beta(\sigma-1)}{1-\beta}} + N_j^s\,\tau^{\frac{1-\sigma}{1-\beta}}\right]^{\frac{\alpha}{\sigma-1}-1}\right\} \geq 0 \tag{C-58}$$

It also follows from Assumption 5 that:

$$\beta < \frac{1}{\sigma} \quad \Longrightarrow \quad \frac{\beta(\sigma-1)}{1-\beta} - 1 < 0 \quad \Longrightarrow \quad \frac{\mathrm{d}}{\mathrm{d}g_i}\left\{N_i^s\big(r_i(g_i)\big)^{\frac{\beta(\sigma-1)}{1-\beta}-1} + \pi_i\,N_j^s\,\tau^{\frac{1-\sigma}{1-\beta}}\right\} \geq 0 \tag{C-59}$$

In addition, we have that $\frac{\mathrm{d}\bar{x}_{ji}}{\mathrm{d}g_i} = -\pi_i < 0$, and thus:

$$\frac{\alpha\beta}{1-\beta} - 1 < \frac{\frac{\alpha}{\sigma}}{1-\frac{1}{\sigma}} - 1 = \frac{\alpha}{\sigma-1} - 1 < 0 \quad \Longrightarrow \quad \frac{\mathrm{d}}{\mathrm{d}g_i}\left\{\big(\bar{x}_{ji}(g_i)\big)^{\frac{\alpha\beta}{1-\beta}-1}\right\} > 0 \tag{C-60}$$

As a positive scalar multiple of these functions (C-57), MC_i^s is strictly increasing in g_i.

<u>MB_i^s **is a decreasing function of** g_i**:**</u> Use the definition of the marginal benefit of regulation (C-52) to write:

$$\begin{aligned} MB_i^s(g_i) &\triangleq \psi_i\left(\bar{x}_{ii}\,N_i^s + \pi_i\,\bar{x}_{ji}\,N_j^s - \frac{1}{2}\Big(\bar{x}_{ii}^2 - \bar{x}_{ji}^2\Big)\underbrace{\frac{\mathrm{d}N_i^s}{\mathrm{d}g_i}}_{=0}\right) \\ &= \psi_i \cdot \bar{x}_{ji}(g_i)\,\cdot\big(r_i(g_i)\,N_i^s + \pi_i N_j^s\big) \end{aligned} \tag{C-61}$$

Each term is positive. Since r_i is decreasing by (C-54), and $\bar{x}_{ji} = 1 - \pi_i g_i$ is strictly decreasing for all $\pi_i \in (0,1]$, MB_i^s is itself strictly decreasing in g_i as their product.

C.7.2 Long Run Welfare Maximisation

We begin by establishing an intermediary result.

Lemma 1 (Concavity of N_i^ℓ in g_i)**.** *The function $g_i \mapsto N_i^\ell(g_i)$ is decreasing and concave. Moreover, if $\pi_i < 1$, it is strictly decreasing and strictly concave:*

$$\frac{\mathrm{d}N_i^\ell}{\mathrm{d}g_i}(g_i;\pi_i) \leq 0 \quad \textit{and} \quad \frac{\mathrm{d}^2N_i^\ell}{\mathrm{d}g_i^2}(g_i;\pi_i) \leq 0, \quad \textit{with equality if and only if } \pi_i = 1. \tag{C-62}$$

Proof. From Proposition 2.4:

$$\frac{\mathrm{d}N_i^\ell}{\mathrm{d}\bar{x}_{ii}} = \frac{\beta(\sigma-1)E_j}{(1-\beta)\sigma w f}\cdot\frac{\bar{x}_{ii}^{\frac{\beta(\sigma-1)}{1-\beta}-1}\bar{x}_{ji}^{\frac{\beta(\sigma-1)}{1-\beta}}\tau^{\frac{1-\sigma}{1-\beta}}}{\Delta_i^2} \qquad \text{and} \qquad \frac{\mathrm{d}N_i^\ell}{\mathrm{d}\bar{x}_{ji}} = -\frac{\beta(\sigma-1)E_j}{(1-\beta)\sigma w f}\cdot\frac{\bar{x}_{ji}^{\frac{\beta(\sigma-1)}{1-\beta}-1}\bar{x}_{ii}^{\frac{\beta(\sigma-1)}{1-\beta}}\tau^{\frac{1-\sigma}{1-\beta}}}{\Delta_i^2} \tag{C-63}$$

Hence:

$$\begin{aligned} \frac{\mathrm{d}N_i^\ell}{\mathrm{d}g_i} &= \frac{\mathrm{d}N_i^\ell}{\mathrm{d}\bar{x}_{ii}}\frac{\mathrm{d}\bar{x}_{ii}}{\mathrm{d}g_i} + \frac{\mathrm{d}N_i^\ell}{\mathrm{d}\bar{x}_{ji}}\frac{\mathrm{d}\bar{x}_{ji}}{\mathrm{d}g_i} \\ &= -\frac{\beta(\sigma-1)E_j\tau^{\frac{1-\sigma}{1-\beta}}}{(1-\beta)\sigma w f\Delta_i^2}\left[\bar{x}_{ii}^{\frac{\beta(\sigma-1)}{1-\beta}-1}\bar{x}_{ji}^{\frac{\beta(\sigma-1)}{1-\beta}} - \pi_i\,\bar{x}_{ji}^{\frac{\beta(\sigma-1)}{1-\beta}-1}\bar{x}_{ii}^{\frac{\beta(\sigma-1)}{1-\beta}}\right] \end{aligned}$$

$$= -\frac{\beta(\sigma-1)E_j\tau^{\frac{1-\sigma}{1-\beta}}}{(1-\beta)\sigma w f} \cdot [\Delta_i(g_i)]^{-2} \cdot \left(\bar{x}_{ji}(g_i)\right)^{2\left(\frac{\beta(\sigma-1)}{1-\beta}-1\right)} \cdot \left(r_i(g_i)\right)^{\frac{\beta(\sigma-1)}{1-\beta}-1} \cdot \underbrace{[1-\pi_i]}_{\geq 0} \tag{C-64}$$

Case 1: $\pi_i = 1$ $\frac{\mathrm{d}N_i^\ell}{\mathrm{d}g_i} = 0$ for all $g_i \in [0, \bar{g}_i]$ and the result follows.

Case 2: $\pi_i \in (0,1)$ Under Assumption 5, $\frac{\beta(\sigma-1)}{1-\beta} - 1 < 0$, while $\frac{\mathrm{d}\bar{x}_{ji}}{\mathrm{d}g_i} = -\pi_i < 0$ and $\frac{\mathrm{d}r_i}{\mathrm{d}g_i} \leq 0$ for all $\pi_i \in (0,1]$ by construction (C-54). Therefore:

$$\frac{\mathrm{d}}{\mathrm{d}g_i}\left\{\left(\bar{x}_{ii}(g_i)\right)^{2\left(\frac{\beta(\sigma-1)}{1-\beta}-1\right)}\right\} > 0 \quad \text{and} \quad \frac{\mathrm{d}}{\mathrm{d}g_i}\left\{\left(\left(r_i(g_i)\right)^{\frac{\beta(\sigma-1)}{1-\beta}-1}\right)^{\frac{\alpha\beta}{1-\beta}-1}\right\} \geq 0 \tag{C-65}$$

Moreover, using the definition of Δ_i (22), we have:

$$[\Delta_i(g_i)]^{-2} = \left([\bar{x}_{ii}(g_i)]^{\frac{\beta(\sigma-1)}{1-\beta}} - \tau^{\frac{1-\sigma}{1-\beta}}\,[\bar{x}_{ji}(g_i)]^{\frac{\beta(\sigma-1)}{1-\beta}}\right)^{-2} = [\bar{x}_{ji}(g_i)]^{-2}\left([r_i(g_i)]^{\frac{\beta(\sigma-1)}{1-\beta}} - \tau^{\frac{1-\sigma}{1-\beta}}\right)^{-2} \tag{C-66}$$

Note that both terms are increasing in g_i and positive, since $\Delta_i > 0$ in any determinate equilibrium by Proposition 2.1. Hence, $\frac{\mathrm{d}N_i^\ell}{\mathrm{d}g_i}$ can be expressed as a negative scalar multiple of the product of positive-valued, increasing functions of g_i, at least one of which strictly. Therefore, $\frac{\mathrm{d}N_i^\ell}{\mathrm{d}g_i}$ is negative-valued and strictly decreasing in g_i. Moreover, it is differentiable as the product of differentiable functions, and it follows that:

$$\pi_i \in (0,1) \quad \Longrightarrow \quad \frac{\mathrm{d}^2 N_i^\ell}{\mathrm{d}g_i^2} < 0 \quad \Longleftrightarrow \quad N_i^\ell \text{ is strictly concave in } g_i. \tag{C-67}$$

□

$MC_i^\ell : [0, \bar{g}_i] \mapsto \mathbb{R}_{>0}$ is an increasing function of g_i: Proceeding similarly to the short run case, notice the comparative statics results from Proposition 2.3 give:

$$\frac{1}{P_i^\ell}\frac{\mathrm{d}P_i^\ell}{\mathrm{d}g_i} = \frac{\beta}{1-\beta}\frac{1}{\Delta}\left(\bar{x}_{ii}^{\frac{\beta(\sigma-1)}{1-\beta}-1}\bar{x}_{jj}^{\frac{\beta(\sigma-1)}{1-\beta}} - \pi_i\,\bar{x}_{ji}^{\frac{\beta(\sigma-1)}{1-\beta}-1}\bar{x}_{ij}^{\frac{\beta(\sigma-1)}{1-\beta}}\tau^{\frac{2(1-\sigma)}{1-\beta}}\right) \tag{C-68}$$

Moreover, we already know that the solution for long-term price index (B-42) is:

$$(P_i^\ell)^{\sigma-1} = \frac{q_0 p_0^\sigma \Delta_j}{E_i \varphi^{\frac{\beta(\sigma-1)}{1-\beta}}\Delta} \quad \Longrightarrow \quad (P_i^\ell)^{-\alpha} = \left(\frac{E_i\varphi^{\frac{\beta(\sigma-1)}{1-\beta}}\Delta}{q_0 p_0^\sigma \Delta_j}\right)^{\frac{\alpha}{\sigma-1}} \tag{C-69}$$

Note that, under Assumptions 3 and 5, we have:

$$\begin{aligned}
\Delta_j > 0 \quad &\Longleftrightarrow \quad \bar{x}_{jj}^{\frac{\beta(\sigma-1)}{1-\beta}} > \tau^{\frac{1-\sigma}{1-\beta}}\bar{x}_{ij}^{\frac{\beta(\sigma-1)}{1-\beta}} (> 0) \\
&\Longrightarrow \quad \bar{x}_{jj}^{\frac{\beta(\sigma-1)}{1-\beta}} > \underbrace{\pi_i\tau^{\frac{1-\sigma}{1-\beta}}}_{<1}\cdot\tau^{\frac{1-\sigma}{1-\beta}}\bar{x}_{ij}^{\frac{\beta(\sigma-1)}{1-\beta}} \\
&\Longrightarrow \quad \bar{x}_{ii}^{\frac{\beta(\sigma-1)}{1-\beta}-1}\bar{x}_{jj}^{\frac{\beta(\sigma-1)}{1-\beta}} > \pi_i\,\bar{x}_{ji}^{\frac{\beta(\sigma-1)}{1-\beta}-1}\bar{x}_{ij}^{\frac{\beta(\sigma-1)}{1-\beta}}\tau^{\frac{2(1-\sigma)}{1-\beta}} \\
&\qquad \text{since} \quad \bar{x}_{ii} \leq \bar{x}_{ji} \quad \text{and thus} \quad \bar{x}_{ii}^{\frac{\beta(\sigma-1)}{1-\beta}-1} \geq \bar{x}_{ji}^{\frac{\beta(\sigma-1)}{1-\beta}-1} \quad \text{whenever } \beta < \frac{1}{\sigma}.
\end{aligned} \tag{C-70}$$

Analogous rewritings in terms of r_i (C-54), omitted for brevity, suggest the following rewriting:

$$\begin{aligned}
MC_i^\ell(g_i) &\triangleq \alpha L_i w^\alpha (P_i^\ell)^{-\alpha} \cdot \frac{1}{P_i^\ell}\frac{\mathrm{d}P_i}{\mathrm{d}g_i} \\
&= \frac{\alpha\beta L_i w^\alpha}{1-\beta}\left[\frac{E_i\varphi^{\frac{\beta(\sigma-1)}{1-\beta}}}{q_0 p_0^\sigma \Delta_j}\right]^{\frac{\alpha}{\sigma-1}} \left[\Delta(g_i)\right]^{\frac{\alpha}{\sigma-1}-1} \underbrace{\left[\left(r_i(g_i)\right)^{\frac{\beta(\sigma-1)}{1-\beta}-1}\bar{x}_{jj}^{\frac{\beta(\sigma-1)}{1-\beta}} - \pi_i \bar{x}_{ij}^{\frac{\beta(\sigma-1)}{1-\beta}}\tau^{\frac{2(1-\sigma)}{1-\beta}}\right]}_{>0 \text{ by (C-70)}} \left[\bar{x}_{ji}(g_i)\right]^{\frac{\beta(\sigma-1)}{1-\beta}-1} \\
&> 0
\end{aligned} \tag{C-71}$$

where, by definition (23):

$$\begin{aligned}
\Delta(g_i) &\triangleq \bar{x}_{ii}^{\frac{\beta(\sigma-1)}{1-\beta}}\bar{x}_{jj}^{\frac{\beta(\sigma-1)}{1-\beta}} - \bar{x}_{ij}^{\frac{\beta(\sigma-1)}{1-\beta}}\bar{x}_{ji}^{\frac{\beta(\sigma-1)}{1-\beta}}\tau^{\frac{2(1-\sigma)}{1-\beta}} \\
&= \left(\bar{x}_{ji}(g_i)\right)^{\frac{\beta(\sigma-1)}{1-\beta}}\left(\left(r_i(g_i)\right)^{\frac{\beta(\sigma-1)}{1-\beta}}\bar{x}_{jj}^{\frac{\beta(\sigma-1)}{1-\beta}} - \bar{x}_{ij}^{\frac{\beta(\sigma-1)}{1-\beta}}\tau^{\frac{2(1-\sigma)}{1-\beta}}\right) > 0 \quad \text{by Assumption 3.}
\end{aligned} \tag{C-72}$$

Hence, MC_i^ℓ maps $[0, \bar{g}_i]$ into $\mathbb{R}_{>0}$, as stated in Proposition 3.2. Again, we proceed by showing that the above rewriting (C-71) amounts to expressing MC_i^ℓ as the product of positive-valued, increasing functions of g_i.

Consider first the term $\left(\Delta(g_i)\right)^{\frac{\alpha}{\sigma-1}-1}$. Since $\bar{x}_{ji} > 0$, the equilibrium determinacy condition $\Delta(g_i) > 0$ implies that both inner terms in (C-72) are strictly positive. Moreover, we have $\frac{\mathrm{d}\bar{x}_{ji}}{\mathrm{d}g_i} = -\pi_i < 0$ and $\frac{\mathrm{d}\bar{r}_i}{\mathrm{d}g_i} \leq 0$, so both components are decreasing in g_i. It follows that their product $\Delta(g_i)$ is also decreasing, hence $\frac{\mathrm{d}\Delta}{\mathrm{d}g_i} < 0$. Combined with the parameter restrictions in Assumption 5, the chain rule then yields:

$$\sigma > 1+\alpha \quad \Longrightarrow \quad \frac{\mathrm{d}}{\mathrm{d}g_i}\left\{\left(\Delta(g_i)\right)^{\frac{\alpha}{\sigma-1}-1}\right\} > 0 = \underbrace{\left(\frac{\alpha}{\sigma-1}-1\right)}_{<0} \cdot \underbrace{\Delta^{\frac{\alpha}{\sigma-1}-2}}_{>0} \cdot \underbrace{\frac{\mathrm{d}\Delta}{\mathrm{d}g_i}}_{\leq 0} \geq 0 \tag{C-73}$$

Moving to the second term in (C-71), the same Assumption 5 gives:

$$\frac{\mathrm{d}}{\mathrm{d}g_i}\left\{\left(r_i(g_i)\right)^{\frac{\beta(\sigma-1)}{1-\beta}-1}\bar{x}_{jj}^{\frac{\beta(\sigma-1)}{1-\beta}} - \pi_i \bar{x}_{ij}^{\frac{\beta(\sigma-1)}{1-\beta}}\tau^{\frac{2(1-\sigma)}{1-\beta}}\right\} = \underbrace{\left(\frac{\beta(\sigma-1)}{1-\beta}-1\right)}_{<0} \cdot \underbrace{r_i^{\frac{\beta(\sigma-1)}{1-\beta}-2}}_{>0} \cdot \underbrace{\frac{\mathrm{d}r_i}{\mathrm{d}g_i}}_{\leq 0} \geq 0 \tag{C-74}$$

Finally, $g_i \mapsto \bar{x}_{ji}(g_i)$ is strictly decreasing for each $\pi_i \in (0,1]$, and we have that:

$$\frac{\mathrm{d}}{\mathrm{d}g_i}\left\{\bar{x}_{ji}(g_i)^{\frac{\beta(1-\sigma)}{1-\beta}-1}\right\} = \underbrace{\left(\frac{\beta(\sigma-1)}{1-\beta}-1\right)}_{<0} \cdot \underbrace{\frac{\mathrm{d}\bar{x}_{ji}}{\mathrm{d}g_i}}_{<0} > 0 \tag{C-75}$$

Collecting our results, we conclude that MC_i^ℓ is a strictly increasing function of g_i as it can be expressed as the product of positive-valued and increasing function g_i, at least one of which strictly.

MB_i^ℓ is a decreasing function of g_i: Using the definition of MB_i^ℓ (C-52):

$$MB_i^\ell(g_i) = \psi_i\left(\pi_i\,\bar{x}_{ji}(g_i)\,N + \left(\bar{x}_{ii}(g_i) - \pi_i\,\bar{x}_{ji}(g_i)\right)N_i^\ell - \frac{1}{2}\left[\left(\bar{x}_{ii}(g_i)\right)^2 - \left(\bar{x}_{ji}(g_i)\right)^2\right]\left[\frac{\mathrm{d}N_i^\ell}{\mathrm{d}g_i}(g_i)\right]\right) \tag{C-76}$$

Case 1: $\pi_i = 1$: When $\pi_i = 1$, $\frac{\mathrm{d}N_i^\ell}{\mathrm{d}g_i} = 0$ from equation (C-64). Hence, N_i^ℓ is constant and $\frac{\mathrm{d}\bar{x}_{ii}}{\mathrm{d}g_i} = \frac{\mathrm{d}\bar{x}_{ji}}{\mathrm{d}g_i} = -1$ directly implies that MB_i^ℓ decreases with g_i.

Case 2: $\pi_i \in (0, 1)$: By total differentiation of :

$$\frac{\mathrm{d}MB_i^\ell}{\mathrm{d}g_i} = \psi_i \left\{ \underbrace{\frac{\mathrm{d}\bar{x}_{ji}}{\mathrm{d}g_i} \left[\pi_i \, N + \left(\frac{1}{\pi} - \pi_i \right) N_i^\ell \right]}_{<0} + 2 \left[\bar{x}_{ii} - \pi_i \, \bar{x}_{ji} \right] \underbrace{\frac{\mathrm{d}N_i^\ell}{\mathrm{d}g_i}}_{<0} + \underbrace{\frac{1}{2} (\bar{x}_{ji}^2 - \bar{x}_{ii}^2)}_{>0} \underbrace{\frac{\mathrm{d}^2 N_i^\ell}{\mathrm{d}g_i^2}}_{<0} \right\} \tag{C-77}$$

Notice that the sign characterisation of $\frac{\mathrm{d}N_i^\ell}{\mathrm{d}g_i}$ and $\frac{\mathrm{d}^2 N_i^\ell}{\mathrm{d}g_i^2}$ follows from Lemma 1. A sufficient condition for $\frac{\mathrm{d}MB_i^\ell}{\mathrm{d}g_i} < 0$ for all $g_i \in [0, \bar{g}_i]$, and each $\pi_i \in (0, 1)$:

$$\underbrace{(1 - g_i)}_{\bar{x}_{ii}} - \pi_i \underbrace{(1 - \pi_i g_i)}_{\bar{x}_{ji}} \geq 0 \iff \bar{g}_i \leq \frac{1}{1 + \pi_i} \quad \text{which holds under Assumption 7.} \tag{C-78}$$

□

C.8 Proof of Proposition 3.3 (Regulation-Retention Trade-off)

Given that Δ is already used as a variable in the model, we denote by D the difference operator, defined as:

$$D[x^h] \triangleq x^\ell - x^s \tag{C-79}$$

Fix $(\bar{x}_{ji}, \bar{x}_{jj})$ such that $\Delta_j > 0$ for equilibrium determinacy and consider the problem faced by government i as it introduces a residency-based privacy regulation, starting from a state of legal vacuum. That is, letting:

$$N_i^\ell(g_i) \triangleq N_i^\ell(\mathbf{X}), \quad \text{with} \quad \mathbf{X} = \begin{pmatrix} 1 - g_i & 1 - \pi_i g_i \\ \bar{x}_{ji} & \bar{x}_{jj} \end{pmatrix} \tag{C-80}$$

Then, short-run firm mass in country i coincides with the long-run solution in the unregulated baseline:

$$N_i^s = N_i^\ell(0) \tag{C-81}$$

We aim to show that, for all $g_i \in [0, \bar{g}_i]$ and $\pi_i \in (0, 1]$:

$$D[MC_i^h](g_i) \geq 0 \quad \text{with equality if and only if } \pi_i = 1, \tag{C-82}$$

$$\text{and} \quad D[MB_i^h](g_i) \leq 0 \quad \text{with equality if and only if } g_i = 0 \text{ or } \pi_i = 1. \tag{C-83}$$

Accordingly, one must have $g_i^\ell \leq g_i^s$ with strict inequality whenever $g_i^h \in (0, \bar{g}_i)$ for some h.

$D[MC_i^h](g_i) \geq 0$ with equality iff $\pi_i = 1$ Setting $\bar{x}_{ii} = \bar{x}_{ji} = 1$ in the solution for firm masses (27), the short-run number of Tech firms active in each country before the enactment of the regulation is given by:

$$N_i^s = \frac{E_i \left(1 - \tau^{\frac{1-\sigma}{1-\beta}}\right) \bar{x}_{jj}^{\frac{\beta(\sigma-1)}{1-\beta}} - E_j \, \Delta_j \, \tau^{\frac{1-\sigma}{1-\beta}}}{\sigma \, w \, f \left(1 - \tau^{\frac{1-\sigma}{1-\beta}}\right) \Delta_j}, \quad \text{and} \quad N_j^s = \frac{E_j \, \Delta_j - E_i \left(1 - \tau^{\frac{1-\sigma}{1-\beta}}\right) \tau^{\frac{1-\sigma}{1-\beta}} \, \bar{x}_{ij}^{\frac{\beta(\sigma-1)}{1-\beta}}}{\sigma \, w \, f \left(1 - \tau^{\frac{1-\sigma}{1-\beta}}\right) \Delta_j} \tag{C-84}$$

From Proposition 2.1, we have $MC_i^h(g_i) > 0$ for all g_i and each h. Hence, the expressions for $MC_i^s(g_i)$ (C-57) and $MC_i^\ell(g_i)$ (C-71) may be used to state the following equivalence:

$$D[MC_i^h](g_i) = MC_i^\ell(g_i) - MC_i^s(g_i) \geq 0 \quad \iff \quad T(g_i) \triangleq \frac{MC_i^\ell(g_i)}{MC_i^s(g_i)} = \left(\frac{P_i^\ell}{P_i^s}\right)^{-\alpha} \cdot \frac{\frac{1}{P_i^\ell}\frac{\mathrm{d}P_i^\ell}{\mathrm{d}g_i}}{\frac{1}{P_i^s}\frac{\mathrm{d}P_i^s}{\mathrm{d}g_i}} \geq 1 \tag{C-85}$$

We begin by deriving an expression for $T(g_i)$. Using the characterisation of total differentials in Proof 3.2, we substitute the firm masses from (C-84) into equation (C-55), obtaining:

$$\begin{aligned}
\frac{1}{P_i^s}\frac{\mathrm{d}P_i^s}{\mathrm{d}g_i} &= \frac{\beta}{1-\beta}\cdot\frac{1}{\bar{x}_{ji}}\cdot\left[\frac{N_i^s r_i^{\frac{\beta(\sigma-1)}{1-\beta}-1} + \pi_i \tau^{\frac{1-\sigma}{1-\beta}} N_j^s}{N_i^s r_i^{\frac{\beta(\sigma-1)}{1-\beta}} + \tau^{\frac{1-\sigma}{1-\beta}} N_j^s}\right] \\
&= \frac{\beta}{1-\beta}\cdot\frac{1}{\bar{x}_{ji}}\cdot\frac{\left[\bar{x}_{jj}^{\frac{\beta(\sigma-1)}{1-\beta}} r_i^{\frac{\beta(\sigma-1)}{1-\beta}-1} - \pi_i \tau^{\frac{2(1-\sigma)}{1-\beta}} \bar{x}_{ij}^{\frac{\beta(\sigma-1)}{1-\beta}}\right] + \frac{E_j\Delta_j\tau^{\frac{1-\sigma}{1-\beta}}}{E_i\left(1-\tau^{\frac{1-\sigma}{1-\beta}}\right)}\left[\pi_i - r_i^{\frac{\beta(\sigma-1)}{1-\beta}-1}\right]}{\left[\bar{x}_{jj}^{\frac{\beta(\sigma-1)}{1-\beta}} r_i^{\frac{\beta(\sigma-1)}{1-\beta}} - \tau^{\frac{2(1-\sigma)}{1-\beta}} \bar{x}_{ij}^{\frac{\beta(\sigma-1)}{1-\beta}}\right] + \frac{E_j\Delta_j\tau^{\frac{1-\sigma}{1-\beta}}}{E_i\left(1-\tau^{\frac{1-\sigma}{1-\beta}}\right)}\left[1 - r_i^{\frac{\beta(\sigma-1)}{1-\beta}}\right]}
\end{aligned} \tag{C-86}$$

Similarly, eliminating firm masses from the short-run price index expression (C-56) gives:

$$\begin{aligned}
(P_i^s)^{-\alpha} &= \left[\frac{p_0}{\varphi^{\frac{\beta}{1-\beta}}}\right]^{-\alpha}\cdot \bar{x}_{ji}^{\frac{\alpha\beta}{1-\beta}}\cdot\left[N_i^s r_i^{\frac{\beta(\sigma-1)}{1-\beta}} + N_j^s \tau^{\frac{1-\sigma}{1-\beta}}\right]^{\frac{\alpha}{\sigma-1}} \\
&= \left(\frac{E_i\varphi^{\frac{\beta(\sigma-1)}{1-\beta}}}{p_0^\sigma q_0\Delta_j}\right)^{\frac{\alpha}{\sigma-1}}\cdot \bar{x}_{ji}^{\frac{\alpha\beta}{1-\beta}}\cdot\left[\bar{x}_{jj}^{\frac{\beta(\sigma-1)}{1-\beta}} r_i^{\frac{\beta(\sigma-1)}{1-\beta}} - \tau^{\frac{2(1-\sigma)}{1-\beta}}\bar{x}_{ij}^{\frac{\beta(\sigma-1)}{1-\beta}} + \frac{E_j\Delta_j\tau^{\frac{1-\sigma}{1-\beta}}}{E_i(1-\tau^{\frac{1-\sigma}{1-\beta}})}\left[1-r_i^{\frac{\beta(\sigma-1)}{1-\beta}}\right]\right]^{\frac{\alpha}{\sigma-1}}
\end{aligned} \tag{C-87}$$

Moreover, combining equations (C-68) and (C-56) immediately gives:

$$\begin{aligned}
(P_i^\ell)^{-\alpha}\cdot\frac{1}{P_i^\ell}\frac{\mathrm{d}P_i^\ell}{\mathrm{d}g_i} = \left(\frac{E_i\varphi^{\frac{\beta(\sigma-1)}{1-\beta}}}{q_0 p_0^\sigma\Delta_j}\right)^{\frac{\alpha}{\sigma-1}} &\frac{\beta}{1-\beta}\cdot\bar{x}_{ji}^{\frac{\beta\alpha}{1-\beta}-1}\cdot\left[r_i^{\frac{\beta(\sigma-1)}{1-\beta}}\bar{x}_{jj}^{\frac{\beta(\sigma-1)}{1-\beta}} - \bar{x}_{ij}^{\frac{\beta(\sigma-1)}{1-\beta}}\tau^{\frac{2(1-\sigma)}{1-\beta}}\right]^{\frac{\alpha}{\sigma-1}-1} \\
&\times\left[r_i^{\frac{\beta(\sigma-1)}{1-\beta}-1}\bar{x}_{jj}^{\frac{\beta(\sigma-1)}{1-\beta}} - \pi_i\bar{x}_{ij}^{\frac{\beta(\sigma-1)}{1-\beta}}\tau^{\frac{2(1-\sigma)}{1-\beta}}\right]
\end{aligned} \tag{C-88}$$

Collecting the above results and simplifying yields:

$$\begin{aligned}
T(g_i) &= \left(\frac{P_i^\ell}{P_i^s}\right)^{-\alpha}\cdot\frac{\frac{1}{P_i^\ell}\frac{\mathrm{d}P_i^\ell}{\mathrm{d}g_i}}{\frac{1}{P_i^s}\frac{\mathrm{d}P_i^s}{\mathrm{d}g_i}} \\
&= \underbrace{\left(1 - \frac{\frac{E_j\Delta_j\tau^{\frac{1-\sigma}{1-\beta}}}{E_i\left(1-\tau^{\frac{1-\sigma}{1-\beta}}\right)}\left(1-\left(r_i(g_i)\right)^{\frac{\beta(\sigma-1)}{1-\beta}}\right)}{\left(r_i(g_i)\right)^{\frac{\beta(\sigma-1)}{1-\beta}}\bar{x}_{jj}^{\frac{\beta(\sigma-1)}{1-\beta}} - \tau^{\frac{2(1-\sigma)}{1-\beta}}\bar{x}_{ij}^{\frac{\beta(\sigma-1)}{1-\beta}} + \frac{E_j\Delta_j\tau^{\frac{1-\sigma}{1-\beta}}}{E_i\left(1-\tau^{\frac{1-\sigma}{1-\beta}}\right)}\left(1-\left(r_i(g_i)\right)^{\frac{\beta(\sigma-1)}{1-\beta}}\right)}\right)^{\frac{\alpha}{\sigma-1}-1}}_{\triangleq T_1(g_i)^{\frac{\alpha}{\sigma-1}-1}}
\end{aligned} \tag{C-89}$$

$$
\times \underbrace{\left[1+\frac{\frac{E_j\Delta_j\,\tau^{\frac{1-\sigma}{1-\beta}}}{E_i\left(1-\tau^{\frac{1-\sigma}{1-\beta}}\right)}\left[r_i(g_i)^{\frac{\beta(\sigma-1)}{1-\beta}-1}-\pi_i\right]}{r_i(g_i)^{\frac{\beta(\sigma-1)}{1-\beta}-1}\bar{x}_{jj}^{\frac{\beta(\sigma-1)}{1-\beta}}-\pi_i\,\tau^{\frac{2(1-\sigma)}{1-\beta}}\,\bar{x}_{ij}^{\frac{\beta(\sigma-1)}{1-\beta}}+\frac{E_j\Delta_j\,\tau^{\frac{1-\sigma}{1-\beta}}}{E_i\left(1-\tau^{\frac{1-\sigma}{1-\beta}}\right)}\left[r_i(g_i)^{\frac{\beta(\sigma-1)}{1-\beta}-1}-\pi_i\right]}\right]}_{\triangleq T_2(g_i)} \tag{C-90}
$$

We claim that $T : g_i \mapsto T_1(g_i)^{\frac{\alpha}{\sigma-1}-1} \times T_2(g_i)$ satisfies $T(g_i) = 1$ for all $g_i \in [0, \bar{g}_i]$ when $\pi_i = 1$, and that $T(g_i) > 1$ for all $g_i \in [0, \bar{g}_i]$ otherwise. From expression (C-85), the latter is equivalent to proving claim (C-82).

<u>**Case 1:** $\pi_i = 1$</u> When $\pi_i = 1$, it is readily seen from the definitions of r_i (C-54) and T (C-90) that:

$$
\begin{aligned}
r_i(g_i)\big|_{\pi_i=1} = \frac{1-g_i}{1-g_i} = 1 \quad &\Longrightarrow \quad \big(r_i(g_i)\big)^{\frac{\beta(\sigma-1)}{1-\beta}-1} - \pi_i\big|_{\pi_i=1} = 1-1 = 0 \\
&\Longrightarrow \quad T(g_i) = 1 \quad \text{for all } \; g_i \in [0,\bar{g}_i].
\end{aligned} \tag{C-91}
$$

<u>**Case 2:** $\pi_i \in (0,1)$</u> In the definition of T_1, the numerator is proportional to $r_i \mapsto 1 - r_i^{\frac{\beta(\sigma-1)}{1-\beta}}$, which is a decreasing function of r_i for admissible parameters. Meanwhile, the denominator is positive in any determinate equilibrium (see Assumption 3) and further includes an increasing term in r_i. Consequently, the entire fraction is decreasing in r_i, and it follows that T_1 is increasing in r_i as negative affine transformation of this fraction.

Further recalling that r_i is strictly decreasing in g_i for all $\pi_i \in (0,1)$ by construction (C-54), we have that:

$$
\sigma > 1+\alpha \quad \text{under Assumption 5} \quad \Longrightarrow \quad \frac{\mathrm{d}}{\mathrm{d}g_i}\left\{T_1(g_i)^{\frac{\alpha}{\sigma-1}-1}\right\} = \underbrace{\left(\frac{\alpha}{\sigma-1}-1\right)}_{<0}\cdot\underbrace{\frac{\mathrm{d}T_1}{\mathrm{d}r_i}}_{>0}\cdot\underbrace{\frac{\mathrm{d}r_i}{\mathrm{d}g_i}}_{<0} > 0 \tag{C-92}
$$

Moreover, observe that:

$$
r_i(0) = \frac{1-0}{1-\pi_i\cdot 0} = 1 \quad \Longrightarrow \quad 1-\big(r_i(0)\big)^{\frac{\beta(\sigma-1)}{1-\beta}} = 0 \quad \Longrightarrow \quad T_1(0)^{\frac{\alpha}{\sigma-1}-1} = 1 \tag{C-93}
$$

These two facts together imply that, for any $\pi_i \in (0,1)$:

$$
T_1(g_i)^{\frac{\alpha}{\sigma-1}-1} \geq 1 \quad \text{for all } \; g_i \in [0,\bar{g}_i]\,\text{, with equality if and only if } g_i = 0\text{, and strictly increasing in } g_i. \tag{C-94}
$$

Turning to T_2, straightforward differentiation gives:

$$
\frac{\mathrm{d}T_2}{\mathrm{d}r_i} = \frac{\frac{E_j\Delta_j\,\tau^{\frac{1-\sigma}{1-\beta}}}{E_i\left(1-\tau^{\frac{1-\sigma}{1-\beta}}\right)}\left(\frac{\beta(\sigma-1)}{1-\beta}-1\right)r_i^{\frac{\beta(\sigma-1)}{1-\beta}-2}\pi_i\left[\bar{x}_{jj}^{\frac{\beta(\sigma-1)}{1-\beta}}-\tau^{\frac{2(1-\sigma)}{1-\beta}}\,\bar{x}_{ij}^{\frac{\beta(\sigma-1)}{1-\beta}}\right]}{\left(r_i^{\frac{\beta(\sigma-1)}{1-\beta}-1}\,\bar{x}_{jj}^{\frac{\beta(\sigma-1)}{1-\beta}}-\pi_i\,\tau^{\frac{2(1-\sigma)}{1-\beta}}\,\bar{x}_{ij}^{\frac{\beta(\sigma-1)}{1-\beta}}+\frac{E_j\Delta_j\,\tau^{\frac{1-\sigma}{1-\beta}}}{E_i\left(1-\tau^{\frac{1-\sigma}{1-\beta}}\right)}\left[\pi_i-r_i^{\frac{\beta(\sigma-1)}{1-\beta}-1}\right]\right)^2}. \tag{C-95}
$$

Under Assumption 5, we have that:

$$
\frac{\beta(\sigma-1)}{1-\beta}-1<0 \quad \Longrightarrow \quad \frac{\mathrm{d}T_2}{\mathrm{d}r_i} \propto -\left(\bar{x}_{jj}^{\frac{\beta(\sigma-1)}{1-\beta}}-\tau^{\frac{2(1-\sigma)}{1-\beta}}\,\bar{x}_{ij}^{\frac{\beta(\sigma-1)}{1-\beta}}\right) < -\left(\bar{x}_{jj}^{\frac{\beta(\sigma-1)}{1-\beta}}-\tau^{\frac{1-\sigma}{1-\beta}}\,\bar{x}_{ij}^{\frac{\beta(\sigma-1)}{1-\beta}}\right) = -\Delta_j < 0
$$

$$\implies \quad \frac{\mathrm{d}T_2}{\mathrm{d}g_i} = \underbrace{\frac{\mathrm{d}T_2}{\mathrm{d}r_i}}_{<0} \cdot \underbrace{\frac{\mathrm{d}r_i}{\mathrm{d}g_i}}_{<0} > 0 \tag{C-96}$$

Furthermore, we know that $r_i(g_i) \leq 1$ for all $g_i \in [0, \bar{g}_i]$, and the same Assumption 5 immediately gives that for any $\pi_i \in (0,1)$:

$$r_i(g_i)^{\frac{\beta(\sigma-1)}{1-\beta}-1} \geq 1 > \pi_i \quad \implies \quad T_2(g_i) > 1 \quad \text{for all} \;\; g_i \in [0, \bar{g}_i] \;\; \text{and is strictly increasing in } g_i. \tag{C-97}$$

Combining facts (C-94), and (C-97), we shall conclude that:

$$\pi_i < 1 \quad \implies \quad T(g_i) = T_1(g_i)^{\frac{\alpha}{\sigma-1}-1} \cdot T_2(g_i) > 1 \quad \text{for all} \;\; g_i \in [0, \bar{g}_i] \; . \tag{C-98}$$

Note that since T is strictly increasing for $\pi_i \in (0,1)$, as shown above, it follows that the gap between long-run and short-run marginal costs of regulation increases with the level of regulation, as claimed.

$D[MB_i^h]$ is decreasing for $\pi_i \in (0,1)$ and $D[MB_i^h](g_i) \leq 0$ with equality iff $\pi_i = 1$ or $g_i = 0$.

Use the definition of the marginal benefit of regulation (C-52) to write:

$$\begin{aligned} D[MB_i^h](g_i) &\triangleq MB_i^\ell(g_i) - MB_i^s(g_i) \\ &= \psi_i \left\{ \left[\pi_i \, \bar{x}_{ji} \, (N - N_i^\ell) + \bar{x}_{ii} \, N_i^\ell - \frac{1}{2}\left(\bar{x}_{ii}^2 - \bar{x}_{ji}^2\right) \frac{\mathrm{d}N_i^\ell}{\mathrm{d}g_i} \right] - \left[\bar{x}_{ii} \, N_i^s + \pi_i \, \bar{x}_{ji} \, (N - N_i^s) \right] \right\} \\ &= \psi_i \left[\underbrace{\left(\bar{x}_{ii} - \pi_i \, \bar{x}_{ji}\right)}_{\geq 0} D[N_i^h] + \underbrace{\frac{1}{2}\left(\bar{x}_{ji}^2 - \bar{x}_{ii}^2\right) \frac{\mathrm{d}N_i^\ell}{\mathrm{d}g_i}}_{\leq 0 \text{ with equality iff } \pi_i = 1 \text{ or } g_i = 0.} \right] \quad \text{for all} \;\; g_i \in [0, \bar{g}_i] \end{aligned} \tag{C-99}$$

We have already shown in Proof C.7 that the first term is positive at every g_i under Assumption 7, while the characterisation of the second term follows from $\bar{x}_{ji}^2 \leq \bar{x}_{ii}^2$ with equality if and only if $\pi_i = 1$ or $g_i = 0$, and $\frac{\mathrm{d}N_i^\ell}{\mathrm{d}g_i} \leq 0$ with equality if and only if $\pi_i = 1$ by (C-64). From rewriting (C-99), a sufficient condition for $D[MB_i^h](g_i) \leq 0$ is thus $D[N_i^h](g_i) \leq 0$. However, notice that it is a direct corollary from $\frac{\mathrm{d}N_i^\ell}{\mathrm{d}g_i} \leq 0$ that:

$$\begin{aligned} D[N_i^h](g_i) \triangleq N_i^\ell(g_i) - N_i^s &= N_i^\ell(g_i) - N_i^\ell(0) \\ &= \int_0^{g_i} \frac{\mathrm{d}N_i^\ell}{\mathrm{d}g_i}(v) \, \mathrm{d}v \leq 0 \quad \text{with equality if and only if } \pi_i = 1 \text{ or } g_i = 0. \end{aligned} \tag{C-100}$$

Hence, we conclude that $D[MB_i^h](g_i) \leq 0$ with equality if and only if $\pi_i = 1$ or $g_i = 0$, proving claim (C-83).

C.9 Proof of Proposition 3.4 (Existence of a Best Response Function in Privacy Regulation)

In a strategic environment where each government unilaterally sets the optimal regulation within a residency-based framework, the relevant optimisation problem for country i becomes:

$$\begin{aligned} \max_{g_i \in [0, \bar{g}_i]} \quad & \mathscr{W}_i^h \left(\bar{x}_{ii}, \, \bar{x}_{ij}; \bar{x}_{ji}, \bar{x}_{jj}\right) \quad \text{as defined in (C-31)}, \\ \text{s.t.} \quad & \bar{x}_{ii} \; = \; 1 - g_i, \quad \bar{x}_{ji} \; = \; 1 - \pi_i g_i \end{aligned} \tag{C-101}$$

$$\bar{x}_{jj} \;=\; 1 - g_j, \quad \bar{x}_{ij} \;=\; 1 - \pi_j g_j, \quad g_j \in [0, \bar{g}_j] \text{ given.}$$

Note that this setup corresponds to a special case of Problem (C-50), where foreign policy g_j determines data usage caps $(\bar{x}_{jj}(g_j), \bar{x}_{ij}(g_j))$. Since Proposition C.7 establishes strict concavity of $\mathscr{W}_i^h$ in g_i whenever $\Delta_j > 0$, this property extends to the strategic environment provided that this determinacy condition holds for all admissible g_j. Assumption 7 ensures that this condition is met, when g_j is restricted to the interval $[0, \bar{g}_j]$, since:

$$\begin{aligned} g_j \leq \bar{g}_j < \min\left\{\frac{1-\tau^{-\beta}}{1-\tau^{-\beta}\pi_i}, \frac{1}{1+\pi_i}\right\} \quad &\Longrightarrow \quad g_j < \frac{1-\tau^{-\beta}}{1-\tau^{-\beta}\pi_i} \\ &\Longleftrightarrow \quad \Delta_j = \bar{x}_{jj}^{\frac{\beta(\sigma-1)}{1-\beta}} - \tau^{\frac{1-\sigma}{1-\beta}} \bar{x}_{jj}^{\frac{\beta(\sigma-1)}{1-\beta}} > 0 \end{aligned} \tag{C-102}$$

To argue that Problem (C-101) gives rise to a continous best response function, it suffices to notice that $\mathscr{W}_i^h$ is itself continuous in both g_i and g_j, while the feasible set $[0, \bar{g}_i]$ is compact. Hence, invoking Berge's Maximum Theorem, we conclude that the solution correspondence:

$$\mathscr{B}_i(g_j) \equiv \arg \max_{g_i \in [0, \bar{g}_i]} \mathscr{W}_i^h\left(\bar{x}_{ii},\, \bar{x}_{ij}; \bar{x}_{ji}, \bar{x}_{jj}\right)$$

is nonempty, compact-valued, and upper hemicontinuous in g_j. Moreover, the strict concavity of $\mathscr{W}_i^h$ in g_i, for each given $g_j \in [0, \bar{g}_j]$, guarantees that this correspondence is single-valued. Since a single-valued, upper hemicontinuous correspondence is a continuous function, so is $\mathscr{B}_i(g_j)$. □

C.10 Proof of Proposition 3.5 (Existence and Uniqueness of a Symmetric Nash Equilibrium in Privacy Regulation)

Consider the symmetric formulation of the privacy regulation game, where all countries $i \in \mathscr{C}$ share identical structural parameters, including enforcement capacity $\pi \in (0, 1]$. A symmetric Nash equilibrium is a regulation level $g^* \in [0, \bar{g}]$ such that, if both governments implement g^*, neither has an incentive to deviate. Formally, g^* is a fixed point of the symmetric mapping $g \mapsto \mathscr{B}(g)$, satisfying:

$$g^* = \mathscr{B}(g^*) \tag{C-103}$$

We want to show that such a regulation level g^* always exists and is unique.

Existence By Proposition 3.4, $\mathscr{B}_i$ is continuous. With symmetric countries, $\mathscr{B} : [0, \bar{g}] \mapsto [0, \bar{g}]$ maps a compact and convex set into itself. By Brouwer's fixed point theorem, it must have at least one fixed point.

Uniqueness In any symmetric equilibrium, $g_i = g_j \equiv g$ and $\bar{g}_i = \bar{g}_j \equiv \bar{g}$. Accordingly, define the diagonal restrictions:

$$\widetilde{MB} : g \mapsto MB^\ell(g; g) \quad \text{and} \quad \widetilde{MC} : g \mapsto MC^\ell(g; g) \tag{C-104}$$

A sufficient condition for uniqueness of the symmetric equilibrium is thus:

$$\begin{aligned} g' > g \quad &\implies \quad \widetilde{MB}(g) > \widetilde{MB}(g') \quad \text{and} \quad \widetilde{MC}(g') < \widetilde{MC}(g) \\ &\implies \quad \frac{\partial \mathscr{W}_i^\ell}{\partial g_i}(g';g') \equiv \widetilde{MB}(g') - \widetilde{MC}(g') < \widetilde{MB}(g) - \widetilde{MC}(g) = \frac{\partial \mathscr{W}_i^\ell}{\partial g_i}(g;g) \end{aligned} \tag{C-105}$$

In words, $\partial_{g_i}\mathscr{W}_i^\ell$ is strictly decreasing in g along the diagonal where $(g_i, g_j) = (g, g)$ and can change sign at most once on $[0, \bar{g}]$. By continuity, if it crosses zero in $(0, \bar{g})$, that crossing pins down the unique symmetric equilibrium. If it never crosses at some interior point, then $\partial_{g_i}\mathscr{W}_i^\ell(g;g)$ has constant sign over $[0,\bar{g}]$ and the unique equilibrium lies at the corner $g^* = 0$ when the FOC is negative everywhere, or $g^* = \bar{g}$ when it is positive everywhere. We now establish that both $\widetilde{MB}$ and $\widetilde{MC}$ have the desired properties. (C-105)

$\widetilde{MC}$ is strictly increasing in g: Evaluating the marginal cost of regulation $MC^\ell(g;g)$ (C-71) along the diagonal where $(g_i, g_j) = (g, g)$ gives:

$$\begin{aligned} \widetilde{MC}(g) &= \frac{\alpha\beta L\, w^\alpha}{1-\beta}\left(\frac{E\,\varphi^{\frac{\beta(\sigma-1)}{1-\beta}}}{q_0\, p_0^\sigma\, \Delta_j}\right)^{\frac{\alpha}{\sigma-1}} \cdot \Delta^{\frac{\alpha}{\sigma-1}-1} \cdot \left[\bar{x}_{ii}^{\frac{\beta(\sigma-1)}{1-\beta}-1}\bar{x}_{jj}^{\frac{\beta(\sigma-1)}{1-\beta}} - \pi\,\bar{x}_{ji}^{\frac{\beta(\sigma-1)}{1-\beta}-1}\bar{x}_{ij}^{\frac{\beta(\sigma-1)}{1-\beta}}\,\tau^{\frac{2(1-\sigma)}{1-\beta}}\right] \\ &= \frac{\alpha\beta L\, w^\alpha}{1-\beta}\left(\frac{E\,\varphi^{\frac{\beta(\sigma-1)}{1-\beta}}}{q_0\, p_0^\sigma}\right)^{\frac{\alpha}{\sigma-1}} \cdot \left[\frac{\Delta}{\Delta_j}\right]^{\frac{\alpha}{\sigma-1}-1} \cdot \left[\frac{\bar{x}_{ii}^{\frac{\beta(\sigma-1)}{1-\beta}-1}\bar{x}_{jj}^{\frac{\beta(\sigma-1)}{1-\beta}} - \pi\,\bar{x}_{ji}^{\frac{\beta(\sigma-1)}{1-\beta}-1}\bar{x}_{ij}^{\frac{\beta(\sigma-1)}{1-\beta}}\,\tau^{\frac{2(1-\sigma)}{1-\beta}}}{\Delta_j}\right] \\ &\text{with} \quad \Delta \triangleq \bar{x}_{ii}^{\frac{\beta(\sigma-1)}{1-\beta}}\bar{x}_{jj}^{\frac{\beta(\sigma-1)}{1-\beta}} - \bar{x}_{ij}^{\frac{\beta(\sigma-1)}{1-\beta}}\bar{x}_{ji}^{\frac{\beta(\sigma-1)}{1-\beta}}\tau^{\frac{2(1-\sigma)}{1-\beta}} \quad \text{and} \quad \Delta_j = \bar{x}_{jj}^{\frac{\beta(\sigma-1)}{1-\beta}} - \tau^{\frac{1-\sigma}{1-\beta}}\bar{x}_{ij}^{\frac{\beta(\sigma-1)}{1-\beta}} \end{aligned} \tag{C-106}$$

In any symmetric equilibrium, g^* solves $\bar{x}_{ii} = 1 - g^* = \bar{x}_{jj}$ and $\bar{x}_{ij} = 1 - \pi g^* = \bar{x}_{ji}$. In other words, there exists a unique ratio $r^* = \frac{\bar{x}_{jj}}{\bar{x}_{ij}} = \frac{\bar{x}_{ii}}{\bar{x}_{ji}}$, recalling our earlier notation (C-54). By substituting the definitions of Δ and Δ_j, expression (C-106) simplifies, up to an irrelevant positive constant factor, to:

$$\widetilde{MC}(g) \propto \left[\frac{\bar{x}_{ii}^{\frac{2\beta(\sigma-1)}{1-\beta}} - \bar{x}_{ji}^{\frac{2\beta(\sigma-1)}{1-\beta}}\tau^{\frac{2(1-\sigma)}{1-\beta}}}{\bar{x}_{ii}^{\frac{\beta(\sigma-1)}{1-\beta}} - \tau^{\frac{1-\sigma}{1-\beta}}\bar{x}_{ji}^{\frac{\beta(\sigma-1)}{1-\beta}}}\right]^{\frac{\alpha}{\sigma-1}-1}\left[\frac{\bar{x}_{ii}^{\frac{2\beta(\sigma-1)}{1-\beta}-1} - \pi\bar{x}_{ji}^{\frac{2\beta(\sigma-1)}{1-\beta}-1}\tau^{\frac{2(1-\sigma)}{1-\beta}}}{\bar{x}_{ii}^{\frac{\beta(\sigma-1)}{1-\beta}} - \tau^{\frac{1-\sigma}{1-\beta}}\bar{x}_{ji}^{\frac{\beta(\sigma-1)}{1-\beta}}}\right] \tag{C-107}$$

$$\iff \quad \widetilde{MC}(g) \propto \bar{x}_{ji}^{\frac{\alpha\beta}{1-\beta}-1}\left[\frac{\left(r^{\frac{\beta(\sigma-1)}{1-\beta}}\right)^2 - \left(\tau^{\frac{1-\sigma}{1-\beta}}\right)^2}{r^{\frac{\beta(\sigma-1)}{1-\beta}} - \tau^{\frac{1-\sigma}{1-\beta}}}\right]^{\frac{\alpha}{\sigma-1}-1}\left[\frac{r^{\frac{2\beta(\sigma-1)}{1-\beta}-1} - \pi\tau^{\frac{2(1-\sigma)}{1-\beta}}}{r^{\frac{\beta(\sigma-1)}{1-\beta}} - \tau^{\frac{1-\sigma}{1-\beta}}}\right] \tag{C-108}$$

By difference of squares, $\left(r^{\frac{\beta(\sigma-1)}{1-\beta}}\right)^2 - \left(\tau^{\frac{1-\sigma}{1-\beta}}\right)^2 = \left(r^{\frac{\beta(\sigma-1)}{1-\beta}} - \tau^{\frac{1-\sigma}{1-\beta}}\right)\left(r^{\frac{\beta(\sigma-1)}{1-\beta}} + \tau^{\frac{1-\sigma}{1-\beta}}\right)$, and we thus obtain:

$$\widetilde{MC}(g) \propto \left[\bar{x}_{ji}(g)\right]^{\frac{\alpha\beta}{1-\beta}-1}\left\{\left[r(g)\right]^{\frac{\beta(\sigma-1)}{1-\beta}} + \tau^{\frac{1-\sigma}{1-\beta}}\right\}^{\frac{\alpha}{\sigma-1}-1}\left\{\frac{\left[r(g)\right]^{\frac{2\beta(\sigma-1)}{1-\beta}-1} - \pi\tau^{\frac{2(1-\sigma)}{1-\beta}}}{\left[r(g)\right]^{\frac{\beta(\sigma-1)}{1-\beta}} - \tau^{\frac{1-\sigma}{1-\beta}}}\right\} \tag{C-109}$$

We aim to show that the product on the right-hand side is strictly increasing in g. Notice that first term is strictly increasing in g for all $\pi \in (0, 1]$ since, under the range of parameters admissible parameters, we have:

$$\frac{\mathrm{d}}{\mathrm{d}g}\left\{\left[\bar{x}_{ji}(g)\right]^{\frac{\alpha\beta}{1-\beta}-1}\right\} = \underbrace{\left(\frac{\alpha\beta}{1-\beta} - 1\right)}_{<0 \text{ by (C-60)}} \cdot \underbrace{\frac{\mathrm{d}\bar{x}_{ji}(g)}{\mathrm{d}g}}_{-\pi<0} > 0 \tag{C-110}$$

Recalling that r is weakly decreasing in g (C-54), a similar reasoning gives that, under Assumption 5:

$$\frac{\alpha}{\sigma-1} < 1 \quad \Longrightarrow \quad \frac{\mathrm{d}}{\mathrm{d}g}\left\{ \left(\left[r(g)\right]^{\frac{\beta(\sigma-1)}{1-\beta}} + \tau^{\frac{1-\sigma}{1-\beta}} \right)^{\frac{\alpha}{\sigma-1}-1} \right\} \geq 0 \tag{C-111}$$

Turning to the last term in (C-109), pose:

$$\frac{N(r)}{D(r)} \equiv \frac{r^{\frac{2\beta(\sigma-1)}{1-\beta}-1} - \pi\tau^{\frac{2(1-\sigma)}{1-\beta}}}{r^{\frac{\beta(\sigma-1)}{1-\beta}} - \tau^{\frac{1-\sigma}{1-\beta}}} \tag{C-112}$$

Recall from Proposition 2.1 that, in any determinate equilibrium:

$$\Delta_i = \bar{x}_{ii}^{\frac{\beta(\sigma-1)}{1-\beta}} - \tau^{\frac{1-\sigma}{1-\beta}}\bar{x}_{ji}^{\frac{\beta(\sigma-1)}{1-\beta}} > 0 \quad \Longleftrightarrow \quad r^{\frac{\beta(\sigma-1)}{1-\beta}} > \tau^{\frac{1-\sigma}{1-\beta}} \Longrightarrow \quad D(r) > 0 \tag{C-113}$$

Moreover, we know that $r \leq 1$ by construction (C-54), and we also deduce from Assumption 5 that:

$$\begin{aligned} \frac{\beta(\sigma-1)}{1-\beta} < 1 \quad \Longrightarrow \quad \frac{2\beta(\sigma-1)}{1-\beta} - 1 < \frac{\beta(\sigma-1)}{1-\beta} \quad &\Longrightarrow \quad r^{\frac{2\beta(\sigma-1)}{1-\beta}-1} \geq r^{\frac{\beta(\sigma-1)}{1-\beta}} > \tau^{\frac{1-\sigma}{1-\beta}} > \pi\tau^{\frac{2(1-\sigma)}{1-\beta}} \\ &\Longrightarrow \quad N(r) > D(r) > 0 \end{aligned} \tag{C-114}$$

Hence, we may apply a log transformation and claim that:

$$\frac{\mathrm{d}}{\mathrm{d}g}\left\{\frac{N\big(r(g)\big)}{D\big(r(g)\big)}\right\} = \frac{\mathrm{d}}{\mathrm{d}r}\left\{\frac{N(r)}{D(r)}\right\}\underbrace{\frac{\mathrm{d}r_i}{\mathrm{d}g}}_{\leq 0} \geq 0 \quad \Longleftrightarrow \quad \frac{\mathrm{d}\ln D}{\mathrm{d}r} = \frac{D'(r)}{D(r)} \geq \frac{N'(r)}{N(r)} = \frac{\mathrm{d}\ln N}{\mathrm{d}r}$$

However, one also deduces from (C-114) that:

$$\frac{\mathrm{d}\ln D}{\mathrm{d}r} = \frac{\frac{\beta(\sigma-1)}{1-\beta}\, r^{\frac{\beta(\sigma-1)}{1-\beta}-1}}{r^{\frac{\beta(\sigma-1)}{1-\beta}} - \tau^{\frac{1-\sigma}{1-\beta}}} > \frac{\frac{\beta(\sigma-1)}{1-\beta}\, r^{\frac{\beta(\sigma-1)}{1-\beta}-1}}{r^{\frac{\beta(\sigma-1)}{1-\beta}}} = \frac{\beta(\sigma-1)}{(1-\beta)\, r} > \frac{\frac{2\beta(\sigma-1)}{1-\beta}-1}{r} \tag{C-115}$$

while, at the same time, we have:

$$\frac{\mathrm{d}\ln N}{\mathrm{d}r} = \frac{\left(\frac{2\beta(\sigma-1)}{1-\beta}-1\right) r^{\frac{2\beta(\sigma-1)}{1-\beta}-2}}{r^{\frac{2\beta(\sigma-1)}{1-\beta}-1} - \pi\,\tau^{\frac{2(1-\sigma)}{1-\beta}}} < \frac{\left(\frac{2\beta(\sigma-1)}{1-\beta}-1\right) r^{\frac{2\beta(\sigma-1)}{1-\beta}-2}}{r^{\frac{2\beta(\sigma-1)}{1-\beta}-1}} < \frac{\frac{2\beta(\sigma-1)}{1-\beta}-1}{r} \quad \Longrightarrow \quad \frac{\mathrm{d}\ln D}{\mathrm{d}r} > \frac{\mathrm{d}\ln N}{\mathrm{d}r} \tag{C-116}$$

Hence, the third term in (C-109) is non-decreasing in g and it follows that $\widetilde{MC}$ is strictly increasing in g as the product of positive-valued, increasing functions of g, at least one of which strictly.

$\widetilde{MB}$ is strictly decreasing in g: By Proposition 2.5, the total mass of Tech firms in the world economy is independent of data regulation. Hence, in any symmetric equilibrium:

$$N_H(g,g) = N_F(g,g) = \frac{N}{2}, \quad N = \frac{2E}{\sigma w f} \tag{C-117}$$

Substituting into the definition of MB_i^ℓ (C-52) and using our preceding characterisation of $\frac{\mathrm{d}N_i}{\mathrm{d}g_i}$ (C-64), we may write that, along the diagonal where $g_i = g_j = g$:

$$\widetilde{MB}(g) \triangleq \psi_i \left(\frac{N}{2} \Big[\bar{x}_{ii}(g_i)\big|_{g_i=g} + \pi\, \bar{x}_{ji}(g_i)\big|_{g_i=g} \Big] - \frac{1}{2} \left(\Big[\bar{x}_{ii}(g_i)\big|_{g_i=g} \Big]^2 - \Big[\bar{x}_{ji}(g_i)\big|_{g_i=g} \Big]^2 \right) \cdot \frac{\mathrm{d}N_i^\ell}{\mathrm{d}g_i}(g_i)\bigg|_{g_i=g} \right) \tag{C-118}$$

$$\frac{\mathrm{d}N_i^\ell}{\mathrm{d}g_i}(g_i)\bigg|_{g_i=g} = -\frac{\beta(\sigma-1)E_j\tau^{\frac{1-\sigma}{1-\beta}}}{(1-\beta)\sigma w f} \cdot \Big[\Delta_i(g_i)\big|_{g_i=g} \Big]^{-2} \cdot \Big[\bar{x}_{ji}(g_i)\big|_{g_i=g} \Big]^{2\left(\frac{\beta(\sigma-1)}{1-\beta}-1\right)} \cdot \Big[r_i(g_i)\big|_{g_i=g} \Big]^{\frac{\beta(\sigma-1)}{1-\beta}-1} \cdot [1-\pi_i]$$

We write each object as a function of g_i, and evaluate at $g_i = g$, to make explicit that the resulting expression is independent of g_j. Hence, the same arguments used to establish that MB_i^h is strictly decreasing in the g_i direction in Proof C.7 can be adapted here to show that $\widetilde{MB}$ is decreasing in g. □

C.11 Proof of Proposition 3.6 (Effect of Extraterritorial Enforcement on Best Responses)

We want to show that:

$$0 < \pi_i < \pi_i' \leq 1 \quad \Longrightarrow \quad \mathscr{B}_i(g_j\,; \pi_i') \geq \mathscr{B}_i(g_j\,; \pi_i), \quad \text{with strict inequality whenever } \mathscr{B}_i(g_j\,; \pi_i) \in (0, \bar{g}_i) \tag{C-119}$$

By Proposition C.7, the marginal benefit of regulation MB_i^ℓ is strictly decreasing in g_i, while the marginal cost MC_i^ℓ is strictly increasing in g_i. It is therefore sufficient to show that:

$$0 < \pi_i < \pi_i' \leq 1 \quad \Longrightarrow \quad MC_i^\ell(g_i, g_j; \pi_i') \leq MC_i^\ell(g_i, g_j; \pi_i) \tag{C-120}$$

$$\Longrightarrow \quad MB_i^\ell(g_i, g_j; \pi_i') \geq MB_i^\ell(g_i, g_j; \pi_i) \tag{C-121}$$

with at least one strict inequality.

In fact, we can prove that both of these inequalities hold strictly, in which case a straightforward application of the implicit function theorem to the first-order condition (C-52) which pins down $\mathscr{B}_i(g_j; \pi_i)$ in the interior case immediately implies that, for any $\pi_i' > \pi_i$, and some $\varepsilon > 0$:

$$\frac{\mathrm{d}\mathscr{B}_i}{\mathrm{d}\pi_i} = \underbrace{\left(-\frac{\partial^2 \mathscr{W}_i^\ell}{\partial g_i^2} \right)^{-1}}_{>0 \text{ by concavity}} \cdot \underbrace{\frac{\partial}{\partial \pi_i} \left\{ MB_i^\ell - MC_i^\ell \right\}}_{>0} > 0 \quad \Longrightarrow \quad \mathscr{B}_i(g_j\,; \pi_i') - \mathscr{B}_i(g_j\,; \pi_i) \geq \int_{\pi_i}^{\pi_i+\varepsilon} \frac{\mathrm{d}\mathscr{B}_i}{\mathrm{d}\pi_i}(g_j; v)\,\mathrm{d}v > 0$$

MC_i^ℓ is strictly decreasing in π_i: Fix a pair (g_i, g_j). Treating MC_i^ℓ (C-71): as a function of π_i, we may write that, up to a strictly positive constant that is independent of π_i:

$$MC^\ell(\pi_i) \propto \big[\Delta(\pi_i)\big]^{\frac{\alpha}{\sigma-1}-1} \cdot \left\{ \bar{x}_{ii}^{\frac{\beta(\sigma-1)}{1-\beta}-1} \bar{x}_{jj}^{\frac{\beta(\sigma-1)}{1-\beta}} - \pi_i \left[\bar{x}_{ji}(\pi_i) \right]^{\frac{\beta(\sigma-1)}{1-\beta}-1} \bar{x}_{ij}^{\frac{\beta(\sigma-1)}{1-\beta}} \tau^{\frac{2(1-\sigma)}{1-\beta}} \right\} \tag{C-122}$$

$$\text{with} \quad \Delta \triangleq \bar{x}_{ii}^{\frac{\beta(\sigma-1)}{1-\beta}} \bar{x}_{jj}^{\frac{\beta(\sigma-1)}{1-\beta}} - \bar{x}_{ij}^{\frac{\beta(\sigma-1)}{1-\beta}} \left[\bar{x}_{ji}(\pi_i) \right]^{\frac{\beta(\sigma-1)}{1-\beta}} \tau^{\frac{2(1-\sigma)}{1-\beta}}$$

Since $\bar{x}_{ji} = 1 - \pi_i g_i$, it is immediate that Δ (23) is weakly increasing in π_i. Meanwhile, from Assumption 5:

$$\alpha < \sigma - 1 \quad \Longrightarrow \quad \frac{\mathrm{d}}{\mathrm{d}\pi_i} \left\{ \big[\Delta(\pi_i)\big]^{\frac{\alpha}{\sigma-1}-1} \right\} = \left(\underbrace{\frac{\alpha}{\sigma-1} - 1}_{<0} \right) \underbrace{\Delta_i'(\pi_i)}_{\geq 0} \leq 0 \tag{C-123}$$

Moreover, under the same Assumption 5, the second term is also decreasing in π_i since:

$$\frac{\beta(\sigma-1)}{1-\beta}-1<0 \quad \Longrightarrow \quad \frac{\mathrm{d}}{\mathrm{d}\pi_i}\left\{\left[\bar{x}_{ji}(\pi_i)\right]^{\frac{\beta(\sigma-1)}{1-\beta}-1}\right\} = \underbrace{\frac{\mathrm{d}}{\mathrm{d}\bar{x}_{ji}}\left\{\bar{x}_{ji}^{\frac{\beta(\sigma-1)}{1-\beta}-1}\right\}}_{\leq 0} \cdot \underbrace{\frac{\mathrm{d}\bar{x}_{ji}}{\mathrm{d}\pi_i}}_{\leq 0} \geq 0$$

$$\Longrightarrow \quad \frac{\mathrm{d}}{\mathrm{d}\pi_i}\left\{\bar{x}_{ii}^{\frac{\beta(\sigma-1)}{1-\beta}-1}\bar{x}_{jj}^{\frac{\beta(\sigma-1)}{1-\beta}} - \pi_i\left[\bar{x}_{ji}(\pi_i)\right]^{\frac{\beta(\sigma-1)}{1-\beta}-1}\bar{x}_{ij}^{\frac{\beta(\sigma-1)}{1-\beta}}\tau^{\frac{2(1-\sigma)}{1-\beta}}\right\}<0 \tag{C-124}$$

Combining facts (C-123) and (C-124) directly implies the claim (C-120), since $MC^{\ell}(\pi_i)$ can be written as the product of positive-valued, decreasing functions of π_i, at least one of which strictly.

MB_i^{ℓ} is strictly increasing in π_i: Using the definition of MB_i^{ℓ} (C-52):

$$MB_i^{\ell}(\pi_i) \triangleq \psi_i\left(\underbrace{\pi_i\,\bar{x}_{ji}(\pi_i)}_{z(\pi_i)}N_j^{\ell}(\pi_i)+\bar{x}_{ii}N_i^{\ell}(\pi_i)-\frac{1}{2}\left(\bar{x}_{ii}^2-[\bar{x}_{ji}(\pi_i)]^2\right)\frac{\mathrm{d}N_i^{\ell}}{\mathrm{d}g_i}(\pi_i)\right) \tag{C-125}$$

Defining $z(\pi_i)=\pi_i\,\bar{x}_{ji}(\pi_i)$. Notice that:

$$z'(\pi_i)=\frac{\mathrm{d}}{\mathrm{d}\pi_i}\left\{\pi_i(1-\pi_i g_i)\right\}=1-2\pi_i g_i\geq 0 \quad \Longleftrightarrow \quad g_i\leq\frac{1}{2\pi_i} \tag{C-126}$$

Under Assumption 7, the above inequality clearly holds since:

$$g_i\leq\bar{g}_i=\min\left\{\frac{1-\tau^{-\beta}}{1-\tau^{-\beta}\pi_i},\frac{1}{1+\pi_i}\right\}\leq\frac{1}{1+\pi_i}\leq\frac{1}{2\pi_i} \quad \text{with strict inequality for any } \pi_i\in(0,1)$$

$$\Longrightarrow \quad z'(\pi_i)\geq 0 \quad \text{with strict inequality for any } \pi_i\in(0,1). \tag{C-127}$$

Moreover, we know from Proposition 2.5 that the total number of firms in the world economy is independent of data regulation. Hence, $\frac{\mathrm{d}N_j^{\ell}}{\mathrm{d}\pi_i}=-\frac{\mathrm{d}N_i^{\ell}}{\mathrm{d}\pi_i}\leq 0$, and it follows from Proposition 2.4 that:

$$\frac{\mathrm{d}}{\mathrm{d}\pi_i}\left\{z(\pi_i)\cdot N_j^{\ell}(\pi_i)+\bar{x}_{ii}N_i^{\ell}(\pi_i)\right\}=\underbrace{\underbrace{\frac{\mathrm{d}N_i^{\ell}}{\mathrm{d}\bar{x}_{ji}}}_{<0}\underbrace{\frac{\mathrm{d}\bar{x}_{ji}}{\mathrm{d}\pi_i}}_{\leq 0}}_{\geq 0}\cdot\left(\underbrace{\bar{x}_{ii}-z(\pi_i)}_{>0\text{ by (C-78)}}\right)+z'(\pi_i)\cdot N_j^{\ell}$$

$$\geq 0 \quad \text{with strict inequality for any } \pi_i\in(0,1). \tag{C-128}$$

Turning to the last term, observe that equation (C-64) can be rearranged as:

$$-\frac{\mathrm{d}N_i^{\ell}}{\mathrm{d}g_i}=\underbrace{\frac{\beta(\sigma-1)E_j\tau^{\frac{1-\sigma}{1-\beta}}}{(1-\beta)\sigma wf\left[\Delta_i(\pi_i)\right]^2}\cdot\left(\bar{x}_{ji}(\pi_i)\right)^{\frac{\beta(\sigma-1)}{1-\beta}-1}\cdot\bar{x}_{ii}^{\frac{\beta(\sigma-1)}{1-\beta}-1}}_{>0\text{ for all }g_i\in[0,\bar{g}_i]}\cdot\underbrace{[1-\pi_i]}_{\geq 0}\geq 0$$

$$\propto\left[\underbrace{\bar{x}_{ii}-\tau^{\frac{1-\sigma}{1-\beta}}\bar{x}_{ji}(\pi_i)}_{\Delta_i}\right]^{-2}\cdot\left(\bar{x}_{ji}(\pi_i)\right)^{\frac{\beta(\sigma-1)}{1-\beta}-1}\cdot[1-\pi_i]\geq 0 \tag{C-129}$$

The above shows that $-\frac{\mathrm{d}N_i^{\ell}}{\mathrm{d}g_i}$ decreases with π_i, as it can be written as the product of positive-valued, decreasing functions of π_i. Equivalently, $\frac{\mathrm{d}N_i^\ell}{\mathrm{d}g_i}$ is negative and increasing in π_i. An analogous reasoning gives:

$$\bar{x}_{ii}^2 - \bar{x}_{ji}^2 = (1-g_i)^2 - (1-\pi_i g_i)^2 \leq 0 \quad \text{and} \quad \frac{\mathrm{d}}{\mathrm{d}\pi_i}\left\{\bar{x}_{ii}^2 - \bar{x}_{ji}^2\right\} = 2(1-\pi_i g_i) \geq 0. \tag{C-130}$$

Hence, $\bar{x}_{ii}^2 - \bar{x}_{ji}^2$ is also negative and increasing in π_i. It follows that the product:

$$\begin{aligned} &\left\{\bar{x}_{ii}^2 - \left[\bar{x}_{ji}(\pi_i)\right]^2\right\} \cdot \frac{\mathrm{d}N_i^\ell}{\mathrm{d}g_i}(\pi_i) \quad \text{is positive and decreases with } \pi_i. \\ \Longrightarrow \quad & -\frac{1}{2}\left[\bar{x}_{ii}^2 - \left(\bar{x}_{ji}(\pi_i)\right)^2\right] \cdot \frac{\mathrm{d}N_i^\ell}{\mathrm{d}g_i}(\pi_i) \quad \text{increases with } \pi_i. \end{aligned} \tag{C-131}$$

Facts (C-128) and (C-131) together imply that, for any $\pi_i \in (0,1)$ and π_i' such that $\pi_i < \pi_i' \leq 1$:

$$MB_i^\ell(g_i, g_j; \pi_i') > MB_i^\ell(g_i, g_j; \pi_i)$$

□

C.12 Proof of Proposition 3.7 (Weak Extraterritorial Reach Crowds Out Privacy)

We proceed in two parts. First, we show that when $\pi \in (0,1)$ and $g^*(\pi) \in (0, \bar{g})$, then it is strictly increasing in extraterritorial enforcement π. Second, we demonstrate that in the knife-edge case of perfect enforcement ($\pi_i = \pi_j = 1$), best responses are constant functions that are independent of the opponent's policy choice.

C.12.1 When $\pi \in (0,1), \pi \mapsto g^*(\pi) \in (0, \bar{g})$ is strictly increasing

Recall from the notation introduced in the proof of Proposition 3.5 that an interior Nash equilibrium regulation $g^* \in (0, \bar{g})$ is the unique solution to the implicit equation:

$$\widetilde{MB}(g^*; \pi) - \widetilde{MC}(g^*; \pi) = 0 = \left.\frac{\partial \mathcal{W}_i^\ell}{\partial g_i}(g_i; g_j, \pi)\right|_{g_i = g_j = g^*} \tag{C-132}$$

where $\widetilde{MB}$ and $\widetilde{MC}$ respectively denote the restrictions of the marginal benefit and marginal cost functions to the diagonal where $g_i = g_j = g^*$. By implicit differentiation, the equilibrium regulation level g^* is increasing in extraterritorial enforcement π if and only if:

$$\frac{\mathrm{d}g^*}{\mathrm{d}\pi} = -\frac{\frac{\partial}{\partial \pi}\left\{\widetilde{MB}(g^*; \pi) - \widetilde{MC}(g^*; \pi)\right\}}{\frac{\partial}{\partial g}\left\{\widetilde{MB}(g^*; \pi) - \widetilde{MC}(g^*; \pi)\right\}} > 0 \quad \Longleftrightarrow \quad \frac{\partial}{\partial \pi}\left\{\widetilde{MB}(g^*; \pi) - \widetilde{MC}(g^*; \pi)\right\} > 0 \tag{C-133}$$

where the latter equivalence follows from $\widetilde{MB} - \widetilde{MC}$ being strictly decreasing in g, by our earlier characterisation (C-105). Again, we proceed separately by showing that both $\widetilde{MB}$ and $-\widetilde{MC}$ are increasing in π.

<u>**$\widetilde{MC}$ is strictly decreasing in π:**</u> Fix any g and use equation (C-107) to write $\widetilde{MC}$ as a function on π:

$$\widetilde{MC}(\pi) \propto \left[\frac{\bar{x}_{ii}^{\frac{2\beta(\sigma-1)}{1-\beta}} - \left[\bar{x}_{ji}(\pi)\right]^{\frac{2\beta(\sigma-1)}{1-\beta}} \tau^{\frac{2(1-\sigma)}{1-\beta}}}{\bar{x}_{ii}^{\frac{\beta(\sigma-1)}{1-\beta}} - \tau^{\frac{1-\sigma}{1-\beta}}\left[\bar{x}_{ji}(\pi)\right]^{\frac{\beta(\sigma-1)}{1-\beta}}}\right]^{\frac{\alpha}{\sigma-1}-1} \left[\frac{\bar{x}_{ii}^{\frac{2\beta(\sigma-1)}{1-\beta}-1} - \pi\left[\bar{x}_{ji}(\pi)\right]^{\frac{2\beta(\sigma-1)}{1-\beta}-1} \tau^{\frac{2(1-\sigma)}{1-\beta}}}{\bar{x}_{ii}^{\frac{\beta(\sigma-1)}{1-\beta}} - \tau^{\frac{1-\sigma}{1-\beta}}\left[\bar{x}_{ji}(\pi)\right]^{\frac{\beta(\sigma-1)}{1-\beta}}}\right]$$

The first term can be simplified by difference of squares to yield:

$$\frac{\bar{x}_{ii}^{\frac{2\beta(\sigma-1)}{1-\beta}} - [\bar{x}_{ji}(\pi)]^{\frac{2\beta(\sigma-1)}{1-\beta}}\tau^{\frac{2(1-\sigma)}{1-\beta}}}{\bar{x}_{ii}^{\frac{\beta(\sigma-1)}{1-\beta}} - [\bar{x}_{ji}(\pi)]^{\frac{\beta(\sigma-1)}{1-\beta}}\tau^{\frac{1-\sigma}{1-\beta}}} = \frac{(\bar{x}_{ii}^{\frac{\beta(\sigma-1)}{1-\beta}})^2 - ([\bar{x}_{ji}(\pi)]^{\frac{\beta(\sigma-1)}{1-\beta}}\tau^{\frac{1-\sigma}{1-\beta}})^2}{\bar{x}_{ii}^{\frac{\beta(\sigma-1)}{1-\beta}} - [\bar{x}_{ji}(\pi)]^{\frac{\beta(\sigma-1)}{1-\beta}}\tau^{\frac{1-\sigma}{1-\beta}}} = \bar{x}_{ii}^{\frac{\beta(\sigma-1)}{1-\beta}} + [\bar{x}_{ji}(\pi)]^{\frac{\beta(\sigma-1)}{1-\beta}}\tau^{\frac{1-\sigma}{1-\beta}} \tag{C-134}$$

which is strictly decreasing in $\bar{x}_{ji}$, and thus weakly increasing in π at each $g \in [0, \bar{g}]$ since $\bar{x}_{ji}(\pi) = 1 - \pi g$. Turning to the second term, note that the numerator can be rewritten as:

$$\frac{\bar{x}_{ii}^{\frac{2\beta(\sigma-1)}{1-\beta}-1} - \pi[\bar{x}_{ji}(\pi)]^{\frac{2\beta(\sigma-1)}{1-\beta}-1}\tau^{\frac{2(1-\sigma)}{1-\beta}}}{\bar{x}_{ii}^{\frac{\beta(\sigma-1)}{1-\beta}} - \tau^{\frac{1-\sigma}{1-\beta}}[\bar{x}_{ji}(\pi)]^{\frac{\beta(\sigma-1)}{1-\beta}}} = \frac{\bar{x}_{ii}^{\frac{2\beta(\sigma-1)}{1-\beta}-1} - z(\pi)\cdot[\bar{x}_{ji}(\pi)]^{2\left(\frac{\beta(\sigma-1)}{1-\beta}-1\right)}\tau^{\frac{2(1-\sigma)}{1-\beta}}}{\bar{x}_{ii}^{\frac{\beta(\sigma-1)}{1-\beta}} - \tau^{\frac{1-\sigma}{1-\beta}}[\bar{x}_{ji}(\pi)]^{\frac{\beta(\sigma-1)}{1-\beta}}} \tag{C-135}$$

$$\text{with } z(\pi) = \pi\bar{x}_{ji}(\pi) \tag{C-136}$$

As already shown earlier (C-126), the function $z : \pi \mapsto \pi\bar{x}_{ji}(\pi)$ is strictly increasing for any $\pi \in (0, 1)$ under Assumption 7. Meanwhile, because $\frac{\beta(\sigma-1)}{1-\beta} - 1 < 0$ under Assumption 5, $[\bar{x}_{ji}(\pi)]^{2\left(\frac{\beta(\sigma-1)}{1-\beta}-1\right)}$ is non-decreasing in g, and it follows that the product $-z(\pi)\cdot[\bar{x}_{ji}(\pi)]^{2\left(\frac{\beta(\sigma-1)}{1-\beta}-1\right)}$ is strictly decreasing in π, and so is the numerator of term (C-136). Moreover, because the denominator is decreasing in $\bar{x}_{ji}(\pi)$, it is weakly increasing in π, and it follows that the fraction (C-136) is unambiguously decreasing in π, strictly. Hence, we have shown that:

$$\frac{\partial}{\partial\pi}\left\{\frac{\bar{x}_{ii}^{\frac{2\beta(\sigma-1)}{1-\beta}} - [\bar{x}_{ji}(\pi)]^{\frac{2\beta(\sigma-1)}{1-\beta}}\tau^{\frac{2(1-\sigma)}{1-\beta}}}{\bar{x}_{ii}^{\frac{\beta(\sigma-1)}{1-\beta}} - [\bar{x}_{ji}(\pi)]^{\frac{\beta(\sigma-1)}{1-\beta}}\tau^{\frac{1-\sigma}{1-\beta}}}\right\} \leq 0 \quad \text{and} \quad \frac{\partial}{\partial\pi}\left\{\frac{\bar{x}_{ii}^{\frac{2\beta(\sigma-1)}{1-\beta}-1} - \pi[\bar{x}_{ji}(\pi)]^{\frac{2\beta(\sigma-1)}{1-\beta}-1}\tau^{\frac{2(1-\sigma)}{1-\beta}}}{\bar{x}_{ii}^{\frac{\beta(\sigma-1)}{1-\beta}} - \tau^{\frac{1-\sigma}{1-\beta}}[\bar{x}_{ji}(\pi)]^{\frac{\beta(\sigma-1)}{1-\beta}}}\right\} < 0 \tag{C-137}$$

$$\Longrightarrow \quad \frac{\partial\widetilde{MC}}{\partial\pi} < 0$$

$\widetilde{MB}$ is strictly increasing in π: Fix $g \in [0, \bar{g}]$, and use equation (C-118) to write $\widetilde{MB}$ as a function of π:

$$\widetilde{MB}_i^{\ell}(\pi_i) \triangleq \psi_i\left\{\frac{N}{2}\left[\underbrace{\pi\,\bar{x}_{ji}(\pi)}_{z(\pi)} + \bar{x}_{ii}\right] - \frac{1}{2}\left(\bar{x}_{ii}^2 - [\bar{x}_{ji}(\pi)]^2\right)\frac{\mathrm{d}N_i^{\ell}}{\mathrm{d}g_i}(\pi)\right\} \tag{C-138}$$

Since $z : \pi \mapsto \pi\bar{x}_{ji}(\pi)$ is strictly increasing at any $\pi \in (0, 1)$ by (C-126), so is the first term. As for the second term, one can check from equation (C-125) that it is independent of g_j, and it follows that the same arguments used to establish its monotonicity in the g_i direction in Proof C.7 may be directly adapted under the diagonal restriction $g_i = g_j = g^*$. Accordingly, we shall conclude that:

$$\frac{\partial\widetilde{MB}}{\partial\pi} > 0$$

C.12.2 The best response function $\mathcal{B}_i(\cdot\,; \pi = 1)$ is constant

First note that, when $\pi \equiv \pi_H = \pi_F = 1$, we have:

$$\begin{aligned} \bar{x}_i(g_i) \equiv \bar{x}_{ii} = \bar{x}_{ji} &= 1 - g_i \\ \Longrightarrow \quad \Delta_i(g_i) = \bar{x}_{ii} - \tau^{\frac{1-\sigma}{1-\beta}}\bar{x}_{ji} &= [\bar{x}_i(g_i)]^{\frac{\beta(\sigma-1)}{1-\beta}}\left(1 - \tau^{\frac{1-\sigma}{1-\beta}}\right) \end{aligned}$$

$$\implies \quad \Delta(g_i;g_j) = \bar{x}_{ii}^{\frac{\beta(\sigma-1)}{1-\beta}}\,\bar{x}_{jj}^{\frac{\beta(\sigma-1)}{1-\beta}} - \tau^{\frac{2(1-\sigma)}{1-\beta}}\,\bar{x}_{ji}^{\frac{\beta(\sigma-1)}{1-\beta}}\,\bar{x}_{ij}^{\frac{\beta(\sigma-1)}{1-\beta}} \;=\; [\bar{x}_i(g_i)]^{\frac{\beta(\sigma-1)}{1-\beta}} \cdot [\bar{x}_j(g_j)]^{\frac{\beta(\sigma-1)}{1-\beta}} \cdot \left(1-\tau^{\frac{2(1-\sigma)}{1-\beta}}\right) \tag{C-139}$$

We want to show that, when $\pi = 1$, the best response $\mathscr{B}_i(\cdot)$ solving Problem (46) is strategically independent of g_j. Equivalently, for any g_j, there exists a constant solution to the implicit equation:

$$MB_i^\ell(g_i;g_j,\pi=1) - MC_i^\ell(g_i;g_j,\pi=1) = 0 = \frac{\partial \mathscr{W}_i^\ell}{\partial g_i}(g_i;g_j,\pi=1) \tag{C-140}$$

Hence, it suffices to show that, when $\pi_H = \pi_F = \pi = 1$, both the marginal benefit MB_i^ℓ and marginal cost MC_i^ℓ of regulation are independent of privacy regulation in the other country.

<u>$MC_i^\ell(\cdot\,;\pi=1)$ **is independent of** g_j</u> Evaluating our earlier characterisation of MC_i^ℓ (C-120) at $\pi = 1$ yields an expression for MC_i^ℓ that is independent of g_j:

$$MC_i^\ell(g_i;\pi=1) = \frac{\alpha\beta\, L_i\, w^\alpha}{1-\beta}\left(\frac{E_i\,\varphi^{\frac{\beta(\sigma-1)}{1-\beta}}\left(1+\tau^{\frac{1-\sigma}{1-\beta}}\right)}{q_0\, p_0^\sigma}\right)^{\frac{\alpha}{\sigma-1}} [\bar{x}_i(g_i)]^{\frac{\alpha\beta}{1-\beta}-1} \tag{C-141}$$

<u>$MB_i^\ell(\cdot\,;\pi=1)$ **is independent of** g_j</u> Recall from (C-64) that $\left.\frac{\mathrm{d}N_i^\ell}{\mathrm{d}g_i}\right|_{\pi_i=1} = 0$. It thus follows from evaluating MB_i^ℓ (C-125) at $\pi = 1$ and setting $\bar{x}_i \equiv \bar{x}_{ii} = \bar{x}_{ji} 1 - g$ that:

$$\begin{aligned} MB_i^\ell(g_i) \triangleq \left\{\psi_i\left[\pi_i\,\bar{x}_{ji}\,N_j^\ell + \bar{x}_{ii}N_i^\ell - \frac{1}{2}\left(\bar{x}_{ii}^2 - \bar{x}_{ji}^2\right)\frac{\mathrm{d}N_i^\ell}{\mathrm{d}g_i}\right]\right\}\Bigg|_{\pi_i=\pi_j=1} &= \psi_i \cdot \left[\,N_j^\ell + N_i^\ell\right]\cdot \bar{x}_i(g_i) \\ &= \psi_i \cdot N \cdot \bar{x}_i(g_i) \end{aligned} \tag{C-142}$$

where N is the constant solution for the mass of Tech firms in the world economy, as stated in Proposition 2.5.

<u>**Solution for the constant best response** $\mathscr{B}_i(\pi=1)$</u> Using equations (C-141) and (C-142), one can solve explicitly for the constant best response under perfect enforcement, given by $\mathscr{B}_i(g_j;\pi=1) = g_i^{\ell*} = g_i^{s*}$ for all $g_j \in [0,\bar{g}_j]$. The resulting regulation level coincides with the unique solution to Problem (C-50), which solves the privacy–consumption trade-off in country i for any policy horizon $h \in \{s,\ell\}$. Straightforward algebra yields:

$$g_i^* = 1 - \left[\frac{\alpha\beta\, L_i\, w^\alpha}{\psi_i\, N\,(1-\beta)}\left(\frac{E_i\,\varphi^{\frac{\beta(\sigma-1)}{1-\beta}}\left(1+\tau^{\frac{1-\sigma}{1-\beta}}\right)}{q_0\, p_0^\sigma}\right)^{\frac{\alpha}{\sigma-1}}\right]^{\frac{1}{2-\frac{\alpha\beta}{1-\beta}}} \tag{C-143}$$

□

C.13 Proof of Proposition 3.8 (Slope of the Symmetric Best Response Function)

Any point in the interior of the image set of the best response mapping $\mathscr{B}_i : [0,\bar{g}] \to [0,\bar{g}]$ is the unique solution to the first-order condition of Problem (C-101):

$$\frac{\partial \mathscr{W}_i^\ell}{\partial g_i}(g_i;g_j) = 0 \quad\iff\quad MB_i^\ell(\mathscr{B}_i(g_j);g_j) - MC_i^\ell(\mathscr{B}_i(g_j);g_j) = 0 \tag{C-144}$$

By implicit differentiation, the slope of $\mathscr{B}_i$ is positive if and only if:

$$\frac{\mathrm{d}\mathscr{B}_i}{\mathrm{d}g_j} = \underbrace{\left(-\frac{\partial^2 \mathscr{W}_i^\ell}{\partial g_i^2}\right)^{-1}}_{>0 \text{ by strict concavity}} \frac{\partial^2 \mathscr{W}_i^\ell}{\partial g_i g_j} > 0 \quad \Longleftrightarrow \quad \frac{\partial MB_i^\ell}{\partial g_j}(g_i; g_j) > \frac{\partial MC_i^\ell}{\partial g_j}(g_i; g_j) \tag{C-145}$$

We now argue that both partial effects must have the same sign in the symmetric game with imperfect extraterritorial reach. Accordingly, the slope of $\mathscr{B}_i$ is theoretically ambiguous. In particular, we claim that:

$$\pi \in (0,1) \quad \Longrightarrow \quad \frac{\partial MB_i^\ell}{\partial g_j}(g_i; g_j) > 0 \quad \text{and} \quad \frac{\partial MC_i^\ell}{\partial g_j}(g_i; g_j) > 0 \tag{C-146}$$

Fix a candidate regulation g_i. With similar derivations to (C-52), and writing all terms as functions of g_j where relevant, the FOC of the government of country i in the symmetric game might be rewritten as:

$$\begin{aligned}
0 &= -\alpha\, L\, w^\alpha\, (P_i^\ell)^{-\alpha} \frac{\mathrm{d}\ln P_i^\ell}{\mathrm{d}g_i} + \psi\left[-N_i^\ell\, \bar{x}_{ii}\, \frac{\mathrm{d}\bar{x}_{ii}}{\mathrm{d}g_i} - N_j^\ell\, \bar{x}_{ji}\, \frac{\mathrm{d}\bar{x}_{ji}}{\mathrm{d}g_i} - \frac{1}{2}\, \left(\bar{x}_{ii}^2 - \bar{x}_{ji}^2\right)\, \frac{\mathrm{d}N_i^\ell}{\mathrm{d}g_i}\right] \\
&= \underbrace{\psi\left[N_i^\ell(g_j)\, [\bar{x}_{ii} - \pi_i\, \bar{x}_{ji}] + \pi_i \bar{x}_{ji} N + \frac{1}{2}\, \left(\bar{x}_{ji}^2 - \bar{x}_{ii}^2\right)\, \frac{\mathrm{d}N_i^\ell}{\mathrm{d}g_i}\right]}_{MB_i^\ell(g_j)} \\
&\quad - \underbrace{\alpha\, L\, w^\alpha\, [P_i^\ell(g_j)]^{-\alpha}\left[-\frac{\partial \ln P_i^\ell}{\partial \bar{x}_{ii}}(g_j) - \pi_i \frac{\partial \ln P_i^\ell}{\partial \bar{x}_{ji}}(g_j) + \frac{\mathrm{d}\ln P_i^\ell}{\mathrm{d}N_i^\ell}\frac{\mathrm{d}N_i^\ell}{\mathrm{d}g_i}\right]}_{MC_i^\ell(g_j)}
\end{aligned} \tag{C-147}$$

Note that $\frac{\mathrm{d}N_i^\ell}{\mathrm{d}g_i}$ does not depend on $\bar{x}_{jj}$ or $\bar{x}_{ij}$, as verifiable from (C-64). Moreover, Lemma 1 showed that:

$$\pi \in (0,1) \quad \Longrightarrow \quad \frac{\mathrm{d}N_i^\ell}{\mathrm{d}g_i} < 0 \quad \Longrightarrow \quad \frac{\mathrm{d}N_j^\ell}{\mathrm{d}g_i} > 0 \quad \Longrightarrow \quad \frac{\mathrm{d}N_i^\ell}{\mathrm{d}g_j} > 0 \tag{C-148}$$

where the first implication follows from the total number of firms in the world economy being independent of data regulation by Proposition 2.5, while the second inequality uses symmetry through index inversion.

Hence, we can use an earlier characterisation of the sign of the term multiplying N_i^ℓ (C-78) to immediately conclude that MB_i^ℓ must rise in the g_j-direction:

$$\bar{x}_{ii} - \pi_i\, \bar{x}_{ji} > 0 \quad \forall g_i \in [0, \bar{g}] \quad \Longrightarrow \quad \frac{\partial MB_i^\ell}{\partial g_j} = \underbrace{\frac{\partial MB_i^\ell}{\partial N_i^\ell}}_{>0} \underbrace{\frac{\mathrm{d}N_i^\ell}{\mathrm{d}g_j}}_{>0} > 0 \tag{C-149}$$

Moving to MC_i^ℓ, a direct application of the comparative statics results from Propostion 2.2 gives that:

$$-\frac{\partial \ln P_i}{\partial \bar{x}_{ii}} - \pi_i \frac{\partial \ln P_i}{\partial \bar{x}_{ji}} = \frac{\beta}{1-\beta} \cdot \frac{N_i^\ell \bar{x}_{ii}^{\frac{\beta(\sigma-1)}{1-\beta}-1} + \pi_i (N - N_i^\ell) \tau_{ji}^{\frac{1-\sigma}{1-\beta}} \bar{x}_{ji}^{\frac{\beta(\sigma-1)}{1-\beta}-1}}{N\, \tau_{ji}^{\frac{1-\sigma}{1-\beta}}\, \bar{x}_{ji}^{\frac{\beta(\sigma-1)}{1-\beta}} \;+\; N_i^\ell\, \left[\bar{x}_{ii}^{\frac{\beta(\sigma-1)}{1-\beta}} - \tau_{ji}^{\frac{1-\sigma}{1-\beta}}\, \bar{x}_{ji}^{\frac{\beta(\sigma-1)}{1-\beta}}\right]} \tag{C-150}$$

$$= \frac{\beta}{1-\beta} \cdot \frac{\pi_i N\, \tau_{ji}^{\frac{1-\sigma}{1-\beta}}\, \bar{x}_{ji}^{\frac{\beta(\sigma-1)}{1-\beta}-1} + N_i^\ell(g_j) \left[\bar{x}_{ii}^{\frac{\beta(\sigma-1)}{1-\beta}-1} - \pi_i \tau_{ji}^{\frac{1-\sigma}{1-\beta}}\, \bar{x}_{ji}^{\frac{\beta(\sigma-1)}{1-\beta}-1}\right]}{N\, \tau_{ji}^{\frac{1-\sigma}{1-\beta}}\, \bar{x}_{ji}^{\frac{\beta(\sigma-1)}{1-\beta}} \;+\; N_i^\ell(g_j)\, \left[\bar{x}_{ii}^{\frac{\beta(\sigma-1)}{1-\beta}} - \tau_{ji}^{\frac{1-\sigma}{1-\beta}}\, \bar{x}_{ji}^{\frac{\beta(\sigma-1)}{1-\beta}}\right]} \tag{C-151}$$

Notice that both the numerator and denominator are affine in N_i^ℓ. Writing:

$$h(x) \;=\; \frac{a+b\,x}{c+d\,x}, \quad \text{one can compute} \quad h'(x) \;=\; \frac{b\,c \;-\; a\,d}{\left(c+d\,x\right)^2} \tag{C-152}$$

In our case, it is readily checked that:

$$b\,c-a\,d = N\,\tau_{ji}^{\frac{1-\sigma}{1-\beta}}\,\bar{x}_{ii}^{\frac{\beta(\sigma-1)}{1-\beta}-1}\,\bar{x}_{ji}^{\frac{\beta(\sigma-1)}{1-\beta}-1}\left(\bar{x}_{ji}-\pi_i\,\bar{x}_{ii}\right) = N\,\tau_{ji}^{\frac{1-\sigma}{1-\beta}}\,\bar{x}_{ii}^{\frac{\beta(\sigma-1)}{1-\beta}-1}\,\bar{x}_{ji}^{\frac{\beta(\sigma-1)}{1-\beta}-1}\left(1-\pi\right) > 0 \tag{C-153}$$

By the chain rule, one therefore concludes that:

$$\frac{\partial}{\partial g_j}\left\{-\frac{\partial \ln P_i}{\partial \bar{x}_{ii}} - \pi_i\frac{\partial \ln P_i}{\partial \bar{x}_{ji}}\right\} = \underbrace{\frac{\partial}{\partial N_i}\left\{-\frac{\partial \ln P_i}{\partial \bar{x}_{ii}} - \pi_i\frac{\partial \ln P_i}{\partial \bar{x}_{ji}}\right\}}_{>0}\underbrace{\frac{\mathrm{d}N_i}{\mathrm{d}g_j}}_{>0} > 0 \tag{C-154}$$

Finally, use the solution for the price index in terms of firm masses (B-44) and the fact that N_i and N_j sum to a constant N by Proposition 2.5 to write:

$$P_i^\ell(g_j) = \left(\frac{p_0}{\varphi^{\frac{\beta}{1-\beta}}}\right)\left(\sum_{k\in C} N_k^\ell \tau_{ki}^{\frac{1-\sigma}{1-\beta}}\,\bar{x}_{ki}^{\frac{\beta(\sigma-1)}{1-\beta}}\right)^{\frac{1}{1-\sigma}} = \left(\frac{p_0}{\varphi^{\frac{\beta}{1-\beta}}}\right)\left[N_i^\ell\underbrace{\left(\bar{x}_{ii}^{\frac{\beta(\sigma-1)}{1-\beta}} - \tau^{\frac{1-\sigma}{1-\beta}}\bar{x}_{ji}^{\frac{\beta(\sigma-1)}{1-\beta}}\right)}_{=\Delta_i>0}\right]^{\frac{1}{1-\sigma}} \tag{C-155}$$

Hence, by partial differentiation with respect to N_i^ℓ, we have:

$$\frac{\mathrm{d}\ln P_i^\ell}{\mathrm{d}N_i^\ell} = -\frac{1}{\sigma-1}\;\frac{\Delta_i}{N\,\tau_{ji}^{\frac{1-\sigma}{1-\beta}}\,\bar{x}_{ji}^{\frac{\beta(\sigma-1)}{1-\beta}} \;+\; N_i^\ell(g_j)\Delta_i} < 0 \tag{C-156}$$

Notice that the whole expression is negative and decreases with g_j, since the denominator increases with $N_i(g_j)$, which is itself an increasing transformation g_j. Because $\frac{\mathrm{d}N_i}{\mathrm{d}g_i}$ is a negative constant with respect to g_j by (C-64), we shall conclude that:

$$\frac{\partial}{\partial g_j}\left\{\frac{\mathrm{d}\ln P_i^\ell}{\mathrm{d}N_i^\ell}\frac{\mathrm{d}N_i^\ell}{\mathrm{d}g_i}\right\} > 0 \tag{C-157}$$

By facts (C-154) and (C-157), the whole term in parentheses in the definition of MC_i^ℓ (C-147) is positive and strictly increasing in the g_j-direction. Furthermore, we can immediately conclude from characterisations (C-148) and (C-156) that the factor $\alpha\,L\,w^\alpha\,[P_i^\ell(g_j)]^{-\alpha}$ is also strictly increasing in g_j, since:

$$\frac{\partial}{\partial g_j}\left\{\alpha\,L\,w^\alpha\left[P_i^\ell(g_j)\right]^{-\alpha}\right\} \propto \frac{\partial}{\partial g_j}\left\{\ln\left[P_i^\ell(g_j)^{-\alpha}\right]\right\} = -\alpha\cdot\underbrace{\frac{\mathrm{d}\ln P_i^\ell}{\mathrm{d}N_i^\ell}}_{<0}\cdot\underbrace{\frac{\mathrm{d}N_i^\ell}{\mathrm{d}g_j}}_{>0} > 0 \tag{C-158}$$

Combining our results establishes claim (C-146). Accordingly, the sign of the slope of the symmetric best response function when $\pi \in (0,1)$ is theoretically ambiguous without further investigation, since a change $\mathrm{d}g_j$ increases both MB_i^ℓ and MC_i^ℓ.

□

C.14 Proof of Proposition 3.9 (Weak Extraterritorial Reach Crowds Out Welfare)

Write the equilibrium welfare function as $\mathcal{W}_i^\ell\left(g_i(\pi); g_j(\pi), \pi\right)$ (C-31). Totally differentiating $\mathcal{W}_i^\ell$ with respect to π and evaluating the resulting expression at an arbitrary, interior, symmetric Nash equilibrium $g^* \in (0, \bar{g})$ gives:

$$\left.\frac{\mathrm{d}\mathcal{W}_i^\ell}{\mathrm{d}\pi}\right|_{g_i=g_j=g^*} = \left.\left\{\underbrace{\frac{\partial \mathcal{W}_i^\ell}{\partial g_i}}_{=0}\frac{\mathrm{d}g_i}{\mathrm{d}\pi} + \frac{\partial \mathcal{W}_i^\ell}{\partial g_j}\frac{\mathrm{d}g_j}{\mathrm{d}\pi} + \frac{\partial \mathcal{W}_i^\ell}{\partial \pi}\right\}\right|_{g_i=g_j=g^*} = \left.\left\{\frac{\partial \mathcal{W}_i^\ell}{\partial g_j}\frac{\mathrm{d}g_j}{\mathrm{d}\pi} + \frac{\partial \mathcal{W}_i^\ell}{\partial \pi}\right\}\right|_{g_i=g_j=g^*} \tag{C-159}$$

The last equality follows from the envelope condition. We claim that:

$$\left.\frac{\mathrm{d}\mathcal{W}_i^\ell}{\mathrm{d}\pi}\right|_{g_i=g_j=g^*} > 0 \tag{C-160}$$

Consider the first term in (C-159). Because the domestic price index P_i^ℓ depends on g_j solely through induced adjustments in firm masses, we immediately deduce from (C-148) and (C-156) that:

$$\frac{\mathrm{d}N_i^\ell}{\mathrm{d}g_j} > 0, \quad \frac{\mathrm{d}P_i^\ell}{\mathrm{d}g_j} < 0 \quad \Longrightarrow \quad \frac{\mathrm{d}P_i^\ell}{\mathrm{d}g_j} = \frac{\mathrm{d}P_i^\ell}{\mathrm{d}N_i^\ell}\frac{\mathrm{d}N_i^\ell}{\mathrm{d}g_j} < 0 \tag{C-161}$$

Hence, by chain rule expansion, and using that the equilibrium mapping $\pi \mapsto g^*(\pi)$ is strictly increasing at any interior $g^*(\pi)$, for each $\pi \in (0,1)$ by Proposition 3.7, we conclude that:

$$\begin{aligned}
\frac{\partial \mathcal{W}_i^\ell}{\partial g_j} &= \underbrace{-\alpha L w^\alpha (P_i^\ell)^{-\alpha-1}}_{<0}\underbrace{\frac{\mathrm{d}P_i^\ell}{\mathrm{d}g_j}}_{<0} + \underbrace{\frac{\psi}{2}\Big([1-\pi g]^2 - [1-g]^2\Big)}_{>0}\underbrace{\frac{\mathrm{d}N_i^\ell}{\mathrm{d}g_j}}_{>0} > 0 \\
\Longrightarrow \quad \left.\left\{\frac{\partial \mathcal{W}_i^\ell}{\partial g_j}\frac{\mathrm{d}g_j}{\mathrm{d}\pi}\right\}\right|_{g_i=g_j=g^*} &= \underbrace{\left.\frac{\partial \mathcal{W}_i^\ell}{\partial g_j}\right|_{g_i=g_j=g^*}}_{>0} \cdot \underbrace{\frac{\mathrm{d}g^*}{\mathrm{d}\pi}}_{>0} > 0
\end{aligned} \tag{C-162}$$

To establish claim (C-160), it thus suffices to prove that the partial derivative of $\mathcal{W}_i^\ell$ with respect to π is non-negative at the symmetric equilibrium $g^* \in (0, \bar{g})$. Specifically, since a symmetric rise in π affects welfare through the cross-country data usage caps, $\bar{x}_{ji} = 1 - \pi g$ and $\bar{x}_{ij} = 1 - \pi g$, we aim to show that:

$$\left.\frac{\partial \mathcal{W}_i^\ell}{\partial \pi}\right|_{g_i=g_j=g^*} = \left.\left\{\frac{\partial \mathcal{W}_i^\ell}{\partial \bar{x}_{ji}}\frac{\mathrm{d}\bar{x}_{ji}}{\mathrm{d}\pi} + \frac{\partial \mathcal{W}_i^\ell}{\partial \bar{x}_{ij}}\frac{\mathrm{d}\bar{x}_{ij}}{\mathrm{d}\pi}\right\}\right|_{g_i=g_j=g^*} \geq 0 \tag{C-163}$$

For this purpose, it is useful to note first that, starting from a symmetric equilibrium with $N_i = N_j = \frac{N}{2}$, a symmetric change in enforcement capacity π induces another symmetric equilibrium and must therefore leave the spatial distribution of Tech activity unchanged. Accordingly, firm masses may be treated as constant, and partially differentiating $\mathcal{W}_i^\ell$ as defined in (C-31) with respect to π yields:

$$\left.\frac{\partial \mathcal{W}_i^\ell}{\partial \pi}\right|_{g_i=g_j=g^*} > 0 \quad \Longleftrightarrow \quad \left.\left\{-\alpha L w^\alpha (P_i^\ell)^{-\alpha-1}\frac{\partial P_i^\ell}{\partial \bar{x}_{ji}} - \frac{\psi N}{2}\bar{x}_{ji}\right\}\right|_{g_i=g_j=g^*} \leq 0 \tag{C-164}$$

We claim that the above inequality holds. By (C-147), any interior equilibrium $g^* \in (0, \bar{g})$ must solve the implicit equation:

$$0 = \left[-\alpha L w^{\alpha} \left(P_i^{\ell}\right)^{-\alpha-1} \underbrace{\frac{\partial P_i^{\ell}}{\partial \bar{x}_{ii}}}_{<0} - \frac{\psi N}{2} \bar{x}_{ii} \right] \underbrace{\frac{\mathrm{d}\bar{x}_{ii}}{\mathrm{d}g_i}}_{<0} + \left[-\alpha L w^{\alpha} \left(P_i^{\ell}\right)^{-\alpha-1} \underbrace{\frac{\partial P_i^{\ell}}{\partial \bar{x}_{ji}}}_{<0} - \frac{\psi N}{2} \bar{x}_{ji} \right] \underbrace{\frac{\mathrm{d}\bar{x}_{ji}}{\mathrm{d}g_i}}_{<0}$$

$$+ \left[\underbrace{-\alpha L w^{\alpha} \left(P_i^{\ell}\right)^{-\alpha-1} \underbrace{\frac{\mathrm{d}P_i^{\ell}}{\mathrm{d}N_i^{\ell}}}_{<0 \text{ by (C-156)}}}_{>0} + \frac{\psi}{2} \underbrace{\left((1-\pi g^*)^2 - (1-g^*)^2\right)}_{>0 \text{ for any } \pi \in (0,1)} \right] \underbrace{\frac{\mathrm{d}N_i^{\ell}}{\mathrm{d}g_i}}_{<0} \tag{C-165}$$

Notice that the last product is always negative. For the purpose of contradiction, suppose that condition (C-164) does not hold at the Nash symmetric equilibrium. Accordingly, the second product is also negative. But, then, we must have:

$$\begin{aligned}
\bar{x}_{ii} = 1 - g^* < 1 - \pi g^* = \bar{x}_{ij} \implies & \left| \frac{\partial P_i^{\ell}}{\partial \bar{x}_{ii}} \right| = \left| \frac{\beta P_i}{1-\beta} \frac{\bar{x}_{ii}^{\frac{\beta(\sigma-1)}{1-\beta}-1}}{E} \right| > \left| \frac{\beta P_i}{1-\beta} \frac{\tau_{ji}^{\frac{1-\sigma}{1-\beta}} \bar{x}_{ji}^{\frac{\beta(\sigma-1)}{1-\beta}-1}}{E} \right| = \left| \frac{\partial P_i^{\ell}}{\partial \bar{x}_{ji}} \right| \\
\implies & -\alpha L w^{\alpha} \left(P_i^{\ell}\right)^{-\alpha-1} \frac{\partial P_i^{\ell}}{\partial \bar{x}_{ii}} - \frac{\psi N}{2} \bar{x}_{ii} < -\alpha L w^{\alpha} \left(P_i^{\ell}\right)^{-\alpha-1} \frac{\partial P_i^{\ell}}{\partial \bar{x}_{ji}} - \frac{\psi N}{2} \bar{x}_{ji} < 0 \\
\implies & \left. \frac{\partial \mathscr{W}_i^{\ell}}{\partial g_i} \right|_{g_i = g_j = g^*} < 0
\end{aligned} \tag{C-166}$$

which contradicts that the first order condition is satisfied. Hence, inequality (C-164) holds at the symmetric equilibrium and, combining this observation with fact (C-162) establishes our main claim (C-160).

C.15 Proof of Proposition 3.10 (Effects of Unbiased Privacy Preference Shocks)

The proof of each claim follows directly from implicit differentiation of the first-order condition that characterises the best response function $\mathscr{B}_i$, and, in the symmetric case, from the fixed-point condition that defines the equilibrium level g^* when the same first-order condition is evaluated along the diagonal where $g_i = g_j$.

$\psi_i \mapsto \mathscr{B}_i(\cdot; \psi_i) \subset (0, \bar{g}_i)$ **is strictly increasing:** The best response $\mathscr{B}_i(g_j)$, when interior, solves:

$$\frac{\partial \mathscr{W}_i^{\ell}}{\partial g_i}(\mathscr{B}_i(g_j); g_j) = 0 = MB^{\ell}(\mathscr{B}_i(g_j); g_j, \psi_i) - MC^{\ell}(\mathscr{B}_i(g_j); g_j, \psi_i) \tag{C-167}$$

We recall that MB_i^{ℓ} was defined in (C-52) and that MC_i^{ℓ} is independent of ψ_i. From Proposition 3.2, $\mathscr{W}_i^{\ell}$ is strictly concave in g_i and thus $\frac{\partial^2 \mathscr{W}_i^{\ell}}{\partial g_i^2} < 0$. In particular, invoking the implicit function theorem yields:

$$\frac{\mathrm{d}\mathscr{B}_i}{\mathrm{d}\psi_i} = -\frac{\frac{\partial}{\partial \psi_i}\left\{MB^{\ell}\right\} - 0}{\frac{\partial^2 \mathscr{W}_i^{\ell}}{\partial g_i^2}} = (-1) \cdot \underbrace{\left[\frac{\partial^2 \mathscr{W}_i^{\ell}}{\partial g_i^2}\right]^{-1}}_{<0} \cdot \underbrace{\frac{MB^{\ell}(\mathscr{B}_i(g_j); g_j)}{\psi_i}}_{>0} > 0 \tag{C-168}$$

The sign characterisation is immediate from the mapping $MC_i^{\ell} : [0, \bar{g}_i] \to \mathbb{R}_{>0}$, as established in Proposition 3.2, which directly implies that $MB^\ell(\mathcal{B}_i(g_j); g_j) = MC^\ell(\mathcal{B}_i(g_j); g_j) > 0$ whenever the FOC (C-167) is satisfied.

<u>$\psi \mapsto g^*(\psi) \subset (0, \bar{g})$ **is strictly increasing:**</u> By Proposition 3.5, $\widetilde{MB} - \widetilde{MC}$ is strictly decreasing in g, and a similar argument yields:

$$\frac{\mathrm{d}g^*}{\mathrm{d}\psi_i} = (-1) \cdot \underbrace{\left(\frac{\partial}{\partial g} \left\{ \widetilde{MB}(g) - \widetilde{MC}(g) \right\} \right)^{-1}}_{<0} \cdot \underbrace{\frac{\widetilde{MB}(g^*)}{\psi}}_{>0} > 0 \tag{C-169}$$

$\square$

C.16 Proof of Proposition 3.11 (Asymmetric Unbiased Preference Shock and the Brussels Effect)

We begin by establishing that a Nash equilibrium (g_i', g_j') of the modified game always exists under the conditions stated in Proposition 3.11, before proceeding to the comparative statics results.

C.16.1 Existence of a Nash equilibrium with asymmetric, unbiased privacy preferences

Define the joint best-response operator:

$$T\colon [0, \bar{g}]^2 \longrightarrow [0, \bar{g}]^2, \qquad T(g_i, g_j) = \Big(\mathcal{B}_i(g_j; \psi_i, \pi),\ \mathcal{B}_j(g_i; \psi_j, \pi) \Big) \tag{C-170}$$

For each $\pi \in (0, 1)$ and each $k \in \{i, j\}$, Assumption 8 ensures that the best–response map $\mathcal{B}_k(\cdot; \psi_k, \pi)\colon [0, \bar{g}] \to [0, \bar{g}]$ is strictly increasing in the rival's policy g_{-k}, hence order–preserving. Consequently, T is isotone on the complete lattice $[0, \bar{g}]^2$. By Tarski's fixed point theorem, which we state below, T admits at least one fixed point, which, by construction, is a Nash equilibrium.

Theorem 1 (Tarski (1955))**.** *Let $(L, \leq)$ be a complete lattice and let $T\colon L \to L$ be isotone (i.e. $x \leq y \implies T(x) \leq T(y)$ for all $x, y \in L$). Then:*

(i) $\mathrm{Fix}(T) \ = \ \{\, x \in L : T(x) = x \}$ *is a non-empty complete lattice.*

(ii) $\min \mathrm{Fix}(T) \ = \ \sup\{\, x \in L : x \leq T(x)\}, \quad \max \mathrm{Fix}(T) \ = \ \inf\{\, x \in L : T(x) \leq x\}.$

An almost immediate corollary, useful for the next part of the analysis, is the following comparison result:

Lemma 2 (Ordering of Fixed Points)**.** *Let $(L, \leq)$ be a complete lattice. Let $F, G : L \to L$ be isotone self-maps such that $F \leq G$ point-wise. If F has a unique fixed point x^*, then $x^* \leq y$ for every y such that $G(y) = y$. Similarly, if G has a unique fixed point y^*, then every fixed point x of F satisfies $x \leq y^*$.*

Proof. Put $S := \{z \in L : F(z) \leq z\}$. Assume F has the unique fixed point x^* and let $y \in L$ satisfy $G(y) = y$. By assumption, $F(y) \leq G(y) = y$ and thus $y \in S$. By Tarski *(ii)*, $\max \mathrm{Fix}(F) = \inf S$, and uniqueness of x^* forces $x^* = \inf S \leq y$. Now let $R := \{z \in L : z \leq G(z)\}$ and suppose G has the unique fixed point y^*. For any fixed point of F, we have $x = F(x) \leq G(x)$ and thus $x \in R$. Uniqueness of y^* implies $x \leq \sup R = y^*$. $\square$

C.16.2 Any post-shock equilibrium involves $g_i' > g_j' > g^*$ for small enough shocks

Define:

$$T_{\text{pre}}(g_i, g_j) \triangleq \big(\mathcal{B}_i(g_j; \psi, \pi), \mathcal{B}_j(g_i; \psi, \pi)\big), \quad \text{and} \quad T_{\text{post}}(g_i, g_j) \triangleq \big(\mathcal{B}_i(g_j; \psi', \pi), \mathcal{B}_j(g_i; \psi, \pi)\big) \tag{C-171}$$

Again, both operators are isotone self-maps on the complete lattice $[0, \bar{g}]^2$ under Assumption 8. Moreover, $T_{\text{post}}(g_i, g_j) \geq T_{\text{pre}}(g_i, g_j)$ in the usual order for every (g_i, g_j), because the first component is evaluated at $\psi_i = \psi' > \psi$ and $\mathcal{B}_i(g_j; \psi', \pi) \geq \mathcal{B}_i(g_j; \psi, \pi)$ by Proposition 3.10. Specifically, take $F = T_{\text{pre}}$, $G = T_{\text{post}}$, $L = [0, \bar{g}]^2$ and $x^* = (g^*, g^*)$, which is indeed the unique fixed point of F by uniqueness of the symmetric Nash equilibrium, from Proposition 3.5. Lemma 2 then implies that for any post-shock equilibrium (g_i', g_j'):

$$(g^*, g^*) \leq (g_i', g_j'), \quad \text{i.e.} \quad g_i',\, g_j' \geq g^*. \tag{C-172}$$

Now, because $g^* \in (0, \bar{g})$, we have $\mathcal{B}_k(g^*; \psi', \pi) > \mathcal{B}_k(g^*; \psi, \pi)$ by Proposition 3.10. We have just shown that $g_i', g_j' \geq g^*$ necessarily, and now claim that the inequality must hold as strict. Indeed, under Assumption 8, the best response map $\mathcal{B}_k(\cdot; \psi_k, \pi)$ is strictly increasing for each $\pi \in (0, 1)$, and we deduce that:

$$g_j' \geq g^* \quad \implies \quad g_i' = \mathcal{B}_i(g_j'; \psi', \pi) \geq \mathcal{B}_i(g^*; \psi', \pi) > \mathcal{B}_i(g^*; \psi, \pi) = g^*. \tag{C-173}$$

But using the above result, the same use of Assumption 8 gives:

$$g_i' > g^* \quad \implies \quad g_j' = \mathcal{B}_j(g_i'; \psi, \pi) > \mathcal{B}_j(g^*; \psi, \pi) = g^* \tag{C-174}$$

Hence, we have shown that $g_i', g_j' > g^*$ and now claim that $g_i' > g_j'$ for small enough shocks. Specifically, by continuity of B_k, established in Proposition 3.6, one can pick ψ' sufficiently close to ψ that every Nash equilibrium of the post-shock game remains interior, i.e $(g_i', g_j') \in (0, \bar{g})^2$ necessarily. Indeed, let $x = (g_i, g_j) \in \mathbb{R}^2$, $H(x, \psi_i) = T(x; \psi_i) - x$ with $T(x; \psi_i) = \big(\mathcal{B}_i(g_j;\, \psi_i, \pi), \mathcal{B}_j(g_i;\, \psi, \pi)\big)$. Because the symmetric equilibrium $x^* = (g^*, g^*)$ is unique, it is isolated in $[0, \bar{g}]$. Accordingly, there exists an open ball $U_\eta = (g^* - \eta, g^* + \eta)^2 \subset [0, \bar{g}]^2$ around x^* such that:

$$H(x, \psi) \neq 0, \quad \text{for all } \; x \in [0, \bar{g}] \setminus U_\eta \tag{C-175}$$

However, notice that $[0, \bar{g}]^2 \setminus U_\eta = [0, \bar{g}]^2 \cap (U_\eta)^c$ is compact as a closed and bounded subset of $\mathbb{R}^2$. Hence, by the Heine-Cantor theorem, H is uniformly continuous over $[0, \bar{g}] \setminus U_\eta$ and we shall conclude that:

$$\begin{aligned}
\forall \varepsilon > 0,\, \exists\, \delta > 0 \text{ s.t.} \quad |\psi - \psi'| < \delta \quad &\implies \quad \|H(x, \psi') - H(x, \psi)\| < \varepsilon \text{ for all } x \in [0, \bar{g}] \setminus U_\eta \\
&\implies \quad H(x, \psi') \neq 0 \text{ for all } x \in [0, \bar{g}] \setminus U_\eta \text{ when } \varepsilon \text{ is chosen small enough.}
\end{aligned}$$

In other words, the arbitrariness of ε ensures the existence of $\delta > 0$ such that, for all $\psi' \in [\psi, \psi + \delta)$, every root of $H(\cdot, \psi')$ over $[0, \bar{g}]^2$ lies within the interior neighbourhood $U_\eta \subset (0, \bar{g})^2$, and so does every fixed point of $T_{\text{post}} = T(\cdot; \psi')$. Accordingly, the map $\psi \mapsto \mathcal{B}_k(g_{-k}; \psi, \pi)$ is locally strictly increasing, by Proposition 3.10.

Now, choose ε small enough and assume towards a contradiction, that $g_i' \leq g_j'$. Monotonicity of $\mathcal{B}_j(\cdot\,; \psi, \pi)$

gives $g_j' = \mathcal{B}_j(g_i'; \psi, \pi) \leq \mathcal{B}_j(g_j'; \psi, \pi)$. But then, we have:

$$g_i' = \mathcal{B}_i(g_j'; \psi', \pi) > \mathcal{B}_j(g_j'; \psi, \pi) \geq g_j', \quad \text{which contradicts } g_i' \leq g_j'. \tag{C-176}$$

Collecting our results establishes that $g_i' > g_j' > g^*$ for any $\pi \in (0,1)$ whenever $\psi' \in [\psi, \psi + \delta)$.

C.16.3 The new spatial distribution for Tech activity is such that $N_j^\ell > \frac{N}{2} > N_i^\ell$

Write N_i^ℓ as a function of (g_i, g_j). By total differentiation, we have:

$$\mathrm{d}N_i^\ell = \frac{\partial N_i^\ell}{\partial g_i}\,\mathrm{d}g_i + \frac{\partial N_i^\ell}{\partial g_j}\,\mathrm{d}g_j, \tag{C-177}$$

Integrating first in g_i from g^* to g_i' at $g_j = g^*$, and then in g_j from g^* to g_j' at $g_i = g_i'$, one obtains:

$$N_i^\ell(g_i', g_j') - N_i^\ell(g^*, g^*) = \int_{g^*}^{g_i'} \frac{\partial N_i^\ell}{\partial g_i}(u, g^*)\,du + \int_{g^*}^{g_j'} \frac{\partial N_i^\ell}{\partial g_j}(g_i', v)\,dv. \tag{C-178}$$

Recall from our earlier characterisation (C-64) that:

$$\frac{\partial N_i^\ell}{\partial g_i} = -\frac{\beta(\sigma-1)E\tau^{\frac{1-\sigma}{1-\beta}}}{(1-\beta)\sigma w f} \cdot [\Delta_i(g_i)]^{-2} \cdot \big(\bar{x}_{ji}(g_i)\big)^{2\left(\frac{\beta(\sigma-1)}{1-\beta}-1\right)} \cdot \big(r_i(g_i)\big)^{\frac{\beta(\sigma-1)}{1-\beta}-1} \cdot \underbrace{[1-\pi]}_{\geq 0} \tag{C-179}$$

which is independent of g_j. By symmetry, $\frac{\partial N_j^\ell}{\partial g_j}$ is analogously independent of g_i, and we further have $\frac{\partial N_i^\ell}{\partial g_j} = -\frac{\partial N_j^\ell}{\partial g_j}$ by Proposition 2.5. In particular, we deduce that:

$$\forall x \in [g^*, g_j'], \quad \frac{\partial N_i^\ell}{\partial g_i}(x, g^*) = -\frac{\partial N_i^\ell}{\partial g_j}(g_i', x) \tag{C-180}$$

Hence, since $g_i' > g_j' > g^*$, we may rewrite (C-178) as:

$$N_i^\ell(g_i', g_j') - N_i^\ell(g^*, g^*) = \int_{g_j'}^{g_i'} \frac{\partial N_i^\ell}{\partial g_i}(u, g^*)\,\mathrm{d}u + \underbrace{\int_{g^*}^{g_j'} \frac{\partial N_i^\ell}{\partial g_i}(u, g^*)\,\mathrm{d}u + \int_{g^*}^{g_j'} \frac{\partial N_i^\ell}{\partial g_j}(g_i', v)\,\mathrm{d}v}_{=0}$$

$$< 0 \quad \text{since} \quad g_i' - g_j' > 0 \quad \text{and} \quad \frac{\partial N_i^\ell}{\partial g_i}(x, g^*) < 0 \quad \text{for all } x \in [g_j', g_i'] \tag{C-181}$$

Accordingly, we conclude that:

$$N_i^\ell(g_i', g_j') < N_i^\ell(g^*, g^*) = \frac{N}{2} < N - N_i^\ell(g_i', g_j') = N_j^\ell(g_i', g_j') \tag{C-182}$$

□

C.17 Proof of Proposition 3.12 (Effect of Biased Privacy Preference Shocks)

$\exists k_1 \in (0,1)$ such that $b_i \mapsto \mathcal{B}_i(\cdot\,; b_i, \pi_i) \subset (0, \bar{g}_i)$ is strictly decreasing for all $\pi_i \in (0, k_1)$ Let the profile of consumer i be of the form $(\psi_{ii}, \psi_{ji}) = (\psi_i, \psi_i + b_i)$. By reasoning analogous to that used in Proof C.15, implicit differentiation of the government's first-order condition gives that the map $b_i \mapsto \mathcal{B}_i(\cdot; b_i, \pi_i)$ decreases if:

$$\frac{\mathrm{d}\mathcal{B}_i}{\mathrm{d}b_i} = (-1) \cdot \underbrace{\left[\frac{\partial^2 \mathcal{W}_i^\ell}{\partial g_i^2}\right]^{-1}}_{<0} \cdot \frac{\partial MB^\ell}{\partial b_i} < 0 \quad \Longrightarrow \quad \frac{\partial MB^\ell}{\partial b_i} < 0 \tag{C-183}$$

In other words, the directional response of the best reply is entirely governed by the shift in the marginal benefit, which, under biased privacy preferences $(\psi_{ii}, \psi_{ji}) = (\psi_i, \psi_i + b_i)$, is given by:

$$MB_i^\ell = \psi_i \bar{x}_{ii} N_i^\ell + (\psi_i + b_i)\pi_i \bar{x}_{ji} N_j^\ell - \frac{1}{2}\Big(\psi_i [\bar{x}_{ii}]^2 - (\psi_i + b_i)[\bar{x}_{ji}]^2\Big)\frac{\mathrm{d}N_i^\ell}{\mathrm{d}g_i} \tag{C-184}$$

$$\text{with} \quad \frac{\mathrm{d}N_i^\ell}{\mathrm{d}g_i} = -\frac{\beta(\sigma-1)E_j \tau^{\frac{1-\sigma}{1-\beta}}}{(1-\beta)\sigma w f} \cdot \Delta_i^{-2} \cdot \bar{x}_{ji}^{2\left(\frac{\beta(\sigma-1)}{1-\beta}-1\right)} \cdot r_i^{\frac{\beta(\sigma-1)}{1-\beta}-1} \cdot [1-\pi_i] < 0$$

Straightforward differentiation of (C-184) with respect to b_i gives:

$$F(g_i; g_j, \pi_i) \equiv \frac{\partial MB_i^\ell}{\partial b_i} = \pi_i \bar{x}_{ji}(g_i; \pi_i) N_j^\ell(g_i; g_j, \pi_i) \; + \; \frac{1}{2}\, \bar{x}_{ji}(g_i; \pi_i)^2 \, \frac{\mathrm{d}N_i^\ell}{\mathrm{d}g_i}(g_i; \pi_i) \tag{C-185}$$

Fix a pair (g_i, g_j). Notice that:

$$F(g_i; g_j, 1) = \bar{x}_{ji}\, N_j^\ell > 0 > \frac{1}{2} \cdot \bar{x}_{ji}^2 \, \underbrace{\frac{\mathrm{d}N_i^\ell}{\mathrm{d}g_i}}_{<0} = F(g_i; g_j, 0) \quad \text{for every} \quad (g_i, g_j) \in [0, \bar{g}_i] \times [0, \bar{g}_j]. \tag{C-186}$$

By continuity of F, the intermediate value theorem guarantees at least one root $\pi_i^*(g_i, g_j) \in (0, 1)$ for each (g_i, g_j). Towards the derivation of a uniform threshold, observe that, also using the continuity of F, we have:

$$\{0\}^c = (-\infty, 0) \cup (0, \infty) \quad \text{is open} \quad \Longrightarrow \quad F^{-1}(\{0\}) \subset [0, \bar{g}_i] \times [0, \bar{g}_j] \times [0, 1] \quad \text{is closed.} \tag{C-187}$$

As a closed subset of a compact set, the set $F^{-1}(\{0\})$ is compact. Define the projection map:

$$P : F^{-1}(\{0\}) \mapsto (0, 1) \subset \mathbb{R}, \quad P(\pi_i^*, g_i, g_j) = \pi_i^* \tag{C-188}$$

As the continuous image of a non-empty, compact set, the set $P(F^{-1}(\{0\}))$ is also non-empty and compact in $\mathbb{R}$. By Weierstrass' theorem, it admits a minimum within that set:

$$k_1 \triangleq \min P\big(F^{-1}(\{0\})\big), \quad \text{with} \quad k_1 \in (0, 1) \tag{C-189}$$

Then, by construction, we have $0 < k_1 < 1$ and for every $\pi_i < k_1$, all (g_i, g_j), we have:

$$\frac{\partial MB_i^\ell}{\partial b_i}(g_i; g_j, \pi_i) < 0 \quad \Longrightarrow \quad \frac{\mathrm{d}\mathcal{B}_i}{\mathrm{d}b_i} < 0 \tag{C-190}$$

$\exists k_2 \in (0, 1)$ **such that** $b \mapsto g^*(b, \pi) \subset (0, \bar{g}_i)$ **is strictly decreasing for all** $\pi \in (0, k_2)$ Let $g^*(b, \pi) \in (0, \bar{g})$ denote a symmetric interior Nash equilibrium, when all countries k share the same preference profile $(\psi_{kk}, \psi_{-kk}) = (\psi, \psi + b)$. By reasoning analogous to that employed in Proof C.15, implicit differentiation of the equilibrium condition $\widetilde{MB}(g) - \widetilde{MC}(g) = 0$ gives that the map $b \mapsto g^*(b, \pi)$ decreases if:

$$\frac{\mathrm{d}g^*}{\mathrm{d}b} = (-1) \cdot \underbrace{\left(\frac{\partial}{\partial g}\left\{\widetilde{MB} - \widetilde{MC}\right\}\right)^{-1}}_{<0 \text{ by Proposition 3.5}} \cdot \frac{\partial \widetilde{MB}}{\partial b} \propto \frac{\partial \widetilde{MB}}{\partial b} < 0 \tag{C-191}$$

where, we recall, $\widetilde{MB}$ denote the marginal benefit function evaluated along the diagonal when $g_i = g_j = g$. Adapting expression (C-184) in this direction, we have that for each country k, $-k = \mathcal{I} \setminus k$:

$$\widetilde{MB} = \frac{N}{2}\left[(\psi + b)\pi\, \bar{x}_{-kk} + \psi \bar{x}_{kk}\right] - \frac{1}{2}\left(\psi \bar{x}_{kk}^2 - (\psi + b)\bar{x}_{-kk}^2\right)\frac{\mathrm{d}N_i^\ell}{\mathrm{d}g_k} \tag{C-192}$$

Partially differentiating the above with respect to b and evaluating the resulting expression at the symmetric Nash equilibrium gives:

$$G(\pi) \equiv \frac{\partial \widetilde{MB}}{\partial b}(g^*(b,\pi),\pi) = \frac{\bar{x}_{-kk}(g^*(b,\pi),\pi)}{2} \cdot \left(N\,\pi + \bar{x}_{-kk}(g^*(b,\pi),\pi)\,\frac{\mathrm{d}N_i^\ell}{\mathrm{d}g_k}(g^*(b,\pi),\pi)\right) \tag{C-193}$$

The map $\pi \mapsto g^*(b,\pi)$ inherits the continuity of the best response function, established in Proposition 3.6. Hence, G is continuous in π as the composition of continuous functions. Moreover, notice that, at the bounds of the unit interval, we have:

$$G(1) = \frac{\bar{x}_{-kk}\big(g^*(b,1),1\big) \cdot N}{2} > 0 > \frac{\bar{x}_{-kk}(g^*(b,0),0)}{2} \cdot \frac{\mathrm{d}N_i^\ell}{\mathrm{d}g_k}\big(g^*(b,0),0\big) = G(0) \tag{C-194}$$

By the intermediate value theorem, the set:

$$G^{-1}(\{0\}) := \left\{\pi \in (0,1) \;:\; \frac{\partial \widetilde{MB}}{\partial b}\big(g^*(b,\pi),\pi\big) = 0\right\} \quad \text{is not empty.} \tag{C-195}$$

Moreover, $G^{-1}(\{0\})$ is closed in the compact set $[0,1]$, as the continuous pre-image of a closed set. Hence:

$$\exists k_2 \triangleq \min G^{-1}(\{0\}), \quad \text{with} \quad k_2 \in (0,1) \tag{C-196}$$

By construction, $0 < k_2 < 1$ and for every $\pi < k_2$ we have:

$$\frac{\mathrm{d}}{\mathrm{d}b}\left\{g^*(b,\pi)\right\} < 0 \quad \text{for all } \pi \in (0,k_2). \tag{C-197}$$

Hence $b \mapsto g^*(b,\pi)$ is strictly decreasing whenever $\pi \in (0,k_2)$, as claimed.

Since neither k_1 nor k_2 is explicitly characterised, set $\pi^* \triangleq \min\{k_1,k_2\}$ for expositional clarity, so that for all $\pi_i, \pi \in (0,\pi^*)$, both claims hold simultaneously, as in the statement of Proposition 3.12.

□

C.18 Proof of Proposition 3.13 (Asymmetric Biased Preference Shock and the Race to the Bottom)

The proof mirrors that of the more detailed Proof C.16, with minor adjustments due to the best-response function shifting downward rather than upward in response to the perturbation. We refer to it where appropriate.

C.18.1 Existence of a Nash equilibrium with asymmetric, biased privacy preferences

Along similar lines as in Proof C.16, define the best-response operator $T(g_i, g_j) = \big(\mathcal{B}_i(g_j; b, \pi), \mathcal{B}_j(g_i; 0, \pi)\big)$. Isotonicity of T over the complete lattice $[0, \bar{g}]^2$ then guarantees a fixed point by Tarski's theorem.

C.18.2 Any post-shock equilibrium involves $g_i' < g_j' \leq g^*$ for small enough shocks

Let (g_i', g_j') be any Nash equilibrium of the post-shock game and define:

$$T_{\text{pre}}(g_i, g_j) \triangleq \big(\mathcal{B}_i(g_j; 0, \pi), \mathcal{B}_j(g_i; 0, \pi)\big), \quad \text{and} \quad T_{\text{post}}(g_i, g_j) \triangleq \big(\mathcal{B}_i(g_j; b, \pi), \mathcal{B}_j(g_i; 0, \pi)\big) \tag{C-198}$$

The argument again parallels that of Proof 3.11. In addition to the isotonicity of both operators, we observe that $T_{\text{post}}(g_i, g_j) \leq T_{\text{pre}}(g_i, g_j)$ component-wise for all (g_i, g_j), since the first component of T_{post} is evaluated at $b > 0$ and the map $b' \mapsto \mathcal{B}_i(\cdot; b', \pi)$ is decreasing when $\pi \in (0, \pi^*)$, as shown in Proposition 3.12. Moreover, T_{pre} admits a unique fixed point by Proposition 3.5. Setting $F = T_{\text{post}}$ and $G = T_{\text{pre}}$, all the conditions of Lemma 2 are satisfied, so every fixed point of T_{post} lies below the unique fixed point of T_{pre} in the product order.

$$(g_i', g_j') \leq (g^*, g^*) \iff g_i', g_j' \leq g^* \tag{C-199}$$

Next, observe that the combination of strategic complementarities, $g_j' \leq g^*$, together with $\mathcal{B}_i(g^*; b, \pi) < \mathcal{B}_i(g^*; 0, \pi)$ since $g^* \in (0, \bar{g})$ by Proposition 3.12 gives:

$$g_i' = \mathcal{B}_i(g_j'; b, \pi) \leq \mathcal{B}_i(g^*; b, \pi) < \mathcal{B}_i(g^*; 0, \pi) = g^* \implies g_j' = \mathcal{B}_j(g_i'; 0, \pi) < \mathcal{B}_j(g^*; 0, \pi) = g^* \tag{C-200}$$

Hence, we have shown that $g_i', g_j' < g^*$, and it only remains to establish that $g_i' < g_j'$ for sufficiently small perturbations. By a uniform continuity argument analogous to that in Proof C.16, every post-shock Nash equilibrium (g_i', g_j') remains interior for small b, and the best-response map is locally strictly decreasing in b by Proposition 3.12. Supposing for contradiction purposes that $g_i' \geq g_j'$, we obtain:

$$g_i' = \mathcal{B}_i(g_j'; b, \pi) < \mathcal{B}_j(g_j'; 0, \pi) \leq \mathcal{B}_j(g_i'; 0, \pi) = g_j', \quad \text{which forces} \quad g_j' > g_i'. \tag{C-201}$$

C.18.3 The new spatial distribution for Tech activity is such that $N_i^\ell > \frac{N}{2} > N_j^\ell$

The proof of this last claim is directly analogous to the one in Proof C.16, so we omit it. Writing N_i^ℓ as a function of (g_i, g_j) and using that $g_i' - g^* < g_j' - g^* < 0$, it suffices to integrate the changes to conclude that:

$$N_i^\ell(g_i', g_j') > N_i^\ell(g^*, g^*) = \frac{N}{2} = N_j^\ell(g^*, g^*) > N_j^\ell(g_i', g_j'). \tag{C-202}$$

Thus, country i's Tech sector expands while country j hosts fewer firms than in the symmetric equilibrium.

□

C.19 Proof of Proposition 3.14 (Multiple Rationales for Maximal Cross-Border Restrictions)

One simply checks that for $\mathcal{O}_i \in \left\{PS_i^s,\ \hat{X}_i^h,\ D_{ji}^h,\ N_i^\ell,\ CS_i^\ell,\ \mathcal{W}_i^\ell\right\}$, the objective is strictly increasing in the instrument g_i. Continuity over the compact set $[0, 1]$ then implies that the solution to Problem (63) is $g_i^* = 1$.

C.19.1 Maximisation of short-run producer surplus or rents: $\mathbb{O}_i = PS_i^s$

Recall from (B-28) that the profit function of a typical firm ω operating from country i is given by:

$$\Pi(\omega) = \frac{w}{\sigma - 1} \cdot \sum_{k \in \mathcal{C}} \left(\frac{c_k(\omega)\tau_{ik}}{\tilde{\varphi}_k(\omega)} \right) - wf \tag{C-203}$$

In equilibrium, all Tech firms operating from the same country behave symmetrically. The aggregate rent earned by country i's Tech sector can thus be written as:

$$PS_i^s \triangleq \int_{\Omega_i^s} \Pi(\omega)\, \mathrm{d}\omega = N_i^s \left[\left(\frac{c_{ii}^s(P_i^s(g_i))}{\tilde{\varphi}_{ii}} \right) + \left(\frac{c_{ij}^s \tau}{\tilde{\varphi}_{ij}} \right) - wf \right] \tag{C-204}$$

which depends on g_i only via c_{ii}^s. Combining the consumption schedule derived in equation (B-10) with the short-run price adjustment characterised in Proposition 2.2 then yields:

$$c_{ii}^s(g_i) = E_i \cdot [P_i^s(g_i)]^{\sigma-1} \cdot p_{ii}^{-\sigma} \quad \Longrightarrow \quad \frac{\mathrm{d}\mathbb{O}_i}{\mathrm{d}g_i} = \frac{\mathrm{d}PS_i^s}{\mathrm{d}g_i} \propto \underbrace{\frac{\mathrm{d}c_{ii}^s}{\mathrm{d}P_i^s}}_{>0} \cdot \underbrace{\frac{\mathrm{d}P_i^s}{\mathrm{d}\bar{x}_{ji}}}_{<0} \cdot \underbrace{\frac{\mathrm{d}\bar{x}_{ji}}{\mathrm{d}g_i}}_{<0} > 0 \tag{C-205}$$

C.19.2 Minimisation of foreign-held resident data in the short- or long-run: $\mathbb{O}_i = -D_{ji}^h$

Using the definition of the stock of foreign-held resident data D_{ji}^h, together with the consumption schedule (B-10) and the expression for the data input $d_{ji}^h(\omega) = \bar{x}_{ji}(\omega)c_{ji}^h(\omega)$, we obtain that, in equilibrium:

$$D_{ji}^h \triangleq \int_{\Omega_j^h} d_i(\omega)\, \mathrm{d}\omega = N_j^h \bar{x}_{ji} c_{ji}^h = N_j^h \bar{x}_{ji} p_{ji}^{-\sigma} (P_i^h)^{\sigma-1} E_i$$

<u>For $h = s$</u> Consider first the short run case, where firm masses are fixed and thus independent of g_i. More precisely, writing all dependencies explicitly gives:

$$D_{ji}^s(g_i) = N_j^s \bar{x}_{ji} [p_{ji}(g_i)]^{-\sigma} (P_i^s)^{\sigma-1} E_i \tag{C-206}$$

We claim that:

$$\begin{aligned} \frac{\mathrm{d}\mathbb{O}_i}{\mathrm{d}g_i} = \frac{\mathrm{d}(-D_{ji}^s)}{\mathrm{d}g_i} > 0 \quad &\iff \quad -\frac{\mathrm{d}\ln D_{ji}^s}{\mathrm{d}g_i} > 0 \\ &\iff \quad \frac{\mathrm{d}\ln D_{ji}^s}{\mathrm{d}\bar{x}_{ji}} = \underbrace{\frac{\mathrm{d}\ln \bar{x}_{ji}}{\mathrm{d}\bar{x}_{ji}}}_{>0} + \sigma \underbrace{\left(-\frac{\mathrm{d}\ln p_{ji}}{\mathrm{d}\bar{x}_{ji}} \right)}_{>0} + (\sigma - 1) \underbrace{\frac{\mathrm{d}\ln P_i^s}{\mathrm{d}\bar{x}_{ji}}}_{<0} > 0 \\ &\qquad \text{given} \quad \mathrm{d}\bar{x}_{ji} = -\pi_i \mathrm{d}g_i. \end{aligned} \tag{C-207}$$

Using the pricing rule (B-27), we obtain:

$$p_{ji} = \frac{\sigma}{\sigma - 1} \cdot \frac{w\, \tau^{\frac{1}{1-\beta}}}{(\varphi\, \bar{x}_{ji}^{\beta})^{\frac{1}{1-\beta}}} \quad \Longrightarrow \quad \frac{\mathrm{d}\ln p_{ji}}{\mathrm{d}\bar{x}_{ji}} = -\frac{\beta}{\bar{x}_{ji}(1 - \beta)} \tag{C-208}$$

Moreover, the short-run comparative statics established in Proposition 2.2 imply that:

$$0 > \frac{\mathrm{d}\ln P_i^s}{\mathrm{d}\bar{x}_{ji}} = -\frac{\beta}{1-\beta} \cdot \frac{N_j^s\, \tau^{\frac{1-\sigma}{1-\beta}}\, \bar{x}_{ji}^{\frac{\beta(\sigma-1)}{1-\beta}-1}}{\sum_{k\in\mathcal{C}} N_k^s\, \tau_{ki}^{\frac{1-\sigma}{1-\beta}}\, \bar{x}_{ki}^{\frac{\beta(\sigma-1)}{1-\beta}}} = -\frac{\beta}{\bar{x}_{ji}(1-\beta)} \cdot \frac{N_j^s\, \tau^{\frac{1-\sigma}{1-\beta}}\, \bar{x}_{ji}^{\frac{\beta(\sigma-1)}{1-\beta}}}{\sum_{k\in\mathcal{C}} N_k^s\, \tau_{ki}^{\frac{1-\sigma}{1-\beta}}\, \bar{x}_{ki}^{\frac{\beta(\sigma-1)}{1-\beta}}}$$
$$> -\frac{\beta}{\bar{x}_{ji}(1-\beta)} \qquad \text{(C-209)}$$

Characterisations (C-208) and (C-209), together with (C-207), then gives:

$$\frac{\mathrm{d}\ln D_{ji}^s}{\mathrm{d}\bar{x}_{ji}} > \sigma\left(\underbrace{\left(-\frac{\mathrm{d}\ln p_{ji}}{\mathrm{d}\bar{x}_{ji}}\right) + \frac{\mathrm{d}\ln P_i^s}{\mathrm{d}\bar{x}_{ji}}}_{>0}\right) - \frac{\mathrm{d}\ln P_i^s}{\mathrm{d}\bar{x}_{ji}} > -\frac{\mathrm{d}\ln P_i^s}{\mathrm{d}\bar{x}_{ji}} > 0 \quad \Longrightarrow \quad \frac{\mathrm{d}\mho_i}{\mathrm{d}g_i} > 0 \quad \text{by (C-207).} \qquad \text{(C-210)}$$

For $h = \ell$ Turning to the long run case, we may write:

$$D_{ji}^\ell = N_j^\ell(g_i) \cdot \bar{x}_{ji}(g_i) \cdot [p_{ji}(g_i)]^{-\sigma} \cdot [P_i^\ell(g_i)]^{\sigma-1} \cdot E_i \qquad \text{(C-211)}$$

Using that $\mathrm{d}\bar{x}_{ji} = -\pi_i \mathrm{d}g_i$, it is immediate that:

$$-\frac{\mathrm{d}\ln D_{ji}^\ell}{\mathrm{d}g_i} \propto \frac{\mathrm{d}\ln D_{ji}^\ell}{\mathrm{d}\bar{x}_{ji}} = \underbrace{\frac{\mathrm{d}\ln N_j^\ell}{\mathrm{d}\bar{x}_{ji}}}_{>0} + \underbrace{\frac{\mathrm{d}\ln \bar{x}_{ji}}{\mathrm{d}\bar{x}_{ji}}}_{>0} + \sigma\underbrace{\left(-\frac{\mathrm{d}\ln p_{ji}}{\mathrm{d}\bar{x}_{ji}}\right)}_{>0} + (\sigma-1)\underbrace{\frac{\mathrm{d}\ln P_i^\ell}{\mathrm{d}\bar{x}_{ji}}}_{>0} > 0 \qquad \text{(C-212)}$$

Hence, we conclude that:

$$\frac{\mathrm{d}\mho_i}{\mathrm{d}g_i} = \frac{\mathrm{d}(-D_{ji}^\ell)}{\mathrm{d}g_i} > 0, \quad \text{as claimed.} \qquad \text{(C-213)}$$

C.19.3 Maximisation of net Tech exports in the short- or long-run: $\mho_i = \hat{X}_i^h$

In equilibrium, the net exports of Tech from country i simplify to:

$$\hat{X}_i^h \triangleq \int_{\Omega_i^h} p_j(\omega)c_j(\omega)\,\mathrm{d}\omega - \int_{\Omega_j^h} p_i(\omega)c_i(\omega)\,\mathrm{d}\omega = N_i^h p_{ij} c_{ij}^h - N_j^h p_{ji} c_{ji}^h \qquad \text{(C-214)}$$

For $h = s$ In the short run, firm masses are held fixed. Moreover, since g_i only affects the data usage of firms from country j when serving consumers in country i, i.e., $\bar{x}_{ji}$, it follows that market conditions in destination j remain unchanged since data usage is siloed across markets. Writing all dependencies explicitly then gives:

$$\hat{X}_i^s = N_i^s p_{ij} c_{ij}^s - N_j^s \cdot p_{ji}(g_i) \cdot c_{ji}^s\big(p_{ji}(g_i), P_i^s(g_i)\big) \qquad \text{(C-215)}$$

Hence:

$$\frac{\mathrm{d}\mho_i}{\mathrm{d}g_i} = \frac{\mathrm{d}\hat{X}_i^s}{\mathrm{d}g_i} \propto -\left(\frac{\mathrm{d}\ln p_{ji}}{\mathrm{d}g_i} + \frac{\mathrm{d}\ln c_{ji}^s}{\mathrm{d}g_i}\right) \propto \left(\frac{\mathrm{d}\ln p_{ji}}{\mathrm{d}\bar{x}_{ji}} + \frac{\mathrm{d}\ln c_{ji}^s}{\mathrm{d}\bar{x}_{ji}}\right)$$
$$\propto \left(\frac{\mathrm{d}\ln p_{ji}}{\mathrm{d}\bar{x}_{ji}} + \sigma\left(-\frac{\mathrm{d}\ln p_{ji}}{\mathrm{d}\bar{x}_{ji}}\right) + (\sigma-1)\frac{\mathrm{d}\ln P_i^s}{\mathrm{d}\bar{x}_{ji}}\right)$$

$$\propto (\sigma - 1)\underbrace{\left[\left(-\frac{\mathrm{d}\ln p_{ji}}{\mathrm{d}\bar{x}_{ji}}\right) + \frac{\mathrm{d}\ln P_i^s}{\mathrm{d}\bar{x}_{ji}}\right]}_{>0 \text{ by (C-208) and (C-209)}} \tag{C-216}$$

For $h = \ell$ In the long run, market conditions in country j are indirectly influenced by the endogenous reallocation of Tech firms across locations. Specifically, we may write:

$$\hat{X}_i^\ell = N_i^\ell(g_i) \cdot p_{ij} \cdot c_{ij}^\ell(P_j^\ell(g_i)) - N_j^\ell(g_i) \cdot p_{ji}(g_i) \cdot c_{ji}^\ell\big(p_{ji}(g_i), P_i^\ell(g_i)\big) \tag{C-217}$$

Using a forthcoming characterisation (C-220), together with the fact that changes in firm masses induced by global data governance must sum to zero by Proposition 2.5, we have:

$$\frac{\mathrm{d}N_i^\ell}{\mathrm{d}g_i} > 0 \quad \Longrightarrow \quad \frac{\mathrm{d}N_j^\ell}{\mathrm{d}g_i} < 0 \tag{C-218}$$

Hence, the total effect of a change in g_i on country i's net exports of Tech, $\hat{X}_i^\ell$, is necessarily bounded below by the corresponding partial effect under fixed firm masses. Applying the comparative statics from Proposition 2.3, and proceeding along similar algebraic steps as in the short-run case (C-216), we obtain:

$$\begin{aligned}
\frac{\mathrm{d}\circledcirc_i}{\mathrm{d}g_i} = \frac{\mathrm{d}\hat{X}_i^\ell}{\mathrm{d}g_i} &> N_i^\ell p_{ij} \cdot \underbrace{\frac{\mathrm{d}c_{ij}^\ell}{\mathrm{d}P_j^\ell}}_{>0} \cdot \underbrace{\frac{\mathrm{d}P_j^\ell}{\mathrm{d}\bar{x}_{ji}}}_{<0} \cdot \underbrace{\frac{\mathrm{d}\bar{x}_{ji}}{\mathrm{d}g_i}}_{<0} - N_j^\ell \frac{\mathrm{d}}{\mathrm{d}g_i}\left\{p_{ji}(g_i) \cdot c_{ji}^\ell\big(p_{ji}(g_i), P_i^\ell(g_i)\big)\right\} \\
&> -N_j^\ell \frac{\mathrm{d}}{\mathrm{d}g_i}\left\{p_{ji}(g_i) \cdot c_{ji}^\ell\big(p_{ji}(g_i), P_i^\ell(g_i)\big)\right\} \\
&\propto -\left(\frac{\mathrm{d}\ln p_{ji}}{\mathrm{d}g_i} - \frac{\mathrm{d}\ln c_{ji}^\ell}{\mathrm{d}g_i}\right) \\
&\propto (\sigma - 1)\left[\underbrace{\left(-\frac{\mathrm{d}\ln p_{ji}}{\mathrm{d}\bar{x}_{ji}}\right)}_{>0} + \underbrace{\frac{\mathrm{d}\ln P_i^\ell}{\mathrm{d}\bar{x}_{ji}}}_{>0}\right] \\
&> 0, \quad \text{as claimed.}
\end{aligned} \tag{C-219}$$

C.19.4 Maximisation of the long-run number of locally established Tech firms: $\circledcirc_i = N_i^\ell$

Since $\mathrm{d}\bar{x}_{ji} = -\pi_i \, \mathrm{d}g_i$, we immediately deduce from Proposition 2.4 that:

$$\frac{\mathrm{d}\circledcirc_i}{\mathrm{d}g_i} = \frac{\mathrm{d}N_i^\ell}{\mathrm{d}g_i} = -\frac{1}{\pi_i} \cdot \underbrace{\frac{\mathrm{d}N_i^\ell}{\mathrm{d}\bar{x}_{ji}}}_{<0} > 0, \quad \text{as claimed.} \tag{C-220}$$

C.19.5 Maximisation of long-run consumer surplus: $\circledcirc_i = CS_i^\ell$

Recalling the definition of the consumer surplus:

$$CS_i^\ell \triangleq \left(\frac{w}{P_i^\ell}\right)^\alpha L_i \tag{C-221}$$

Using the characterisation of long-run price index adjustments in Proposition 2.3 then gives:

$$\frac{\mathrm{d}\mathbb{O}_i}{\mathrm{d}g_i} = \frac{\mathrm{d}CS_i^\ell}{\mathrm{d}g_i} = \alpha \left(\frac{w}{P_i^\ell}\right)^\alpha \frac{\mathrm{d}\ln P_i^\ell}{\mathrm{d}g_i} \propto \alpha \left(\frac{w}{P_i^\ell}\right)^\alpha \left(\underbrace{-\frac{\mathrm{d}\ln P_i^\ell}{\mathrm{d}\bar{x}_{ji}}}_{>0}\right) > 0 \tag{C-222}$$

C.19.6 Maximisation of long-run national welfare, under Assumption 10: $\mathbb{O}_i = \mathbb{W}_i^\ell$

Adapting the consumer value function (B-12) to the present setting, and using that the global Tech firm mass is constant, so that $N_i^\ell(g_i) = N - N_j^\ell(g_i)$ by Proposition 2.5, we can express long-run welfare as:

$$\mathbb{W}_i^\ell(g_i) = CS_i^\ell(g_i) - \frac{1}{2}\Big[\psi_{ii}\,\bar{x}_{ii}^2\,N + (\psi_{ji}\,\bar{x}_{ji}(g_i)^2 - \psi_{ii}\,\bar{x}_{ii}^2)\,N_j^\ell(g_i)\Big] \tag{C-223}$$

Treating all variables as functions of g_i and differentiating $\mathbb{W}_i^h$, we obtain that:

$$\begin{aligned}\frac{\mathrm{d}\mathbb{O}_i}{\mathrm{d}g_i} = \frac{\mathrm{d}\mathbb{W}_i^h}{\mathrm{d}g_i} &= \frac{\mathrm{d}CS_i^\ell}{\mathrm{d}g_i} - \psi_{ji}\,\bar{x}_{ji}\,\frac{\mathrm{d}\bar{x}_{ji}}{\mathrm{d}g_i}\,N_j^\ell - \frac{1}{2}\,(\psi_{ji}\,\bar{x}_{ji}^2 - \psi_{ii}\,\bar{x}_{ii}^2)\,\frac{\mathrm{d}N_j^\ell}{\mathrm{d}g_i} \\ &= \underbrace{\frac{\mathrm{d}CS_i^\ell}{\mathrm{d}g_i}}_{>0 \text{ by (C-222)}} + \pi_i\psi_{ji}\,\bar{x}_{ji}\,\frac{\mathrm{d}\bar{x}_{ji}}{\mathrm{d}g_i}\,N_j^\ell + \frac{1}{2}\,\underbrace{(\psi_{ji}\,\bar{x}_{ji}^2 - \psi_{ii}\,\bar{x}_{ii}^2)}_{>0 \text{ under Assumption 10}} \cdot \underbrace{\left(-\frac{\mathrm{d}N_j^\ell}{\mathrm{d}g_i}\right)}_{>0 \text{ by (C-220)}} > 0\end{aligned} \tag{C-224}$$

□

C.20 Proof of Proposition 3.15 (Short-Run Welfare Constraints Curb Discrimination)

Fix a global data governance $\mathbf{X}$ for the world economy, and let $g_i^0 \in [0,1)$ denote the underlying status quo regulation in the associated equilibrium. Given some strictly increasing objective $\mathbb{O}_i$, the government chooses a restriction level $g_i \in [0,1]$ to solve the augmented maximisation problem:

$$\max_{g_i \in [0,1]} \quad \mathbb{O}_i\,(g_i) \tag{C-225}$$

$$\text{s.t.} \quad \mathbb{W}_i^s(g_i) \geq \mathbb{W}_i^s(g_i^0) \tag{C-226}$$

$\mathbb{W}_i^s$, the short-run welfare function with firm masses held fixed, can be written as a function of the level of cross-border data restrictions as:

$$\mathbb{W}_i^s(g_i;\psi_{ji}) = \left(\frac{w}{P_i^s(g_i)}\right)^\alpha L_i - \frac{1}{2}\Big[\psi_{ii}\,\bar{x}_{ii}^2\,N + (\psi_{ji}\,\bar{x}_{ji}(g_i)^2 - \psi_{ii}\,\bar{x}_{ii}^2)\,N_j^s\Big] \tag{C-227}$$

Letting g_i^* denote an optimal solution to Problem (C-225), we claim that:

$$\exists \psi_{ji}^* > 0 \quad \text{s.t.} \quad \psi_{ji} < \psi_{ji}^* \quad \Longrightarrow \quad \exists! g_i^* \in [0,1) \tag{C-228}$$

In words, the constraint (C-226) binds for sufficiently weak privacy concerns regarding foreign data processing, and the resulting maximiser $g_i^* \in [0,1)$ is unique.

Existence of ψ_{ji}^* Observe that differentiating (C-227) gives:

$$\frac{\mathrm{d}\mathscr{W}_i^s}{\mathrm{d}g_i} = \alpha w^\alpha (P_i^s)^{-\alpha} L_i \underbrace{\left(-\frac{\mathrm{d}\ln P_i^s}{\mathrm{d}\bar{x}_{ji}}\frac{\mathrm{d}\bar{x}_{ji}}{\mathrm{d}g_i}\right)}_{<0 \text{ by Proposition 2.2}} + \pi_i \psi_{ji} N_j^s \bar{x}_{ji} \tag{C-229}$$

Notice that:

$$\lim_{\psi_{ji}\to 0^+} \frac{\mathrm{d}\mathscr{W}_i^s}{\mathrm{d}g_i} < 0,\ \forall g_i \in [0,1] \quad \Longrightarrow \quad \mathscr{W}_i^s(g_i^0;0) - \mathscr{W}_i^s(1;0) > 0 \tag{C-230}$$

Hence, by continuity, there exists ψ_{ji}^* such that inequality (C-230) is preserved for all $\psi_{ji} \in [0, \psi_{ji}^*)$:

$$\psi_{ji} < \psi_{ji}^* \quad \Longrightarrow \quad \mathscr{W}_i^s(g_i^0;\psi_{ji}) > \mathscr{W}_i^s(1;\psi_{ji}) \quad \Longrightarrow \quad g_i = 1 \quad \text{is infeasible in Problem (C-225).} \tag{C-231}$$

Existence and uniqueness of $g_i^* \in [0,1)$ Fix any $\psi_{ji} < \psi_{ji}^*$ and define:

$$W_0 \triangleq \mathscr{W}_i^s(g_i^0;\psi_{ji}) \quad \text{and} \quad \bar{W} \triangleq \max_{g_i' \in [0,1]} \mathscr{W}_i^s(g_i';\psi_{ji}) \tag{C-232}$$

Because $\mathscr{W}_i^s(\cdot;\psi_{ji})$ is continuous on the compact interval $[0,1]$, such $\bar{W}$ necessarily exists by Weierstrass' theorem and satisfies $\bar{W} < \infty$. Moreover, the interval $[W_0, \bar{W}]$ is a closed subset of $\mathbb{R}$. By continuity of $\mathscr{W}_i^s$, its pre-image is a closed subset of $[0,1]$, which is itself compact. Hence:

$$C := \left\{ g_i \in [0,1] \ : \ \mathscr{W}_i^s(g_i;\psi_{ji}) \in [W_0, \bar{W}] \right\} \quad \text{is compact.} \tag{C-233}$$

By assumption, $\mathscr{O}_i$ is continuous on $C \subset [0,1]$. Moreover, since $g_i^0 \in C$ by construction, the set C is nonempty. It follows from Weierstrass' theorem that $\mathscr{O}_i$ attains a maximum on C, which lies within the set. Furthermore, since $\mathscr{O}_i$ is strictly increasing, the maximum is uniquely achieved at the largest element of C:

$$g_i^* = \max C, \quad \text{with } g_i^* < 1 \text{ by (C-231).} \tag{C-234}$$

□

C.21 Proof of Proposition 3.16 (Existence of an Asymmetric Markov Perfect Equilibrium)

The dynamic policy problem faced by government $i \in \{H, F\}$ can be compactly written as:

$$\begin{aligned}
\max_{\{g_{i,t}\}_{t\geq 1}} \quad & \sum_{t=1}^{\infty} \delta^{t-1}\, \mathscr{O}_i\big(g_{i,t}, N_{i,t}^s; g_{j,t}\big), \quad \delta \in (0,1) \\
& g_{i,t} \in [0,1] \quad \text{for all } t, \\
& N_{i,t+1}^s = (1-\rho)\, N_{i,t}^s + \rho\, N_i^\ell(g_{i,t}; g_{j,t}), \quad N_1^s(g_0) \text{ given}, \\
\text{and} \quad & \mathscr{W}_i^s(g_{i,t}, N_{i,t}^s) \geq (1-\gamma_t)\cdot \mathscr{W}_i^s(g_{i,t-1}, N_{i,t}^s), \quad \text{if } i = H.
\end{aligned} \tag{C-235}$$

Well-posedness Because the payoff function $z \mapsto \mathcal{O}_i(z\,; g_{j,t})$ is continuous over the compact action space $[0,1]$, it is bounded. In particular, there exists a real number $M < \infty$, such that:

$$\mathcal{O}_i(g_{i,t}\,; g_{j,t}) \leq M,\ \forall t \quad \Longrightarrow \quad \lim_{T\to\infty} \sum_{t=1}^{T} \delta^{t-1}\, \mathcal{O}_i(g_{i,t}, N^s_{i,t}; g_{j,t}) \leq \sum_{t=1}^{\infty} \delta^{t-1} M = \frac{M}{1-\delta} < \infty$$
$$\Longrightarrow \quad \sum_{t=1}^{\infty} \delta^{t-1}\, \mathcal{O}_i(g_{i,t}, N^s_{i,t}; g_{j,t}) \quad \text{converges absolutely.} \tag{C-236}$$

Equilibrium structure Since $N^s_{F,t} = N - N^s_{H,t}$ for some constant N by Proposition 2.5, the cross-country distribution of Tech activity is fully summarised by $N^s_{H,t}$. The Home government also inherits its previous regulation level $g_{H,t-1}$, which determines the status quo welfare level $\mathcal{W}^s_H(g_{H,t-1}, N^s_{H,t})$. In each period, it faces a time-varying political constraint, represented by a sequence $\{\gamma_t\}_{t\geq 1}$, that limits the extent to which policy can reduce welfare relative to the status quo. Specifically, γ_t denotes the maximum allowable percentage decrease in welfare compared to $\mathcal{W}^s_H(g_{H,t-1}, N^s_{H,t})$.

Hence, we may define the payoff-relevant state space $\mathcal{S}$ as:

$$\mathcal{S} := \left\{ (N^s_H, g_{H,-1}, \gamma) \in \mathbb{R}^3 \;\middle|\; N^s_H \in [N_{\min}, N_{\max}] \subset [0, N],\ g_{H,-1} \in [0,1],\ \gamma \in [0,1) \right\}. \tag{C-237}$$

In this context, a Markov strategy for country $i \in \{H, F\}$ is a rule that assigns a feasible regulation level to each possible state $\mathcal{s} \in \mathcal{S}$, depending only on the payoff-relevant variables at that point in time:

$$s_i : \mathcal{S} \to [0,1]. \tag{C-238}$$

Proof outline We establish the existence of a subgame-perfect Markov equilibrium in which:

(*i*) Foreign plays the trivial strategy:

$$s_F(\mathcal{s}) = 1, \quad \forall \mathcal{s} \in \mathcal{S} \tag{C-239}$$

(*ii*) Home follows the constrained Markov strategy:

$$s_H(\mathcal{s}) = \arg\max_{g_H \in [0,1]} \left\{ g_H \mid \mathcal{W}^s_H(g_H, N^s_H) \geq (1-\gamma)\, \mathcal{W}^s_H(g_{H,-1}, N^s_H) \right\}, \quad \forall \mathcal{s} \in \mathcal{S}, \tag{C-240}$$

For any sequence $\{\gamma_t\}_{t\geq 1}$ satisfying $\sum_{t=1}^{\infty} \gamma_t < S(g_0)$ for some finite bound $S(g_0)$ to be characterised, this rule generates an increasing sequence of policies $\{g_{H,t}\}$ satisfying:

$$g_{H,t} \leq g_{H,t+1} < 1, \quad g_{H,\infty} = g_{\max} < 1. \tag{C-241}$$

(*iii*) In the steady state, the policy gap $g_{H,\infty} < 1 = g_{F,\infty}$ leads to an asymmetric distribution of firm masses, with Foreign hosting a larger Tech industry:

$$N^{\ell}_F(\mathbf{X}_\infty) > \frac{N}{2} > N^{\ell}_H(\mathbf{X}_\infty). \tag{C-242}$$

C.21.1 Preliminaries

We first establish two technical lemmas required for the equilibrium construction. Lemma 3 verifies that the short-run welfare function satisfies the regularity conditions needed to characterise the optimality of Home's strategy. It also proves the existence of the threshold ψ^*_{FH} referenced in Assumption 12, ensuring that our results apply to a non-empty subset of the parameter space. Lemma 4 then shows that Home's strategy is well-defined and monotone in firm mass, which underpins the government's incentives to gradually tighten restrictions.

Lemma 3 (Properties of the Short-Run Welfare Function). *The short-run welfare function $\mathscr{W}^s_i(g_i, N^s_i)$ satisfies the following properties for all $(g_i, N^s_i) \in [0,1] \times [N_{\min}, N_{\max}]$, $N_{\min} > 0$ by Assumption 2 and $N_{\max} = N - N_{\min}$:*

(i) Under Assumption 5, $\mathscr{W}^s_i(g_i, N^s_i)$ is strictly concave in g_i:

$$\frac{\partial^2 \mathscr{W}^s_i}{\partial g_i^2}(g_i, N^s_i) < 0; \tag{C-243}$$

(ii) Under Assumption 5, there exists a threshold $\psi^*_{ji} > 0$ such that, for all $\psi_{ji} \in (0, \psi^*_{ji})$, the short-run welfare function $\mathscr{W}^s_i(g_i, N^s_i)$ is strictly supermodular (submodular) in (g_i, N^s_i) (in (g_i, N^s_j)) and attains a unique global minimum at $g_i = 1$:

$$\frac{\partial^2 \mathscr{W}^s_i}{\partial g_i\, \partial N^s_i}(g_i, N^s_i; \psi_{ji}) > 0 \quad \text{and} \qquad g_H < 1 \quad \Longrightarrow \quad \mathscr{W}^s_H(g_H, N^s_H) > \mathscr{W}^s_H(1, N^s_H) \tag{C-244}$$

(iii) Under Assumption 10, $\mathscr{W}^s_i(g_i, N^s_i)$ is strictly increasing in N^s_i:

$$\frac{\partial \mathscr{W}^s_i}{\partial N^s_i}(g_i, N^s_i) > 0. \tag{C-245}$$

Proof. Writing $\mathscr{W}^s_i$ as a function of (g_i, N^s_i), equation (C-229) then gives:

$$\frac{\partial \mathscr{W}^s_i}{\partial g_i}(g_i, N^s_i) = (-1) \cdot \alpha w^\alpha \big(P^s_i(g_i, N^s_i)\big)^{-\alpha} L_i \cdot \left[\frac{\mathrm{d} \ln P^s_i}{\mathrm{d} g_i}(g_i, N^s_i)\right] + \pi_i \psi_{ji} \cdot N^s_j(N^s_i) \cdot \bar{x}_{ji}(g_i) \tag{C-246}$$

<u>(i) $\mathscr{W}^s_i$ **is strictly concave in** g_i</u> Fix any $N^s_i > 0$. We claim that the mapping $g_i \mapsto \frac{\partial \mathscr{W}^s_i}{\partial g_i}(g_i, N^s_i)$ (C-246) is strictly decreasing. Since the term $\pi_i \psi_{ji} N^s_j \bar{x}_{ji}(g_i)$ is strictly increasing in $\bar{x}_{ji} = 1 - \pi_i g_i$, and therefore strictly decreasing in g_i for all $\pi_i \in (0,1)$, it suffices to show that the first term in (C-246) is also decreasing in g_i. Using the comparative statics results of Proposition 2.2, and writing every dependence on g_i explicitly, we obtain:

$$\frac{\mathrm{d} \ln P^s_i}{\mathrm{d} g_i}(g_i) = \frac{\mathrm{d} \ln P^s_i}{\mathrm{d} \bar{x}_{ji}} \cdot \frac{\mathrm{d} \bar{x}_{ji}}{\mathrm{d} g_i} = \frac{\pi_i \beta}{1-\beta} \cdot \frac{N^s_j \tau^{\frac{1-\sigma}{1-\beta}} [\bar{x}_{ji}(g_i)]^{\frac{\beta(\sigma-1)}{1-\beta}-1}}{N^s_i \bar{x}_{ii}^{\frac{\beta(\sigma-1)}{1-\beta}} + N^s_j \tau^{\frac{1-\sigma}{1-\beta}} [\bar{x}_{ji}(g_i)]^{\frac{\beta(\sigma-1)}{1-\beta}}} > 0 \tag{C-247}$$

Moreover, recall that, for given firm masses, the short run price index (C-56) is given by:

$$[P^s_i]^{-\alpha}(g_i) = \left[\frac{p_0}{\varphi^{\frac{\beta}{1-\beta}}}\right]^{-\alpha} \cdot \left[N^s_i \bar{x}_{ii}^{\frac{\beta(\sigma-1)}{1-\beta}} + N^s_j \tau^{\frac{1-\sigma}{1-\beta}} \cdot [\bar{x}_{ji}(g_i)]^{\frac{\beta(\sigma-1)}{1-\beta}}\right]^{\frac{\alpha}{\sigma-1}} \tag{C-248}$$

Hence, omitting all constants for clarity, we may write that:

$$\left(P_i^s(g_i)\right)^{-\alpha} \cdot \left[\frac{\mathrm{d}\ln P_i^s}{\mathrm{d}g_i}(g_i)\right] \propto N_j^s \tau^{\frac{1-\sigma}{1-\beta}} [\bar{x}_{ji}(g_i)]^{\frac{\beta(\sigma-1)}{1-\beta}-1} \cdot \left[N_i^s \bar{x}_{ii}^{\frac{\beta(\sigma-1)}{1-\beta}} + N_j^s \tau^{\frac{1-\sigma}{1-\beta}} \cdot [\bar{x}_{ji}(g_i)]^{\frac{\beta(\sigma-1)}{1-\beta}}\right]^{\frac{\alpha}{\sigma-1}-1} \tag{C-249}$$

Assumption 5 ensures that both factors on the right-hand side are strictly decreasing in $\bar{x}_{ji}$. Indeed, under our parameter restrictions, we have:

$$\frac{\beta(\sigma-1)}{1-\beta} < 1 \quad \Longrightarrow \quad \frac{\mathrm{d}}{\mathrm{d}\bar{x}_{ji}} \left\{ N_j^s \tau^{\frac{1-\sigma}{1-\beta}} \bar{x}_{ji}^{\frac{\beta(\sigma-1)}{1-\beta}-1} \right\} = \underbrace{\left(\frac{\beta(\sigma-1)}{1-\beta} - 1\right)}_{<0} N_j^s \tau^{\frac{1-\sigma}{1-\beta}} \bar{x}_{ji}^{\frac{\beta(\sigma-1)}{1-\beta}-2} < 0 \tag{C-250}$$

Similarly, we obtain:

$$\alpha < \sigma - 1 \quad \Longrightarrow \quad \frac{\mathrm{d}}{\mathrm{d}\bar{x}_{ji}} \left\{ \left[N_i^s \bar{x}_{ii}^{\frac{\beta(\sigma-1)}{1-\beta}} + N_j^s \tau^{\frac{1-\sigma}{1-\beta}} \cdot [\bar{x}_{ji}(g_i)]^{\frac{\beta(\sigma-1)}{1-\beta}} \right]^{\frac{\alpha}{\sigma-1}-1} \right\} \propto \frac{\alpha}{\sigma-1} - 1 < 0 \tag{C-251}$$

Since both components of expression (C-249) are strictly positive and strictly decreasing in $\bar{x}_{ji}$, their product is also strictly decreasing in $\bar{x}_{ji}$. Using the fact that $\mathrm{d}\bar{x}_{ji} \propto -\mathrm{d}g_i$, it follows that:

$$\begin{aligned} &\alpha w^\alpha \left(P_i^s(g_i)\right)^{-\alpha} L_i \cdot \left[\frac{\mathrm{d}\ln P_i^s}{\mathrm{d}g_i}(g_i)\right] \quad \text{increases with } g_i, \\ \Longleftrightarrow \quad &(-1) \cdot \alpha w^\alpha \left(P_i^s(g_i)\right)^{-\alpha} L_i \cdot \left[\frac{\mathrm{d}\ln P_i^s}{\mathrm{d}g_i}(g_i)\right] \quad \text{decreases with } g_i \quad \Longrightarrow \quad \frac{\partial \mathscr{W}_i^s}{\partial^2 g_i} < 0, \quad \text{as claimed.} \end{aligned} \tag{C-252}$$

<u>*(ii)* $\exists \psi_{ji}^*$ **s.t.** $\forall \psi_{ji} \in (0, \psi_{ji}^*)$, $\mathscr{W}_i^s$ **is supermodular in** (g_i, N_i^s) **and** $\mathscr{W}_H^s(g_H, N_H^s) > \mathscr{W}_H^s(1, N_H^s), \forall g_H < 1$</u> Fix any $g_i \in [0,1]$. Substituting $N_j^s = N - N_i^s$ and recalling that $\Delta_i = \bar{x}_{ii}^{\frac{\beta(\sigma-1)}{1-\beta}} - \tau^{\frac{1-\sigma}{1-\beta}} \bar{x}_{ji}^{\frac{\beta(\sigma-1)}{1-\beta}} > 0$, expression (C-249) can be rewritten as a function of N_i^s as follows:

$$\left(P_i^s(N_i^s)\right)^{-\alpha} \cdot \left[\frac{\mathrm{d}\ln P_i^s}{\mathrm{d}g_i}(N_i^s)\right] \propto (N - N_i^s) \tau^{\frac{1-\sigma}{1-\beta}} \bar{x}_{ji}^{\frac{\beta(\sigma-1)}{1-\beta}-1} \cdot \left[N_i^s \Delta_i + N \tau^{\frac{1-\sigma}{1-\beta}} \cdot \bar{x}_{ji}^{\frac{\beta(\sigma-1)}{1-\beta}}\right]^{\frac{\alpha}{\sigma-1}-1} \tag{C-253}$$

The first term is clearly decreasing in N_i^s. As for the second term, a similar application of Assumption 5, combined with the fact that $\Delta_i > 0$ in any determinate equilibrium by Proposition 2.1, implies that:

$$\alpha < \sigma - 1 \quad \Longrightarrow \quad \frac{\mathrm{d}}{\mathrm{d}\bar{x}_{ji}} \left\{ \left[N_i^s \Delta_i + N \tau^{\frac{1-\sigma}{1-\beta}} \cdot \bar{x}_{ji}^{\frac{\beta(\sigma-1)}{1-\beta}} \right]^{\frac{\alpha}{\sigma-1}-1} \right\} \propto \Delta_i \left(\frac{\alpha}{\sigma-1} - 1\right) < 0 \tag{C-254}$$

Hence, holding all else constant, expression (C-253) is the product of two strictly positive functions, each decreasing in N_i^s. It follows that:

$$\begin{aligned} &\alpha w^\alpha \left(P_i^s(N_i^s)\right)^{-\alpha} L_i \cdot \left[\frac{\mathrm{d}\ln P_i^s}{\mathrm{d}g_i}(N_i^s)\right] \quad \text{decreases with } N_i^s, \\ \Longleftrightarrow \quad &(-1) \cdot \alpha w^\alpha \left(P_i^s(N_i^s)\right)^{-\alpha} L_i \cdot \left[\frac{\mathrm{d}\ln P_i^s}{\mathrm{d}g_i}(N_i^s)\right] \quad \text{increases with } N_i^s \end{aligned} \tag{C-255}$$

Hence, partially differentiating equation (C-246) with respect to N_i^s, and applying (C-255), we may write:

$$\frac{\partial \mathcal{W}_i^s}{\partial g_i \partial N_i^s} = \underbrace{\frac{\partial}{\partial N_i^s}\left\{(-1)\alpha w^\alpha \big(P_i^s(g_i, N_i^s)\big)^{-\alpha} L_i \cdot \left[\frac{\mathrm{d}\ln P_i^s}{\mathrm{d}g_i}(g_i, N_i^s)\right]\right\}}_{>0} - \pi_i \psi_{ji} \bar{x}_{ji}(g_i) \tag{C-256}$$

The sign of $\frac{\partial \mathcal{W}_i^s}{\partial g_i \partial N_i^s}$ is generally ambiguous. However, as $\psi_{ji} \to 0^+$, the second term vanishes and the expression becomes strictly positive. By continuity, this inequality must hold in an open neighbourhood of zero. In particular, for each (g_i, N_i^s), we have:

$$\exists \varepsilon_1(g_i, N_i^s) > 0 \quad \text{s.t.} \quad \frac{\partial \mathcal{W}_i^s}{\partial g_i \partial N_i^s} > 0 \quad \text{for all} \quad \psi_{ji} \in \big(0, \varepsilon_1(g_i, N_i^s)\big). \tag{C-257}$$

By analogous reasoning, letting $\psi_{ji} \to 0^+$ in equation (C-246) as in Proof C.20, we deduce that:

$$\exists \varepsilon_2(g_i, N_i^s) > 0 \quad \text{s.t.} \quad \frac{\partial \mathcal{W}_i^s}{\partial g_i} < 0 \quad \text{for all} \quad \psi_{ji} \in \big(0, \varepsilon_2(g_i, N_i^s)\big) \tag{C-258}$$

Invoking compactness and Weierstrass' theorem, it then suffices to define:

$$\psi_{ji}^* \;=\; \min_{(g_i, N_i^s) \in [0,1] \times [N_{\min}, N_{\max}]} \big\{\varepsilon_1(g_i, N_i^s),\, \varepsilon_2(g_i, N_i^s)\big\} \;>\; 0. \tag{C-259}$$

Then, by construction, both $\frac{\partial \mathcal{W}_i^s}{\partial g_i\, \partial N_i^s} > 0$ and $\frac{\partial \mathcal{W}_i^s}{\partial g_i} < 0$ hold uniformly over the domain for all $\psi_{ji} \in (0, \psi_{ji}^*)$. To get the equivalent statement that $\mathcal{W}_i^s$ is submodular in (g_i, N_j^s), simply use the chain rule to deduce that:

$$\frac{\partial \mathcal{W}_i^s}{\partial g_i\, \partial N_i^s} > 0 \quad \Longrightarrow \quad \frac{\partial \mathcal{W}_i^s}{\partial g_i\, \partial N_j^s} = \frac{\partial \mathcal{W}_i^s}{\partial g_i\, \partial N_i^s} \underbrace{\frac{\mathrm{d}N_i^s}{\mathrm{d}N_j^s}}_{<0} < 0 \tag{C-260}$$

<u>*(iii)* $\mathcal{W}_i^s$ **is strictly increasing in** N_i^s</u> Partial differentiation of (C-246) with to N_i^s immediately gives:

$$\frac{\partial \mathcal{W}_i^s}{\partial N_i^s} = (-1) \cdot \alpha w^\alpha \big(P_i^s\big)^{-\alpha-1} L_i \cdot \underbrace{\frac{\partial P_i^s}{\partial N_i^s}}_{<0 \text{ by (C-156)}} + \underbrace{\big(\psi_{ii}\, \bar{x}_{ii}^2 - \psi_{ji}\, \bar{x}_{ji}^2\big)}_{>0 \text{ under Assumption 10}} > 0 \tag{C-261}$$

□

Lemma 4 (Properties of Home's Strategy)**.** *Under Assumptions 5, 10, and 12, the candidate Markov strategy* $s_H(N_H^s, g_{H,-1}, \gamma)$ (C-240) *is well-defined satisfies the following properties:*

(i) s_H *is inflationary in the* $g_{H,-1}$ *coordinate:*

$$s_H(N_H^s, g_{H,-1}, \gamma) \geq g_{H,-1}. \tag{C-262}$$

(ii) s_H *is non-decreasing in* N_H^s*:*

$$\tilde{N}_H^s > N_H^s \quad \Longrightarrow \quad s_H(\tilde{N}_H^s, g_{H,-1}, \gamma) \geq s_H(N_H^s, g_{H,-1}, \gamma). \tag{C-263}$$

Proof. Since $g_H \mapsto \mathscr{W}^s_H(g_H, N^s_H)$ is strictly concave in g_H on $[0,1]$ by Lemma 3, it is strictly quasi-concave. Equivalently, the super-level sets of $\mathscr{W}^s_H$ are closed intervals. In particular, the Home's government set of politically feasible policies given state $\jmath = (N^s_H, g_{H,-1}, \gamma) \in \mathcal{S}$ can be written as:

$$L(\jmath) = \left\{ g_H \in [0,1] \;\middle|\; \mathscr{W}^s_H(g_H, N^s_H) \geq (1-\gamma)\, \mathscr{W}^s_H(g_{H,-1}, N^s_H) \right\} \equiv \left[g^{\min}_H,\, g^{\max}_H\right] \subseteq [0,1] \tag{C-264}$$

with $0 \leq g^{\min}_H \leq g^{\max}_H \leq 1$. By definition of Home's rule (C-240), we then have:

$$s_H(\jmath) = \max L(\jmath) \equiv g^{\max}_H(N^s_H), \quad \text{which is well-defined.} \tag{C-265}$$

<u>*(i)* $s_H(N^s_H, g_{H,-1}, \gamma) \geq g_{H,-1}$</u> By construction, the status quo level of regulation is always feasible. We may therefore trivially conclude that $g_{H,-1} \leq g^{\max}_H = s_H(\jmath)$.

<u>*(ii)* $N^s_H \mapsto g^{\max}_H(N^s_H)$ **is a non-decreasing function**</u> Pick $N^s_H \in (N_{\min}, N_{\max})$, $\tilde{N}^s_H \in (N^s_H, N_{\max}]$ and define the associated state $\tilde{\jmath} = (\tilde{N}^s_H, g_{H,-1}, \gamma)$. By Lemma 3, $\mathscr{W}^s_H$ is strictly increasing in N^s_H and supermodular in (g_H, N^s_H). Accordingly, we deduce that:

$$\begin{aligned}
\mathscr{W}^s_H\big(g^{\max}_H(N^s_H),\, \tilde{N}^s_H\big) - \mathscr{W}^s_H\big(g_{H,-1},\, \tilde{N}^s_H\big) >\;& \mathscr{W}^s_H\big(g^{\max}_H(N^s_H),\, N^s_H\big) - \mathscr{W}^s_H\big(g_{H,-1},\, N^s_H\big) \quad \text{(by supermodularity)} \\
\geq\;& (1-\gamma)\, \mathscr{W}^s_H\big(g_{H,-1},\, N^s_H\big) - \mathscr{W}^s_H\big(g_{H,-1},\, N^s_H\big) \quad \big(\text{since } g^{\max}_H(N^s_H) \in L(\jmath)\big) \\
=\;& -\gamma\, \mathscr{W}^s_H\big(g_{H,-1},\, N^s_H\big) \\
\geq\;& -\gamma\, \mathscr{W}^s_H\big(g_{H,-1},\, \tilde{N}^s_H\big) \quad \text{(by monotonicity in } N^s_H\text{)}
\end{aligned} \tag{C-266}$$

Hence, $g^{\max}_H(N^s_H) \in L(\tilde{\jmath})$, and it follows that:

$$g^{\max}_H(N^s_H) \leq \max L(\tilde{\jmath}) = g^{\max}_H(\tilde{N}^s_H), \quad \text{as claimed.} \tag{C-267}$$

□

C.21.2 Foreign plays the trivial strategy: $s_F(\jmath) = 1$, for all $\jmath \in \mathcal{S}$.

Fix an arbitrary state $\jmath = (N^s_H,\, g_{H,-1},\, \gamma) \in \mathcal{S}$ and let t be the date at which $\jmath$ occurs for concreteness. Consider the one–shot deviation:

$$g_{F,\tau} = 1 \;(\forall \tau), \qquad g'_{F,\tau} = \begin{cases} y < 1, & \tau = t, \\ 1, & \tau > t. \end{cases} \tag{C-268}$$

Let $N^s_{F,\tau}$ be the equilibrium path and $N^{s,\prime}_{F,\tau}$ the deviated path of Foreign's short–run mass. Using the law of motion of motion of N^s_F (C-235), the change in N^s_F immediately after the deviation is:

$$\hat{N}^s_{F,t+1} \triangleq N^{s,\prime}_{F,t+1} - N^s_{F,t+1} = \rho\big(N^\ell_F(1,y) - N^\ell_F(1,1)\big) < 0. \tag{C-269}$$

writing $N^\ell_F(g_H, g_F)$, and using that local firm presence is increasing in the level of outbound data restrictions by Proposition 2.4. By a simple induction, for all $\tau \geq t+1$, we then get:

$$\hat{N}^s_{F,\tau} = (1-\rho)^{\tau-(t+1)}\,\hat{N}^s_{F,t+1} < 0. \tag{C-270}$$

By Proposition 2.5, we have $\hat{N}^s_{H,\tau} = -\hat{N}^s_{F,\tau} > 0$. Moreover, by Lemma 4, the mapping $N^s_H \mapsto s_H(N^s_H;\, g_{H,-1};\, \gamma)$ is non-decreasing, and it follows that for all $\tau \geq t+1$:

$$g'_{H,\tau} = s_H(N^s_H + \hat{N}^s_{H,\tau},\, g_{H,-1},\, \gamma) \geq s_H(N^s_H,\, g_{H,-1},\, \gamma) = g_{H,\tau} \tag{C-271}$$

Next, writing the difference in continuation values, and using that $\mathcal{O}_F$ is both non-decreasing in N^s_F, and non-increasing in g_H, it is immediate that:

$$\begin{aligned} V'_F - V_F &= \sum_{\tau=t}^{\infty} \delta^{\tau-t}\Big[\mathcal{O}_F\big(1, N^{s,\prime}_{F,\tau}; g'_{H,\tau}\big) - \mathcal{O}_F\big(1, N^s_{F,\tau}; g_{H,\tau}\big)\Big] \\ &\leq \underbrace{\mathcal{O}_F\big(y, N^s_{F,\tau}; g_{H,\tau}\big) - \mathcal{O}_F\big(1, N^s_{F,\tau}; g_{H,\tau}\big)}_{<0} + \sum_{\tau=t+1}^{\infty} \delta^{\tau-t}\,\underbrace{\Big[\mathcal{O}_F\big(1, N^{s,\prime}_{F,\tau}; g'_{H,\tau}\big) - \mathcal{O}_F\big(1, N^s_{F,\tau}; g_{H,\tau}\big)\Big]}_{\leq 0} \\ &< 0 \end{aligned} \tag{C-272}$$

Thus no one–shot deviation by Foreign can increase its payoff and s_F is optimal.

C.21.3 Home plays s_H (C-240) and $\exists\, S(g_0) > 0$ s.t. $\{\gamma_t\}_{t\geq 1}$ with $\sum_t \gamma_t \leq S(g_0) \implies g_{H,\infty} = g^* < 1$.

Optimality of s_H An argument entirely analogous to the one just given for Foreign shows that, because any deviation y must be feasible, for any state $\mathcal{s} \in \mathcal{S}$ one must have:

$$y \in L(\mathcal{s}) \quad \implies \quad g'_H < \max L(\mathcal{s}) = s_H(\mathcal{s}) \tag{C-273}$$

Hence, by similar reasoning as in (C-270), firm presence in Home declines in the subsequent period under the deviated path, which, by Lemma 4, also induces a weakly lower policy choice.

$$N^{s,\prime}_{H,t+1} > N^s_{H,t+1} \quad \implies \quad g'_{H,t+1} = s_H\big(N^{s,\prime}_{H,t+1}, y, \gamma_{t+1}\big) \leq s_H\big(N^s_{H,t+1}, g_{H,t}, \gamma_{t+1}\big) = g_{H,t+1}. \tag{C-274}$$

By straightforward induction, the post-deviation sequences satisfy $g'_{H,\tau} \leq g_{H,\tau}$ and $N^{s,\prime}_{H,\tau} \leq N^s_{H,\tau}$ for all $\tau \geq t+1$. Since $\mathcal{O}_H$ is strictly increasing in g_H and non-decreasing in N^s_H, it follows that:

$$V'_H = \mathcal{O}_H\big(y, N^s_{H,\tau}; 1\big) + \sum_{\tau=t+1}^{\infty} \delta^{\tau-t}\,\mathcal{O}_H\big(g'_{H,\tau}, N^{s,\prime}_{H,\tau}; 1\big) \tag{C-275}$$

$$< \mathcal{O}_H\big(y, N^s_{H,\tau}; 1\big) + \sum_{\tau=t+1}^{\infty} \delta^{\tau-t}\,\mathcal{O}_H\big(g'_{H,\tau}, N^{s,\prime}_{H,\tau}; 1\big) \tag{C-276}$$

$$\begin{aligned} &\leq \mathcal{O}_H\big(s_H(\mathcal{s}), N^s_{H,\tau}; 1\big) + \sum_{\tau=t+1}^{\infty} \delta^{\tau-t}\,\mathcal{O}_H\big(g_{H,\tau}, N^s_{H,\tau}; 1\big) \\ &= V_H \end{aligned} \tag{C-277}$$

Derivation of the political budget: $S(g_0) > 0$ By Lemma 4, the Home policy sequence satisfies $g_{H,t-1} \leq g_{H,t} \leq 1$ for all t. In other words, it is monotone non-decreasing and bounded above, and therefore converges to a limit, denoted $g_{H,\infty} \leq 1$. Our goal is to show that, for any initial condition $g_0 < 1$, one can construct a bound $S(g_0) > 0$ such that this inequality is strict:

By continuity of the mapping $g_H \mapsto N_H^\ell(g_H, g_F)$, and by global convergence of firm masses to their long-run values (B-48), together with the fact that Foreign's strategy implies $g_{F,t} = 1$ for all $t \geq 1$, we obtain:

$$\lim_{T\to\infty} N_H^\ell(g_{H,T}, g_{F,T}) = N_H^\ell(g_{H,\infty}, 1) \quad \Longrightarrow \quad N_{H,\infty}^s = N_H^\ell(g_{H,\infty}, 1) \tag{C-278}$$

Moreover, because the mapping $(g_H, g_F) \mapsto N_H^\ell(g_H, g_F)$ is constant along the diagonal where $g_H = g_F$ by Proposition 2.5, and strictly increasing in its first argument, we have:

$$g_{H,\infty} \leq 1 \quad \Longrightarrow \quad N_{H,\infty}^s = N_H^\ell(g_{H,\infty}, 1) \leq N_H^\ell(g_0, g_0) = N_{H,1}^s, \quad \text{with equality iff } g_{H,\infty} = 1 \tag{C-279}$$

We now claim that $N_{H,t}^s \leq N_{H,1}^s$ for all $t \geq 2$. We proceed by induction on t. From the law of motion (C-235), we have:

$$\begin{aligned} N_{H,2}^s &= (1-\rho)\, N_{H,1}^s + \rho\, N_H^\ell\big(g_{H,1}, 1\big) \\ &\leq (1-\rho)\, N_{H,1}^s + \rho\, N_H^\ell\big(g_\infty, 1\big) \\ &\leq (1-\rho)\, N_{H,1}^s + \rho\, N_{H,1}^s \\ &= N_{H,1}^s. \end{aligned} \tag{C-280}$$

where the second line uses monotonicity of $g_{H,t} \leq g_\infty$. Now, assume that $N_{H,t}^s \leq N_{H,1}^s$ for some t. Then, an analogous chain of inequalities gives:

$$N_{H,t+1}^s \leq (1-\rho)\, N_{H,t}^s + \rho\, N_H^\ell\big(g_{H,\infty}, 1\big) \leq N_{H,1}^s \tag{C-281}$$

We build on inequality (C-281) to derive an intermediate result which is useful in the subsequent analysis.

Lemma 5 (Inertia of Home's Markov Strategy Along the Equilibrium Path)**.** *Fix any initial condition $s_1 = (N_{H,1}^s, g_0, \gamma_1)$, and consider the sequence $\{g_{H,t}, N_{H,t}^s\}_{t\geq 1}$ induced by Home's Markov strategy s_H (C-240). For all $t \geq 1$, if $N_{H,t}^s \leq N_{H,1}^s$ the following holds:*

(i) The implemented policy $g_{H,t}$ lies on the decreasing region of the short-run welfare function $g \mapsto \mathcal{W}_H^s(g, N_{H,t}^s)$, and so does the subsequent status quo policy $g_{H,t}$ in period $t+1$. Formally:

$$\left.\frac{\partial \mathcal{W}_H^s(g, N_{H,t}^s)}{\partial g}\right|_{g=g_{H,t}} \leq 0, \quad \textit{and} \quad \left.\frac{\partial \mathcal{W}_H^s(g, N_{H,t+1}^s)}{\partial g}\right|_{g=g_{H,t}} \leq 0. \tag{C-282}$$

(ii) If $\gamma_t = 0$, then $g_{H,t} = g_{H,t-1}$ for any $t \geq 2$. If instead $\gamma_t > 0$ and $g_{H,t-1} < 1$, then $g_{H,t} > g_{H,t-1}$ for any $t \geq 1$. In the latter case, the inequalities in (C-282) hold strictly at time t and in all subsequent periods.

Proof. Towards a proof of claim (i), define:

$$g_{\text{peak}}(N_H^s) \triangleq \arg\max_{g\in[0,1]} \mathcal{W}_H^s(g, N_H^s) \tag{C-283}$$

Note that, under Assumption 12, we have $g_{\text{peak}}(N_{H,1}^s) < 1$. By definition of the maximum and convexity of the super-level sets of $\mathcal{W}_H^s$, it is immediate that:

$$\gamma \geq 0 \quad \Longrightarrow \quad \mathcal{W}_H^s(g_{\text{peak}}, N_H^s) \geq \mathcal{W}_H^s(g_{H,-1}, N_H^s) \geq (1-\gamma)\,\mathcal{W}_H^s(g_{H,-1}, N_H^s), \tag{C-284}$$

Hence, $g_{\text{peak}}(N_H^s)$, which is unique by strict concavity of $\mathcal{W}_H^s$, is always feasible, and it follows that $s_H(\mathcal{s}) \geq g_{\text{peak}}(N_H^s)$. In particular, $g_{H,1} \geq g_{\text{peak}}(N_{H,1}^s)$ must hold.

Now, from the supermodularity of $\mathcal{W}_H^s$ in $(g; N_H^s)$, the mapping $N_H^s \mapsto g_{\text{peak}}(N_H^s)$ is non-decreasing by Topkis' theorem. Since the policy path $\{g_{H,t}\}_{t\geq 1}$ is non-decreasing and $N_{H,t}^s \leq N_{H,1}^s$ for all t by (C-281), it follows that:

$$\forall t \geq 1, \quad N_{H,t}^s \leq N_{H,1}^s \quad \Longrightarrow \quad g_{\text{peak}}(N_{H,t}^s) \leq g_{\text{peak}}(N_{H,1}^s) \leq g_{H,1} \leq g_{H,t-1} \leq g_{H,t} \tag{C-285}$$

Hence, the implemented policy remains on the decreasing region of $\mathcal{W}_H^s$ in every period. The derivative characterisation presented in the statement follows directly from the strict concavity of $\mathcal{W}_H^s$ in its first argument.

Claim (ii) is an almost immediate corollary. By construction, the political feasibility constraint corresponds to a super-level set of $\mathcal{W}_H^s$ which, by strict quasi-concavity, takes the form of an interval:

$$L(\mathcal{s}) = \left\{\, g_H \in [0,1] \;\middle|\; \mathcal{W}_H^s(g_H, N_H^s) \geq (1-\gamma)\,\mathcal{W}_H^s(g_{H,-1}, N_H^s) \,\right\} \equiv \left[\, a,\, b \,\right] \subseteq [0,1] \tag{C-286}$$

If $\gamma_t = 0$, then feasibility at time t reduces to $\mathcal{W}_H^s(g_{H,t}, N_{H,t}^s) \geq \mathcal{W}_H^s(g_{H,t-1}, N_{H,t}^s)$. But since $g_{\text{peak}}(N_{H,t}^s) \leq g_{H,t-1}$ for all $t \geq 2$, strict concavity implies that no $g > g_{H,t-1}$ can satisfy this inequality. Hence, $g_{H,t} = g_{H,t-1}$.

As for the case $\gamma_t > 0$, suppose towards a contradiction that $g_{H,t} = \max L(\mathcal{s}) = g_{H,t-1}$ for some $g_{H,t-1} < 1$. Then feasibility at the solution implies:

$$\mathcal{W}_H^s(g_{H,t}, N_{H,t}^s) \geq (1-\gamma_t)\,\mathcal{W}_H^s(g_{H,t-1}, N_{H,t}^s) \quad \Longleftrightarrow \quad 0 \geq -\gamma_t\,\mathcal{W}_H^s(g_{H,t-1}, N_{H,t}^s), \tag{C-287}$$

which holds strictly. Hence, $g_{H,t-1}$ is an interior point of $L(\mathcal{s})$, which contradicts that $g_{H,t} = \max L(\mathcal{s}) = g_{H,t-1}$.

To connect this result to claim (i), it suffices to rewrite (C-285) with a strict inequality and invoke monotonicity of the policy path, characterised in Lemma 4. In particular, if $\gamma_t > 0$ for some t, then we have:

$$g_{H,t+k} \geq g_{H,t} > g_{H,1} \geq g_{\text{peak}}(N_{H,t}^s),\, g_{\text{peak}}(N_{H,t+k}^s), \tag{C-288}$$

so that both $g_{H,t}$ and $g_{H,t+k}$ lie strictly on the decreasing region of $\mathcal{W}_H^s$. $\square$

Towards the derivation of $S(g_0)$, pick any $t \geq 1$, and invoke the mean-value theorem to write that, for some $\xi_t \in (g_{H,t-1},\, g_{H,t})$:

$$\mathcal{W}_H^s\big(g_{H,t}, N_{H,t}^s\big) - \mathcal{W}_H^s\big(g_{H,t-1}, N_{H,t}^s\big) = \frac{\partial \mathcal{W}_H^s\big(\xi_t, N_{H,t}^s\big)}{\partial g_H}\big[g_{H,t} - g_{H,t-1}\big] \leq \frac{\partial \mathcal{W}_H^s\big(\xi_t, N_{H,1}^s\big)}{\partial g_H}\big[g_{H,t} - g_{H,t-1}\big] \tag{C-289}$$

where the inequality follows from the supermodularity of $\mathcal{W}_H^s$ in $(g_{H,t}, N_{H,t}^s)$ and $N_{H,t}^s \leq N_{H,1}^s$ (C-281). Meanwhile, under Assumption 12, the regularity condition $\mathcal{W}_H^s(g_H, N_H^s) > 0$ holds for all (g_H, N_H^s), and the feasibility constraint (C-235), together with the fact that $N_H^s \mapsto \mathcal{W}_H^s(g_H, N_H^s)$ increases by Lemma 3, gives the lower bound:

$$\mathcal{W}_H^s\big(g_{H,t}, N_{H,t}^s\big) - \mathcal{W}_H^s\big(g_{H,t-1}, N_{H,t}^s\big) \geq -\gamma_t \mathcal{W}_H^s\big(g_{H,t-1}, N_{H,t}^s\big) \geq -\gamma_t \mathcal{W}_H^s\big(g_{H,t-1}, N_{H,1}^s\big) \tag{C-290}$$

Hence, combining (C-289) and (C-290) and summing from 1 to T large enough, we may write that:

$$\begin{aligned}
\sum_{t=1}^{T} \frac{\partial \mathcal{W}_H^s(\xi_t, N_{H,1}^s)}{\partial g_H} (g_{H,t} - g_{H,t-1}) \;&\geq\; -\sum_{t=1}^{T} \gamma_t\, \mathcal{W}_H^s(g_{H,t-1}, N_{H,1}^s) \\
&\geq -\sum_{t=1}^{T} \gamma_t W_H^s(g_{\text{peak}}(N_{H,1}^s), N_{H,1}^s), \quad \text{for } g_{\text{peak}} < 1 \text{ by Assumption 12.} \\
&\equiv -M(g_0) \sum_{t=1}^{T} \gamma_t, \quad \text{since } N_{H,1}^s \text{ is fully determined by } g_0.
\end{aligned} \tag{C-291}$$

Now, notice that a direct application of Lemma 5 gives that:

$$\frac{\partial \mathcal{W}_H^s(\xi_t, N_{H,1}^s)}{\partial g_H}(g_{H,t} - g_{H,t-1}) \leq 0 \quad \text{for all } t, \text{ with strict inequality whenever } \gamma_t > 0. \tag{C-292}$$

Now, we consider two cases.

If $\sum_t \gamma_t = 0$ Lemma 5 immediately gives $g_{H,t} = g_{H,1}$ for all $t \geq 2$, with $g_{H,1} < 1$ since $1 \notin L(\mathscr{s}_0)$ under Assumption 12. It follows that $g_{H,\infty} < 1$.

If $\sum_t \gamma_t > 0$ Assume that $\{\gamma_t\}_{t \geq 1}$ is summable and pose $t_0 \triangleq \min\{t \geq 1 : \; \gamma_t > 0\} < \infty$. Strict concavity of $g_H \mapsto \mathcal{W}_H^s(g_H, N_{H,1}^s)$ implies its partial derivative with respect to g_H is strictly decreasing. Moreover, we have $\xi_t \in (g_{H,t-1}, g_{H,t})$ by definition, which forces $\xi_{t'} \geq \xi_{t_0}$ for all $t' \geq t_0$ by path monotonicity, and we thus get that:

$$t' \geq t_0 \quad \Longrightarrow \quad 0 < -\frac{\partial \mathcal{W}_H^s}{\partial g_H}(\xi_{t_0}, N_{H,1}^s) \leq -\frac{\partial \mathcal{W}_H^s}{\partial g_H}(\xi_{t'}, N_{H,1}^s) \tag{C-293}$$

Pose $-m(g_0) \triangleq \frac{\partial \mathcal{W}_H^s}{\partial g_H}(\xi_{t_0}, N_{H,1}^s) > 0$, which is indeed an exogenous quantity. For every $1 \leq t < t_0$ we have $\gamma_t = 0$, and it follows from (C-292) that:

$$\begin{aligned}
M \sum_{t=1}^{T} \gamma_t &\geq \sum_{t=1}^{T} \left[-\frac{\partial \mathcal{W}_H^s(\xi_t, N_{H,1}^s)}{\partial g_H}\right] (g_{H,t} - g_{H,t-1}) \\
&= \sum_{t=t_0}^{T} \left[-\frac{\partial \mathcal{W}_H^s(\xi_t, N_{H,1}^s)}{\partial g_H}\right] (g_{H,t} - g_{H,t-1}) \\
&\geq m \sum_{t=t_0}^{T} (g_{H,t} - g_{H,t-1}) \\
&= m(g_{H,T} - g_0)
\end{aligned} \tag{C-294}$$

Finally, letting $T \to \infty$, we conclude that:

$$0 \le \sum_{t=1}^{\infty} \gamma_t \; < \; \frac{m}{M}\,(1-g_0) \triangleq S(g_0) \quad \implies \quad g_{H,\infty} \le g_0 + \frac{M}{m}\sum_{t=1}^{\infty}\gamma_t < 1 \tag{C-295}$$

It is then immediate from (C-279) that, in such a case, we have:

$$N_H^{\ell}(\mathbf{X}_\infty) < N_{H,1} = \frac{N}{2} < N - N_H^{\ell}(\mathbf{X}_\infty) = N_F^{\ell}(\mathbf{X}_\infty) \tag{C-296}$$

□

C.22 Proof of Proposition 3.17 (Relocation Speed and Long-Run Retaliation Capacity in Home)

Let $\rho' > \rho$, and denote by $\mathbf{e}_t(\rho) = \{g_{H,t}(\rho),\, N^s_{H,t}(\rho)\}_{t\ge 1}$ the joint sequence of policies and firm masses in Home under convergence to the long-run equilibrium in the Tech sector at rate ρ. Note that $g_{H,1} = \max L(\mathcal{s}_1) < 1$, under Assumption 12, is fully determined by the initial condition $(g_0, N^s_{H,1})$ and thus remains invariant to changes in ρ. It follows that any divergence between the two paths must occur from period $t \ge 2$ onwards.

More generally, pose:

$$t_0 \triangleq \min\{\, t \ge 2 \;:\; \gamma_t > 0\,\}, \quad \text{which is well-defined for} \sum_{t=1}^{\infty}\gamma_t \in (0, S(g_0)) \tag{C-297}$$

By Lemma 5, for all t with $1 \le t < t_0$ and every $\rho \in (0,1)$, we have $g_{H,t} = g_{H,1}$. Hence, the policy paths are in fact identical up to period t_0, so that $\{g_{H,t}(\rho)\}_{1\le t<t_0} \equiv \{g_{H,t}\}_{1\le t<t_0}$ is independent of ρ.

Now, consider the policy problem faced by the Home government in period $t_0 \ge 2$. We start by showing, that, by this point, the sequences $\{N^s_{H,t}(\rho)\}$ and $\{N^s_{H,t}(\rho')\}$ must have already diverged as a result of the regulatory asymmetry introduced in period $t = 1$. Indeed, recursively applying the law of motion (C-235) gives:

$$\begin{aligned} N^s_{H,t} &= (1-\rho)^{\,t-1}\, N^s_{H,1} + \rho \sum_{k=1}^{t-1} (1-\rho)^{\,t-1-k}\, N^{\ell}_H\big(g_{H,k}(\rho), 1\big) \\ &\equiv w_0(\rho) N^s_{H,1} + \sum_{k=1}^{t-1} w_k(\rho)\, N^{\ell}_H\!\big(g_{H,k}(\rho), 1\big), \quad \text{with} \quad w_0(\rho) \;+\; \sum_{k=1}^{t-1} w_k(\rho) \;=\; 1 \end{aligned} \tag{C-298}$$

Using that the weights $\{w_k\}$ sum to 1, we have:

$$t \ge 2 \quad \implies \quad w_0'(\rho) = \frac{\mathrm{d}}{\mathrm{d}\rho}\big\{(1-\rho)^{\,t-1}\big\} < 0 \tag{C-299}$$

$$\implies \quad \frac{\mathrm{d}}{\mathrm{d}\rho}\left\{\sum_{k=1}^{t-1} w_k(\rho)\right\} = -w_0'(\rho) > 0 \tag{C-300}$$

Moreover, because the policy path $\{g_{H,t}(\rho)\}$ is non-decreasing by Lemma 4, with limit $g_{H,\infty} < 1$ whenever $\sum_t \gamma_t < S(g_0)$, while $N^{\ell}_H(g_H, g_F)$ is strictly increasing in g_H and constant along the diagonal, we deduce that:

$$\frac{\mathrm{d}N^s_{H,t}(\rho)}{\mathrm{d}\rho} = w_0'(\rho)\, N^s_{H,1} + \sum_{k=1}^{t-1} w_k'(\rho)\, N^{\ell}_H\big(g_{H,k}(\rho), 1\big) \; + \sum_{k=1}^{t-1} w_k(\rho)\, \frac{\mathrm{d}N^{\ell}_H}{\mathrm{d}g_{H,k}}\, \frac{\mathrm{d}g_{H,k}(\rho)}{\mathrm{d}\rho}$$

$$
\begin{aligned}
&\leq w_0'(\rho)\, N^s_{H,1} + N^\ell_H\big(g_{H,t-1}(\rho),1\big) \sum_{k=1}^{t-1} w_k'(\rho) \; + \sum_{k=1}^{t-1} w_k(\rho)\, \frac{\mathrm{d}N^\ell_H}{\mathrm{d}g_{H,k}} \frac{\mathrm{d}g_{H,k}(\rho)}{\mathrm{d}\rho} \\
&= \underbrace{w_0'(\rho)}_{<0} \quad \underbrace{\Big[N^\ell_H\big(g_0,g_0\big) - N^\ell_H\big(g_{H,t-1}(\rho),1\big)\Big]}_{>0 \text{ since } g_{H,t-1} \leq g_{H,\infty} < 1 \text{ when } \sum_t \gamma_t < S(g_0).} \quad + \sum_{k=1}^{t-1} w_k(\rho) \underbrace{\frac{\mathrm{d}N^\ell_H}{\mathrm{d}g_{H,k}}}_{>0} \frac{\mathrm{d}g_{H,k}(\rho)}{\mathrm{d}\rho}
\end{aligned}
\tag{C-301}
$$

In particular, because Home's policy path is independent of ρ for all $t < t_0$, we have:

$$
\frac{\mathrm{d}N^s_{H,t_0}(\rho)}{\mathrm{d}\rho} \leq \underbrace{w_0'(\rho)}_{<0} \underbrace{\Big[N^\ell_H\big(g_0,g_0\big) - N^\ell_H\big(g_{H,t_0-1}(\rho),1\big)\Big]}_{>0} + \sum_{k=1}^{t_0-1} w_k(\rho)\, \frac{\mathrm{d}N^\ell_H}{\mathrm{d}g_{H,k}} \underbrace{\frac{\mathrm{d}g_{H,k}(\rho)}{\mathrm{d}\rho}}_{=0} < 0 \tag{C-302}
$$

An application of the fundamental theorem of calculus then immediately gives:

$$
N^s_{H,t_0}(\rho') - N^s_{H,t_0}(\rho) = \int_\rho^{\rho'} \frac{\mathrm{d}N^s_{H,t_0}}{\mathrm{d}\rho}(u)\, \mathrm{d}u < 0 \tag{C-303}
$$

To deduce that $g_{H,t_0}(\rho') < g_{H,t_0}(\rho)$, it suffices to recall from Lemma 5 that, along the equilibrium path, the rule s_H selects a point g_H whose image lies on the strictly decreasing region of $\mathscr{W}^s_H$ whenever $\gamma > 0$. Specifically, a straightforward application of the implicit function theorem to the binding constraint strengthens Lemma 4 by showing that the map $N^s_H \mapsto s_H(N^s_H, g_{H,-1}, \gamma)$ is strictly increasing whenever the state $\mathscr{s}$ features $\gamma > 0$.

$$
\frac{\mathrm{d}s_H}{\mathrm{d}N^s_H} = - \underbrace{\left[\frac{\partial \mathscr{W}^s_H}{\partial g_H}(g_H, N^s_H)\right]^{-1}}_{<0 \text{ by concavity and } g_H > g_{\text{peak}}} \left[\underbrace{\frac{\partial \mathscr{W}^s_H}{\partial N^s_H}(g_H, N^s_H) - \frac{\partial \mathscr{W}^s_H}{\partial N^s_H}(g_{H,-1}, N^s_H)}_{\geq 0 \text{ by supermodularity in } (g_H, N^s_H)} + \underbrace{\gamma\, \frac{\partial \mathscr{W}^s_H}{\partial N^s_H}(g_{H,-1}, N^s_H)}_{>0 \text{ by monotonicity in } N^s_H} \right] > 0 \tag{C-304}
$$

Since t_0 was precisely chosen so that $\gamma_{t_0} > 0$, combining the strict monotonicity of s_H in N^s_H (C-304) with the fact that $N^s_{H,t_0}(\rho) > N^s_{H,t_0}(\rho')$ (C-303) allows us to conclude that:

$$
\rho' > \rho \quad \Longrightarrow \quad g_{H,t_0}(\rho') = s_H(N^s_{H,t_0}(\rho'), g_{H,t_0-1}, \gamma_{t_0}) < s_H(N^s_{H,t_0}(\rho), g_{H,t_0-1}, \gamma_{t_0}) = g_{H,t_0}(\rho) \tag{C-305}
$$

We now claim that, for all $t \geq t_0$, we have both $g_{H,t}(\rho) - g_{H,t}(\rho') > 0$ and $N^s_{H,t}(\rho) - N^s_{H,t}(\rho') > 0$. We have just shown that the base step holds, and the induction hypothesis gives:

$$
\begin{aligned}
\forall k \leq t, \quad \rho' > \rho \quad &\Longrightarrow \quad g_{H,k}(\rho) > g_{H,k}(\rho') \\
&\Longrightarrow \quad \rho \mapsto g_{H,k}(\rho) \quad \text{is strictly decreasing} \quad \Longleftrightarrow \quad \frac{\mathrm{d}g_{H,k}}{\mathrm{d}\rho} < 0
\end{aligned}
\tag{C-306}
$$

Moreover, the differential characterisation (C-302) gives:

$$
\frac{\mathrm{d}N^s_{H,t+1}(\rho)}{\mathrm{d}\rho} \leq \sum_{k=1}^{t} \underbrace{w_k(\rho)}_{>0} \underbrace{\frac{\mathrm{d}N^\ell_H}{\mathrm{d}g_{H,k}}}_{>0} \underbrace{\frac{\mathrm{d}g_{H,k}(\rho)}{\mathrm{d}\rho}}_{<0} < 0 \tag{C-307}
$$

By monotonicity of integration, we then have:

$$N^s_{H,t+1}(\rho') - N^s_{H,t+1}(\rho) \leq \int_\rho^{\rho'} \sum_{k=1}^{t} w_k(u) \, \frac{\mathrm{d}N^\ell_H\big(g_{H,k}(u)\big)}{\mathrm{d}g_{H,k}} \, \frac{\mathrm{d}g_{H,k}(u)}{\mathrm{d}\rho} \, \mathrm{d}u < 0 \tag{C-308}$$

Finally, to complete the inductive step, we claim that:

$$g_{H,t+1}(\rho') = s_H(N^s_{H,t+1}(\rho'), g_{H,t}(\rho'), \gamma_{t+1}) < s_H(N^s_{H,t+1}(\rho), g_{H,t}(\rho), \gamma_{t+1}) = g_{H,t+1}(\rho) \tag{C-309}$$

If $\underline{\gamma_{t+1} = 0}$ A direct application of Lemma 5 gives that $g_{H,t+1}(\rho) = g_{H,t}(\rho)$ for all $\rho \in (0,1)$. The conclusion then follows immediately from the induction hypothesis.

If $\underline{\gamma_{t+1} > 0}$ We have already established that $N^s_{H,t+1}(\rho) > N^s_{H,t+1}(\rho')$ (C-308), and we know that the mapping $N^s_H \mapsto s_H(N^s_H, g_{H,-1}, \gamma)$ is strictly increasing along the equilibrium path (C-304). To conclude that inequality (C-309) holds, it is therefore sufficient to show that the mapping $g_{H,-1} \mapsto s_H(N^s_H, g_{H,-1}, \gamma)$ is non-decreasing in the relevant region, since the induction hypothesis implies that $g_{H,t}(\rho) > g_{H,t}(\rho')$.

From Lemma 5, we know that both $g_{H,t}$ and $g_{H,t-1}$ lie on the decreasing portion of $\mathscr{W}^s_H$ at time $t \geq t_0 \geq 2$. A straightforward application of the implicit function theorem to the binding feasibility constraint then gives:

$$\forall t \geq t_0, \quad \left.\frac{\mathrm{d}s_H}{\mathrm{d}g_{H,-1}}\right|_{g_{H,-1}=g_{H,t-1}} = \frac{(1-\gamma_t)\,\partial_g \mathscr{W}^s_H(g_{H,t-1}, N^s_{H,t+1})}{\partial_g \mathscr{W}^s_H(g_{H,t}, N^s_{H,t+1})} > 0, \quad \text{which implies claim (C-309).} \tag{C-310}$$

Hence, for all $t \geq t_0$, we have both $g_{H,t}(\rho) > g_{H,t}(\rho')$ and $N^s_{H,t}(\rho) > N^s_{H,t}(\rho')$. Moreover, since the sequence $\{\gamma_t\}$ has only finitely many non-zero terms by assumption, there exists a finite terminal date $T < \infty$ such that:

$$T \triangleq \max\{t \geq t_0 \ : \ \gamma_t > 0\} < \infty. \tag{311}$$

In particular, because $\gamma_t = 0$ for all $t \geq T$, Lemma 5 implies that:

$$\forall \rho \in (0,1), \quad g_{H,t}(\rho) = g_{H,t-1}(\rho) = \cdots = g_{H,T}(\rho) \quad \Longrightarrow \quad g_{H,\infty}(\rho') = g_{H,T}(\rho') < g_{H,T}(\rho) = g_{H,\infty}(\rho). \tag{312}$$

$\square$

C.23 Proof of Proposition 3.18 (Welfare Implications of Asymmetric Escalation)

The steady-state welfare function might be written as a mapping $(g_i, g_j) \mapsto \mathscr{W}_i(g_i, N_i(g_i, g_j))$. Pose $\mathscr{W}_{i,0} \triangleq \mathscr{W}_i(g_0, N^\ell_i(g_0, g_0))$ and $\mathscr{W}_{i,\infty} \triangleq \mathscr{W}_i(g_{i,\infty}, N^\ell_i(g_{i,\infty}, g_{j,\infty}))$.

$\underline{\mathscr{W}_{H,\infty} < \mathscr{W}_{H,0}}$ Because $g \mapsto N^\ell_H(g,g)$ is constant as a straightforward corollary of Proposition 2.5, we have:

$$\frac{\mathrm{d}}{\mathrm{d}g}\left\{\mathscr{W}_H\big(g,\, N^\ell_H(g,g)\big)\right\} = \frac{\partial}{\partial g}\left\{\mathscr{W}_H\big(g,\, N^\ell_H(g,g)\big)\right\} \tag{313}$$

Hence, we have $N^\ell_H(g_{H,\infty}, g_{H,\infty}) = N^\ell_H(g_0, g_0) \triangleq N^s_{H,1}$ in particular, and we may write that:

$$\mathscr{W}_{H,\infty} - \mathscr{W}_{H,0} = \Big[\mathscr{W}_H\big(g_{H,\infty},\, N^\ell_H(g_{H,\infty}, g_{H,\infty})\big) - \mathscr{W}_H\big(g_0, N^\ell_H(g_0, g_0)\big)\Big]$$

$$
\begin{aligned}
&+ \int_{g_{H,\infty}}^{1} \frac{\partial \mathcal{W}_H}{\partial N_H^\ell}(g_{H,\infty},\, N_H^\ell(g_{H,\infty}, g_F))\, \frac{\partial N_H^\ell}{\partial g_F}(g_F)\, \mathrm{d}g_F \\
= &\Big[\mathcal{W}_H\big(g_{H,\infty}, N_{H,1}^s\big) - \mathcal{W}_H(g_0, N_{H,1}^s)\Big] \\
&+ \int_{g_{H,\infty}}^{1} \frac{\partial \mathcal{W}_H}{\partial N_H^\ell}(g_{H,\infty},\, N_H^\ell(g_{H,\infty}, g_F))\, \frac{\partial N_H^\ell}{\partial g_F}(g_F)\, \mathrm{d}g_F
\end{aligned} \tag{314}
$$

Suppose first that $g_{H,\infty} < 1 = g_{F,\infty}$. The feasibility constraint (C-235), together with Lemma 5, give:

$$
g_{\text{peak}}(N_{H,1}^s) \le g_{H,1} \le g_{H,\infty} \implies \mathcal{W}_H\big(g_{H,\infty},\, N_{H,1}^s\big) - \mathcal{W}_H(g_0, N_{H,1}^s) \le \mathcal{W}_H\big(g_{H,1},\, N_{H,1}^s\big) - \mathcal{W}_H(g_0, N_{H,1}^s) \le 0 \tag{315}
$$

Hence, using (314), we conclude that:

$$
\mathcal{W}_{H,\infty} - \mathcal{W}_{H,0} \le \int_{g_{H,\infty}}^{1} \underbrace{\frac{\partial \mathcal{W}_H}{\partial N_H^\ell}(g_{H,\infty},\, N_H^\ell(g_{H,\infty}, g_F))}_{>0 \text{ by Lemma 3}} \quad \underbrace{\frac{\partial N_H^\ell}{\partial g_F}(g_F)}_{<0 \text{ by Proposition 2.4}} \quad \mathrm{d}g_F < 0
$$

where the strict inequality follows by integration over a set of strictly positive measure. If in contrast, $g_{H,\infty} = 1$, then evaluating (314) accordingly yields:

$$
\mathcal{W}_{H,\infty} - \mathcal{W}_{H,0} = \Big[\mathcal{W}_H\big(1,\, N_{H,1}^s\big) - \mathcal{W}_H(g_0, N_{H,1}^s)\Big] < 0, \quad \text{by Assumption 12.} \tag{316}
$$

<u>$\mathcal{W}_{F,\infty} \gtreqless \mathcal{W}_{F,0} \iff g_{H,\infty} \lesseqgtr g_H^*(g_0)$</u> By a similar argument, we have $N_F^\ell(1,1) = N_F^\ell(g_0, g_0) \triangleq N_{F,1}^s$ and thus:

$$
\mathcal{W}_{F,\infty} - \mathcal{W}_{F,0} = \underbrace{\big[\mathcal{W}_F(1, N_{F,1}^s) - \mathcal{W}_F(g_0, N_{F,1}^s)\big]}_{<0 \text{ by extending Assumption 12 to } \mathcal{W}_F} - \int_{g_{H,\infty}}^{1} \frac{\partial \mathcal{W}_F}{\partial N_F^\ell}(1, N_F^\ell(1, g_H))\, \frac{\partial N_F^\ell}{\partial g_H}(g_H)\, \mathrm{d}g_H \tag{317}
$$

Letting $g_{H,\infty} \to 1$, it is clear that $\mathcal{W}_{F,\infty} - \mathcal{W}_{F,0}\big|_{g_{H,\infty}=1} < 0$, as the right integral term vanishes. Meanwhile, for $g_{H,\infty} = g_0$, one may invoke the fundamental theorem of calculus to write that:

$$
\begin{aligned}
\mathcal{W}_{F,\infty} - \mathcal{W}_{F,0}\big|_{g_{H,\infty}=g_0} = \mathcal{W}_F(1, N_F^\ell(1, g_0)) - \mathcal{W}_F(g_0, N_F^\ell(g_0, g_0)) &= \int_{g_0}^{1} \underbrace{\frac{\mathrm{d}\mathcal{W}_F}{\mathrm{d}g_F}\big(u,\, N_F^\ell(u, g_0)\big)}_{>0 \text{ under Assumption 10}} \mathrm{d}u \\
&> 0
\end{aligned} \tag{318}
$$

where we use that the steady-state welfare function coincides with long-run welfare in the static model, which is increasing in own restriction level on outbound data flows by Proposition 3.14. Finally, observe that:

$$
\frac{\partial}{\partial g_{H,\infty}}\big(\mathcal{W}_{F,\infty} - \mathcal{W}_{F,0}\big) = \underbrace{\frac{\partial \mathcal{W}_F}{\partial N_F^\ell}(1,\, N_F^\ell(1, g_{H,\infty}))}_{>0} \; \underbrace{\frac{\partial N_F^\ell}{\partial g_H}(g_{H,\infty})}_{<0} < 0 \tag{319}
$$

Because $\mathcal{W}_{F,\infty} - \mathcal{W}_{F,0}\big|_{g_{H,\infty}} > 0$ and $\mathcal{W}_{F,\infty} - \mathcal{W}_{F,0}\big|_{g_{H,\infty}=1} < 0$, while the map $g_{H,\infty} \mapsto (\mathcal{W}_{F,\infty} - \mathcal{W}_{F,0})(g_{H,\infty})$ is strictly monotonic, there exists a unique $g_H^*(g_0) \in (g_0, 1)$ such that $\mathcal{W}_{F,\infty} - \mathcal{W}_{F,0} = 0$, establishing the claim.

□